

Dynamics of Structures

Steen Krenk

Department of Mechanical Engineering
Technical University of Denmark
2018

Professor Steen Krenk
Department of Mechanical Engineering
Technical University of Denmark
Kgs. Lyngby
Denmark
E-mail: sk@mek.dtu.dk

Copyright © Steen Krenk, 2013, 2018

Contents

1	Basic Dynamic Properties	1
1.1	Free Vibrations	1
1.1.1	Undamped vibrations	1
1.1.2	Damped vibrations	2
1.1.3	Logarithmic decrement	4
1.1.4	State vector and phase-plane	6
1.2	Harmonic Load and Response	8
1.2.1	Harmonic Load on Undamped Structures	8
1.2.2	Harmonic Load on Damped Structures	11
1.2.3	Base Excitation	15
1.3	Transient Response	18
1.3.1	Load Impulse	18
1.3.2	Response to impulse load	19
1.3.3	Response to Transient Load	21
1.3.4	Relation between time and frequency analysis	25
1.4	Damping Mechanisms	25
1.4.1	Energy loss factor	26
1.4.2	Equivalent viscous damping	27
1.4.3	Relaxation damping	34
1.4.4	Typical damping values	34
1.5	Exercises	35
2	Continuous Systems	43
2.1	Dynamics of a taut cable	43
2.1.1	The wave equation	43
2.1.2	Vibrations of a taut cable	45
2.1.3	Dynamics of an elastic bar	49
2.2	Dynamics of Beams	52
2.2.1	Equation of transverse motion	52
2.2.2	Bending waves of beams	54
2.2.3	Bending vibrations of beams	55
2.2.4	Damping of transverse beam vibrations	60
2.3	Exercises	61

3	Finite Element Method	63
3.1	Finite element formulation for cables and bars	63
3.1.1	Virtual work formulation	64
3.1.2	Discretized equations of motion	66
3.1.3	Estimating the highest natural frequency	71
3.2	Finite element formulation for beams	72
3.2.1	Virtual work formulation	72
3.2.2	The beam bending element	75
3.3	Exercises	76
4	Multi-Degree-of-Freedom Systems	79
4.1	Equations of motion	80
4.1.1	Energy balance	82
4.2	Free vibrations	83
4.2.1	Vibration modes and frequencies	84
4.2.2	Modal representation	93
4.3	Modal response analysis	97
4.3.1	Response analysis	97
4.3.2	Frequency response	99
4.4	Sensitivity analysis and modal damping	99
4.4.1	Sensitivity analysis	100
4.4.2	Rayleigh damping	102
4.5	System reduction	105
4.5.1	Master-slave constraints	105
4.5.2	Quasi-static condensation	107
4.5.3	Reduced set of dynamic modes	109
4.6	Exercises	112
5	External Damping of Structures	117
5.1	Damping of Cables	118
5.1.1	Taut cable with a local damper	119
5.1.2	Effect of damper stiffness and mass	126
5.1.3	Non-linear damper properties	126
5.1.4	Cable with sag	127
5.2	Local Dampers on Structures	131
5.2.1	Introduction of local damper	131
5.2.2	Two-component subspace approximation	133
5.2.3	Placement and sizing of multiple dampers	136
5.3	Resonant vibration absorbers	140
5.3.1	Basic theory of the tuned mass absorber	141
5.3.2	Harmonic response	142
5.3.3	Frequency analysis	143
5.4	Tuned mass absorbers on flexible structures	149
5.4.1	Modal representation of the structure	149

5.4.2	Modal representation of tuned mass absorbers	150
5.4.3	Design for resonant loads	151
5.5	Exercises	154
6	Time Integration by Newmark methods	155
6.1	Newmark integration of linear structures	156
6.1.1	The basis of the Newmark procedure.....	157
6.1.2	Numerical implementation	158
6.1.3	Numerical stability	160
6.1.4	Numerical accuracy	164
6.2	Generalized α -procedure	168
6.2.1	Generalized α -procedure for linear problems	169
6.2.2	Non-linear generalized α -procedure	175
6.3	The Newmark method revisited	178
6.3.1	Energy relations of the Newmark method.....	178
6.3.2	Stability of the Newmark method	179
6.3.3	Effect of modified stiffness	181
6.4	Exercises	184
7	Conservative Time Integration.....	187
7.1	Momentum Conserving Algorithm	187
7.1.1	Discrete state-space format	187
7.1.2	Energy balance equation	188
7.1.3	Damping in state space algorithms	190
7.1.4	Linear algorithm with energy dissipation	191
7.2	Nonlinear Dynamics	192
7.2.1	Discrete state-space format	192
7.2.2	Non-linear algorithm with energy dissipation.....	202
7.3	Exercises	204
	References	207
	Index	211



1.1 Free Vibrations

1.1.1 Undamped vibrations

Equation of motion by balance of *inertial force* $m\ddot{x}$ and *elastic force* kx with *mass* m and stiffness k ,

$$m\ddot{x} + kx = 0 \quad (1.1)$$

Obtain *normalized equation* by division by m

$$\ddot{x} + \omega_0^2 x = 0 \quad (1.2)$$

where the *natural angular frequency* ω_0 is defined by

$$\omega_0 = \sqrt{\frac{k}{m}} \quad (1.3)$$

Note that the (angular) frequency is determined by

$$\omega_0^2 = \frac{\text{'stiffness'}}{\text{'mass'}}$$

In many problems 'stiffness' and 'mass' can be estimated by use of energy methods.

Solution Solution requires *initial conditions* in terms of initial position x_0 and initial velocity $v_0 = \dot{x}_0$.

$$x(t) = x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t) \quad (1.4)$$

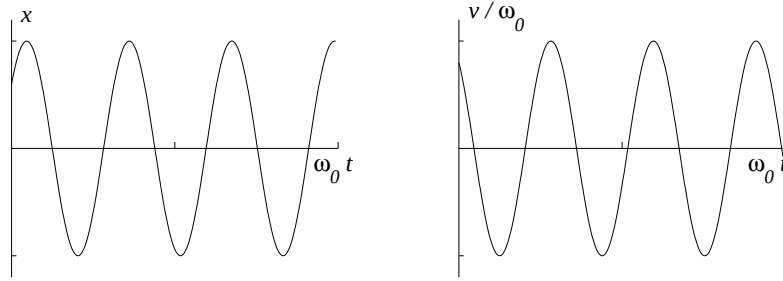


Fig. 1.1: a) displacement $x(t)$, and b) velocity $\dot{x}(t)$ as function of time t .

The *period* T requires the phase $\omega_0 t$ to increase by 2π , Note that the (angular) frequency is determined by

$$T = \frac{2\pi}{\omega_0} \quad (1.5)$$

The *natural frequency* f is the number of oscillations per time unit, whereby

$$f = \frac{1}{T} = \frac{\omega_0}{2\pi} \quad (1.6)$$

1.1.2 Damped vibrations

Include *energy dissipation* through damping force $c\dot{x}$,

$$m \ddot{x}(t) + c \dot{x}(t) + k x(t) = 0 \quad (1.7)$$

Note, that all three parameters m , c and k are positive, when the forces are restoring.

Normalize equation by division by m

$$\ddot{x} + 2\zeta\omega_0 \dot{x} + \omega_0^2 x = 0 \quad (1.8)$$

using the *natural angular frequency* ω_0 ,

$$\omega_0 = \sqrt{\frac{k}{m}} \quad (1.9)$$

and the *damping ratio* ζ ,

$$\zeta = \frac{c}{2\sqrt{km}} \quad (1.10)$$

The characteristics of the free vibration behavior depends on the magnitude of the damping ratio ζ .

Solutions The free vibration solution is of exponential type and is found by substitution of the partial solution $x(t) = Ae^{\lambda t}$ into the homogeneous equation of motion, yielding the characteristic equation

$$\lambda^2 + 2\zeta\omega_0\lambda + \omega_0^2 = 0 \quad (1.11)$$

There are three different cases of damped vibrations, depending on the magnitude of the damping ratio ζ . The solutions are given below for initial conditions $(x, \dot{x})_{t=0} = (x_0, v_0)$.

UNDERDAMPED, $0 < \zeta < 1$:

Two complex roots of characteristic equation, $\lambda = \omega_0(-\zeta \pm i\sqrt{1-\zeta^2})$.

Imaginary part is expressed by defining the *damped natural frequency*

$$\omega_d = \omega_0\sqrt{1-\zeta^2} \quad (1.12)$$

This leads to complex partial solutions of the form

$$x(t) = \underbrace{A e^{-\zeta\omega_0 t}}_{\text{amplitude}} \underbrace{e^{\pm i\omega_d t}}_{\text{oscillation}} \quad (1.13)$$

The last factor can be expressed in terms of $\sin(\omega_d t)$ and $\cos(\omega_d t)$.

$$x(t) = x_0 e^{-\zeta \omega_0 t} \left[\cos(\omega_d t) + \frac{\zeta \omega_0}{\omega_d} \sin(\omega_d t) \right] + \frac{v_0}{\omega_d} e^{-\zeta \omega_0 t} \sin(\omega_d t) \quad (1.14)$$

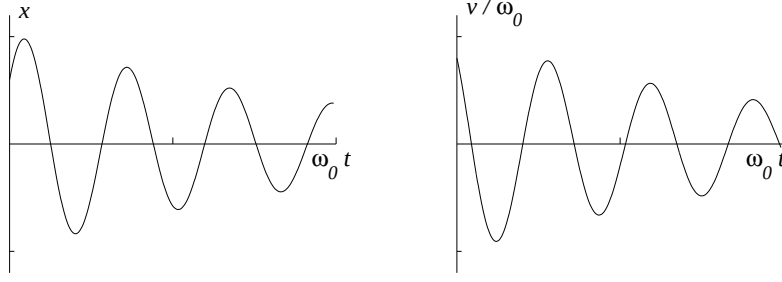


Fig. 1.2: a) displacement $x(t)$, and b) velocity $\dot{x}(t)$ for $\zeta = 0.05$.

The period of the oscillatory factor in damped free oscillations is $T_d = 2\pi/\omega_d$.

CRITICALLY DAMPED, $\zeta = 1$:

The characteristic equation has the real double root, $\lambda = -\omega_0$, and in this case the amplitude A is replaced by a linear function $A + Bt$.

$$x(t) = [x_0 + (\omega_0 x_0 + v_0)t] e^{-\omega_0 t} \quad (1.15)$$

OVERDAMPED, $1 < \zeta$:

The characteristic equation has two real roots $\lambda = \omega_0(-\zeta \pm \sqrt{\zeta^2 - 1})$. Introducing the parameter $\omega_d = \omega_0 \sqrt{\zeta^2 - 1} < \zeta \omega_0$ the solution is expressed in terms of hyperbolic functions as

$$x(t) = x_0 e^{-\zeta \omega_0 t} \left[\cosh(\omega_d t) + \frac{\zeta \omega_0}{\omega_d} \sinh(\omega_d t) \right] + \frac{v_0}{\omega_d} e^{-\zeta \omega_0 t} \sinh(\omega_d t) \quad (1.16)$$

1.1.3 Logarithmic decrement

Underdamping can also be characterized by the decrease in amplitude from one maximum to the next. The ratio between any two maximum values fol-

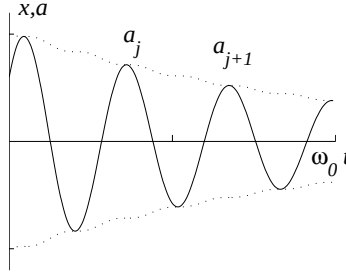


Fig. 1.3: Displacement record with maxima a_j and a_{j+1} for $\zeta = 0.05$.

lowing each other is constant, and the *logarithmic decrement* is defined as

$$\delta = \ln \left(\frac{a_j}{a_{j+1}} \right) = \ln \left(e^{\zeta \omega_0 T_d} \right) = \frac{2\pi \zeta}{\sqrt{1 - \zeta^2}} \quad (1.17)$$

or for lightly damped systems

$$\delta \simeq 2\pi \zeta \quad \text{for} \quad \zeta \ll 1 \quad (1.18)$$

NOTE: Lightly damped structures, like cables, may have ζ as low as 0.001.

Energy balance

Form the *rate of work* by multiplication of the equation of motion with $\dot{x}$,

$$\dot{x}(t) \left(m \ddot{x}(t) + c \dot{x}(t) + k x(t) \right) = 0 \quad (1.19)$$

Rewrite as time derivative,

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} k x^2 \right) = -c \dot{x}^2 \leq 0 \quad (1.20)$$

This defines the *mechanical energy* E as

$$E = E_{kin} + E_{el} \quad (1.21)$$

with *kinetic energy*

$$E_{kin} = \frac{1}{2} m \dot{x}^2 \quad (1.22)$$

and *elastic energy*

$$E_{el} = \frac{1}{2} k x^2 \quad (1.23)$$

The mechanical energy of undamped free vibrations is constant.

When the motion can be described (estimated) by a single degree of freedom, the equation of undamped motion can be determined from the time derivative of the mechanical energy.

1.1.4 State vector and phase-plane

Time histories of displacement $x(t)$ and velocity $v(t) = \dot{x}(t)$ against time t .

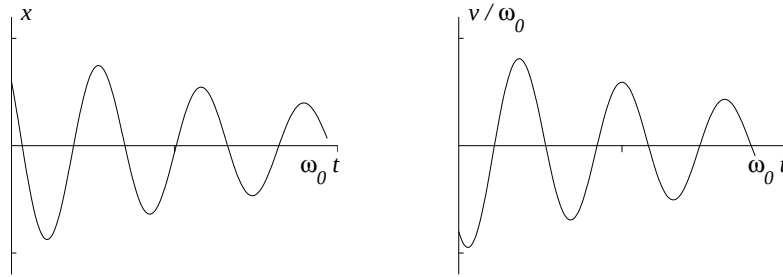


Fig. 1.4: a) displacement $x(t)$, and b) velocity $\dot{x}(t)$ as function of time t .

Combination into a single three-dimensional graph as shown in Fig. 1.5, with time t along the first axis.

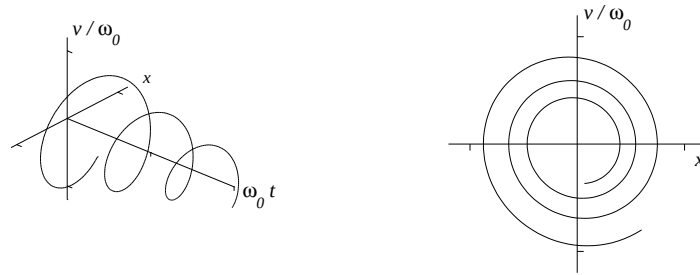


Fig. 1.5: a) response path and b) phase-plane diagram.

Projection of the state vector $(x(t), \dot{x}(t))$ on the phase-plane. Use of normalized coordinates $x, \dot{x}/\omega_0$ reduces to 'near-circle'.

Discrete time increment – Undamped system Undamped equation of free vibrations, $\zeta = 0$,

$$\ddot{x} + \omega_0^2 x = f(t)$$

Free oscillations described entirely in terms of initial conditions,

$$\begin{aligned} x &= x_0 \cos(\omega_0 t) + \dot{x}_0 \omega_0^{-1} \sin(\omega_0 t) \\ \dot{x} &= -x_0 \omega_0 \sin(\omega_0 t) + \dot{x}_0 \cos(\omega_0 t) \end{aligned}$$

State vector $(x, \dot{x})_1$ at time $t = \Delta t$ in terms of initial state vector $(x, \dot{x})_0$.

Recurrence relation for any pair of state vectors with time separation Δt . The time separation only appears in the form of the non-dimensional parameter

$$\alpha = \omega_0 \Delta t$$

Non-dimensional form, when $\dot{x}$ is replaced with $\Delta t \dot{x}$.

$$\begin{bmatrix} x \\ \Delta t \dot{x} \end{bmatrix}_{i+1} = \begin{bmatrix} \cos \alpha & \alpha^{-1} \sin \alpha \\ -\alpha \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ \Delta t \dot{x} \end{bmatrix}_i$$

Discrete time increment – Damped system Normalized equation for free vibrations

$$\ddot{x} + 2\zeta \omega_0 \dot{x} + \omega_0^2 x = 0$$

Damped natural angular frequency,

$$\omega_d = \omega_0 \sqrt{1 - \zeta^2}$$

Free damped oscillations given in terms of the initial conditions

$$\begin{aligned} x &= x_0 e^{-\zeta \omega_0 t} \left(\cos(\omega_d t) + \frac{\zeta \omega_0}{\omega_d} \sin(\omega_d t) \right) + \frac{\dot{x}_0}{\omega_d} e^{-\zeta \omega_0 t} \sin(\omega_d t) \\ \dot{x} &= -x_0 \frac{\omega_0^2}{\omega_d} e^{-\zeta \omega_0 t} \sin(\omega_d t) + \dot{x}_0 e^{-\zeta \omega_0 t} \left(\cos(\omega_d t) - \frac{\zeta \omega_0}{\omega_d} \sin(\omega_d t) \right) \end{aligned}$$

Two non-dimensional time scales, conveniently defined as

$$\alpha = \omega_d \Delta t \quad , \quad \beta = \zeta \omega_0 \Delta t$$

Recurrence relations for time increment Δt ,

$$\begin{bmatrix} x \\ \Delta t \dot{x} \end{bmatrix}_{i+1} = \frac{e^{-\beta}}{\alpha} \begin{bmatrix} \alpha \cos \alpha + \beta \sin \alpha & \sin \alpha \\ -(\alpha^2 + \beta^2) \sin \alpha & \alpha \cos \alpha - \beta \sin \alpha \end{bmatrix} \begin{bmatrix} x \\ \Delta t \dot{x} \end{bmatrix}_i$$

Time-stepping algorithm The exact solution $(x_n, \dot{x}_n)$ can be obtained at discrete times $0, \Delta t, 2\Delta t, \dots$ by starting at the initial conditions $(x_0, \dot{x}_0)$ and multiplying with the matrix $\mathbf{A}$, found above, in each step.

$$\begin{aligned} \mathbf{y}_0 &= (x_0, \Delta t \dot{x}_0)^T \\ \text{for } i &= 0 : n-1 \\ \mathbf{y}_{i+1} &= \mathbf{A} \mathbf{y}_i \\ (\mathbf{x}, \Delta t \dot{\mathbf{x}}) &= \mathbf{y}^T \end{aligned}$$

Table 1.1: Direct time-stepping algorithm.

NOTE: The simplicity of the algorithm is obtained because the natural frequency and damping ratio are known. For multi-degree-of-freedom systems, this procedure requires a modal analysis, described later.

1.2 Harmonic Load and Response

Harmonic excitation of structures serves as an individual load case, e.g. in connection with loads from rotating machinery or water waves, but also as a basic component in a Fourier representation of a general load.

1.2.1 Harmonic Load on Undamped Structures

Equation of motion by balance of the inertial force $m\ddot{x}$, the elastic force kx and the external harmonic force $f_0 \cos(\omega t)$,

$$m\ddot{x} + kx = f_0 \cos(\omega t) \quad (1.24)$$

Obtain normalized equation by division with m ,

$$\boxed{\ddot{x} + \omega_0^2 x = \omega_0^2 \tilde{f}_0 \cos(\omega t)} \quad (1.25)$$

with $\tilde{f}_0 = f_0/k$ denoting the force normalized with respect to the stiffness k . Hereby the normalized force $\tilde{f}_0$ corresponds to the quasi-static displacement that would be obtained in the limit of vanishing frequency.

Solution The full solution contains a *particular solution* with the excitation frequency ω ,

$$x_f(t) = \frac{\omega_0^2}{\omega_0^2 - \omega^2} \tilde{f}_0 \cos(\omega t) \quad (1.26)$$

When written in this way the last two factors give the time variation of an equivalent quasi-static response, while the first factor defines the *dynamic amplification*.

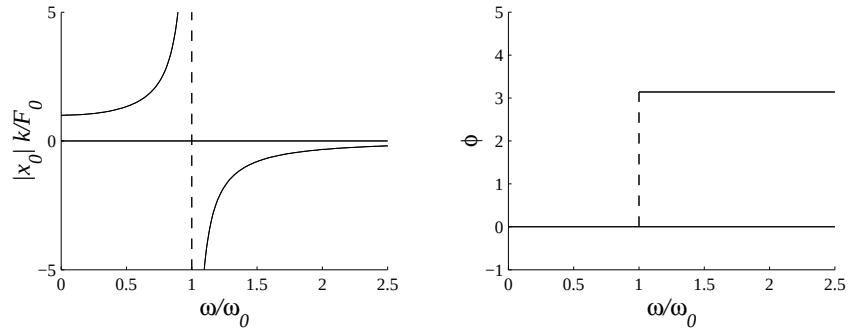


Fig. 1.6: a) dynamic amplification, and b) phase lag.

Note, that the response is in phase for excitation frequencies $\omega < \omega_0$ and has phase lag π for excitation frequencies $\omega > \omega_0$. The transition happens at the resonance frequency ω_0

The homogeneous part of the solution has the natural frequency ω_0 ,

$$x_h(t) = x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t) \quad (1.27)$$

This is to be combined with the non-homogeneous solution with excitation and response frequency ω

$$x_f(t) = \frac{\omega_0^2}{\omega_0^2 - \omega^2} \tilde{f}_0 \cos(\omega t) \quad (1.28)$$

The full solution with initial conditions x_0, v_0 is

$$\begin{aligned}
 x(t) = & \frac{\omega_0^2 \tilde{f}_0}{\omega_0^2 - \omega^2} \cos(\omega t) + \\
 & \left(x_0 - \frac{\omega_0^2 \tilde{f}_0}{\omega_0^2 - \omega^2} \right) \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t)
 \end{aligned}
 \tag{1.29}$$

Note, that the first term has the excitation frequency ω , while the two last terms have the natural frequency ω_0 .

‘Beats’ for excitation near the natural frequency

If the excitation frequency ω is close to the natural frequency ω_0 of the oscillator, the response has the character of oscillations close to the natural frequency but with amplitude modulated by the frequency difference $\omega - \omega_0$. Homogeneous initial conditions $x_0 = 0$ and $v_0 = 0$ lead to

$$\begin{aligned}
 x(t) &= \frac{\omega_0^2 \tilde{f}_0}{\omega_0^2 - \omega^2} \left[\cos(\omega t) - \cos(\omega_0 t) \right] \\
 &= \frac{2\omega_0^2 \tilde{f}_0}{\omega_0^2 - \omega^2} \underbrace{\sin\left(\frac{\omega_0 - \omega}{2} t\right)}_{\text{‘slow’}} \underbrace{\sin\left(\frac{\omega_0 + \omega}{2} t\right)}_{\text{‘fast’}}
 \end{aligned}
 \tag{1.30}$$

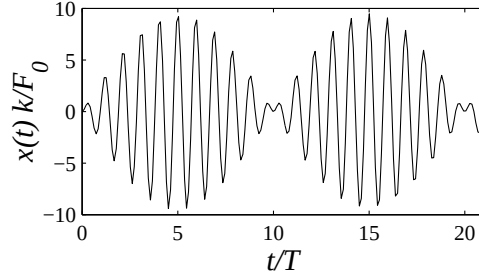
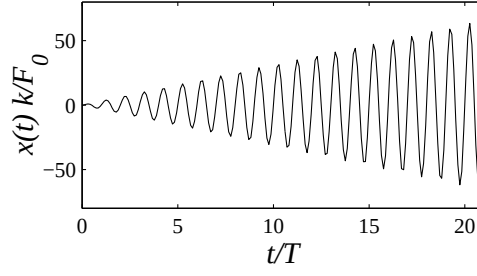


Fig. 1.7: Beats for $\omega = 1.1\omega_0$.

In the limit $\omega \rightarrow \omega_0$ resonance generates infinite linear growth,

$$x(t) = \frac{\omega_0^2 \tilde{f}_0}{\omega_0 + \omega} \frac{\sin\left(\frac{\omega_0 - \omega}{2} t\right)}{\frac{\omega_0 - \omega}{2}} \sin\left(\frac{\omega_0 + \omega}{2} t\right) \rightarrow 2\tilde{f}_0 (\omega_0 t) \sin(\omega_0 t)
 \tag{1.31}$$

Fig. 1.8: Linear growth of resonance for $\omega = \omega_0$.

1.2.2 Harmonic Load on Damped Structures

Equation of motion by balance of *inertial force* $m\ddot{x}$, *damping force* $c\dot{x}$, *elastic force* kx , and *external harmonic force* $f_0 \cos(\omega t)$,

$$m\ddot{x} + c\dot{x} + kx = f_0 \cos(\omega t) \quad (1.32)$$

Obtain *normalized equation* by division by m

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \omega_0^2 \tilde{f}_0 \cos(\omega t) \quad (1.33)$$

with normalized force $\tilde{f}_0 = f_0/k$. It that the normalized force $\tilde{f}_0$ corresponds to an equivalent quasi-static displacement that would occur in the limit of a very low angular frequency ω of the load.

Solution Particular solution with frequency ω by introducing the complex representation

$$\tilde{f} = \tilde{f}_0 \cos(\omega t) = \text{Re}[\tilde{f}_0 e^{i\omega t}] \quad (1.34)$$

and

$$\begin{aligned} x_f(t) &= \text{Re}[X_0 e^{i\omega t}] = \text{Re}[A e^{-i\varphi} e^{i\omega t}] \\ &= \text{Re}[A e^{i(\omega t - \varphi)}] = A \cos(\omega t - \varphi) \end{aligned} \quad (1.35)$$

The solution is expressed in terms of the response amplitude A and the phase lag φ , representing the lag of the phase of the response behind the phase of the force.

Substitute complex representation into equation of motion,

$$\left(-\omega^2 + 2i\zeta\omega_0\omega + \omega_0^2 \right) X_0 e^{i\omega t} = \omega_0^2 \tilde{f}_0 e^{i\omega t} \quad (1.36)$$

where the complex amplitude $X_0 = A_0 e^{-i\varphi}$ contains the real amplitude A_0 and the phase lag φ .

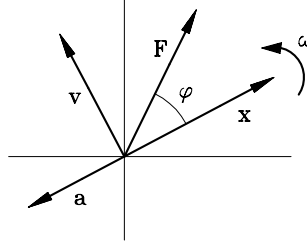


Fig. 1.9: Rotating 'force' diagram.

Solution for the complex amplitude X_0 gives

$$X_0 = H(\omega) \frac{f_0}{k} = \frac{\omega_0^2}{\omega_0^2 - \omega^2 + 2i\zeta\omega_0\omega} \frac{f_0}{k} \quad (1.37)$$

This defines the *frequency response function*

$$H(\omega) = \frac{\omega_0^2}{\omega_0^2 - \omega^2 + 2i\zeta\omega_0\omega} \quad (1.38)$$

representing the response to a unit (normalized) force.

The response amplitude for a unit (normalized) force is

$$A = |H(\omega)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2}} \quad (1.39)$$

The phase angle $-\varphi$ is the complex argument of $H(\omega)$,

$$\varphi = -\text{Arg}[H(\omega)] \Rightarrow \tan \varphi = \frac{2\zeta\omega_0\omega}{\omega_0^2 - \omega^2} \quad (1.40)$$

Pronounced dynamic *resonance* and rapid change of phase occurs around the natural frequency ω_0 for small damping.

Dynamic amplification at $\omega = \omega_0$,

$$\frac{|X_0|_{\omega_0}}{f_0/k} = \frac{1}{2\zeta} \quad (1.41)$$

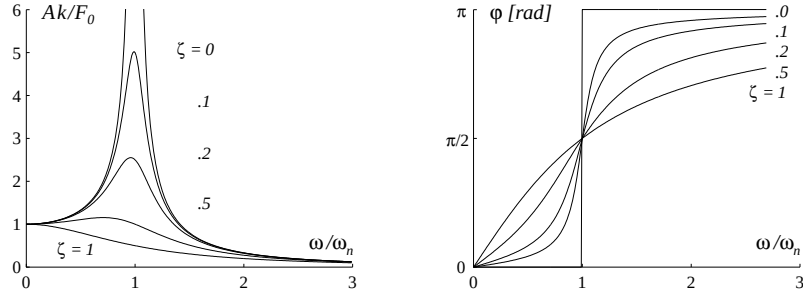


Fig. 1.10: a) Dynamic amplification and b) phase angle for damping ratio $\zeta = 0, 0.1, 0.2, 0.5, 1$.

Dynamic amplification

The normalized *dynamic amplification* is

$$\frac{|X_0|}{f_0/k} = |H(\omega)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2}} \quad (1.42)$$

Introduce the *frequency ratio* $r = \omega/\omega_0$ to get

$$\frac{|X_0|}{f_0/k} = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} \quad (1.43)$$

The maximum occurs for

$$\frac{d}{dr} \left| \frac{X_0}{f_0/k} \right| = \frac{d}{dr} [(1 - r^2)^2 + (2\zeta r)^2]^{-1/2} = 0 \quad (1.44)$$

Calculation of the derivative gives the equation

$$-2(2r)(1 - r^2) + 2(2\zeta)(2\zeta r) = 0 \quad (1.45)$$

with the solution

$$r^2 = 1 - 2\zeta^2 \quad (1.46)$$

Thus, maximum dynamic amplification for $\zeta \leq 1/\sqrt{2}$ is

$$\left| \frac{X_0}{f_0/k} \right|_{\max} = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \quad \text{at} \quad \omega_{\max} = \omega_0\sqrt{1 - 2\zeta^2} \quad (1.47)$$

NOTE: The location of the maximum approaches $\omega = 0$ with damping increasing to $\zeta = 1/\sqrt{2}$.

High and low frequency response

High and low frequency response can be *identified directly* from the equation of motion,

$$\underbrace{m \ddot{x}}_{\text{'high'}} + \underbrace{c \dot{x}}_{\text{'resonance'}} + \underbrace{k x}_{\text{'low'}} = f_0 \cos(\omega t) \quad (1.48)$$

At resonance the inertia and stiffness forces balance each other.

Response in the three regimes

$$|x(t)| \simeq \begin{cases} f_0/k & \text{for } \omega \ll \omega_0 \\ f_0/(\omega c) & \text{for } \omega \simeq \omega_0 \\ f_0/(\omega^2 m) & \text{for } \omega \gg \omega_0 \end{cases} \quad (1.49)$$

At resonance, $\omega = \omega_0$, the elastic and inertial forces cancel, leaving the damping force to balance the external load.

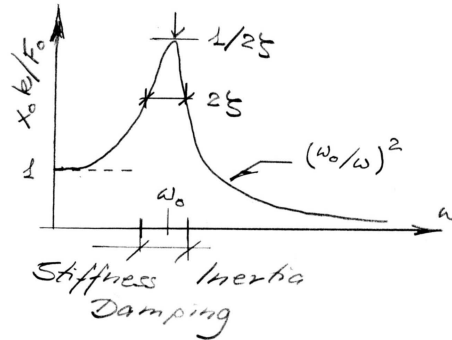


Fig. 1.11: The 'stiffness', 'damping', and 'inertia' regimes of dynamic amplification.

Full solution

The full solution consists of particular solution with frequency ω ,

$$\begin{aligned} x_f(t) &= A \tilde{f}_0 \cos(\omega t - \varphi) \\ &= A \cos(\varphi) \tilde{f}_0 \cos(\omega t) + A \sin(\varphi) \tilde{f}_0 \sin(\omega t) \end{aligned} \quad (1.50)$$

and an attenuating homogeneous solution with damped frequency ω_d , determined by the initial conditions x_0, v_0 .

When combining these two parts the full solution is

$$\begin{aligned}
 x(t) = & A \cos(\varphi) \tilde{f}_0 \cos(\omega t) + A \sin(\varphi) \tilde{f}_0 \sin(\omega t) \\
 & + \left(x_0 - A \cos(\varphi) \tilde{f}_0 \right) e^{-\zeta \omega_0 t} \left[\cos(\omega_d t) + \frac{\zeta \omega_n}{\omega_d} \sin(\omega_d t) \right] \\
 & + \left(\frac{v_0}{\omega_d} - \frac{\omega}{\omega_d} A \sin(\varphi) \tilde{f}_0 \right) e^{-\zeta \omega_0 t} \sin(\omega_d t)
 \end{aligned} \tag{1.51}$$

1.2.3 Base Excitation

Let the support (base) move as described by $y(t)$. The force and damping depends on the *relative motion* of the mass and the base $x_r = x - y$, while the inertial force depends on the absolute motion x .

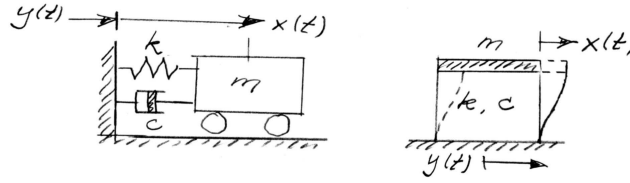


Fig. 1.12: Base motion $y(t)$ and relative motion $x(t)$.

This gives the equation of motion

$$m \ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0 \tag{1.52}$$

Excitation of a ‘structure’ by base motion is important e.g. for earthquake loads and for measuring instruments like the accelerometer and seismograph.

The solution for harmonic base excitation is obtained by substituting the complex representation

$$x(t) = \text{Re}[X e^{i\omega t}] \quad , \quad y(t) = \text{Re}[Y e^{i\omega t}] \tag{1.53}$$

into the equation of motion,

$$(-\omega^2 + 2i\zeta\omega_0\omega + \omega_0^2) X e^{i\omega t} = (2i\zeta\omega_0\omega + \omega_0^2) Y e^{i\omega t} \tag{1.54}$$

This gives the frequency response function,

$$\frac{X}{Y} = \frac{\omega_0^2 + 2i\zeta\omega_0\omega}{\omega_0^2 - \omega^2 + 2i\zeta\omega_0\omega} \quad (1.55)$$

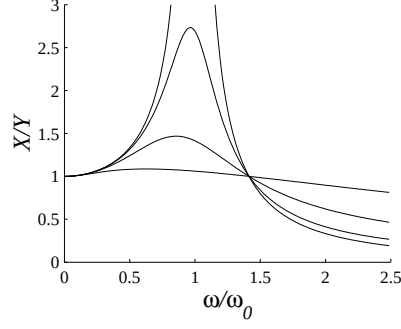


Fig. 1.13: Frequency response X/Y for harmonic base excitation with $\zeta = 0, 0.2, 0.5, 1/\sqrt{2}$.

Accelerometer and seismograph

Measurement devices for the base motion y can be based on the relative motion $x_r = x - y$.

$$m\ddot{x}_r + c\dot{x}_r + kx_r = -m\ddot{y} \quad (1.56)$$

In the low frequency domain, $\omega \ll \omega_0$

$$x_r \simeq -\frac{m}{k}\ddot{y} = -\frac{\ddot{y}}{\omega_0^2} \quad (1.57)$$

This is the principle of an accelerometer, where ω_0^2 is an instrument parameter.

In the high frequency domain, $\omega \gg \omega_0$

$$\ddot{x}_r = -\ddot{y} \quad (1.58)$$

This is the principle of a seismograph, where ω_0 is usually low due to a large suspended mass.

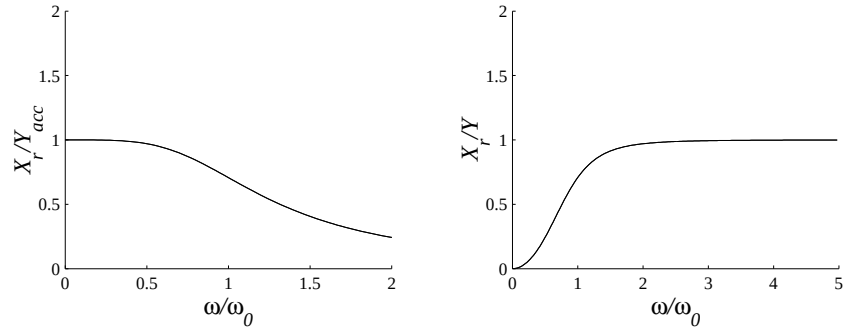
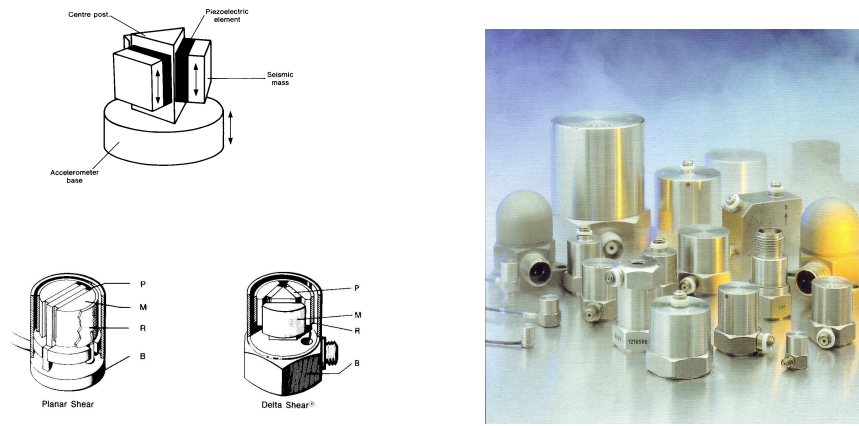


Fig. 1.14: a) Accelerometer and b) seismograph for damping ratio $\zeta = 1/\sqrt{2}$.

Piezoelectric accelerometers



Piezoelectric transducers. A small mass is supported via a piezoelectric layer that is deformed when the mass is accelerated.

Example 1.1. Rotating unbalanced mass. An unbalanced mass m_0 with eccentricity e rotates with angular frequency ω . The *total* mass is m .

Equation of motion - total mass times acceleration -

$$(m - m_0) \ddot{x} + m_0 \frac{d^2}{dt^2} [x + e \sin(\omega t)] = -kx - c\dot{x} \quad (1.59)$$

corresponding to

$$m \ddot{x} + c\dot{x} + kx = e m_0 \omega^2 \sin(\omega t) \quad (1.60)$$

This is normalized to

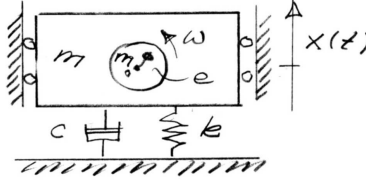


Fig. 1.15: Unbalanced mass m_0 with eccentricity e rotating with angular frequency ω .

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x = e\frac{m_0}{m}\omega^2\sin(\omega t) \quad (1.61)$$

This equation is equivalent to the equation of the relative motion x_r for an equivalent base motion $y_{eq} = -e(m_0/m)\sin(\omega t)$. $\square$

1.3 Transient Response

The response to a load that is not periodic is often called transient. In the case of linear structures the total response may be obtained by *superposition* of effects, either a sequence of *load pulses* or as a summation of *simultaneous harmonic load components*.

1.3.1 Load Impulse

Newton's second law defines force $\mathbf{F}(t)$ on a 'particle' of mass m as the rate of change of momentum $m\mathbf{v}(t)$, where $\mathbf{v}(t)$ is the velocity of the particle,

$$\underbrace{\frac{d}{dt}(m\mathbf{v})}_{\text{momentum}} = \underbrace{\mathbf{F}(t)}_{\text{force}} \quad (1.62)$$

When this relation is integrated between the times t_1 and t_2 ,

$$\underbrace{m\mathbf{v}(t_2) - m\mathbf{v}(t_1)}_{m(v_2 - v_1)} = \int_{t_1}^{t_2} \frac{d}{dt}(m\mathbf{v}) dt = \underbrace{\int_{t_1}^{t_2} \mathbf{F}(t) dt}_{\text{impulse } \hat{F}} \quad (1.63)$$

For a total force $F(t)$ this gives the velocity increment

$$\boxed{\mathbf{v}_2 - \mathbf{v}_1 = \frac{1}{m} \int_{t_1}^{t_2} \mathbf{F}(t) dt} \quad (1.64)$$

where the time integral of the force is called the impulse $\hat{F}$.

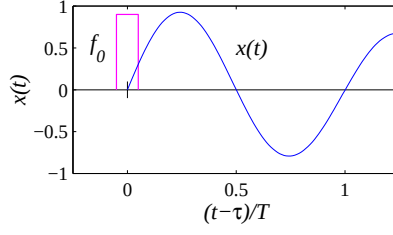


Fig. 1.16: The load impulse at τ and the free response $x(t)$.

1.3.2 Response to impulse load

Equation of motion

$$m \ddot{x}(t) + c \dot{x}(t) + k x(t) = f(t) \quad (1.65)$$

with pulse type load within a ‘small’ time interval $[t_1, t_2]$.

The total force now comprises the external force $f(t)$, the damping force $-c\dot{x}(t)$ and the elastic force $-kx(t)$.

Integrate over the time interval t_1 to t_2 ,

$$m (\dot{x}_2 - \dot{x}_1) + c (x_2 - x_1) + \int_{t_1}^{t_2} k x(t) dt = \int_{t_1}^{t_2} f(t) dt = \hat{f} \quad (1.66)$$

The first term is the velocity difference, known from the free particle. The magnitude of the damping and elastic terms can be estimated.

The mass is at rest prior to t_1 , and the interval of loading is $\Delta t = t_2 - t_1$.

A Taylor expansion of the displacement $x(t)$, the velocity $\dot{x}(t)$ and the acceleration $\ddot{x}(t)$ then leads to the estimate

$$\ddot{x} \simeq a, \quad \dot{x} \simeq a \Delta t, \quad x \simeq \frac{1}{2} a \Delta t^2, \quad t_1 < t < t_2 \quad (1.67)$$

where a is a characteristic magnitude of the acceleration.

For sufficiently small duration, $\Delta t \ll T$, the damping and elastic terms are small (no velocity and displacement have built up yet), and the velocity increment during the load is

$$\Delta v \simeq \frac{\hat{f} \Delta t}{m} \quad (1.68)$$

A load pulse $\hat{f} = f\Delta t$ centered around time τ creates the apparent initial displacement $x_0 \simeq 0$ and velocity $v_0^+ = \hat{f}/m$. By the analysis of free vibrations with these initial conditions this can be expressed as

$$x(t) = \frac{v_0^+}{\omega_d} e^{-\zeta\omega_0(t-\tau)} \sin \omega_d(t-\tau) \quad , \quad t \geq \tau \quad (1.69)$$

The solution is conveniently expressed in terms of the impulse $\hat{f}$ and the *impulse response function* $h(t-\tau)$,

$$\begin{aligned} x(t) &= \frac{\hat{f}}{m} h(t-\tau) \\ h(t) &= \frac{1}{\omega_d} e^{-\zeta\omega_0 t} \sin(\omega_d t) \quad , \quad t \geq 0 \end{aligned} \quad (1.70)$$

The delta function

For 'short' load duration the response only depends on the integral

$$\hat{f}_{\Delta t} = \int_{\Delta t} f(t) dt \quad (1.71)$$

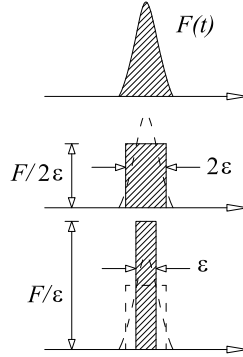


Fig. 1.17: Sequence of increasingly 'narrow' pulses.

$$\begin{aligned} &\text{The delta function } \delta(t-\tau) \text{ is a 'spike' located} \\ &\text{at } \tau \text{ with normalized integral } \int_{-\infty}^{\infty} \delta(t-\tau) dt = 1. \end{aligned} \quad (1.72)$$

An impulse load $F(t)$ at τ is expressed in terms of the delta function as

$$f(t) = \hat{f} \delta(t - \tau) \quad (1.73)$$

1.3.3 Response to Transient Load

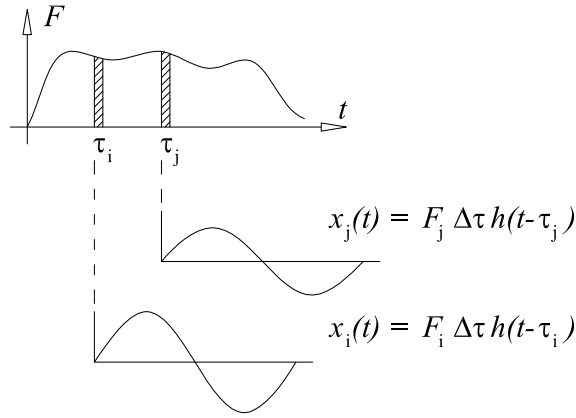


Fig. 1.18: Response from pulses at τ_i and τ_j .

The total response at time t is the sum of the responses to (normalized) pulse loads $f(\tau_i)\Delta\tau_i/m$ at all times τ_i prior to the current time t ,

$$x(t) = \frac{1}{m} \sum_i f(\tau_i) \Delta\tau_i h(t - \tau_i) \quad (1.74)$$

The summation can be converted into an integral by considering the limit $\Delta\tau \rightarrow 0$, whereby

$$x(t) = \frac{1}{m} \int_0^t f(\tau) h(t - \tau) d\tau = \frac{1}{m} \int_0^t f(t - s) h(s) ds \quad (1.75)$$

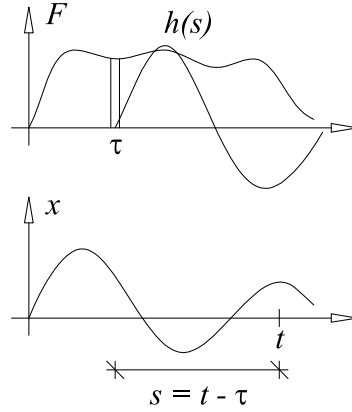


Fig. 1.19: Relation between 'pulse time' τ and 'delay' s .

Compact representation of free vibration solution

The *free vibration solution* can be written in terms of the initial conditions x_0, v_0 and the initial displacement and impulse response functions as

$$\begin{aligned} x(t) &= x_0 g(t) + v_0 h(t) \quad , \quad t \geq 0 \\ g(t) &= e^{-\zeta\omega_0 t} \left[\cos(\omega_d t) + \frac{\zeta\omega_0}{\omega_d} \sin(\omega_d t) \right] \\ h(t) &= \frac{1}{\omega_d} e^{-\zeta\omega_0 t} \sin(\omega_d t) \end{aligned} \quad (1.76)$$

The function $g(t)$ describes the free response from a unit initial displacement with zero velocity, while the function $h(t)$ describes the free response from a unit initial velocity with zero initial displacement. It is easily verified by differentiation that the two free response functions $g(t)$ and $h(t)$ satisfy the relation

$$\frac{d}{dt}g(t) = -\omega_0^2 h(t) \quad (1.77)$$

Example 1.2. Constant load from time $t = 0$. Normalized equation of motion with constant load $\tilde{f}_p = f_p/k$ from $t = 0$,

$$\ddot{x} + 2\zeta\omega_0 \dot{x} + \omega_0^2 x = \omega_0^2 \tilde{f}_p \quad , \quad t \geq 0 \quad (1.78)$$

Full solution as sum of particular and homogeneous solution,

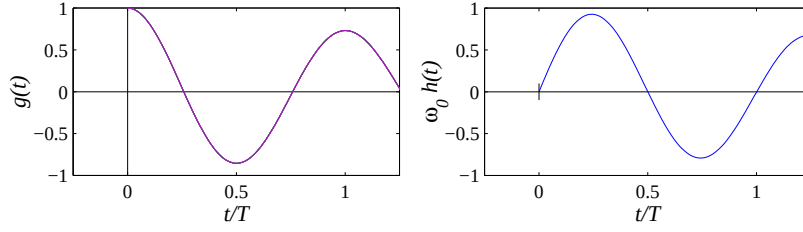


Fig. 1.20: Initial displacement response function $g(t)$ and (normalized) impulse response function $h(t)$.

$$x(t) = \tilde{f}_p + x_* g(t) + v_* h(t) \quad (1.79)$$

Find x_* and v_* from initial conditions $x(0) = 0$ and $v(0) = 0$.

Response to constant load from $t = 0$,

$$x(t) = [1 - g(t)] \tilde{f}_p \quad (1.80)$$

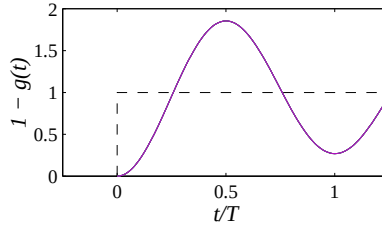


Fig. 1.21: Response to unit step load at $t = 0$.

Note, that the response to a unit step load follows directly from the free unit initial displacement response. For low damping there is nearly a 100 pct. 'overshoot' relative to the static displacement. $\square$

Example 1.3. Box load.

Consider a constant (normalized) load $\tilde{f}_p$ acting in the time interval $[t_1, t_2]$. This load can be considered as the sum a step load $+\tilde{f}_p$ at t_1 and a step load $-\tilde{f}_p$ at t_2 .

The responses $x_1(t)$ and $x_2(t)$ to the two step loads are

$$x_1(t) = [1 - g(t - t_1)] \tilde{f}_p \quad , \quad x_2(t) = -[1 - g(t - t_2)] \tilde{f}_p \quad (1.81)$$

The total response for $t > t_2$ follows from superposition and use of the relation between the functions $g(t)$ and $h(t)$,

$$x(t) = -[g(t - t_1) - g(t - t_2)] \tilde{f}_p \simeq -\frac{\Delta g(t - t_*)}{T_p} \frac{f_p T_p}{k} \simeq h(t - t_*) \underbrace{\frac{f_p T_p}{m}}_{\text{impulse}} \quad (1.82)$$

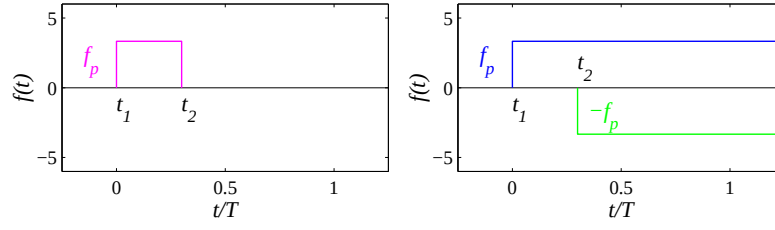


Fig. 1.22: Initial displacement response function $g(t)$ and (normalized) impulse response function $h(t)$.

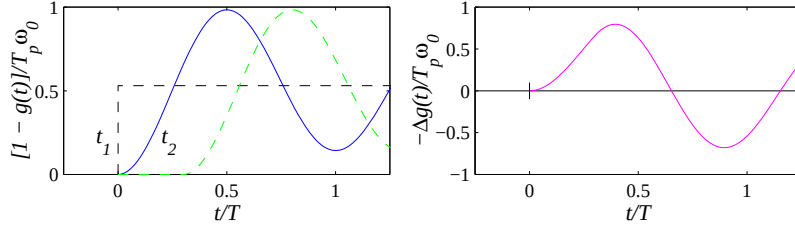


Fig. 1.23: a) Response to the two step loads, b) The total response.

The effect of the duration of a unit pulse is illustrated by the family of unit box pulses $[0, T_p]$ with $T_p/T = 0.1, 0.2, 0.3, 0.4, 0.5$,

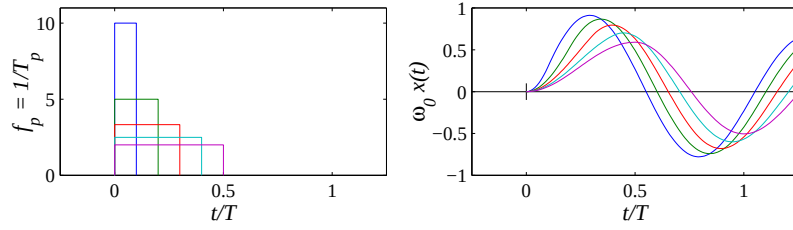


Fig. 1.24: Extending the pulses with common t_1 .

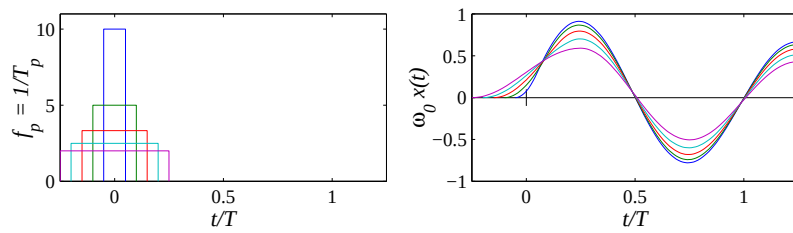


Fig. 1.25: Centered pulses of increasing duration.

Note, that the timing of the response from the center of the box load remains valid in all cases, while the magnitude of the response decreases with duration of the 'pulse'. $\square$

1.3.4 Relation between time and frequency analysis

1.4 Damping Mechanisms

Damping of structures may occur from different mechanisms. A common feature is that the damping must extract energy from the system. The loss of energy due to damping is called *energy dissipation*. First the dissipation condition is identified, and then a graphical representation of the energy loss associated with a closed displacement cycle is presented. Damping often involves nonlinear characteristics. However, in many cases a fair estimate of the effect of damping can be obtained via a linear analysis, using a so-called *equivalent viscous damping* based on identical energy dissipation per cycle in a forced vibration. Three particular non-viscous damping mechanisms are then presented: *friction*, *hysteresis*, and *motion in a fluid*. The characteristic features of the damping mechanism are discussed, and the equivalent viscous damping obtained.

For many structures exposed to wind, wave and earthquake loads dynamic effects are important and damping plays an essential role. For many years equivalent static loads have been used in building codes to represent dynamic effects. However, it is realized increasingly that the use of equivalent static loads leads to focus on strength and not on damping. In recent years more emphasis is placed on dynamic design of structures - advocated by Housner already in the fifties - and many new tall buildings and bridges are equipped with dampers to limit dynamic amplification of the response, see e.g. Soong and Dargush (1997).

Dissipation condition

Consider the equation of motion

$$m \ddot{x} + h(x, \dot{x}) + kx = f(t) \quad (1.83)$$

where the damping is represented by the force $f_d = -h(x, \dot{x})$.

Form the *rate of work* by multiplication of the equation of motion with $\dot{x}$,

$$(m \ddot{x} + h(x, \dot{x}) + kx) \dot{x} = f \dot{x} \quad (1.84)$$

Integrate the inertia and stiffness terms,

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} k x^2 \right) = f \dot{x} - h(x, \dot{x}) \dot{x} \quad (1.85)$$

The *mechanical energy* E is defined as the sum of the *kinetic energy* E_{kin} and the *elastic energy* E_{el} ,

$$E = E_{kin} + E_{el} = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} k x^2 \quad (1.86)$$

Thus, the rate of change of the change of mechanical energy is

$$\boxed{\frac{dE}{dt} = f \dot{x} - h(x, \dot{x}) \dot{x}} \quad (1.87)$$

The energy balance equation has two contributions on the right side, the rate of external work $f \dot{x}$ and the rate of dissipation due to damping $h(x, \dot{x}) \dot{x}$.

$$\frac{dE}{dt} = \underbrace{f \dot{x}}_{\text{rate of external work}} - \underbrace{h(x, \dot{x}) \dot{x}}_{\text{rate of dissipation}} \quad (1.88)$$

While the rate of external work may be either positive or negative, depending on the load and response, the energy dissipation due to passive damping must always extract energy.

Thus, valid passive damping mechanisms must satisfy the inequality

$$\boxed{h(x, \dot{x}) \dot{x} \geq 0} \quad (1.89)$$

If the function $h(x, \dot{x})$ is linear in both x and $\dot{x}$, the only valid form that leads to damping is $h = c \dot{x}$, i.e. the linear viscous damping.

1.4.1 Energy loss factor

Non-viscous damping is often accounted for by a so-called *equivalent viscous damping*, determined from energy equivalence. Transform equation of motion with damping force f_d

$$m \ddot{x} + k x = f_d \quad (1.90)$$

to energy balance equation

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} k x^2 \right) = f_d \dot{x} \quad (1.91)$$

where $f_d \dot{x}$ is the rate of energy dissipation.

Energy loss in a full cycle $T = 2\pi/\omega_0$ is

$$\Delta E = - \int_T \left(f_d \frac{dx}{dt} \right) dt = - \oint f_d dx \quad (1.92)$$

The energy loss per cycle can be used to describe the magnitude of the damping for a variety of damping mechanisms.

Note, that the integral of the elastic force vanishes, and f_d may therefore be replaced by the damping and the elastic force.

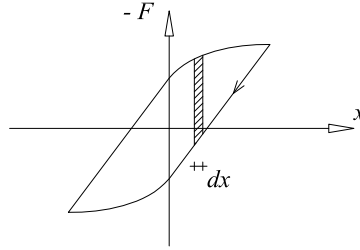


Fig. 1.26: Energy integral of a full cycle.

The *loss factor* η is defined by the energy loss per cycle ΔE relative to the maximum energy in the cycle E_{max} ,

$$\eta = \frac{\Delta E}{2\pi E_{max}} \quad (1.93)$$

1.4.2 Equivalent viscous damping

For viscous damping $f_d = -c\dot{x}$. The energy loss is evaluated by considering a full harmonic cycle,

$$x(t) = X \cos(\omega t) \quad (1.94)$$

The force displacement relation during a viscous load cycle,

$$f = -kX \cos(\omega t) + c\omega X \sin(\omega t) = -kx(t) \pm c\omega \sqrt{X^2 - x(t)^2} \quad (1.95)$$

For viscous damping the curve is an ellipse. This may be seen by rewriting the equation in the quadratic form

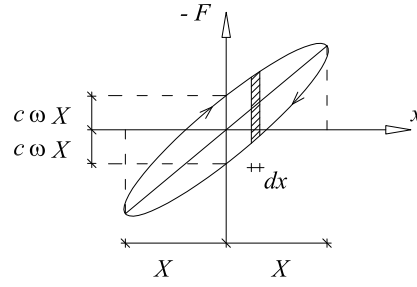


Fig. 1.27: Force displacement ellipse for viscous structure.

$$[kx(t) - F]^2 = (c\omega)^2 [X^2 - x(t)^2] \quad (1.96)$$

From this it follows that $f - kx(t)$ forms an ellipse with maximum value $c\omega X$ for $x = 0$.

The energy loss per cycle follows from the area of the ellipse,

$$\Delta E = \int_T c \dot{x}^2 dt = \frac{1}{2} \omega^2 c X^2 \left(\frac{2\pi}{\omega} \right) = \pi \omega c X^2 \quad (1.97)$$

It is seen to be proportional with frequency.

The loss factor for viscous damping now is obtained as

$$\eta = \frac{\pi \omega c X^2}{2\pi \frac{1}{2} k X^2} = \omega \frac{c}{k} = 2\zeta \frac{\omega}{\omega_0} \quad (1.98)$$

The *equivalent viscous damping* is defined from the energy loss factor,

$$c_{eq} = \eta \frac{k}{\omega}, \quad \zeta_{eq} = \frac{\eta}{2} \frac{\omega_0}{\omega} \quad (1.99)$$

Friction damping

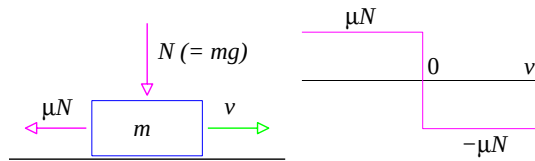


Fig. 1.28: a) Block with 'dry friction', b) Force-displacement relation.

The friction force f_f is independent of x and in opposite direction of the motion,

$$f_f = -\mu N \text{sign}(\dot{x}) = \begin{cases} -\mu N, & \dot{x} > 0 \\ 0, & \dot{x} = 0 \\ \mu N, & \dot{x} < 0 \end{cases} \quad (1.100)$$

NOTE: Satisfies dissipation inequality $-f_f \dot{x} = \mu N |\dot{x}| \geq 0$.

Substitution of spring, friction and external force gives the *non-linear* equation of motion

$$m \ddot{x} + \mu N \text{sign}(\dot{x}) + kx = f(t) \quad (1.101)$$

Table 1.2: Typical coefficients of friction μ .

Metal/metal lubrication	$\simeq$	0.07
Wood/wood	$\simeq$	0.2
Steel/steel (dry)	$\simeq$	0.3
Rubber/steel	$\simeq$	1.0

Free vibrations with friction

Normalized equation of motion

$$\ddot{x} + \mu g \text{sign}(\dot{x}) + \omega_0^2 x = 0 \quad (1.102)$$

Start from rest at $x_0 > \mu N/k$, whereby $\dot{x} \leq 0$,

$$\ddot{x} + \omega_0^2 x = \mu g \quad (1.103)$$

Corresponds to solution of undamped oscillator with constant load,

$$x(t) = x_0 \cos(\omega_0 t) + \frac{\mu g}{\omega_0^2} [1 - \cos(\omega_0 t)] \quad (1.104)$$

The velocity changes sign at $t = \frac{1}{2}T$, where

$$x_{1/2} = -x_0 + 2\mu g/\omega_0^2 = -x_0 + 2\mu N/k \quad (1.105)$$

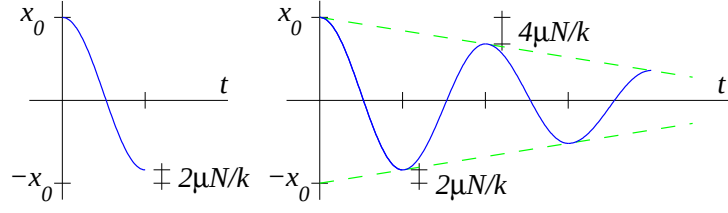


Fig. 1.29: a) The first half-period, b) Linear decrease of peaks.

Similar analysis of the following periods demonstrate that the amplitude (peak) decreases by $4\mu N/k$ per period, as shown in the figure.

Oscillations with friction will stop before the maximum, where the spring force is not sufficiently large to overcome the friction force, i.e. $kx_n < \mu N$. When starting at rest with initial displacement x_0 , the amplitude decreases with $4\mu N/k$ for each completed period. This determines the number of period as

$$n \frac{4\mu N}{k} \simeq x_0 \quad \Rightarrow \quad n \simeq \frac{k x_0}{4\mu N} \quad (1.106)$$

Equivalent viscous damping

The energy loss in a completed cycle with amplitude X is

$$\Delta E = - \oint f_d dx = 4\mu N X \quad (1.107)$$

This gives the energy loss factor

$$\eta = \frac{\Delta E}{2\pi E} = \frac{4\mu N X}{2\pi \frac{1}{2} k X^2} = \frac{4}{\pi} \frac{\mu N}{k X} \quad (1.108)$$

and thereby the equivalent viscous damping parameters

$$c_{eq} = \frac{4}{\pi} \frac{\mu N}{\omega X} \quad , \quad \zeta_{eq} = \frac{2}{\pi} \frac{\mu N}{k X} \frac{\omega_0}{\omega} \quad (1.109)$$

NOTE: The damping ratio depends on frequency and amplitude.

Hysteretic damping

For a harmonic load the viscous damping force $\dot{x} \propto i\omega x$, i.e. proportional with the angular frequency ω . Many structural materials exhibit much less dependence on frequency.

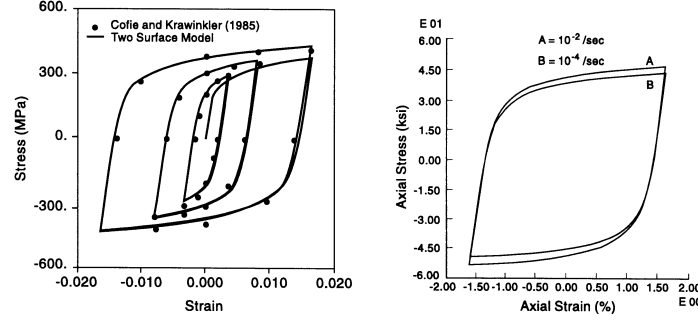


Fig. 1.30: a) Hysteretic response of steel, b) Dependence on strain rate.

Complex damping force

The limiting case of no frequency dependence is called *hysteretic damping*. In the frequency domain this may be described in complex notation. For the linear viscous damping force

$$f_v = -i c \omega x = -c \dot{x} \quad (1.110)$$

and for frequency independent linear hysteretic damping force

$$f_h = -i c \omega_0 x = -i \beta k x \quad (1.111)$$

This defines the hysteretic equation of motion in the frequency domain

$$m \ddot{x} + (1 + i\beta)k x = f_0 e^{i\omega t} \quad (1.112)$$

Formally the real elastic stiffness k is replaced by a complex stiffness $(1+i\beta)k$.

Harmonic response

Division by m leads to the normalized equation of motion for harmonic excitation

$$\ddot{x} + (1 + i\beta) \omega_0^2 x = \omega_0^2 \tilde{f}_0 e^{i\omega t} \quad (1.113)$$

The response is harmonic, in the form $x(t) = X_0 e^{i\omega t}$, with the complex amplitude determined by,

$$X_0 = H(\omega) \tilde{f}_0 = \frac{\tilde{f}_0}{\omega_0^2 - \omega^2 + i\beta\omega_0^2} \quad (1.114)$$

Thus, the hysteretic frequency response function is determined by

$$H(\omega) = \frac{\omega_0^2}{\omega_0^2 - \omega^2 + i\beta\omega_0^2} \quad (1.115)$$

NOTE: The last term in the denominator, representing the hysteretic damping, does not depend on frequency.

Equivalent viscous damping

An equivalent viscous damping can be determined from the energy loss coefficient. However, in the present case it is simpler to compare the hysteretic and the viscous frequency response functions directly. This leads to

$$i\beta\omega_0^2 = 2i\zeta_{eq}\omega_0\omega \quad (1.116)$$

and thereby the equivalent viscous damping ratio

$$\zeta_{eq} = \frac{\beta}{2} \frac{\omega_0}{\omega} \quad (1.117)$$

Thus, hysteretic leads to a smaller damping ratio for harmonic excitation with frequency ω above the natural frequency ω_0 , and conversely.

NOTE: Hysteretic damping, represented by a constant complex stiffness is not fully consistent, when used on a non-harmonic time history, but may be a useful approximation.

Fluid damping

A reference force in fluid problems is introduced via the *stagnation pressure* $p = \frac{1}{2}\rho v^2$ as defined by the Bernoulli equation.

The fluid (drag) force - for laminar flow - is approximated by

$$f_f = -\alpha \dot{x} |\dot{x}| \quad (1.118)$$

Equation of motion with laminar fluid damping

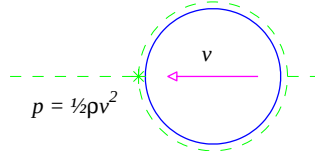


Fig. 1.31: Stagnation pressure at cylinder.

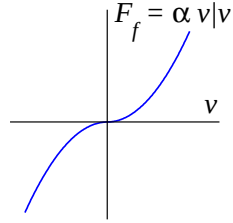


Fig. 1.32: Fluid (drag) force.

$$m \ddot{x} + \alpha \dot{x} |\dot{x}| + k x = f(t) \quad (1.119)$$

Note, that the fluid force is a quadratic function of velocity.

Equivalent viscous damping

Energy loss per cycle

$$\Delta E = \int_0^T \alpha |\dot{x}^3| dt = \alpha \omega^3 X^3 \int_0^T |\sin^3(\omega t)| dt \quad (1.120)$$

and by calculation of the integral

$$\int_0^T |\sin^3(\omega t)| dt = 2 \int_0^{T/2} \sin^3(\omega t) dt = \frac{2}{\omega} \int_0^\pi \sin^3 \varphi d\varphi = \frac{8}{3\omega} \quad (1.121)$$

the energy loss factor becomes

$$\frac{\Delta E}{2\pi E} = \frac{\frac{8}{3} \alpha \omega^2 X^3}{2\pi \frac{1}{2} k X^2} = \frac{8}{3\pi} \frac{\alpha \omega^2}{k} X \quad (1.122)$$

and the equivalent viscous damping

$$c_{eq} = \eta \frac{k}{\omega} = \frac{8}{3\pi} \alpha \omega X \quad (1.123)$$

1.4.3 Relaxation damping

1.4.4 Typical damping values

Pedestrian bridges

Typical damping values for footbridges have been given in H. Bachmann et al. *Vibration Problems in Structures – Practical Guidelines*, 2nd edition, Birkhuser Verlag, Basel, 1997.

Table 1.3: Typical values of damping ratio of footbridges.

Construction type	min.	mean	max.
Reinforced concrete	0.008	0.013	0.020
Prestressed concrete	0.005	0.010	0.017
Composite steel-concrete	0.003	0.006	–
Steel	0.002	0.004	–

Buildings in earthquakes

Damping data for buildings have been estimated from Californian earthquake records by N.M. Newmark and W.J. Hall, *Earthquake Spectra and Design*, Earthquake Engineering Research Institute, Berkeley, California, 1982. They apply to the working stress level below one-half the yield point, and stress levels just below the yield point, respectively. In structures, like bridges, chimneys and towers, the damping may be much lower.

Chimneys, towers and masts

Damping data for chimneys, towers and masts have been collected in I. Langen and R. Sigbjörnsson, *Dynamisk Analyse av Konstruktioner*, Tapir, Trondheim, 1979. A summary is given in the table.

Table 1.4: Typical values of damping of buildings.

Stress level	Type and condition of structure	Damping ratio (%)
Working stress, less than about half yield point	Welded steel, prestressed concrete, well reinforced concrete (slight cracking)	2–3
	Reinforced concrete with considerable cracking	3–5
	Bolted and/or riveted steel, wood structures with nailed or bolted joints	5–7
At or just below yield point	Welded steel, prestressed concrete (without complete loss of prestress)	5–7
	Prestressed concrete with complete loss of prestress	7–10
	Reinforced concrete	7–10
	Bolted and/or riveted steel, wood structures with bolted joints	10–15
	Wood structures with nailed joints	15–20

Table 1.5: Typical values of damping ratio of chimneys, towers and masts.

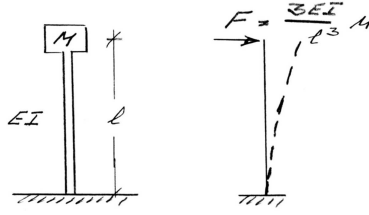
Type	Material	Height (m)	Damping ratio (%)
Chimney	Reinforced concrete	126.5	1.0–2.0
Chimney	Welded steel	45–91	0.5–1.4
TV-tower	Bolted steel	134–216	0.3–2.9
Cable-stayed mast	Welded steel	37–152	0.8

1.5 Exercises

Exercise 1.1. The natural frequency of a single-degree-of-freedom-system depends on the ratio of ‘stiffness’ to ‘mass’. For a simple mass–spring system the relation is $\omega_0^2 = k/m$.

- a) For the beam shown in the figure the force-displacement relation for a transverse force at the end is

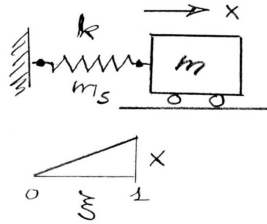
$$F = 3 \frac{EI}{l^3} u$$

Fig. 1.33: Beam supporting a mass M .

What is the natural frequency for transverse vibrations of a heavy mass M fixed to the end of the beam, when the mass of the beam is neglected.

- b) Let the mass $M = 1000 \text{ kg}$ be supported by a beam of length $l = 10 \text{ m}$. What is the bending stiffness EI necessary to give the frequency $f = 2.0 \text{ Hz}$.

Exercise 1.2. Consider the mass m supported by a spring with stiffness k . The motion of the mass m is described by x . The mass of the spring m_s is small relative to the mass m , and therefore the motion of the spring is quasi-static. This implies that the spring extends uniformly. Thus, the left end point is a rest, while the right end point moves x . A non-dimensional coordinate ξ is introduced such that the spring is described by $0 \leq \xi \leq 1$. This implies that the motion of a point described by the position ξ is ξx .

Fig. 1.34: Linear deformation of spring supporting a mass M .

- Find the total kinetic energy of the mass m and the spring in terms of $\dot{x}$.
- Use energy balance to find an expression for the natural angular frequency ω_0 .
- The contribution from the spring can be included as an extra contribution to the *effective mass* m_{eff} ,

$$\omega_0^2 = k/m_{eff} \quad \text{with} \quad m_{eff} = m + (?)m_s$$

Find the coefficient (?) of the spring mass.

Exercise 1.3. The figure shows a water column of total length l and cross-section area A . The mass density is ρ . When the water surface at the left side is displaced the distance x downward, the water surface at the right side is lifted the same distance x and conversely. The system is exposed to downward gravitation with acceleration constant g .

- Express the potential energy E_{pot} and the kinetic energy E_{kin} as function of x and $\dot{x}$, respectively.
- Use an energy balance argument to find an expression of the natural angular frequency ω_0 .
- The angular frequency ω_0 is independent of the mass density ρ . Why?

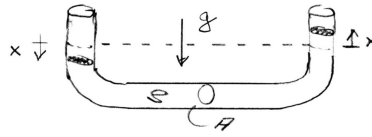


Fig. 1.35: Motion of water column.

The motion of the fluid may be constrained, whereby damping is introduced. This device can be used as damper of ship roll motion.

Exercise 1.4. Consider the case of harmonic base excitation with frequency ω ,

$$x(t) = \text{Re}[X e^{i\omega t}] \quad , \quad y(t) = \text{Re}[Y e^{i\omega t}]$$

The complex amplitude of the transmitted force depends on the relative motion $X - Y$,

$$F = [k + i\omega c] (X - Y)$$

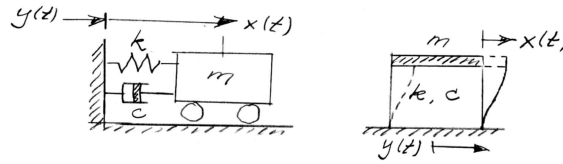


Fig. 1.36: Base motion $y(t)$ and relative motion $x(t)$.

- Find the normalized frequency function of the transmitted force, $F/(kY)$.
- Plot the absolute value of the force frequency function $|F/(kY)|$, and find the value at $\omega = \omega_0$.

- c) Show that there is a ‘neutral point’ for $\omega = \sqrt{2}\omega_0$, at which $|F/(kY)|$ is independent of damping, and determine the value at this frequency.

Summary

- Examples have been given of structures that often exhibit dynamic behavior and must be analyzed for dynamic effects.
- Free vibrations of a single degree-of-freedom system constitute an exchange between potential and kinetic energy. The time scale is characterized by the *natural angular frequency* ω_0 .
- The square of the natural (angular) frequency is determined by the ratio ‘stiffness’/‘mass’, i.e. $\omega_0^2 = k/m$.
- Damping is characterized by the non-dimensional *damping ratio* ζ , describing attenuation per vibration cycle. Vibrations with $\zeta < 1$ are ‘under-damped’ and may have damping ratio as low as 0.001.
- Damping may be measured from attenuation of free vibration response in terms of the *logarithmic decrement* $\delta = 2\pi\zeta$.
- The equation of motion may be considered as the time derivative of an energy balance equation. For simple systems with distributed mass or stiffness the equation of motion may be obtained from the energy balance relation.
- The displacement and velocity may be combined into a *state vector* $(x, \dot{x})$ describing the response of the system. Initial conditions and response are conveniently represented in a phase-plane.

Summary

- Harmonic excitation of a single-degree-of-freedom structure leads to response in phase with load for excitation frequencies $\omega < \omega_0$, and out of phase for $\omega > \omega_0$. At resonance, $\omega = \omega_0$, the response magnitude increases linearly with time, and can become very large.
- For excitation frequencies ω close to the natural frequency ω_0 the response exhibits ‘beats’.
- The response of damped single-degree-of-freedom structures to harmonic excitation is characterized by the *response amplitude* and the *phase lag*. Around resonance the phase lag is around $\frac{1}{2}\pi$, and the dynamic amplification is $1/(2\zeta)$. The width of the resonant frequency range is 2ζ .
- The response may be described by three regimes, depending on excitation frequency: i) a low-frequency regime dominated by stiffness, ii) a resonant regime dominated by damping, and iii) a high-frequency regime dominated by inertial forces.

- Excitation by ‘base motion’ is important in earthquake resistant design and for measurement instruments like accelerometers and seismographs. The problem of a rotating unbalanced mass can be described by the same equation.

Summary

- For a free particle the increase in velocity $\Delta \mathbf{v}$ can be related to the normalized time integral of the force, the so-called impulse of the load, $\hat{\mathbf{f}} = \hat{\mathbf{F}}/m$. This result remains valid as a good approximation for a spring supported mass excited by a pulse of short duration, relative to the natural period of oscillation.
- The response of a single-degree-of-freedom system to an ideal impulse of infinitely short duration is described by the *impulse response function* $h(t)$. This function represents the response to an initial unit velocity and zero initial displacement.
- The response to an arbitrary load history can be expressed as a convolution integral that represents a summation of a sequence of impulses. This result can be used to establish analytic response results for idealized load histories. However, for many cases of practical interest efficient numerical integration techniques based on ‘time-stepping’ constitute an important alternative.
- Some solutions to simple load histories can be obtained by direct integration of the differential equation of motion in combination with appropriate boundary conditions. These cases include time-limited constant load and linear load variation. The effect of initial conditions is conveniently introduced via the unit displacement response $g(t)$ and the unit velocity response $h(t)$.

Summary

- Damping is often represented in the form of a term $h(x, \dot{x})$ in the equation of motion. The corresponding rate of dissipation $h(x, \dot{x})\dot{x}$ must always be non-negative.
- The energy loss in a cycle relative to the available mechanical energy can be represented by the *energy loss factor* $\eta = \Delta E/2\pi E_{max}$.
- The relative energy loss per cycle can be used to define an *equivalent viscous damping ratio* $\zeta_{eq} = \frac{1}{2}\eta\omega_0/\omega$. For non-viscous damping mechanisms the equivalent damping ratio typically depend on frequency and ampli-

tude, and is therefore only valid for oscillations in a certain frequency and amplitude range.

- Special cases of non-viscous damping with explicit equivalent damping properties are:
 - friction damping
 - hysteretic damping
 - fluid damping
- It is difficult, and in most cases impossible, to estimate the damping theoretically, and representative levels of damping must therefore often be estimated from empirical data or obtained by direct measurements on the completed structure. Tables are given, showing damping ratios ζ below 1% for structures like pedestrian bridges, chimneys and towers, while classic buildings have damping ratios in the order of 2-5% in the working stress range and 5-10% around yield.

2.1 Dynamics of a taut cable

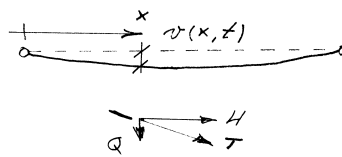


Fig. 2.1: Taut cable with cable force T .

Axial and transverse force components

$$H \simeq T \quad , \quad Q = H \frac{\partial v}{\partial x} \quad (2.1)$$

Equation of transverse motion for displacement component $v(x, t)$ for transverse load intensity $g(x, t)$,

$$\frac{\partial}{\partial x} \left(H \frac{\partial v}{\partial x} \right) + g(x, t) = m \frac{\partial^2 v}{\partial t^2} \quad (2.2)$$

The axial motion is negligible, whereby $H \simeq \text{const}$.

2.1.1 The wave equation

Use $H = \text{const}$ and normalize with H ,

$$\frac{\partial^2 v}{\partial x^2} = \frac{m}{H} \frac{\partial^2 v}{\partial t^2} \quad (2.3)$$

The parameter H/m has dimension $[L^2/T^2]$, i.e. speed squared.

This leads to the wave equation

$$\boxed{\frac{\partial^2 v}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 v}{\partial t^2}} \quad (2.4)$$

with the *wave speed*

$$\boxed{c = \sqrt{\frac{H}{m}}} \quad (2.5)$$

Many dynamic problems are described by the wave equation in 1, 2 or 3 dimensions. Examples of one-dimensional problems are cables, extension of bars, torsion of bars and acoustic pressure waves in pipes. A two-dimensional example is membrane vibrations, and three-dimensional examples include acoustic pressure fields in rooms.

In each particular application the square of the wave speed c represents the ratio of stiffness to specific mass.

d'Alembert's solution

A general integral can be found in the form

$$v(x, t) = w(x - ct) \quad (2.6)$$

where by substitution

$$\frac{\partial^2 v}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 v}{\partial t^2} = w_{xx} - c^{-2} w_{tt} = 0 \quad (2.7)$$

This corresponds to an arbitrary wave travelling at speed c .

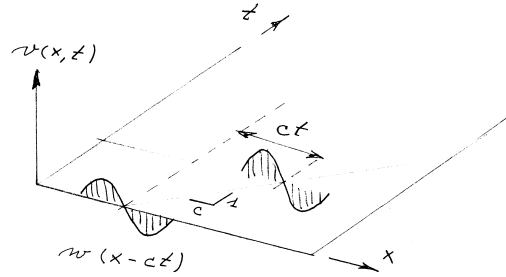


Fig. 2.2: Propagation of wave form with speed c .

Typically the wave speed of any harmonic wave form can be determined. When the wave speed of any harmonic wave form is the same, waves of arbitrary shape retain this shape during propagation. Such waves are called

non-dispersive. In contrast, wave problems in which waves with different wave-length travel at different wave speed are called *dispersive waves*. In a dispersive wave the various wave harmonics will separate during propagation of the wave, leading to dispersion of the original wave form. Examples of dispersive waves are water waves and waves and vibrations in beams.

2.1.2 Vibrations of a taut cable

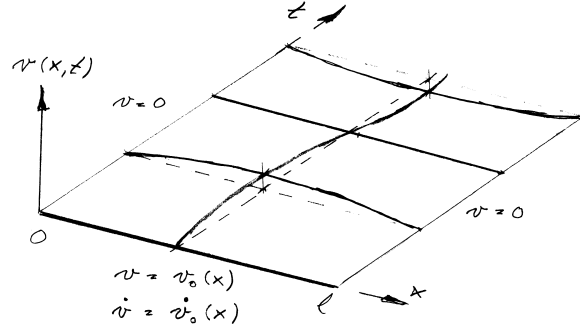


Fig. 2.3: Vibrations of a taut cable - string.

The vibrations of a taut cable or string of length ℓ requires *initial conditions*

$$v(x, 0) = v_0(x) \quad , \quad v_t(x, 0) = \dot{v}_0(x) \quad (2.8)$$

and a ‘kinematic’ or ‘dynamic’ *boundary condition* at each end,

$$v(0, t) = 0 \quad \text{or} \quad H v_x(0, t) = -Q_0(t) \quad (2.9)$$

and

$$v(\ell, t) = 0 \quad \text{or} \quad H v_x(\ell, t) = Q_\ell(t) \quad (2.10)$$

Note, that there are two initial conditions at the initial time $t = 0$, and one boundary condition at each end.

Also note the change of sign of the transverse force Q from the left to the right end due to the sign convention.

Separation of variables

A solution is assumed in factored form as

$$v(x, t) = X(x)T(t) \quad (2.11)$$

Substitution into the homogeneous wave equation gives

$$X''(x)T(t) = \frac{1}{c^2} X(x) \ddot{T}(t) \quad (2.12)$$

This equation is normalized by division with the factors that are not differentiated, i.e. by $X(x)T(t)$,

$$\underbrace{\frac{X''(x)}{X(x)}}_{\text{independent of } t} = \underbrace{\frac{1}{c^2} \frac{\ddot{T}(t)}{T(t)}}_{\text{independent of } x} = -k^2 \quad (2.13)$$

The two terms are independent of t and x , respectively, and thus must be a constant. This constant must be negative to permit meaningful solutions, and is chosen as $-k^2$. By this choice the constant k has the dimension $[L^{-1}]$. k is called the *wave-number*.

First the spatial problem is solved to determine k and $X(x)$, Substitution into the homogeneous wave equation gives

$$X''(x) + k^2 X(x) = 0 \quad , \quad X(0) = X(\ell) = 0 \quad (2.14)$$

This is an eigenvalue problem with infinitely many solutions of the form

$$X(x) = A_n \sin(k_n x) \quad , \quad k_n = n \frac{\pi}{\ell} \quad n = 1, 2, \dots \quad (2.15)$$

Each solution of the eigenvalue problem contributes to the full solution,

$$v(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin(k_n x) \quad n = 1, 2, \dots \quad (2.16)$$

The time-function $T_n(t)$ is determined from the differential equation

$$\ddot{T}_n(t) + k_n^2 c^2 T_n(t) = 0 \quad n = 1, 2, \dots \quad (2.17)$$

The product $k_n c$ has the character of an angular frequency,

$$\boxed{\omega_n = k_n c = n \frac{\pi c}{\ell} \quad n = 1, 2, \dots} \quad (2.18)$$

The time-function $T_n(t)$ is then of the form

$$T_n(t) = A_n \cos(\omega_n t) + B_n \sin(\omega_n t) \quad n = 1, 2, \dots \quad (2.19)$$

The arbitrary constants A_n and B_n are determined from the initial conditions,

$$v_0(x) = \sum_{n=1}^{\infty} A_n \sin(k_n x) \quad (2.20)$$

$$\dot{v}_0(x) = \sum_{n=1}^{\infty} \omega_n B_n \sin(k_n x) \quad (2.21)$$

Use the orthogonality condition of the sine function,

$$\int_0^{\ell} \sin\left(n \frac{\pi x}{\ell}\right) \sin\left(r \frac{\pi x}{\ell}\right) dx = \frac{1}{2} \ell \delta_{nr} \quad (2.22)$$

to determine the constants as

$$\boxed{\begin{aligned} A_n &= \frac{2}{\ell} \int_0^{\ell} \sin(k_n x) v_0(x) dx \\ \omega_n B_n &= \frac{2}{\ell} \int_0^{\ell} \sin(k_n x) \dot{v}_0(x) dx \end{aligned}} \quad (2.23)$$

This gives the full solution as an infinite sum

$$\boxed{v(x, t) = \sum_{n=1}^{\infty} \underbrace{\left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t) \right]}_{T_n(t)} \underbrace{\sin(k_n x)}_{X_n(x)} \quad (2.24)}$$

The wave character of the vibration solution can be illustrated by introducing $\omega_n = k_n c$ and using the trigonometric product formulae,

$$\begin{aligned} v(x, t) &= \sum_{n=1}^{\infty} \frac{1}{2} A_n \left[\sin(k_n(x - ct)) + \sin(k_n(x + ct)) \right] \\ &\quad + \frac{1}{2} B_n \left[\cos(k_n(x - ct)) - \cos(k_n(x + ct)) \right] \end{aligned} \quad (2.25)$$

The sign ‘-’ corresponds to waves moving to the right, while the sign ‘+’ corresponds to waves moving to the left.

The term A_n corresponds to an initial harmonic shape with zero velocity splitting into equal parts moving to the right and the left as illustrated in the figure.

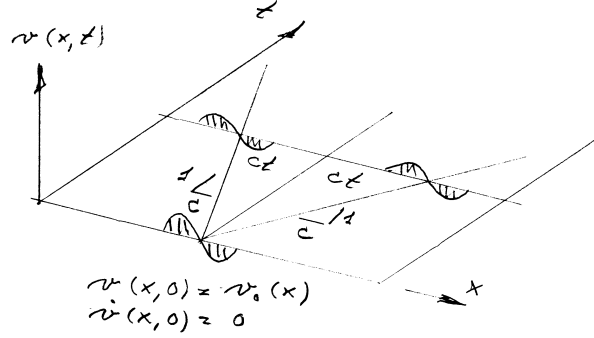


Fig. 2.4: Wave splitting into equal parts moving to the right and left.

Example 2.1. Spring supported cable.

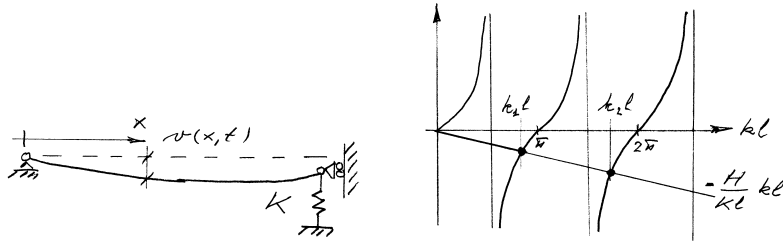


Fig. 2.5: a) Spring supported cable, b) Wave number $k_n \ell$ for spring supported cable.

The spring boundary condition with spring constant K at $x = \ell$ is

$$H v_x(\ell) = -K v(\ell) \quad (2.26)$$

The solution is factored as $v(x, t) = X(x)T(t)$ and the spatial solution is

$$X(x) = A \sin(kx) \quad (2.27)$$

The wave number k is determined by substitution into the boundary condition,

$$H X'(\ell) = -K X(\ell) \quad (2.28)$$

giving the following equation for $k\ell$,

$$\tan(k_n \ell) = -\frac{H}{K \ell} k_n \ell \quad n = 1, 2, \dots \quad (2.29)$$

This equation must be solved by iteration. □

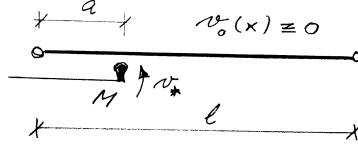
Example 2.2. Impulsive load on string.

Fig. 2.6: Impulsive load on string.

The full solution is

$$v(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t) \right] \sin(k_n x) \quad (2.30)$$

The initial displacement is identically zero, whereby $A_n = 0$, while the initial velocity is determined by the impulse Mv_* .

The initial velocity equation is written in ‘impulse form’

$$m\omega_n B_n = \frac{2}{\ell} \int_0^{\ell} \sin(k_n x) \underbrace{m\dot{v}_0(x)}_{Mv_* \delta(x-a)} dx = \frac{2}{\ell} Mv_* \sin(k_n a) \quad (2.31)$$

Thus, the constants B_n are

$$B_n = 2 \frac{M}{m\ell} \frac{v_*}{\omega_n} \sin(k_n a) \quad (2.32)$$

and the full solution becomes

$$v(x, t) = 2 \frac{M}{m\ell} \sum_{n=1}^{\infty} \frac{v_*}{\omega_n} \sin(\omega_n t) \sin(k_n a) \sin(k_n x) \quad (2.33)$$

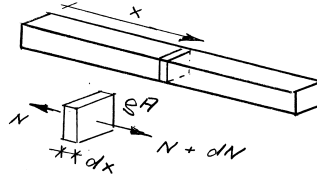
Note, that the contributions from the individual harmonics depend on the value of the corresponding eigenfunction at the point of excitation $x = a$. $\square$

2.1.3 Dynamics of an elastic bar

The acceleration of a slice of unit thickness is determined by the gradient $\partial N / \partial x$ of the normal force N and the distributed axial load $f(x, t)$,

$$\frac{\partial N}{\partial x} + f(x, t) = \rho A \frac{\partial^2 u}{\partial t^2} \quad (2.34)$$

where A is the cross-section area and ρ is the mass density. The axial strain is $\varepsilon = \partial u / \partial x$, and for an elastic bar with elastic modulus E the normal force then is

Fig. 2.7: Elastic bar with normal force N .

$$N = EA\varepsilon = EA \frac{\partial u}{\partial x} \quad (2.35)$$

Differentiation of this relation with respect to x and substitution into the previous equation gives the dynamic equation of an elastic bar,

$$\frac{\partial}{\partial x} \left(EA \frac{\partial u}{\partial x} \right) + f(x, t) = \rho A \frac{\partial^2 u}{\partial t^2} \quad (2.36)$$

Note: This equation is similar to that of transverse motion of a taut cable. In the present case the stiffness EA takes the place of the horizontal cable force and the mass per unit length is ρA .

Wave equation form

For an elastic bar with constant cross section the equation of axial motion may be written as the *wave equation*

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} \quad (2.37)$$

with the *wave speed*

$$c = \sqrt{\frac{E}{\rho}} \quad (2.38)$$

The wave equation implies that the d'Alembert solution format (10.6) and the associated notion of wave propagation with wave speed c illustrated in Fig. 10.2 also applies to axial motion of the elastic bar with constant cross section.

In the case of axial motion of an elastic bar the wave speed is determined by the ratio of the elastic modulus E to the mass density ρ . Thus the wave speed is independent of the cross-section area. The assumption behind the present

theory is that u describes the average axial motion over the cross-section. In order for this average motion to be representative in the elasticity relation transverse deformation must be negligible. Thus the physical assumptions behind the theory restricts its validity to phenomena with wave length λ considerably larger than a typical cross section dimension. Thus, implicitly it is assumed that $\lambda^2 \gg A$. This type of restriction on the validity of technical theories for bars, beams, plates and shells is typical. In practice the restriction implies that the theory may loose its validity for very short wave lengths, typically corresponding to very high frequencies. The infinite frequency limit of the theory may therefore not correspond to a physical result.

Example 2.3. Elastic bar with spring or mass at one end.

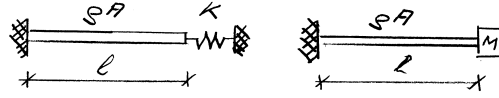


Fig. 2.8: Built-in elastic bar with spring or mass at right end.

An elastic bar with built-in end at $x = 0$ has a solution of the form $u(x, t) = X(x)T(t)$ with

$$X(x) = C \sin(kx) \quad (2.39)$$

Case a: If the right end is supported by a spring with stiffness K , the corresponding boundary condition is

$$EA u_x = -K u \quad (2.40)$$

and upon substitution of the separated solution

$$\tan(k\ell) = -\frac{EA}{K\ell} k\ell \quad (2.41)$$

Case b: If the right end supports a mass M , the corresponding boundary condition is

$$EA u_x = -M \ddot{u} \quad (2.42)$$

and upon substitution of the separated solution with $\ddot{T} = -\omega^2 T$ and $\omega = ck$,

$$\tan(k\ell) = \frac{EA}{M} \frac{k}{\omega^2} = \frac{EA\ell}{Mc^2} \frac{1}{k\ell} \quad (2.43)$$

The wave number $k_n\ell$ of mode n is found as the intersection of the function $\tan(k\ell)$ with a line with negative slope in the case of a spring, and a branch of a hyperbola in the case of a supported mass, as illustrated in the figure.

For a very small mass the branch of the hyperbola moves up towards plus infinity, and for a small spring the slope of the line increases towards minus infinity. In both cases the limit is a spring with a free end at $x = \ell$. In this case the solution is the vertical asymptotes

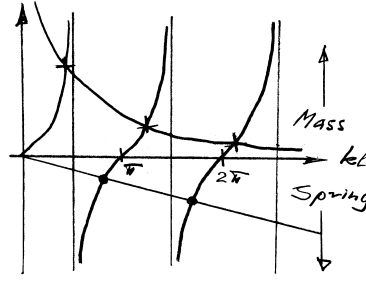


Fig. 2.9: Wave number $k_n \ell$ for elastic bar, $\bullet$ spring support, $\times$ supporting mass.

of $\tan(k\ell)$,

$$k_n \ell = (n - \frac{1}{2})\pi \rightarrow \omega_n = ck_n = (n - \frac{1}{2})\pi \frac{c}{\ell} \quad (2.44)$$

For increasing mass the wave number k_n , and thereby the frequency ω_n , goes down, while for increasing spring stiffness k_n goes up.

In the limit of a very stiff spring the solution also corresponds to the positive intersections of $\tan(k\ell)$ with the x -axis, $k_n \ell = n\pi$. These modes correspond closely to a bar without motion at $x = \ell$.

In the limit of very large mass a single very small root $k_n \ell \ll \pi$ corresponds to nearly free motion of the large mass M . The remaining roots correspond closely to the positive intersections of $\tan(k\ell)$ with the x -axis, $k_n \ell = n\pi$. It is interesting to observe that the case of a very large mass has an 'extra' mode corresponding to nearly homogeneous deformation of the bar at a frequency well below the smallest natural frequency of the bar with any type of homogeneous boundary constraints. In this mode the bar acts like a spring support of the concentrated mass M . $\square$

2.2 Dynamics of Beams

2.2.1 Equation of transverse motion

The equation of transverse motion is obtained by considering the forces necessary to produce the transverse acceleration $\partial^2 w / \partial t^2$ of the beam section with area A and mass density ρ .

It is customary to neglect the inertial effects associated with rotation of the beam section, and it can be shown that this is a higher order correction for slender beams, see e.g. Inman (1996).

The equation of motion is obtained by considering a slice of the beam with transverse displacement $w(x, t)$. The cross-section area is A , the mass density ρ , and the elastic modulus E .

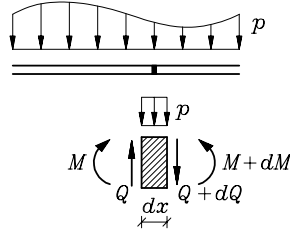


Fig. 2.10: Equations of motion of thin beam section.

The transverse acceleration is generated by the gradient of the transverse force Q and the distributed transverse load $p(x, t)$,

$$\frac{\partial Q}{\partial x} + p = \rho A \frac{\partial^2 w}{\partial t^2} \quad (2.45)$$

If the inertial effects of rotating the section are neglected, the moment balance equation is

$$\frac{\partial M}{\partial x} = Q \quad (2.46)$$

Elimination of the shear force Q between these two equations leads to the equation of motion in the form

$$\frac{\partial^2 M}{\partial x^2} - \rho A \frac{\partial^2 w}{\partial t^2} + p = 0 \quad (2.47)$$

Note, that the acceleration term $-\rho A \partial^2 w / \partial t^2$ appears as an *inertial force*, that is added to the external force $p(x, t)$, – d'Alembert's principle. It enables direct formulation of dynamic equations of motion from the static equilibrium equations for many types of structures like beams, cables, plates and shells.

The elastic properties of the beam enters through the moment-curvature relation

$$M = -EI \frac{\partial^2 w}{\partial x^2} \quad (2.48)$$

When this relation is entered into the equation of transverse motion the resulting equation is

$$\boxed{\frac{\partial^2}{\partial x^2} \left(EI \frac{\partial^2 w}{\partial x^2} \right) + \rho A \frac{\partial^2 w}{\partial t^2} = p} \quad (2.49)$$

This equation can model two types of seemingly different dynamic phenomena in beams, wave propagation and vibrations.

2.2.2 Bending waves of beams

Consider the idealized problem of an infinitely long homogeneous beam without external load. Transverse motion is governed by

$$\boxed{\frac{\partial^4 w}{\partial x^4} + \frac{\rho A}{EI} \frac{\partial^2 w}{\partial t^2} = 0} \quad (2.50)$$

This equation permits solutions of the form

$$w(x, t) = w_0 \sin(\omega t - kx) \quad (2.51)$$

with angular frequency ω and wave number k . The independent variables x and t only appear in the combination

$$\boxed{\varphi = \omega t - kx} \quad (2.52)$$

called the *phase angle*.

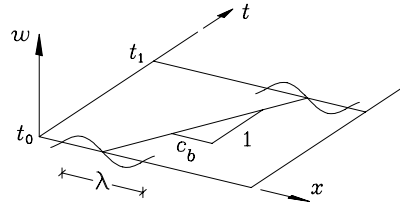


Fig. 2.11: Propagation of harmonic wave in an infinite beam.

The motion is periodic with period T corresponding to an increase of the phase angle by 2π ,

$$\boxed{\omega = \frac{2\pi}{T}} \quad (2.53)$$

At any particular time the solution appears as a sine-wave. The wave length λ corresponds to an increase of the phase angle of 2π ,

$$\boxed{k = \frac{2\pi}{\lambda}} \quad (2.54)$$

Points (x_0, t_0) and $(x_0 + \Delta x, t_0 + \Delta t)$ of equal phase propagate with speed

$$c_b = \frac{\Delta x}{\Delta t} = \frac{\omega}{k} = \frac{\lambda}{T} \quad (2.55)$$

where the subscript b indicates bending waves.

Substitution of the solution into the homogeneous equation of motion of the beam gives

$$\frac{\omega^2}{k^4} = \frac{EI}{\rho A} \quad (2.56)$$

from which the speed of bending waves is

$$c_b = \frac{\omega}{k} = 2\pi \sqrt{\frac{I}{A\lambda^2}} \sqrt{\frac{E}{\rho}} \quad (2.57)$$

In this formula $c_l = \sqrt{E/\rho}$ is the speed of longitudinal waves in the beam. For a rectangular beam of height h , $I = \frac{1}{12}Ah^2$, whereby $c_b = \frac{2\pi}{\sqrt{12}} \frac{h}{\lambda} \sqrt{\frac{E}{\rho}} \simeq 1.8 \frac{h}{\lambda} c_l$. Bending waves are *dispersive* with wave speed decreasing with increasing wave length.

2.2.3 Bending vibrations of beams

The dynamics of *bending vibrations of beams* is described by the differential equation of motion

$$\frac{\partial^2}{\partial x^2} \left(EI \frac{\partial^2 w}{\partial x^2} \right) + \rho A \frac{\partial^2 w}{\partial t^2} = p(x, t) \quad (2.58)$$

supplemented with *boundary conditions*, formulated in terms of

$$w, \quad \frac{\partial w}{\partial x}, \quad M = -EI \frac{\partial^2 w}{\partial x^2}, \quad Q = -\frac{\partial}{\partial x} \left(EI \frac{\partial^2 w}{\partial x^2} \right) \quad (2.59)$$

and *initial conditions*, prescribing displacement and velocity at the initial time t_0 ,

$$w(x, t_0) = w_0(x), \quad \frac{\partial w}{\partial t}(x, t_0) = \dot{w}_0(x) \quad (2.60)$$

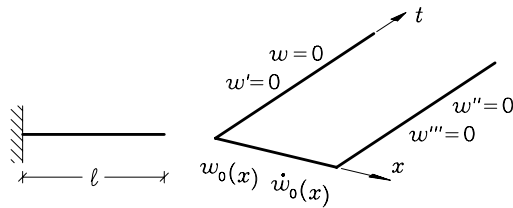


Fig. 2.12: Boundary and initial conditions of beam.

Solutions to beam vibration problems are typically obtained by the method of *separation of variables*.

Separation of variables

A free harmonic vibration with angular frequency ω can be represented in the form

$$w(x, t) = \tilde{w}(x) \cos(\omega t - \varphi_0) \quad (2.61)$$

Substitution of this into the equation of motion with $p(x, t) = 0$ gives the ordinary differential equation

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 \tilde{w}}{dx^2} \right) - \omega^2 \rho A \tilde{w} = 0 \quad (2.62)$$

For a homogeneous beam this equation can be written as

$$\frac{d^4 \tilde{w}}{dx^4} - k^4 \tilde{w} = 0 \quad (2.63)$$

with the parameter k given as

$$k^4 = \omega^2 \frac{\rho A}{EI} \quad (2.64)$$

General solution in terms of trigonometric and hyperbolic functions,

$$\tilde{w}(x) = C_1 \sin(kx) + C_2 \cos(kx) + C_3 \sinh(kx) + C_4 \cosh(kx) \quad (2.65)$$

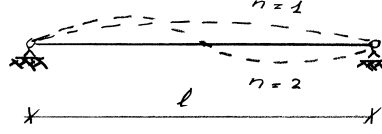
In the case of free vibrations the boundary conditions are homogeneous. A non-trivial solution can therefore only be obtained for particular values of the parameter k . Thus, the solution procedure consists of the determination of these values $k_1, k_2, \dots$, and the corresponding values of the arbitrary constants $C_1, \dots, C_4$. These solutions represent the mode-shapes of free vibrations with frequencies $\omega_1, \omega_2, \dots$.

Example 2.4. Simply supported beam.

Boundary conditions

$$w = 0, \quad w'' = 0 \quad \text{at} \quad x = 0, \quad x = \ell \quad (2.66)$$

The spatial solution satisfying all four boundary conditions is

Fig. 2.13: Vibration modes $n = 1, 2$ of simply supported beam.

$$\tilde{w}_n(x) = C_n \sin(k_n x) = C_n \sin\left(\frac{n\pi}{\ell} x\right), \quad n = 1, 2, \dots \quad (2.67)$$

corresponding to wave numbers $k_n = n\pi/\ell$.

The angular frequencies follow from

$$\omega_n = k_n^2 \sqrt{\frac{EI}{\rho A}} = \left(\frac{n\pi}{\ell}\right)^2 \sqrt{\frac{EI}{\rho A}}, \quad n = 1, 2, \dots \quad (2.68)$$

Note, that while the wave number k_n is proportional to n , the angular frequency ω_n is proportional to n^2 .

General free vibration solution of the simply supported beam,

$$w(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t) \right] \sin(k_n x) \quad (2.69)$$

The constants A_n and B_n are determined from the initial conditions,

$$\begin{aligned} w(x, 0) &= \sum_n A_n \sin(k_n x) = w_0(x) \\ \dot{w}(x, 0) &= \sum_n \omega_n B_n \sin(k_n x) = \dot{w}_0(x) \end{aligned} \quad (2.70)$$

The arbitrary constants A_n and B_n are seen to appear as coefficients in a Fourier sine series expansion of $w_0(x)$ and $\dot{w}_0(x)$, respectively.

The coefficients are determined by use of the orthogonality relation

$$\frac{2}{\ell} \int_0^{\ell} \sin\left(\frac{n\pi}{\ell} x\right) \sin\left(\frac{m\pi}{\ell} x\right) dx = \begin{cases} 1 & \text{for } m = n \\ 0 & \text{for } m \neq n \end{cases} \quad (2.71)$$

Multiplication with $\sin(k_m x)$, followed by integration, then gives the arbitrary constants

$$A_n = \frac{2}{\ell} \int_0^{\ell} \sin(k_n x) w_0(x) dx \quad (2.72)$$

$$\omega_n B_n = \frac{2}{\ell} \int_0^{\ell} \sin(k_n x) \dot{w}_0(x) dx \quad (2.73)$$

This determines the full free vibration solution in terms of the frequencies ω_n , the corresponding mode shapes $\sin(k_n x)$, and the initial displacement $w_0(x)$ and velocity $\dot{w}_0(x)$ distributions. $\square$

Alternative base functions

It is often convenient to use a set of base functions defined by

$$\begin{aligned} S_+(kx) &= \sinh(kx) + \sin(kx) \\ C_+(kx) &= \cosh(kx) + \cos(kx) \\ S_-(kx) &= \sinh(kx) - \sin(kx) \\ C_-(kx) &= \cosh(kx) - \cos(kx) \end{aligned} \quad (2.74)$$

These functions satisfy the circular differentiation rules

$$\begin{aligned} \frac{dS_+(kx)}{dx} &= k C_+(kx), \quad \frac{dC_+(kx)}{dx} = k S_-(kx) \\ \frac{dS_-(kx)}{dx} &= k C_-(kx), \quad \frac{dC_-(kx)}{dx} = k S_+(kx) \end{aligned} \quad (2.75)$$

It is observed, that only the function $C_+(kx)$ does not vanish at $x = 0$.

The general spatial solution for harmonic vibrations of a beam is expressed in terms of these functions as

$$\tilde{w}(x) = C_1 S_+(kx) + C_2 C_+(kx) + C_3 S_-(kx) + C_4 C_-(kx) \quad (2.76)$$

The differentiation rules and the simple values at $x = 0$ makes this representation particularly convenient as illustrated in the following example.

Example 2.5. Vibration of fixed-free beam.

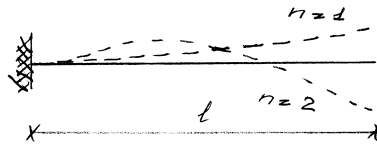


Fig. 2.14: Vibration modes $n = 1, 2$ of fixed-free supported beam.

Boundary conditions at fixed end at $x = 0$,

$$\tilde{w}(0) = 0 \quad , \quad \tilde{w}'(0) = 0 \quad (2.77)$$

From these conditions it follows immediately that $C_2 = 0$ and $C_1 = 0$. Thus the problem to be solved consists of the function

$$\tilde{w}(x) = C_3 S_-(kx) + C_4 C_-(kx) \quad (2.78)$$

Boundary conditions at free end at $x = \ell$,

$$\tilde{w}''(\ell) = 0 \quad , \quad \tilde{w}'''(\ell) = 0 \quad (2.79)$$

Substitution of the representation of the function $\tilde{w}(x)$ leads to

$$\begin{bmatrix} S_+(k\ell) & C_+(k\ell) \\ C_+(k\ell) & S_-(k\ell) \end{bmatrix} \begin{bmatrix} C_3 \\ C_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2.80)$$

This equation system only permits a non-trivial solution, if the determinant vanishes. Thus, the parameter k must satisfy the equation

$$S_+(k\ell)S_-(k\ell) - C_+^2(k\ell) = 0 \quad (2.81)$$

Substitution of the hyperbolic and trigonometric functions gives

$$\begin{aligned} & \left[\sinh^2(k\ell) - \sin^2(k\ell) \right] - \\ & \left[\cosh^2(k\ell) + \cos^2(k\ell) + 2 \cosh(k\ell) \cos(k\ell) \right] = 0 \end{aligned} \quad (2.82)$$

Use of standard identities leads to

$$\cos(k\ell) \cosh(k\ell) + 1 = 0 \quad (2.83)$$

which is conveniently written as

$$\cos k\ell = \frac{-1}{\cosh(k\ell)} \quad (2.84)$$

The solution is the intersection of the graphs of the functions defined by the left and right hand sides of the equation, as illustrated in the figure.

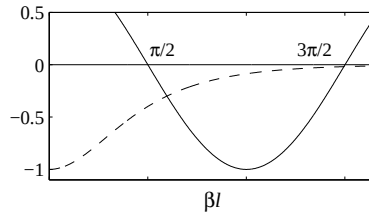


Fig. 2.15: Frequency equation: $\cos(k\ell)$ (—) and $-1/\cosh(k\ell)$ (---).

The roots of the frequency equation are close to the roots of $\cos k\ell = 0$,

$$k_n^{(0)} \ell = (2n-1) \frac{\pi}{2} \quad , \quad n = 1, 2, \dots \quad (2.85)$$

The actual roots found by iteration in the form

$$k_n^{(i+1)}\ell = (2n-1)\frac{\pi}{2} - (-1)^n \sin^{-1}\left(\frac{1}{\cosh(k_n^{(i)}\ell)}\right) \quad (2.86)$$

The first correction gives the approximate solution

$$k_n^{(1)}\ell \simeq (2n-1)\frac{\pi}{2} - \frac{(-1)^n}{\cosh\left((n-\frac{1}{2})\pi\right)} \quad (2.87)$$

The two first steps and the final value of the wave numbers $k_n\ell$ are given below, together with the angular frequency.

n	1	2	3	4
$k_n^{(0)}\ell$	1.5708	4.7124	7.8540	10.9956
$k_n^{(1)}\ell$	1.9693	4.6944	7.8548	10.9955
$k_n\ell$	1.8751	4.6941	7.8548	10.9955
$\omega_n / (EI/\rho A \ell^4)^{1/2}$	3.5160	22.034	61.698	120.901

The table illustrates that, apart from the first two to three values of the wave numbers, the remaining wave numbers are regularly spaced and can be determined with great accuracy without iteration. $\square$

2.2.4 Damping of transverse beam vibrations

Two simple damping mechanisms may be identified directly, one from motion and another from viscoelastic material properties.

A damping term proportional to the velocity can be included in the equation of motion,

$$\frac{\partial Q}{\partial x} + p - \gamma \frac{\partial w}{\partial t} = \rho A \frac{\partial^2 w}{\partial t^2} \quad (2.88)$$

The other term is a viscoelastic contribution to the stress,

$$\sigma = E\varepsilon + \beta E \frac{\partial \varepsilon}{\partial t} = E \left(1 + \beta \frac{\partial}{\partial t}\right) \varepsilon \quad (2.89)$$

whereby the moment curvature relation becomes

$$M = -EI \left(1 + \beta \frac{\partial}{\partial t}\right) \frac{\partial^2 w}{\partial x^2} \quad (2.90)$$

Elimination of Q and M leads to the damped beam equation

$$\boxed{
 \begin{aligned}
 \frac{\partial^2}{\partial x^2} \left(EI \frac{\partial^2 w}{\partial x^2} \right) + \frac{\partial^2}{\partial x^2} \left(\beta EI \frac{\partial^2}{\partial x^2} \left(\frac{\partial w}{\partial t} \right) \right) \\
 + \gamma \frac{\partial w}{\partial t} + \rho A \frac{\partial^2 w}{\partial t^2} = p
 \end{aligned}
 } \quad (2.91)$$

If $\alpha = \gamma/(\rho A)$ is a constant and β is constant, the beam has *Rayleigh damping* with parameters α and β . The equation is then separable with the undamped mode shapes, and the properties of the damping terms and the parameters α and β are similar to the discrete case investigated earlier.

2.3 Exercises

Summary

- The dynamics of a taut cable is governed by a partial differential equation that is second order in both the spatial variable x and in time t . For a homogeneous cable without external loads this is the *wave equation*.
- The wave equation permits wave-like solutions - that is a fixed shape propagating either to the right or to the left with a constant wave speed $c = \sqrt{H/m}$, where H is the tension and m is the mass per unit length. This solution was first found by d'Alembert.
- Cables can also vibrate in periodic motion with frequencies determined by the ratio H/m and the boundary conditions. The vibration solutions are typically found by the mathematical technique called *separation of variables*.
- In *separation of variables* the solution is assumed to be the product format $v(x, t) = X(x)T(t)$. This leads to an eigenvalue problem in the spatial function $X(x)$, whereby the frequency is determined, followed by an initial value problem for the time function $T(t)$.
- The axial motion of an elastic bar is governed by a similar equation with wave speed $c = \sqrt{E/\rho}$, where E is the elastic modulus and ρ is the mass density of the material.
- The transverse motion of a beam is governed by a fourth order differential equation. This implies that waves in beams are dispersive, i.e. the wave speed depends on the wave length. It has been demonstrated that the speed of bending waves c_b is inversely proportional with the wave length λ .
- Beam vibration problems can be solved by the method of separation of variables. For free undamped beam vibrations the time factor is harmonic.
- The wave number k and thereby the frequency ω of beam vibrations are determined from the four boundary conditions of the fourth order spatial differential equation. Apart from the case of a simply supported beam this typically leads to a transcendental frequency equation containing trigonometric and hyperbolic functions. These equations can be solved by fixed point iteration - i.e. by repeated substitution of new iterates.
- The calculation of vibration frequencies and shape functions are often facilitated by use of a set of base functions combining trigonometric and hyperbolic functions.
- Damping of beams can be represented by the two Rayleigh terms relating to motion and straining, respectively.

Most dynamic structural systems are continuous. In principle this implies that they have infinitely many modes and can vibrate with infinitely high frequencies. However, most engineering applications are concerned with problems within a finite frequency range, and in practice the major part of structural dynamics analysis is carried out by analyzing discrete systems, representing an approximation to the original continuous system. Thus, the nature of such approximations and the range of their validity is important in order to use them with confidence.

Within structural and solid mechanics the Finite Element Method - often abbreviated FEM - is by far the most used technique to obtain discretized models. As a consequence there is a vast literature on the Finite Element Method, including its use within the area of Structural Dynamics. The present note uses the simple theories for cables, bars and beams to illustrate some basic features shared by most finite element formulations. These are: the derivation of the finite element equations by use of shape functions and the principle of virtual work, the decrease of accuracy of the finite element model with increasing frequency, and estimating the highest natural frequency of a dynamic system from the free vibrations of the most critical element.

3.1 Finite element formulation for cables and bars

The equation of transverse motion $v(x, t)$ of a taut cable is

$$\frac{\partial}{\partial x} \left(H \frac{\partial v}{\partial x} \right) - m \frac{\partial^2 v}{\partial t^2} + g(x, t) = 0 \quad (3.1)$$

where H is the axial tension component, m is the mass per unit length, and $g(x, t)$ is the transverse load per unit length. A cable vibration problem requires specification of suitable initial conditions $v(x, 0) = v_0(x)$ and $v_t(x, 0) = \dot{v}_0(x)$, as well as a boundary condition at each end of the cable.

The axial motion $u(x, t)$ of an elastic bar is described by the similar equation

$$\frac{\partial}{\partial x} \left(EA \frac{\partial u}{\partial x} \right) - \rho A \frac{\partial^2 u}{\partial t^2} + f(x, t) = 0 \quad (3.2)$$

where EA is the axial elastic stiffness, ρA is the mass per unit length, and $f(x, t)$ is the axial load per unit length. Also vibrations of an elastic bar involves suitable initial conditions $u(x, 0) = u_0(x)$ and $u_t(x, 0) = \dot{u}_0(x)$, as well as a boundary condition at each end of the bar.

It is clear that the mathematical form of these two equations is identical, and therefore the following derivations will be based on the cable equation, as transverse motion conforms most closely to the illustrations of the displacement functions.

3.1.1 Virtual work formulation

The equation of motion is a force balance equation, valid at each point x . It can be transformed into a work balance equation by multiplication with a virtual displacement $\tilde{v}(x)$, followed by integration over the length of the cable, L .

$$\int_0^L \tilde{v} \left\{ \frac{\partial}{\partial x} \left(H \frac{\partial v}{\partial x} \right) - m \frac{\partial^2 v}{\partial t^2} + g \right\} dx = 0 \quad (3.3)$$

The first term is rewritten via integration by parts

$$\int_0^L \tilde{v} \frac{\partial}{\partial x} \left(H \frac{\partial v}{\partial x} \right) dx = \left[\tilde{v} \left(H \frac{\partial v}{\partial x} \right) \right]_0^L - \int_0^L \frac{\partial \tilde{v}}{\partial x} H \frac{\partial v}{\partial x} dx \quad (3.4)$$

When this is substituted into the original equation, and the derivatives are indicated by dash and dot symbols, it takes the form

$$\int_0^L \tilde{v}' H v' dx + \int_0^L \tilde{v} m \ddot{v} dx = \left[\tilde{v} (H v') \right]_0^L + \int_0^L \tilde{v} g dx \quad (3.5)$$

This equation must be satisfied for any choice of the virtual displacement function $\tilde{v}$, that permits calculation of the derivative $\tilde{v}'$.

The variation form is often called ‘weak form’ because it imposes only weak conditions on the functions involved. While the original force balance equation makes use of the second derivative v'' and requires the derivative H' of the stiffness, the variation form only requires the first derivative v' , and does not involve any derivative of the stiffness H explicitly.

Shape functions

The simplest representation of a taut cable via finite elements is shown in Fig. 3.1. The cable of total length L is divided into n equal *elements* of length $l = L/n$. The elements are connected at the $n + 1$ *nodes*, and the motion of the cable is described by interpolation between the motion at the nodes, $v_1(t), v_2(t), \dots, v_{n+1}(t)$.

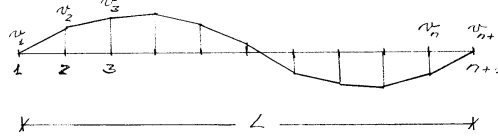


Fig. 3.1: Discrete representation of taut cable by n elements.

The representation of the displacement function in terms of the motion of the nodes has the form

$$v(x, t) = \sum_{j=1}^{n+1} N_j(x) v_j(t) \quad (3.6)$$

In this representation the displacement generated by the displacement $v_j(t)$ at node j is described by the time-independent *shape function* $N_j(x)$.

When the nodal displacements $v_j(t)$ and the shape functions $N_j(x)$ are collected in a column and row vector, respectively,

$$\mathbf{v}(t) = [v_1(t), \dots, v_{n+1}(t)]^T, \quad \mathbf{N}(x) = [N_1(x), \dots, N_{n+1}(x)] \quad (3.7)$$

the time-dependent motion of the cable can be written in the form

$$v(x, t) = \mathbf{N}(x) \mathbf{v}(t) \quad (3.8)$$

where the summation has been replaced by a matrix product.

The virtual displacement field is represented in a similar way, using the same shape functions,

$$\tilde{v}(x) = \mathbf{N}(x) \tilde{\mathbf{v}} = \tilde{\mathbf{v}}^T \mathbf{N}(x)^T \quad (3.9)$$

The second form is obtained by transposing the matrix product, bringing the virtual nodal displacements out in front.

When these representations of the displacement field and the virtual displacement field in terms of the same shape functions are substituted into the virtual work equation (3.5), it takes the following form

$$\tilde{\mathbf{v}}^T \left\{ \mathbf{K} \mathbf{v} + \mathbf{M} \ddot{\mathbf{v}} - \mathbf{g} \right\} = 0 \quad (3.10)$$

where any boundary loads have been included in the load vector $\mathbf{g}$.

In this equation the *stiffness matrix*

$$\mathbf{K} = \int_0^L H \mathbf{N}'^T \mathbf{N}' dx \quad (3.11)$$

and the *mass matrix*

$$\mathbf{M} = \int_0^L m \mathbf{N}^T \mathbf{N} dx \quad (3.12)$$

follow from the left hand side of (3.5). Note, that both stiffness and mass matrix are symmetric $n + 1$ by $n + 1$ matrices in the present case.

The load vector $\mathbf{g}$ corresponds to the right hand side of (3.5). It is noted that in the boundary term in the square brackets Hv' is the transverse force. The boundary loads may therefore be included in the load $g(x, t)$ as concentrated contributions at the ends $x = 0, L$. The simplified form of the *load vector* then is

$$\mathbf{g}(t) = \int_0^L \mathbf{N}^T g(x, t) dx \quad (3.13)$$

3.1.2 Discretized equations of motion

The discretized equations of motion follow from the scalar equation (3.10) by observing that this equation should be satisfied for *any* choice of the virtual displacement vector $\tilde{\mathbf{v}}$. Thus the contribution from each component can be taken one at a time, leading to the following set of $n + 1$ equations

$$\mathbf{K} \mathbf{v} + \mathbf{M} \ddot{\mathbf{v}} = \mathbf{g}(t) \quad (3.14)$$

This is the well known matrix equation system for discrete dynamic systems, analyzed through the initial part of the course.

Note, that the content of the stiffness and mass matrix depends on the shape functions used, as well as the number of elements used in the representation of the structure.

Initial conditions are introduced directly on the motion of the nodes by prescribing $\mathbf{v}(0)$ and $\dot{\mathbf{v}}(0)$. Typical boundary conditions are free, in which case the node act like an internal node, and fully constrained, in which case the corresponding displacement component is eliminated from the system of equations together with the corresponding equation of motion.

In most finite element formulations, like the present, the shape functions are independent of time and frequency, and are therefore typically determined by the simplest interpolation that is consistent with the displacement of the nodes. This gives the method its simplicity, but also limits its accuracy at higher frequencies, where the shape functions may not represent the dynamic modes with sufficient accuracy.

Assembly of structure from individual elements

The representation of the motion of the cable $v(x, t)$ in the form of linear interpolation between the nodes implies that the shape functions introduced in (3.6) and (3.8) are of the form illustrated in Fig. 3.2. These interpolation functions are highly local, involving only the (two) elements connected to the node in question. In principle it is possible to evaluate the stiffness matrix and mass matrix directly from these so-called global shape functions. However, a main feature of the finite element method is the change from global form to a form in which each element is considered separately, and the properties of each individual element is then assembled to form the full structure.

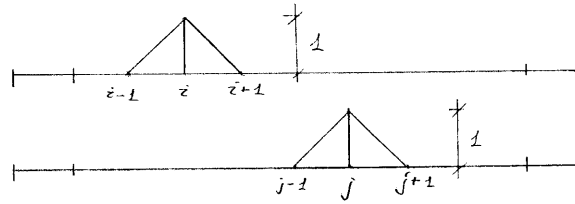


Fig. 3.2: Global linear shape functions $N_i(x)$ and $N_j(x)$.

The basic idea is to form the integrals (3.11)-(3.13) for stiffness, mass, and load for each element separately, and then assemble the element contributions

afterwards. In the case of the mass matrix (3.12) this amounts to considering the sum of element integrals

$$\mathbf{M} = \sum_{\text{elem}} \int_0^l m \mathbf{N}_e^T \mathbf{N}_e dx \quad (3.15)$$

where $\mathbf{N}_e(x)$ denotes the part of the shape function associated with the individual element, and l is the element length.

The local element shape functions N_i^{elem} and N_j^{elem} are shown in Fig. 3.3 for the case where i and j denote the left and right end of the element, whereby $j = i + 1$.

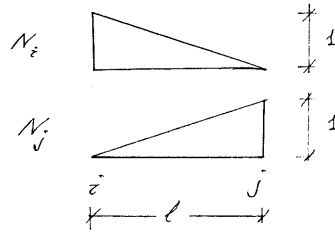


Fig. 3.3: Local element shape functions $N_i(x)$ and $N_j(x)$.

The element contribution to the mass matrix follows by substituting the linear shape functions from Fig. 3.3 into one of the terms in (3.15),

$$\mathbf{M}_{\text{elem}} = \int_0^l m \begin{bmatrix} N_i(x)N_i(x) & N_i(x)N_j(x) \\ N_j(x)N_i(x) & N_j(x)N_j(x) \end{bmatrix} dx = \frac{ml}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (3.16)$$

where ml is the total mass of the element. The numbers follow from well known integrals of products of triangles. Note, that the ‘consistent’ integration of the uniform mass distribution leads to off-diagonal terms in the mass matrix.

The element contribution to the stiffness matrix follows for the similar expression with m replaced by H , and the shape functions $N(x)$ by their derivatives $N'(x)$,

$$\mathbf{K}_{\text{elem}} = \int_0^l H \begin{bmatrix} N'_i(x)N'_i(x) & N'_i(x)N'_j(x) \\ N'_j(x)N'_i(x) & N'_j(x)N'_j(x) \end{bmatrix} dx = \frac{H}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (3.17)$$

The coefficient follow from observing that the derivatives are constants of magnitude $\pm 1/l$.

Assembly of the element matrices is based on the principle, that the displacement v_i at a node connecting two elements is common, while the corresponding force contributions to the equation of motion must be added. This leads to a 'stacking procedure', in which all element contributions involving global node numbers i and j are simply added to the global matrices at positions ii , ij , ji and jj .

In the simple cable problem of Fig. 3.1 all elements have $j = i + 1$, and thus all element matrices are of magnitude 2×2 and are stacked along the matrix diagonal. For n elements there are $n + 1$ nodes, and the complete stiffness matrix is the following $(n + 1) \times (n + 1)$ matrix,

$$\mathbf{K} = \frac{H}{l} \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 1 \end{bmatrix} \quad (3.18)$$

Similarly, the mass matrix computed from the shape functions by (3.16) gives the *consistent mass matrix*

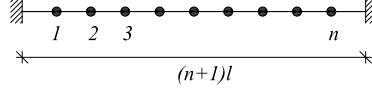
$$\mathbf{M}_c = \frac{ml}{6} \begin{bmatrix} 2 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & 4 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & 2 \end{bmatrix} \quad (3.19)$$

At times the mass matrix is calculated simply by distributing the element mass equally between the ends, the so-called *lumped mass matrix*,

$$\mathbf{M}_l = ml \begin{bmatrix} \frac{1}{2} & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & \frac{1}{2} \end{bmatrix} \quad (3.20)$$

Note, that the matrices given here include all degrees of freedom. When boundary conditions are introduced some of the degrees of freedom are typically eliminated or constrained.

Example 3.1. FEM for cable vibrations. The figure shows a cable with $n + 1$ elements of length l , corresponding to n internal nodes with transverse displacements $v_1, \dots, v_n$.

Fig. 3.4: Cable vibrations with n degrees of freedom

The full system also includes nodes $j = 0$ and $j = n + 1$ at the end points of the cable. The displacement at these points is constrained to $v_0 = v_{n+1} = 0$, and thus these variables as well as the corresponding equations are removed from the system and matrices. This corresponds to removing the first and last row and column from the stiffness and mass matrices. Thus the $n \times n$ stiffness matrix of the cable is

$$\mathbf{K} = \frac{H}{l} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 \end{bmatrix} \quad (3.21)$$

while the consistent and lumped $n \times n$ mass matrices are

$$\mathbf{M}_c = \frac{ml}{6} \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & 4 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & 4 \end{bmatrix}, \quad \mathbf{M}_l = ml \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} \quad (3.22)$$

The angular frequency ω_i of the free vibration modes of the cable were given in (2.18) as

$$\omega_i = i \frac{\pi}{L} \sqrt{\frac{H}{m}}, \quad i = 1, 2, \dots \quad (3.23)$$

where $L = (n + 1)l$ is the total length of the cable.

The accuracy of the finite element model can be estimated by calculating the natural frequencies of the discrete model from the $n \times n$ eigenvalue problem

$$(\mathbf{K} - \omega_i^2 \mathbf{M}) \mathbf{v}_i = \mathbf{0} \quad (3.24)$$

where $\mathbf{M}$ may be either the ‘consistent’ or the lumped mass matrix.

Figure 3.5 shows the normalized natural frequencies ω_i/ω_1 , $i = 1, \dots, n$ for the exact solution, and a 10 element model with a consistent or a lumped mass matrix. It is seen that only the first three frequencies are calculated with reasonable accuracy by the finite element models. A characteristic feature is that the consistent mass matrix leads to overestimation of the natural frequencies, while a lumped mass matrix leads to underestimation. $\square$

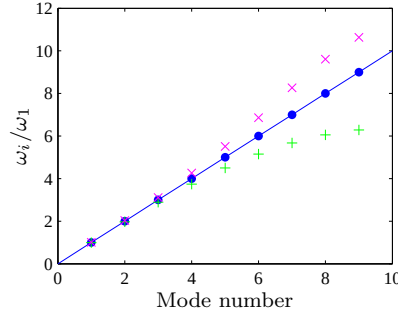


Fig. 3.5: Cable vibration frequencies. • exact, × consistent mass, + lumped mass.

3.1.3 Estimating the highest natural frequency

The highest vibration frequency of a structure is important, when using a conditionally stable integration algorithm. It can be shown that the maximum free vibration frequency is bounded from above by the highest vibration frequency of any of the structures individual elements, when they vibrate without support, see e.g. Hughes et al. (1979) or Hughes (1987). Usually the highest individual element vibration frequency gives a good estimate.

Example 3.2. Bar or cable element. For a simple cable or elastic bar element the vibration modes of a single element without support are $\mathbf{v}_1 = [1, 1]^T$ and $\mathbf{v}_2 = [1, -1]^T$. The mode $\mathbf{v}_1$ corresponds to uniform translation, and as there is no deformation the frequency of this rigid body mode is zero. Thus, the maximum frequency is associated with the mode $\mathbf{v}_2$.

For an element with consistent mass matrix, the element matrices are given by (3.16) and (3.17),

$$\mathbf{K} = \frac{H}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{M}_c = \frac{ml}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (3.25)$$

The lumped element matrix is formed by distributing the total element mass ml equally at the two nodes.

$$\mathbf{M}_l = \frac{ml}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (3.26)$$

As the mode shape vector is known, the frequency can be determined directly from Rayleigh's quotient.

For the consistent mass element

$$\omega_{max}^2 = \frac{\mathbf{v}_2^T \mathbf{K} \mathbf{v}_2}{\mathbf{v}_2^T \mathbf{M}_c \mathbf{v}_2} = \frac{6H}{ml^2} \frac{[1, -1] \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}{[1, -1] \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}} = \frac{12H}{ml^2} \quad (3.27)$$

whereby

$$\omega_{max}^c = \frac{\sqrt{12}}{l} \sqrt{\frac{H}{m}} = \frac{3.464}{l} \sqrt{\frac{H}{m}} \quad (3.28)$$

For the lumped mass element

$$\omega_{max}^2 = \frac{\mathbf{v}_2^T \mathbf{K} \mathbf{v}_2}{\mathbf{v}_2^T \mathbf{M}_l \mathbf{v}_2} = \frac{2H}{ml^2} \frac{\begin{bmatrix} 1, & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}{\begin{bmatrix} 1, & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}} = \frac{4H}{ml^2} \quad (3.29)$$

whereby

$$\omega_{max}^l = \frac{2}{l} \sqrt{\frac{H}{m}} \quad (3.30)$$

Thus, the lumped mass element has a lower vibration frequency in accordance with the results of Fig. 3.5. Therefore the lumped mass element permits longer time steps in conditionally stable time integration algorithms. $\square$

3.2 Finite element formulation for beams

The *bending vibrations of beams* are described by the differential equation of motion

$$\frac{\partial^2}{\partial x^2} \left(EI \frac{\partial^2 w}{\partial x^2} \right) + \rho A \frac{\partial^2 w}{\partial t^2} - p(x, t) = 0 \quad (3.31)$$

where $w(x, t)$ is the transverse displacement field, EI is the elastic bending stiffness, ρA is the mass per unit length, and $p(x, t)$ is the intensity of the transverse load.

This equation requires *initial conditions*, prescribing displacement $w(x, 0) = w_0(x)$ and velocity $w_t(x, 0) = \dot{w}_0(x)$.

In addition, two *boundary conditions* must be prescribed at each end of the beam. The boundary conditions are formulated in terms of

$$w, \quad \theta = -w', \quad M = -EI w'', \quad Q = -(EI w''')' \quad (3.32)$$

where M is the bending moment, and Q is the transverse shear force.

In the following the transverse force balance equation (3.31) for transverse beam vibrations is reformulated as an energy balance equation, the displacement field is represented in terms of shape functions, and the stiffness and mass matrix of a beam element are obtained.

3.2.1 Virtual work formulation

A virtual work equation is obtained by multiplication of the force balance equation (3.31) with a virtual displacement function $\tilde{w}$ and integration over the length of the beam L .

$$\int_0^L \tilde{w} \left\{ (EI w'')'' + \rho A \ddot{w} - p \right\} dx = 0 \quad (3.33)$$

The first term is rewritten by integration by parts,

$$\int_0^L \tilde{w} (EI w'')'' dx = \left[\tilde{w} (EI w'')' \right]_0^L - \int_0^L \tilde{w}' (EI w'')' dx \quad (3.34)$$

The integral on the right hand side is integrated further by parts, leading to the relation

$$\int_0^L \tilde{w} (EI w'')'' dx = \left[\tilde{w} (EI w'')' - \tilde{w}' (EI w'') \right]_0^L + \int_0^L \tilde{w}'' EI w'' dx \quad (3.35)$$

It is noted that the first term in the square brackets can be expressed in terms of the shear force $Q = -(EI w'')'$, and the second term in terms of the bending moment $M = -EI w''$. Introduction of this and the last relation into (3.33) gives the virtual work equation

$$\int_0^L \left\{ \tilde{w}'' EI w'' + \tilde{w} \rho A \ddot{w} \right\} dx = \left[\tilde{w} Q - \tilde{w}' M \right]_0^L + \int_0^L \tilde{w} p dx \quad (3.36)$$

This equation must be satisfied for any choice of the virtual displacement function $\tilde{w}$, and a discrete formulation is obtained by making a finite representation of the displacement $w(x, t)$ as well as of $\tilde{w}$.

Shape function representation

The virtual work formulation assumes the displacement $w(x, t)$ and the rotation $\theta = w'(x, t)$ to be continuous functions of x . Both w and θ are therefore represented explicitly at the nodes. The global shape function representation then is

$$w(x, t) = \sum_j^n \left(N_j^w(x) w_j(t) + N_j^\theta(x) \theta_j(t) \right) \quad (3.37)$$

Also in this case the shape function representation is conveniently written in the form of a matrix product

$$w(x, t) = \mathbf{N}(x) \mathbf{w}(t) \quad (3.38)$$

where the nodal displacements $w_j(t), \theta_j(t)$ are collected in a column vector,

$$\mathbf{w}(t) = [w_1(t), \theta_1(t), \dots, w_n(t), \theta_n(t)]^T \quad (3.39)$$

while the corresponding shape functions are collected in a row vector,

$$\mathbf{N}(x) = [N_1^w(x), N_1^\theta(x), \dots, N_n^\theta(x)] \quad (3.40)$$

Note, that there are now two displacement components $w_j(t), \theta_j(t)$ at each node. This ensures spatial continuity of the displacement $w(x, t)$ and its derivative $w'(x, t)$, and provides for two boundary conditions at each end.

Discretized equations of motion

The discretized equations of motion follow from substitution of the shape function representations of the displacement field $w(x, t)$ and the virtual displacement $\tilde{w}(x)$ into the virtual work equation (3.36). After the usual argument of arbitrary virtual displacement component vector $\tilde{\mathbf{w}}$, the equations of motion are obtained in the form

$$\mathbf{K} \mathbf{w} + \mathbf{M} \ddot{\mathbf{w}} = \mathbf{p}(t) \quad (3.41)$$

which is a matrix equation system of size $2n \times 2n$ for a beam bending problem with n nodes.

In this equation the *stiffness matrix*

$$\mathbf{K} = \int_0^L EI \mathbf{N}''^T \mathbf{N}'' dx \quad (3.42)$$

and the *mass matrix*

$$\mathbf{M} = \int_0^L \rho A \mathbf{N}^T \mathbf{N} dx \quad (3.43)$$

follow from the left hand side of (3.36). Note, that both stiffness and mass matrix are symmetric $2n$ by $2n$ matrices in the present case.

The load vector $\mathbf{p}(t)$ corresponds to the right hand side of (3.36). Any loads at the ends of the beam are included in $p(x, t)$ as concentrated contributions at the ends $x = 0, L$, and the *load vector* can then be written as

$$\mathbf{f}(t) = \int_0^L \mathbf{N}^T p(x, t) dx \quad (3.44)$$

3.2.2 The beam bending element

When using the finite element method, the contributions to mass, stiffness and distributed loads are evaluated at the element level, and the shape functions are therefore required within the individual element. The shape functions of the beam bending element are shown in Fig. 3.6.

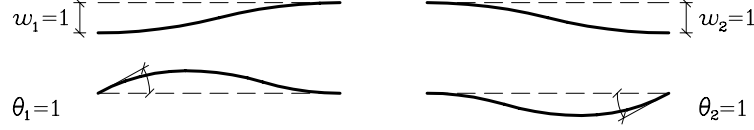


Fig. 3.6: Shape functions for imposed unit end motions $w_1, \theta_1, w_2, \theta_2$.

The shape functions, appearing as factors to $w_1, \theta_1, w_2, \theta_2$ and shown in Fig. 3.6, are given by

$$\begin{aligned} N_1(x) &= \left(1 + 2\frac{x}{l}\right)\left(1 - \frac{x}{l}\right)^2, & N_2(x) &= -\frac{x}{l}\left(1 - \frac{x}{l}\right)^2 l \\ N_3(x) &= \left(\frac{x}{l}\right)^2\left(3 - 2\frac{x}{l}\right), & N_4(x) &= \left(\frac{x}{l}\right)^2\left(1 - \frac{x}{l}\right) l \end{aligned} \quad (3.45)$$

where l is the length of the element.

Substitution of the element shape functions $N_1(x), \dots, N_4(x)$ into (3.42) and (3.43) gives the beam bending element stiffness matrix and mass matrix, respectively.

$$\mathbf{K}_{elem} = \frac{EI}{l^3} \begin{bmatrix} 12 & -6l & -12 & -6l \\ -6l & 4l^2 & 6l & 2l^2 \\ -12 & 6l & 12 & 6l \\ -6l & 2l^2 & 6l & 4l^2 \end{bmatrix} \quad (3.46)$$

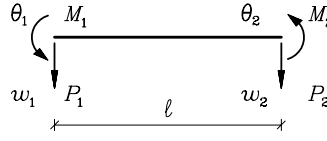
and

$$\mathbf{M}_{elem} = \frac{\rho Al}{420} \begin{bmatrix} 156 & -22l & 54 & 13l \\ -22l & 4l^2 & -13l & -3l^2 \\ 54 & -13l & 156 & 22l \\ 13l & -3l^2 & 22l & 4l^2 \end{bmatrix} \quad (3.47)$$

Note, that the signs occurring in the beam bending matrices depend on the sign convention of the displacement components. The present matrix corresponds to the displacement components defined as shown in Fig. 3.7

Example 3.3. Simple beam vibrations.

Figure 3.8 shows the symmetric vibration of a simply supported beam element. The mode of the beam element is defined by $\mathbf{w}^T = [w_1, \theta_1, w_2, \theta_2] = [0, 1, 0, -1]$.

Fig. 3.7: Homogeneous beam with end loads (P_1, M_1, P_2, M_2) .Fig. 3.8: Simple beam element vibration with $\theta_1 = -\theta_2 = \theta$.

The mode shape is known, and the frequency of vibration can therefore be determined from Rayleigh's quotient,

$$\omega^2 = \frac{\mathbf{w}^T \mathbf{K} \mathbf{w}}{\mathbf{w}^T \mathbf{M} \mathbf{w}} \quad (3.48)$$

The element stiffness and mass matrices are given in (3.46) and (3.47), respectively. In terms of condensed matrices with only the rotation components the calculation is

$$\omega^2 = \frac{420EI}{\rho A l^4} \frac{\begin{bmatrix} 1, & -1 \end{bmatrix} \begin{bmatrix} 4l^2 & 2l^2 \\ 2l^2 & 4l^2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}{\begin{bmatrix} 1, & -1 \end{bmatrix} \begin{bmatrix} 4l^2 & -3l^2 \\ -3l^2 & 4l^2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}} = \frac{120EI}{\rho A l^4} \quad (3.49)$$

Hereby the approximate angular frequency is

$$\omega = \frac{\sqrt{120}}{l^2} \sqrt{\frac{EI}{\rho A}} = \frac{10.95}{l^2} \sqrt{\frac{EI}{\rho A}} \quad (3.50)$$

The exact result has been found previously in (11.24) as

$$\omega = \frac{\pi^2}{l^2} \sqrt{\frac{EI}{\rho A}} = \frac{9.87}{l^2} \sqrt{\frac{EI}{\rho A}} \quad (3.51)$$

The error is about 10%, and this confirms the common observation, that the finite element model should have at least double as many degrees of freedom as the desired number of frequencies. $\square$

3.3 Exercises

Summary

- The Finite Element Method is a method for reducing continuum problems described in terms of differential equations to discrete equations. In Structural Dynamics the Finite Element Method is often used to reduce the spatial part of the problem to discrete form, leaving a finite number of ordinary differential equations in the time variable. These equations describe inertial effects via the mass matrix $\mathbf{M}$ and the stiffness of the structure via the stiffness matrix $\mathbf{K}$.
- Finite element models are obtained by reducing to differential equations to weak form - typically a process analogous to virtual work. The system matrices and load vector then follow from substitution of a shape function representation of the displacement distribution in space.
- While loads follow directly from this formulation, displacement boundary conditions must be introduced by elimination of selected displacement variables, or by introduction of supports by springs.
- The highest frequency of a structure can be estimated from the highest frequency of any single element vibrating without supports. This frequency is important for time step limitations on conditionally stable time integration procedures.
- The vibration modes corresponding to the higher frequencies in a structural model are typically represented with less accuracy than the low frequencies. A rule of thumb is that approximately the lower half of the vibration modes are represented with sufficient accuracy.

The dynamic behavior of most structures involve simultaneous motion of several masses in shapes that are not known before the analysis. Thus, the theory of dynamics of a single degree of freedom must be extended to deal with several masses, and also with systems with distributed mass like beam and frame structures, as well as complete buildings and structures like bridges, towers and offshore platforms.

The first extension of the theory is from one degree of freedom involving a single mass to structures with multiple degrees of freedom, describing the motion of the structure. The first step is to identify the vibration shapes associated with free undamped vibrations. These so-called mode-shapes or vibration modes satisfy orthogonality relations that ensure that vibrations in each of the modes occur independently. The vibration modes are also associated with a characteristic mass and a characteristic stiffness, the modal mass and modal stiffness, respectively.

The next step is the so-called modal analysis, in which damping is introduced and the the structure is loaded. The total load is decomposed into modal components, each associated with one particular mode, and the dynamic problem essentially decomposes into an uncoupled set of modal response problems, each corresponding closely to the response problem for a single degree of freedom system treated in Chapter 1. In addition to providing a method of analysis of the dynamic response of a multi degree of freedom structure the modal analysis also provides a basic understanding of the properties of the structure that determine its dynamic response characteristics.

Computational models of structures often contain thousands of degrees of freedom, and a full modal analysis can then be computationally quite heavy, if all vibration modes are included. Reduction techniques by which the number of modes that are included in the dynamic analysis are therefore an important part of the modal analysis technique. Section 4.5 describes the simple static condensation technique, in which the inertial forces associated with some of the degrees of freedom are neglected, and the generally more accurate quasi-static correction method, in which only the inertial forces associated with high-frequency modes are neglected.

The chapter concludes with a brief discussion of the equations of motion in a rotating coordinate system, where Coriolis forces play a role. In particular a hybrid formulation in terms of local displacement components and global velocity components is described, and the associated vibration problem is discussed.

4.1 Equations of motion

The linear equations of motion of a structure are formed by a balance between an external force represented by the vector $\mathbf{f}$ and internal forces from stiffness, mass and damping. The stiffness contribution is a linear combination of the current displacements $\mathbf{q}$ of the form $\mathbf{K}\mathbf{q}$, and the inertial forces are similarly a linear combination of contributions from the accelerations $\ddot{\mathbf{q}}$ of the form $\mathbf{M}\ddot{\mathbf{q}}$. Both of these contributions are typically represented via the finite element method as discussed in Chapter 3. In addition to these two contributions most applications also require a representation of the structural damping. While elastic and inertial forces can often be represented with reasonable accuracy, it is more difficult to obtain a realistic representation of damping without complicating the equations of motion considerably. As a consequence damping is often introduced in the form of linear viscous damping, proportional to the velocities $\dot{\mathbf{q}}$. The resulting multi-degrees of freedom equations of motion take the form

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{f}(t). \quad (4.1)$$

In addition to the system parameters and the load history, solution of the dynamic response problem also requires specification of two sets of initial conditions, here denoted

$$\mathbf{q}(0) = \mathbf{q}_0, \quad \dot{\mathbf{q}}(0) = \dot{\mathbf{q}}_0, \quad (4.2)$$

where $\mathbf{q}_0$ and $\dot{\mathbf{q}}_0$ represent the initial position and the initial velocity of each degree of freedom.

The structural model is represented by the stiffness, mass and damping matrices $\mathbf{K}$, $\mathbf{M}$ and $\mathbf{C}$. In practice these matrices are most often determined via a finite element model of the structure. However, in order to illustrate the basic structure of the theory it is often convenient to make use of simple qualitative models in which concentrated masses are connected by stiffness elements, represented by springs, and concentrated damping elements in the form of viscous dashpots. Often, this type of model can be formulated directly by simple means and illustrated the components of the problem in a clear intuitive manner, as illustrated by the following example.

Example 4.1. Discrete mass-stiffness model. The equations of motion for a typical multi degree of freedom structure are introduced via the simple examples shown in Fig. 4.1. The figure to the left shows two masses m_1 and m_2 that can move horizontally as described by the coordinates x_1 and x_2 . The mass m_1 is connected to a fixed point with a spring k_1 and connected to the other mass by a spring k_2 . The figure also shows the forces developed by extension/compression of the springs. The figure to the right shows a simple ‘shear building’, represented by two floors with mass m_1 and m_2 , respectively. The motion of the floors is described by the coordinates x_1 and x_2 . The motion of the floors is counteracted by shear forces developed in the frame/walls forming the building. The forces are assumed to have an elastic component proportional with the relative motion between floors and a viscous component proportional with the difference in velocity between the floors as indicated in the lower part of the figure. It is seen that in both models the forces acting on the two masses correspond, and thus the resulting equations of motion are identical for the two systems.

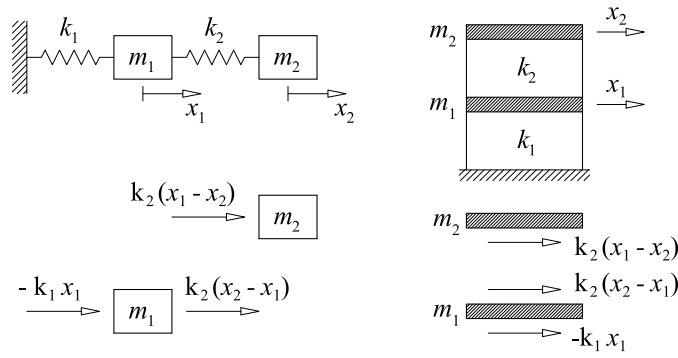


Fig. 4.1: a) Series of masses connected by springs, b) Idealized shear building.

In the present simple illustration the equations of motion are obtained by identifying all force components that contribute to the acceleration of each of the two masses. The acceleration of mass m_1 is given by

$$m_1 \ddot{q}_1 = f_1 - k_1 q_1 + k_2 (q_2 - q_1),$$

while the similar equation for acceleration of mass m_2 is

$$m_2 \ddot{q}_2 = f_2 - k_2 (q_2 - q_1),$$

These two equations of motion can be collected in the form

$$\begin{aligned} m_1 \ddot{q}_1 &+ (k_1 + k_2) q_1 - k_2 q_2 = f_1, \\ m_2 \ddot{q}_2 &- k_2 q_1 + k_2 q_2 = f_2, \end{aligned}$$

where the first term containing the mass is the inertial force, while the second term is the and elastic spring forces. These equations take the form (4.1), when representing the displacements and the by the vectors $\mathbf{q} = [q_1, q_2]^T$ and $\mathbf{f} = [f_1, f_2]^T$, and the system properties by the matrices

$$\mathbf{M} = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}.$$

Various forms of these equations will be used in the following to illustrate dynamic properties of structures. $\square$

4.1.1 Energy balance

The multi-degree of freedom equation of motion (4.1) is associated with an energy balance equation. This relation is found by considering the work performed in the individual terms in an incremental motion $d\mathbf{q} = \dot{\mathbf{q}}dt$. When leaving out the time increment dt this corresponds to pre-multiplication of the equation of motion by the transpose of the velocity vector $\dot{\mathbf{q}}$,

$$\dot{\mathbf{q}}^T (\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q}) = \dot{\mathbf{q}}^T \mathbf{f}. \quad (4.3)$$

The mass and stiffness matrices are symmetric, and it is then easily verified that the inertial and stiffness terms can be integrated. Hereby the energy balance equation takes the form

$$\frac{d}{dt} \left(\frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^T \mathbf{K} \mathbf{q} \right) = \dot{\mathbf{q}}^T \mathbf{f} - \dot{\mathbf{q}}^T \mathbf{C} \dot{\mathbf{q}}. \quad (4.4)$$

The two terms on the left side are recognized as the kinetic and internal elastic energy, respectively,

$$T(\dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}}, \quad U(\mathbf{q}) = \frac{1}{2} \mathbf{q}^T \mathbf{K} \mathbf{q}. \quad (4.5)$$

It follows from these expressions that the mass and stiffness must be positive semi-definite. The mass matrix $\mathbf{M}$ will be only positive semi-definite if the model includes parts, that are represented as mass-less. Similarly the stiffness matrix $\mathbf{K}$ will be fully positive definite if the structure does not contain moving parts, but is fully supported against rigid-body motion. In fact, finite element models are typically formed by first assembling all elements including also supported degrees of freedom. The corresponding stiffness matrix then permits rigid body motion and is only positive semi-definite. When supports are introduced in the model, the resulting stiffness matrix becomes positive definite, when the supports prevent rigid body motion.

The two terms on the right side represent external energy input per time unit W and dissipated energy per time unit D , respectively,

$$W = \dot{\mathbf{q}}^T \mathbf{f}, \quad D(\dot{\mathbf{q}}) = \dot{\mathbf{q}}^T \mathbf{C} \dot{\mathbf{q}}. \quad (4.6)$$

It is seen that only the symmetric part of the matrix $\mathbf{C}$ contributes to the energy dissipation, and the damping matrix $\mathbf{C}$ is therefore defined as being symmetric. For structures in a moving frame of reference there may also be gyroscopic forces associated with the velocity vector via a non-symmetric matrix. This class of problems is discussed further in Section ??.

The energy balance equation (4.4) can now be expressed in the generic form

$$\frac{d}{dt}(T + U) = W - D. \quad (4.7)$$

This form also holds for non-linear structures, and will be discussed in connection with numerical integration in Chapter 6.

In the particular case where the external force $\mathbf{f}$ is conservative, it can be derived from an external potential $V(\mathbf{q})$ in the form

$$\mathbf{f} = -\frac{\partial V}{\partial \mathbf{q}^T}. \quad (4.8)$$

In this formula $\mathbf{f}$ is a column vector, and thus by implication the derivative $\partial V/\partial \mathbf{q}$ is a row vector. This notation is commonly used as it gives the differential increment the simple form

$$dV = \frac{\partial V}{\partial \mathbf{q}} d\mathbf{q}. \quad (4.9)$$

with the derivative $\partial V/\partial \mathbf{q}$ as the first factor in the form of a row vector, followed by the differential increment $d\mathbf{q}$ in the form of a column vector. When all external forces are conservative the external work may be integrated, and the energy balance equation then takes the form

$$\frac{d}{dt}(T + U + V) = -D. \quad (4.10)$$

In this form the energy balance equation expresses that the rate of change of the total energy, consisting of the kinetic energy T the internal energy U plus the external potential energy V , is equal to the rate of dissipation expressed by D .

4.2 Free vibrations

The dynamic properties of a structure are to a large extent characterized by its free vibration characteristics. These include the important property that a linear structure can sustain vibrations at particular frequencies with specific shapes, that are retained during the vibration. Naturally a full description of a structure to a time dependent load includes its time variation and distribution over the structure as well as the damping characteristics of the structure. However, the natural vibration frequencies determine the frequencies at which the structure is susceptible to resonance, and the mode shapes at these frequencies determine the load distributions that will excite these resonances. Thus, the analysis of free vibrations discussed in Section 4.2.1 is a prerequisite to identifying the inherent dynamic properties of the struc-

ture. Furthermore, the dynamic response of the structure can be represented in terms of the mode-shapes, leading to independent vibrations of the individual modes as presented in Section 4.2.2. The damping properties of the structure play an important role in relation to the dynamic response. This is analyzed in Section 4.4, where it is demonstrated that for moderately damped structures the damping may be introduced in the form of a damping ratio for each of the modes. This simplifies the analysis and is often used as a practical approximation.

4.2.1 Vibration modes and frequencies

Free undamped vibrations are governed by the equations of motion (4.1) without loading and in the absence of damping, i.e. $\mathbf{C} = \mathbf{0}$. This corresponds to the homogeneous equations

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}. \quad (4.11)$$

A solution to this equation is now sought in the form of a harmonic oscillation with constant shape. When expressing the time variation in complex notation this assumption amounts to a solution in the form

$$\mathbf{q}(t) = \mathbf{u} e^{i\omega t}, \quad (4.12)$$

where ω is an angular frequency and the vector $\mathbf{u}$ defines the mode-shape, that is independent of time.

Substitution of the assumed form of the solution into the homogeneous equation of motion gives the generalized eigenvalue problem

$$(\mathbf{K} - \omega^2 \mathbf{M}) \mathbf{u} = \mathbf{0}. \quad (4.13)$$

Nontrivial solution $\mathbf{u}$ can only be found when the matrix is singular, i.e. when ω^2 is an eigenvalue of the generalized eigenvalue problem. This implies that the determinant of the system matrix is zero,

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0. \quad (4.14)$$

For $n > 2$ the computation of eigenvalues and the corresponding eigenvectors is performed by one of a number of iterative strategies, depending on the character and size of the problem, see e.g. Géradin and Rixen (1997) or Golub and van Loan (1989). The complete solution to the generalized eigenvalue problem consists of n sets of eigenvalue and eigenvector, arranged as corresponding pairs of natural frequency ω_j and mode-shape vector $\mathbf{u}_j$,

$$\omega_j, \quad \mathbf{u}_j, \quad j = 1, 2, \dots, n \quad (4.15)$$

In the case of double or higher order roots, there will be a similar number of eigenvectors $\mathbf{u}_j$ associated with the corresponding natural frequency. This is quite common in structural dynamics, e.g. due to symmetry of the structure. Traditionally the mode-shapes are ordered with respect to increasing magnitude of the associated natural frequency.

Example 4.2. Two story building. Consider the 2 degree of freedom system represented in Fig. 4.1 in the form of a spring-mass system or a two story shear building. The stiffness and mass matrices were given in (4.1), and the corresponding generalized eigenvalue problem is

$$\left(\begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} - \omega^2 \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \right) \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This equation system permits a non-trivial solution, if the determinant of the combined coefficient matrix is zero,

$$\begin{vmatrix} k_1 + k_2 - \omega^2 m_1 & -k_2 \\ -k_2 & k_2 - \omega^2 m_2 \end{vmatrix} = 0.$$

By evaluation of the determinant this condition becomes

$$m_1 m_2 \omega^4 - (m_1 k_2 + m_2 (k_1 + k_2)) \omega^2 + k_1 k_2 = 0.$$

This is a quadratic equation in ω^2 . The solution is expressed in terms of the characteristic angular frequencies $\omega_a^2 = k_1/m_1$ and $\omega_b^2 = k_2/m_2$ and the mass ratio $\mu = m_2/m_1$.

$$2\omega^2 = \omega_a^2 + (1 + \mu)\omega_b^2 \pm \sqrt{(\omega_a^2 - (1 + \mu)\omega_b^2)^2 + 4\mu\omega_a^2\omega_b^2}.$$

In the particular case $m_2 = m_1$, $k_2 = k_1$ the mass ratio is $\mu = 1$ and the solution takes the simple form

$$\left(\frac{\omega_{1,2}}{\omega_a} \right)^2 = \frac{3 \pm \sqrt{5}}{2} = \begin{cases} 0.382 \\ 2.618 \end{cases}$$

corresponding to the natural frequencies

$$\frac{\omega_{1,2}}{\omega_a} = \frac{\sqrt{5} \mp 1}{2} = \begin{cases} 0.618 & (< 1) \\ 1.618 & (> 1) \end{cases}$$

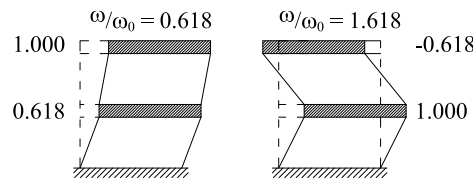


Fig. 4.2: a) Lowest mode $\mathbf{u}_1$, b) Highest mode $\mathbf{u}_2$.

The shape vectors corresponding to these two natural frequencies follow by solving the now singular equations using ω_1 and ω_2 , respectively. The equations are now singular corresponding to proportional, and therefore only determines the ratio between the components of the shape vectors,

$$\frac{u_1}{u_2} = \frac{-1 \pm \sqrt{5}}{2} = \begin{cases} 0.618 \\ -1.618 \end{cases}$$

When normalizing the mode shape vectors such that the largest element is equal to one, they are

$$\mathbf{u}_1 = [0.618, 1.000]^T, \quad \mathbf{u}_2 = [1.000, -0.618]^T$$

These mode shapes are illustrated in Fig. 4.2. Note that the highest frequency corresponds to the most irregular mode shape. $\square$

Orthogonality conditions

The mode-shape vectors $\mathbf{u}_j$ and $\mathbf{u}_k$ satisfy the generalized eigenvalue problem (4.13) with the natural frequencies ω_j and ω_k , respectively. This is expressed by the equations

$$(\mathbf{K} - \omega_j^2 \mathbf{M}) \mathbf{u}_j = \mathbf{0}, \quad (4.16)$$

$$(\mathbf{K} - \omega_k^2 \mathbf{M}) \mathbf{u}_k = \mathbf{0}. \quad (4.17)$$

A set of orthogonality relations with respect to the mass matrix $\mathbf{M}$ can be obtained by pre-multiplication of (4.16) with $\mathbf{u}_k^T$ and (4.17) with $\mathbf{u}_j^T$, followed by subtraction.

$$(\omega_j^2 - \omega_k^2) \mathbf{u}_j^T \mathbf{M} \mathbf{u}_k = 0. \quad (4.18)$$

For distinct eigenvalues, $\omega_k \neq \omega_j$, it follows from this relation that

$$\mathbf{u}_j^T \mathbf{M} \mathbf{u}_k = 0, \quad j \neq k \quad (4.19)$$

If the natural frequencies ω_j and ω_k are identical, the corresponding eigenvectors lie in a subspace from which orthogonal eigenvectors $\mathbf{u}_j^T$ and $\mathbf{u}_k^T$ can be obtained.

A similar orthogonality relation with respect to the stiffness matrix $\mathbf{K}$ is obtained by the same procedure after division by ω^2 .

$$(\omega_j^{-2} - \omega_k^{-2}) \mathbf{u}_j^T \mathbf{K} \mathbf{u}_k = 0 \quad (4.20)$$

By appropriate choice of eigenvectors corresponding to multiple eigenvalues this leads to the orthogonality relation

$$\mathbf{u}_j^T \mathbf{K} \mathbf{u}_k = 0, \quad j \neq k \quad (4.21)$$

Thus, the mode-shape vectors $\mathbf{u}_j, j = 1, 2, \dots, n$ satisfy orthogonality relations with respect to both the mass and the stiffness matrix.

Modal mass and stiffness

The generalized eigenvalue problem does not determine the magnitude of the mode-shape vectors. They can therefore be normalized by multiplication with a suitable scaling factor. It is convenient to introduce the notion of a modal mass, defined for mode j as

$$m_j = \mathbf{u}_j^T \mathbf{M} \mathbf{u}_j \quad (4.22)$$

The modal stiffness for mode j is defined similarly as

$$k_j = \mathbf{u}_j^T \mathbf{K} \mathbf{u}_j \quad (4.23)$$

With these definitions the orthogonality relations are conveniently expressed in terms of the modal mass and stiffness as

$$\begin{aligned} \mathbf{u}_j^T \mathbf{M} \mathbf{u}_k &= m_j \delta_{jk} \\ \mathbf{u}_j^T \mathbf{K} \mathbf{u}_k &= k_j \delta_{jk} \end{aligned} \quad j = 1, 2, \dots, n, \quad (4.24)$$

where δ_{jk} is Kronecker's delta, defined by $\delta_{jk} = 1$ for $j = k$ and $\delta_{jk} = 0$ for $j \neq k$.

Once a mode shape vector $\mathbf{u}_j$ has been determined, the corresponding generalized eigenvalue equation constitutes an equation for the corresponding natural frequency ω_j . Pre-multiplication of the generalized eigenvalue equation (4.16) with $\mathbf{u}_j^T$ gives

$$\mathbf{u}_j^T \mathbf{K} \mathbf{u}_j - \omega_j^2 \mathbf{u}_j^T \mathbf{M} \mathbf{u}_j = 0. \quad (4.25)$$

This is a scalar equation from which ω_j^2 is determined as

$$\omega_j^2 = \frac{\mathbf{u}_j^T \mathbf{K} \mathbf{u}_j}{\mathbf{u}_j^T \mathbf{M} \mathbf{u}_j} = \frac{k_j}{m_j} \quad (4.26)$$

Thus, ω_j^2 is determined as the ratio between the corresponding modal stiffness and modal mass. This is completely analogous to the case of a single degree of freedom.

When computing mode shape vectors their normalization must be chosen explicitly. Many computer programs set each of the modal masses to unity, $m_j = 1$. According to (4.26) this implies that the modal stiffness parameters are ω_j^2 . This procedure has the unintended consequence that the dimension and magnitude of the mode-shape vector components are only determined indirectly. As an alternative it may be advantageous to relate the modal mass to the actual mass M of the structure under consideration, e.g. in the

form $m_j = M$ or $m_j = M/n$, where n is the number of degrees of freedom or the number of elements in the model.

Example 4.3. Discrete cable model. Figure 4.3 shows a set of n identical masses m equally spaced on a taut cable of negligible mass. The distance between neighboring masses is l , and thus the total length of the string is $L = (n + 1)l$. The tension in the string is T . Thus, a unit transverse displacement of one of the masses will lead to a force of magnitude $-2T/l$ on the displaced mass and forces of magnitude T/l on each of the two neighboring masses. This is equivalent to a connection of the masses by springs of stiffness $k = T/l$.

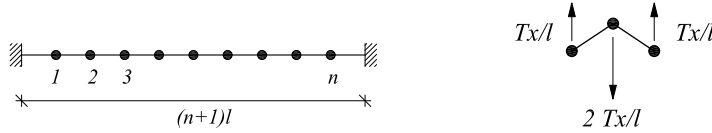


Fig. 4.3: Model of taut cable with concentrated masses

In the absence of external loading the transverse motion of the masses is governed by the homogeneous equation of motion (4.11) with stiffness matrix and mass matrix given by

$$\mathbf{K} = \frac{T}{l} \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{bmatrix}, \quad \mathbf{M} = m \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}.$$

These equations describe the free vibrations of an n degree-of-freedom system. The n mode shapes are described by vectors $\mathbf{u}_r = [u_1, u_2, \dots, u_n]_r$, $r = 1, 2, \dots, n$. These vectors are determined from the eigenvalue problem

$$\left(\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{bmatrix} - \frac{\omega^2}{\omega_0^2} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} \right) \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

where ω_0 denotes a reference angular frequency, determined by

$$\omega_0^2 = \frac{T}{lm}$$

In a similar system generated by springs with stiffness $k = T/l$ this angular frequency corresponds to the natural vibration of a single element.

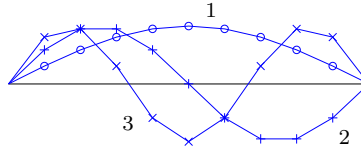


Fig. 4.4: Mode shapes of equally spaced masses on a taut cable

This particular eigenvalue problem is one of the few of size larger than two that can be solved analytically. The solution is obtained demonstrating that an assumed form of the mode shape vectors is actually correct and determines a corresponding set of natural frequencies. It is assumed that the displacement components u_j of mode number r are given by

$$u_{j,r} = \sin\left(\pi \frac{j r}{n+1}\right), \quad j = 1, 2, \dots, n, \quad r = 1, 2, \dots, n$$

The first three mode shapes are illustrated in Fig. 4.4. It is seen that $u_{0,r} = u_{n+1,r} = 0$, and thus all the equations can be written in the common form

$$-u_{j-1,r} + 2u_{j,r} - u_{j+1,r} = \frac{\omega_r^2}{\omega_0^2} u_{j,r}, \quad r = 1, 2, \dots, n$$

The assumed mode shape vector is now substituted into the left hand side of this equation. The first step makes use of the trigonometric formula

$$\sin z_1 - \sin z_2 = 2 \cos\left(\frac{z_1 + z_2}{2}\right) \sin\left(\frac{z_1 - z_2}{2}\right)$$

The left side of the equation is written in single difference form, whereby

$$\begin{aligned} & (-u_{j-1,r} + u_{j,r}) + (u_{j,r} - u_{j+1,r}) \\ &= 2 \sin\left(\pi \frac{r}{2(n+1)}\right) \left[\cos\left(\pi \frac{(j-\frac{1}{2})r}{n+1}\right) - \cos\left(\pi \frac{(j+\frac{1}{2})r}{n+1}\right) \right] \end{aligned}$$

The two cosine terms can be combined by use of the trigonometric formula

$$\cos z_1 - \cos z_2 = -2 \sin\left(\frac{z_1 + z_2}{2}\right) \sin\left(\frac{z_1 - z_2}{2}\right)$$

The equation of motion then reduces to

$$-u_{j-1,r} + 2u_{j,r} - u_{j+1,r} = \left[2 \sin\left(\pi \frac{r}{2(n+1)}\right) \right]^2 \sin\left(\pi \frac{j r}{n+1}\right)$$

The factor in the square brackets is independent of the index j , and represents the frequency ratio ω_r^2/ω_0^2 , while the last factor is $u_{j,r}$. Thus, the assumed mode shape format is a solution to the equation system, and the angular frequency ω_r of mode number r is determined by

$$\frac{\omega_r}{\omega_0} = 2 \sin\left(\pi \frac{r}{2(n+1)}\right)$$

This result can also be written directly in terms of the mass m , the string tension T , and the distance between the masses l , as

$$\omega_r = 2 \sqrt{\frac{T}{lm}} \sin\left(\pi \frac{r}{2(n+1)}\right)$$

Note, that the formula contains the square root of a stiffness divided by a mass, as in the case of a single degree-of-freedom system. The frequencies are illustrated in Fig. 4.5.

When the cable supports a large number of masses, $n \gg 1$, the natural frequency of the first mode can be estimated by expansion of the sine function,

$$\omega_1 = 2 \sqrt{\frac{T}{lm}} \sin\left(\pi \frac{1}{2(n+1)}\right) \simeq \frac{\pi}{n+1} \sqrt{\frac{T}{lm}} = \pi \sqrt{\frac{T}{LM}},$$

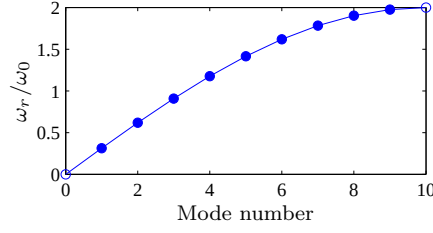


Fig. 4.5: Normalized natural frequencies of 9 masses on string.

where it is used that the total length of the string is $L = (n+1)l$ and its total mass is $M = (n+1)m$. This result is the exact result for the lowest natural angular frequency for the continuous string. Thus, the discrete system provides a good approximation for the lowest natural frequencies of a continuous string, when there are sufficiently many masses. However, the correspondence decreases for the higher modes.

The modal mass m_r corresponding to mode shape r are found as

$$m_r = \sum_{j=1}^n m u_{j,r}^2 = \frac{n+1}{2} m = \frac{1}{2} M$$

This result follows from the fact that summation over equally spaced points over an integer number of half-periods integrate trigonometric functions exactly. In the present case the points $j = 0$ and $j = n+1$ must be included with half weight to make up the half-period. It is noted that M is not directly the sum of the n concentrated masses associated with the moving degrees of freedom in the model. $\square$

Example 4.4. Vibration of portal frame.

In Lecture 5 an analytic solution was found for the mode shapes and natural frequencies of the transverse vibrations of a taut string with n concentrated masses m with equal spacing l . The solution is

$$u_{j,r} = \sin\left(\pi \frac{j r}{n+1}\right) \quad , \quad \omega_r = 2\omega_0 \sin\left(\frac{\pi}{2} \frac{r}{n+1}\right)$$

with reference frequency ω_0 determined by

$$\omega_0^2 = \frac{T}{l m} = \frac{k}{m}$$

where $k = T/l$ is the stiffness of the individual ‘links’. In this solution r denotes the mode number, and j is the number of the mass.

The free vibration solution of the n -story portal frame shown in Fig. 4.6a can be obtained from the solution of the string with $2n$ equally spaced masses shown in Fig. 4.6b. The bottom of the portal frame is fixed, while no force is transmitted on the top of the top floor. These boundary conditions are identical to those of the left part of the symmetric modes of the string with $2n$ masses.

Thus, the solution to the free vibrations of the n -story portal frame can be obtained directly from the solution $u_{j,r}$ in Example 4.3 by changing the number of masses from n to $2n$,

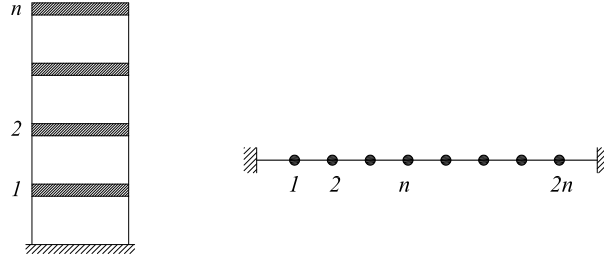


Fig. 4.6: a) n -story portal frame. b) $2n$ equally spaced masses on a string.

and by including only the symmetric modes $1, 3, \dots, 2n-1$, i.e. by numbering the modes as $2r-1$, $r = 1, 2, \dots, n$. With these modifications the solution of the vibration problem of the n -story portal frame is

$$u_{j,r} = \sin\left(\pi j \frac{2r-1}{2n+1}\right), \quad \omega_r = 2\omega_0 \sin\left(\frac{\pi}{2} \frac{2r-1}{2n+1}\right)$$

The mode shapes of a 4-story portal frame are illustrated in Fig. 4.7. The model only recognizes the motion of the discrete masses as shown by the straight connecting lines in the figure. The corresponding normalized natural frequencies are shown in Fig. 4.8.

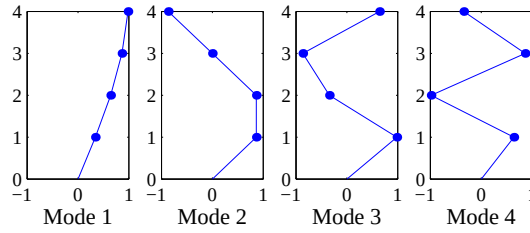


Fig. 4.7: Mode shapes of 4-story portal frame

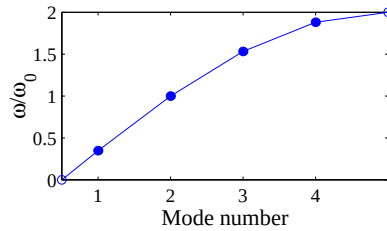


Fig. 4.8: Normalized natural frequencies of 4-story portal frame

The modal mass m_r corresponding to the mode shapes of the shear building are found as

$$m_r = \sum_{j=1}^n m u_{j,r}^2 = \frac{n + \frac{1}{2}}{2} m = \frac{2n + 1}{4} m \quad (4.27)$$

This result follows from the fact that summation over equally spaced points over an integer number of half-periods integrate trigonometric functions exactly. In the present case the support of the point $j = 0$ must be included with half weight to make up the half-period. $\square$

Example 4.5. Natural frequencies of bridge. Figure 4.9 shows a simply supported bridge with three spans of length $l_1 = l_2 = 30.0$ m and $l_3 = 50.0$ m. The bridge deck is represented as a homogeneous beam with cross-section area $A = 0.3$ m² and moment of inertia $I = 0.07$ m⁴. The mass density is $\rho = 1.05 \cdot 10^4$ kg/m³, and the modulus of elasticity is $E = 3.0 \cdot 10^{11}$ N/m².

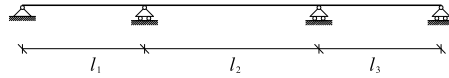


Fig. 4.9: Simply supported three-span bridge

The bridge deck is represented by the simple cubic beam bending elements described in Section 3.2.2. The natural frequency of the first five modes are given in Table 4.1 for three different element discretizations. In the middle configuration there are $n_1 = n_3 = 3$ elements in the side spans and $n_2 = 5$ elements in the center span, while the table also contains results from a coarser model and a more detailed model, respectively.

Table 4.1: Natural angular frequencies of three-span bridge, [rad/s]

$[n_1, n_2, n_3]$	ω_1	ω_2	ω_3	ω_4	ω_5
[2, 3, 2]	15.06	32.36	38.09	56.79	112.2
[3, 5, 3]	15.04	32.22	37.87	55.99	101.2
[6, 10, 6]	15.04	32.18	37.82	55.83	100.4

The five natural frequencies from the refined model are only changed by one unit on the last digit by further refinement, and thus can be taken as the correct result. It is seen that the error of the results of the intermediate model increases monotonically with the mode number from zero for the lowest mode to about 1 pct. for mode number five. This is fully satisfactory, and may in fact be better than the general accuracy of the model. The coarser model still gives the first four natural frequencies to within 1.5 pct. with an error on the fifth frequency of 10 pct., and even this may be satisfactory in practice, as the response of the first few modes may often dominate the total response.

The first five mode shapes are illustrated in Fig. 4.10, connecting the nodes by straight line segments. As the mode number increases the mode shape becomes increasingly irregular, and thus imposes increased requirements on the element size. It is seen that mode 5 has one full wave in both side spans and one and a half wave in the center span. As illustrated in Section 3.2.2 a fair accuracy of the beam vibration frequency requires at least two elements per wave length. This observation is confirmed by the present example where the first four

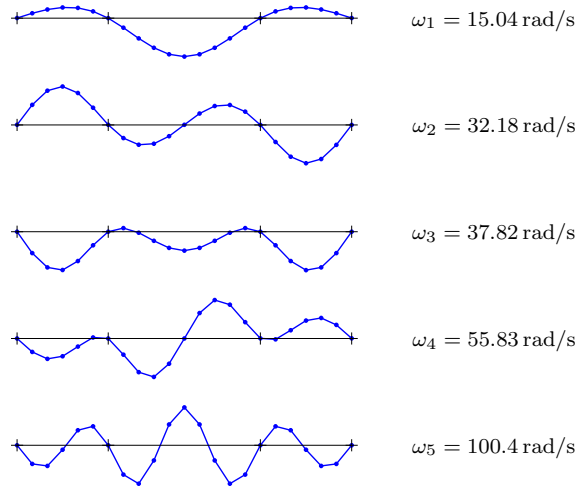


Fig. 4.10: Normalized lowest vibration modes of three-span bridge

modal frequencies are within engineering accuracy in the coarse model, while the fifth modal frequency is too inaccurate due to the irregularity of the mode shape. $\square$

4.2.2 Modal representation

Vibration problems can be formulated and solved by representing the displacements in terms of the mode-shapes in the form

$$\mathbf{q}(t) = \sum_{j=1}^n \mathbf{u}_j r_j(t) \quad (4.28)$$

where $r_j(t)$ is the time history of the motion associated with mode j . This representation can be written in a compact vector format by introducing the modal matrix

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n] \quad (4.29)$$

in which the mode shape vectors $\mathbf{u}_j$ are stacked as columns, and the modal response vector

$$\mathbf{r}(t) = [r_1(t), r_2(t), \dots, r_n(t)]^T \quad (4.30)$$

The modal response components $r_j(t)$ are often called the modal coordinates. The modal representation (4.28) of the response now takes the compact form

$$\mathbf{q}(t) = \mathbf{U} \mathbf{r}(t). \quad (4.31)$$

This form clearly indicates the transformation from the physical components in the vector $\mathbf{q}$ to the modal components contained in $\mathbf{r}$.

Modal energy

The mode shapes play an important role with respect to the energy of a structure. The mechanical energy of the system is the sum of the kinetic energy and the elastic energy,

$$E = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^T \mathbf{K} \mathbf{q}. \quad (4.32)$$

When the representation (4.31) is substituted the energy is expressed in terms of the following two matrices

$$\mathbf{U}^T \mathbf{M} \mathbf{U} = \mathbf{m}, \quad \mathbf{U}^T \mathbf{K} \mathbf{U} = \mathbf{k}. \quad (4.33)$$

It follows from the orthogonality relations (4.19) and (4.21) that $\mathbf{m}$ and $\mathbf{k}$ are diagonal matrices containing the modal mass and stiffness, respectively.

$$\mathbf{m} = \begin{bmatrix} m_1 & & \\ & \ddots & \\ & & m_n \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} k_1 & & \\ & \ddots & \\ & & k_n \end{bmatrix}. \quad (4.34)$$

The mechanical energy can therefore be expressed in terms of the modal mass, stiffness and coordinates as

$$\begin{aligned} E &= \frac{1}{2} \dot{\mathbf{r}}(t)^T \mathbf{m} \dot{\mathbf{r}}(t) + \frac{1}{2} \mathbf{r}(t)^T \mathbf{k} \mathbf{r}(t) \\ &= \sum_{j=1}^n \left[\frac{1}{2} m_j \dot{r}_j(t)^2 + \frac{1}{2} k_j r_j(t)^2 \right]. \end{aligned} \quad (4.35)$$

Thus, the total mechanical energy is simply the sum of the mechanical energy in each of the modes, without any coupling effects.

Rayleigh's quotient

The expression (4.26) for the natural frequency ω_j is often used in an approximate form, called Rayleigh's quotient,

$$\omega_R^2 = \frac{\mathbf{q}^T \mathbf{K} \mathbf{q}}{\mathbf{q}^T \mathbf{M} \mathbf{q}}, \quad (4.36)$$

where the displacement vector $\mathbf{u}$ is an estimate of an exact mode shape vector $\mathbf{u}_j$. The relation between the approximate result ω_R^2 and the desired counterpart can be investigated by expanding the approximate mode-shape vector $\mathbf{u}$ in terms of the complete set of mode-shape vectors $\mathbf{u}_j$.

$$\mathbf{q} = \mathbf{u}_1 r_1 + \mathbf{u}_2 r_2 + \cdots = \sum_{j=1}^n \mathbf{u}_j r_j \quad (4.37)$$

Substitution into Rayleigh's quotient,

$$\omega_R^2 = \frac{\mathbf{q}^T \mathbf{K} \mathbf{q}}{\mathbf{q}^T \mathbf{M} \mathbf{q}} = \frac{\sum_j k_j r_j^2}{\sum_j m_j r_j^2}. \quad (4.38)$$

This expression is now rewritten by introducing the modal frequencies $\omega_j^2 = k_j/m_j$, and re-normalisation in terms of a particular modal frequency ω_k ,

$$\omega_R^2 = \frac{\sum_j \omega_j^2 m_j r_j^2}{\sum_j m_j r_j^2} = \omega_k^2 \frac{\sum_j (\omega_j/\omega_k)^2 m_j r_j^2}{\sum_j m_j r_j^2}. \quad (4.39)$$

When selecting ω_k as the smallest modal frequency, $\omega_k = \omega_1$, the numerator becomes larger than the denominator, whereby $\omega_1 \leq \omega_R$. Furthermore, when selecting ω_k as the largest natural frequency, $\omega_k = \omega_n$, the numerator becomes smaller than the denominator, whereby $\omega_R \leq \omega_n$. Thus the lowest and the highest modal frequency form bounds on the frequency determined by Rayleigh's quotient,

$$\omega_1 \leq \omega_R \leq \omega_n. \quad (4.40)$$

Rayleigh's quotient is useful for estimating the lowest frequency, which often corresponds to a fairly regular shape that can be estimated without detailed calculation. If the lowest natural frequency of a structure is sufficiently high relative to the frequency content of the load processes it may not be necessary to consider dynamic effects.

Example 4.6. Vibration frequency of two story building. Consider the simple shear building shown in Fig. 4.1b, and let $k_1 = k_2 = k$ and $m_1 = m_2 = m$. The lowest frequency corresponds to the swaying mode shown in Fig. 4.2a. If the shape is assumed to increase linearly with height, $\mathbf{u} = [0.5, 1]$, the Rayleigh estimate is

$$\omega_R^2 = \frac{[0.5, 1] \begin{bmatrix} 2k & -k \\ -k & k \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}}{[0.5, 1] \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}} = 0.4 \frac{k}{m}.$$

The correct value was obtained in Example 4.2 as $\omega_1^2 = 0.382k/m$, corresponding to an error in ω_R of 2.3 pct. $\square$

Estimating the highest modal frequency

In connection with numerical time integration of dynamic response it may be needed to obtain an estimate of the highest modal frequency in a structural model. Typically, this frequency will not represent a physical characteristic of the actual structure, as models often rely on assumed shape functions, that become less representative at high frequencies. However, the highest modal frequency represents a property of the model that may be important for the numerical scheme used for the time integration. As it turns out a rather good

estimate of the highest modal frequency of the structural model can often be obtained by isolating and analyzing a small part of the model - typically a single element, Hughes et al. (1979) or Hughes (1987).

The first step is to realize that the maximum vibration frequency of a structure can be expressed as a maximization of Rayleigh's quotient with respect to the displacement vector of the structure $\mathbf{q}$,

$$(\omega_{\max}^s)^2 = \max_{\mathbf{q}} \frac{\mathbf{q}^T \mathbf{K} \mathbf{q}}{\mathbf{q}^T \mathbf{M} \mathbf{q}}. \quad (4.41)$$

This result follows from representing the displacement vector $\mathbf{q}$ in terms of the mode shape vectors and then using the expression (4.39) to identify the maximization condition as $\mathbf{q} \sim \mathbf{u}_n$, where $\mathbf{u}_n$ is the mode shape vector corresponding to the highest natural frequency.

Then a hypothetical structure is considered, in which all the elements of the original structure are considered in unassembled form. The stiffness and mass matrices of this hypothetical structure are expressed in terms of the individual element matrices $\mathbf{K}_j$ and $\mathbf{M}_j$ in block diagonal form as

$$\bar{\mathbf{K}} = \begin{bmatrix} \mathbf{K}_1 & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \mathbf{K}_m \end{bmatrix}, \quad \bar{\mathbf{M}} = \begin{bmatrix} \mathbf{M}_1 & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \mathbf{M}_m \end{bmatrix}. \quad (4.42)$$

The corresponding displacement vector is of the form $\bar{\mathbf{q}}^T = [\mathbf{q}_1^T, \dots, \mathbf{q}_m^T]^T$, where the vectors $\mathbf{q}_j$ contain the displacement components associated with element no. j . Due to the block diagonal form of the matrices $\bar{\mathbf{K}}$ and $\bar{\mathbf{M}}$ the generalized eigenvalue problem uncouples, and the associated eigenvectors and eigenfrequencies consist of eigenvectors and eigenfrequencies of each individual element. Thus, the corresponding Rayleigh quotient is

$$\bar{\omega}_{\max}^2 = \max_{\bar{\mathbf{q}}} \frac{\bar{\mathbf{q}}^T \bar{\mathbf{K}} \bar{\mathbf{q}}}{\bar{\mathbf{q}}^T \bar{\mathbf{M}} \bar{\mathbf{q}}} = \max_e (\omega_{\max}^e)^2. \quad (4.43)$$

The real structure can be re-established from the unassembled structure, by imposing equality of displacements for elements at the same nodes. Thus, the maximum frequency $\omega_{\max}^s$ of the real structure can be found from the Rayleigh quotient of the disassembled structure, if the general displacement vector $\bar{\mathbf{q}}$ is constrained at common nodes. Clearly, constraining the maximization process can not increase the maximum obtained in free maximization, and thus the results (4.41) and (4.43) can be combined to yield the final result

$$\omega_{\max}^s \leq \omega_{\max}^e, \quad (4.44)$$

where $\omega_{\max}^e$ denotes the maximum free vibration frequency in any of the elements of the structure, when released from the structure.

Example 4.7. Highest frequency of cable or bar element. For the cable or bar element there is a unique stiffness matrix $\mathbf{K}$, while the mass matrix can be formed for lumped masses at the element nodes $\mathbf{K}_l$, or by consistent integration of a homogeneous distribution of the mass along the element $\mathbf{K}_c$. The matrices were evaluated in Section 3.1.2 as

$$\mathbf{K} = \frac{T}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{M}_l = \frac{ml}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{M}_c = \frac{ml}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (4.45)$$

The deformation pattern described by the element displacement vector $\mathbf{q}_e$ contains rigid body modes as well as deformation vibrations. The rigid body modes do not produce elastic forces via the stiffness matrix, and therefore have the natural frequency $\omega_r = 0$. In the case of the cable element the rigid body mode is described by identical displacement of the two nodes, corresponding to the displacement vector $\mathbf{q}_e = [1, 1]$. The free vibration mode is described by the displacement vector $\mathbf{q}_e = [1, -1]$. This deformation leads to different frequencies, depending on the assumption of lumped or distributed mass. The corresponding frequencies are determined by Rayleigh's quotient as

$$\omega_l^2 = \frac{\mathbf{q}_e^T \mathbf{K} \mathbf{q}_e}{\mathbf{q}_e^T \mathbf{M}_l \mathbf{q}_e} = \frac{4H}{l^2 m}, \quad \omega_c^2 = \frac{\mathbf{q}_e^T \mathbf{K} \mathbf{q}_e}{\mathbf{q}_e^T \mathbf{M}_c \mathbf{q}_e} = \frac{12H}{l^2 m} \quad (4.46)$$

The correct vibration shape corresponds to half of a sine-wave leading to $\omega_{\max}^2 = \pi^2 H / (l^2 m) = 9.87 H / (l^2 m)$. $\square$

Example 4.8. Highest frequency of beam bending element. $\square$

4.3 Modal response analysis

The dynamic response of a linear system with n degrees of freedom $\mathbf{q}(t) = [q_1(t), q_2(t), \dots, q_n(t)]^T$ is described by the set of second order differential equations

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f}(t) \quad (4.47)$$

where a dot denotes differentiation with respect to time t . The physical system parameters are: the mass matrix $\mathbf{M}$, the viscous damping matrix $\mathbf{C}$, and the stiffness matrix $\mathbf{K}$. The load history is given by the vector $\mathbf{f}(t)$. In addition to the system parameters and the load history, solution of the dynamic response problem also requires specification of two sets of initial conditions,

$$\mathbf{q}(0) = \mathbf{q}_0, \quad \dot{\mathbf{q}}(0) = \dot{\mathbf{q}}_0 \quad (4.48)$$

The objective of modal analysis is the solution of the coupled equations of motion (4.47) by transforming the differential equations into an uncoupled form in terms of the mode-shape vectors introduced in the previous section.

4.3.1 Response analysis

The modal representation (4.31) gives the physical components $\mathbf{u}$ in terms of the modal matrix $\mathbf{U}$ and the modal coordinates $\mathbf{r}$,

$$\mathbf{q}(t) = \mathbf{U} \mathbf{r}(t). \quad (4.49)$$

When this representation is substituted into the equations of motion (4.47), followed by pre-multiplication with the matrix $\mathbf{U}^T$, the resulting equations are

$$(\mathbf{U}^T \mathbf{M} \mathbf{U}) \ddot{\mathbf{r}} + (\mathbf{U}^T \mathbf{C} \mathbf{U}) \dot{\mathbf{r}} + (\mathbf{U}^T \mathbf{K} \mathbf{U}) \mathbf{r} = \mathbf{U}^T \mathbf{f}(t). \quad (4.50)$$

It follows from the orthogonality relations (4.33) that the mass and stiffness terms uncouple. Furthermore, as discussed in Section 4.4, for moderately damped structures it may often be assumed that the modal damping matrix

$$\mathbf{U}^T \mathbf{C} \mathbf{U} = \mathbf{c} \quad (4.51)$$

can be represented by its diagonal elements $c_j = c_{jj}$. With this assumption the equations of motion uncouple and take the form

$$m_j \ddot{r}_j(t) + c_j \dot{r}_j(t) + k_j r_j(t) = p_j(t), \quad j = 1, 2, \dots, n, \quad (4.52)$$

where $p_j(t) = \mathbf{u}_j^T \mathbf{f}(t)$ are the components of the modal load vector

$$\mathbf{p} = \mathbf{U}^T \mathbf{f}(t). \quad (4.53)$$

Each modal equation of motion is similar to the equation of motion of a single-degree-of-freedom oscillator.

The initial conditions of the modal coordinates are found by introducing the modal representation (4.31) into (4.48),

$$\mathbf{q}_0 = \mathbf{U} \mathbf{r}_0, \quad \dot{\mathbf{q}}_0 = \mathbf{U} \dot{\mathbf{r}}_0. \quad (4.54)$$

The initial conditions of the modal coordinates follow by pre-multiplication with $\mathbf{U}^T \mathbf{M}$ and use of the orthogonality relation (4.33a),

$$\mathbf{r}_0 = \mathbf{m}^{-1} \mathbf{U}^T \mathbf{M} \mathbf{q}_0, \quad \dot{\mathbf{r}}_0 = \mathbf{m}^{-1} \mathbf{U}^T \mathbf{M} \dot{\mathbf{q}}_0. \quad (4.55)$$

It is observed that the modal mass matrix is diagonal, and computation of its inverse therefore trivial. Thus, the initial conditions are recast into modal coordinate format without matrix inversion.

The modal equations of motion (4.52) are divided by the modal mass m_j to obtain the normalized form

$$\ddot{r}_j(t) + 2\zeta_j \omega_j \dot{r}_j(t) + \omega_j^2 r_j(t) = \frac{1}{m_j} p_j(t), \quad j = 1, 2, \dots, n \quad (4.56)$$

where the angular frequency and damping ratio of mode j are defined by

$$\omega_j^2 = \frac{k_j}{m_j}, \quad \zeta_j = \frac{c_j}{2\sqrt{k_j m_j}} \quad (4.57)$$

Table 4.2: Computational procedure in a modal analysis.

-
- 1) Solve generalized eigenvalue problem (4.13) for ω_j and $\mathbf{u}_j$.
 - 2) Compute modal masses m_j (4.22), and stiffnesses k_j (4.23).
 - 3) Find modal initial conditions $\mathbf{r}_0$ and $\dot{\mathbf{r}}_0$ by (4.55).
 - 4) Compute the n modal time histories $r_j(t)$ from (4.56).
 - 5) Obtain displacement time histories $\mathbf{q}(t)$ from (4.49).
-

as for the corresponding single-degree-of-freedom system. The general analytical solution to the modal equation of motion consists of terms representing the contributions from the initial conditions and a term representing the load history,

$$\begin{aligned}
 r_j(t) = & r_{j0} e^{-\zeta_j \omega_j t} \left(\cos(\omega_{jd} t) + \frac{\zeta_j \omega_j}{\omega_{jd}} \sin(\omega_{jd} t) \right) + \frac{\dot{r}_{j0}}{\omega_{jd}} e^{-\zeta_j \omega_j t} \sin(\omega_{jd} t) \\
 & + \frac{1}{m_j \omega_{jd}} \int_0^t f_j(\tau) e^{-\zeta_j \omega_j (t-\tau)} \sin(\omega_{jd} (t-\tau)) d\tau
 \end{aligned} \quad (4.58)$$

The modal damped angular frequency is defined as for the corresponding single-degree-of-freedom,

$$\omega_{jd} = \omega_j \sqrt{1 - \zeta_j^2} \quad (4.59)$$

A typical modal analysis can be carried out by the steps indicated in Table 4.2. In practice it is often convenient to evaluate the modal response by numerical integration, e.g. by the simple time-stepping procedure shown in Table 1.1.

4.3.2 Frequency response

Modal analysis plays a central role in the design of damping and control devices for structures. These devices are often designed to target a particular mode or a group of modes spanning a frequency interval. To facilitate the discussion and analysis of modal response in relation to harmonic excitation and damping and control devices it is convenient to give a concise formulation of the response to excitation in the frequency domain. The general format is developed in the section and later applied in Chapter 5.

4.4 Sensitivity analysis and modal damping

4.4.1 Sensitivity analysis

The general homogeneous form of the linear equation of motion with the damping represented in terms of the instantaneous velocity is given by

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}. \quad (4.60)$$

It is important to investigate the influence of the damping matrix $\mathbf{C}$ on the free vibrations. This is conveniently done in the form of a perturbation analysis, where the solution to the corresponding undamped system is used as a reference, and the terms resulting from the damping matrix are obtained by a first order series expansion. In this analysis the effect of changes in the stiffness matrix $\mathbf{K}$ and the mass matrix $\mathbf{M}$ are easily incorporated.

Let the mass matrix, the stiffness matrix, and the damping matrix in the homogeneous equation of motion (4.60) be represented in the form

$$\mathbf{M} = \mathbf{M}_0 + \delta\mathbf{M}, \quad \mathbf{K} = \mathbf{K}_0 + \delta\mathbf{K}, \quad \mathbf{C} = \delta\mathbf{C}. \quad (4.61)$$

In this notation $\delta\mathbf{M}$, $\delta\mathbf{K}$ and $\delta\mathbf{C}$ denote changes in the system matrices, relative to an undamped system with mass matrix $\mathbf{M}_0$ and stiffness matrix $\mathbf{K}_0$. A free vibration solution is assumed in the form

$$\mathbf{q}(t) = \mathbf{u}e^{i\omega t} \quad (4.62)$$

with mode shape vector $\mathbf{u}$ and angular frequency ω . Substitution of this representation into the equation of motion (4.60) leads to the equation

$$(\mathbf{K} - \omega^2\mathbf{M} + i\omega\mathbf{C})\mathbf{u} = \mathbf{0}. \quad (4.63)$$

This is a quadratic eigenvalue problem, because the eigenvalue ω^2 appears both in linear and quadratic form. In principle this eigenvalue problem can be recast into a linear eigenvalue problem in terms of an extended mode shape vector $[\mathbf{u}^T, i\omega\mathbf{u}^T]$, but the present purpose is to relate the solution directly to the limiting case of the corresponding undamped vibration problem with stiffness and mass matrices $\mathbf{K}_0$ and $\mathbf{M}_0$. The solution is therefore represented in the perturbation format

$$\mathbf{u} = \mathbf{u}_j + \delta\mathbf{u}_j, \quad \omega = \omega_j + \delta\omega_j, \quad (4.64)$$

where ω_j , $\mathbf{u}_j$ are the j 'th frequency and mode shape vector of the undamped vibration problem represented by the matrices $\mathbf{K}_0$ and $\mathbf{M}_0$.

Substitution of the assumed solution (4.64) into the quadratic eigenvalue problem (4.63) gives the equation

$$\begin{aligned}
& (\mathbf{K}_0 - \omega_j^2 \mathbf{M}_0) \mathbf{u}_j + (\mathbf{K}_0 - \omega_j^2 \mathbf{M}_0) \delta \mathbf{u}_j + \\
& (\delta \mathbf{K} - \omega_j^2 \delta \mathbf{M} + i\omega_j \delta \mathbf{C} - 2\omega_j \delta \omega_j \mathbf{M}_0) \mathbf{u}_j + \dots = \mathbf{0},
\end{aligned} \tag{4.65}$$

where the dots represent higher order terms. The equation is now pre-multiplied by $\mathbf{u}_j^T$. This is an eigenvector of the original undamped vibration eigenvalue problem, and therefore the contributions from two first square brackets vanish, leaving the first-order result

$$\delta \omega_j \simeq \frac{\mathbf{u}_j (\delta \mathbf{K} - \omega_j^2 \delta \mathbf{M} + i\omega_j \delta \mathbf{C}) \mathbf{u}_j}{2\omega_j \mathbf{u}_j \mathbf{M}_0 \mathbf{u}_j}. \tag{4.66}$$

It follows from this result that to within first order the change of the frequency is given by the changes of the system matrices, when pre- and post-multiplied by the corresponding mode shape vector $\mathbf{u}_j$.

In analogy with the modal stiffness and mass matrices $\mathbf{k}$ and $\mathbf{m}$ introduced in (4.24) it is convenient to introduce a modal damping matrix with elements

$$c_{jk} = \mathbf{u}_j \mathbf{C} \mathbf{u}_k. \tag{4.67}$$

While the modal stiffness and mass matrices are diagonal due to the special eigenvector properties of the mode shapes, the modal damping matrix $\mathbf{c}$ will not in general be diagonal. Nonetheless the asymptotic formula (4.66) indicates that to within a first-order approximation the imaginary part of the angular frequencies, representing the modal damping, only depends on the diagonal terms of the modal damping matrix. Based on this result and the considerable uncertainty about its precise character the damping matrix of lightly damped structures are often assumed to be in uncoupled modal form. The result (4.66) is conveniently expressed in terms of the incremental changes in modal mass, modal stiffness and modal damping components as

$$\delta \omega_j \simeq \frac{\delta k_{jj} - \omega_j^2 \delta m_{jj} + i\omega_j \delta c_{jj}}{2\omega_j m_{jj}}. \tag{4.68}$$

It follows from this expression that an increase of the modal stiffness leads to an increase of the real part of the associated natural frequency, while an increase of the modal mass leads to a decrease.

The approximate solution (4.64) to the perturbed eigenvalue problem also contains a contribution $\delta \mathbf{u}_j$ to the original mode-shape vector $\mathbf{u}_j$. Mode-shape vectors are only determined to within a normalization factor, and if the perturbation $\delta \mathbf{u}_j$ is expanded in terms of the original mode-shape vectors, the original j 'th component can be retained, whereby the representation of $\delta \mathbf{u}_j$ in terms of the remaining mode-shape vectors is

$$\delta \mathbf{u}_j = \sum_{k \neq j} \gamma_{jk} \mathbf{u}_k, \quad (4.69)$$

When this representation is substituted into (4.65), pre-multiplication by $\mathbf{u}_k^T$ with $k \neq j$ gives

$$\mathbf{u}_k^T (\mathbf{K}_0 - \omega_j^2 \mathbf{M}_0) \sum_{l \neq j} \gamma_{jl} \mathbf{u}_l + \mathbf{u}_k^T (\delta \mathbf{K} - \omega_j^2 \delta \mathbf{M} + i \omega_j \delta \mathbf{C}) \mathbf{u}_j \simeq 0, \quad (4.70)$$

In the summation only the term with subscript $l = k$ contributes due to the orthogonality between the original mode shape vectors with respect to the matrices $\mathbf{M}_0$ and $\mathbf{K}_0$. Thus the expansion coefficients are determined as

$$\gamma_{jk} \simeq \frac{\mathbf{u}_k^T (\delta \mathbf{K} - \omega_j^2 \delta \mathbf{M} + i \omega_j \delta \mathbf{C}) \mathbf{u}_j}{\mathbf{u}_k^T (\mathbf{K}_0 - \omega_j^2 \mathbf{M}_0) \mathbf{u}_k}, \quad k \neq j. \quad (4.71)$$

In terms of the corresponding modal matrix elements this formula takes the form

$$\gamma_{jk} \simeq \frac{\delta k_{kj} - \omega_j^2 \delta m_{kj} + i \omega_j \delta c_{kj}}{(\omega_k^2 - \omega_j^2) m_j}, \quad k \neq j. \quad (4.72)$$

The result breaks down if $\omega_k = \omega_j$, but otherwise indicates that the largest influence of perturbation of the system parameters occurs for modes with closely spaced frequencies. As in the case of the frequency perturbation, changes in mass and stiffness leads to real-valued changes, while perturbation of the off-diagonal damping parameters lead to complex-valued mode-shape vectors. An important special case is gyroscopic motion, treated in Section ???. The gyroscopic forces are contained in an anti-symmetric matrix similar to $\mathbf{C}$, and in undamped gyroscopic systems the modal frequencies ω_j remain real-valued, while the mode-shape vectors are complex-valued.

4.4.2 Rayleigh damping

The undamped modes determined by the generalized eigenvalue problem (4.13) lead to uncoupling of the equations of motion if the damping matrix leads to uncoupled modal damping, i.e. if the modal damping matrix $\mathbf{c}$ in (4.51) is diagonal. The simplest instance of this is the so-called Rayleigh damping, where the damping matrix consists of two terms, proportional to the mass and stiffness matrix, respectively.

$$\mathbf{C} = \alpha \mathbf{M} + \beta \mathbf{K}. \quad (4.73)$$

The modal damping matrix follows from (4.51) and the orthogonality relations (4.24) as

$$\mathbf{c} = \mathbf{U}^T \mathbf{C} \mathbf{U} = \alpha \mathbf{m} + \beta \mathbf{k}. \quad (4.74)$$

Thus, Rayleigh damping leads to modal uncoupling, as $\mathbf{m}$ and $\mathbf{k}$ are diagonal.

In a structure with Rayleigh damping, the modal damping ratios ζ_j depend on the corresponding modal frequency ω_j in a simple way. It follows from the modal damping (4.74) and the formulae (4.57) for ω_j and ζ_j that

$$2\zeta_j = \frac{\alpha}{\omega_j} + \beta \omega_j \quad , \quad j = 1, 2, \dots, n. \quad (4.75)$$

Thus, the damping ratio consists of two terms, one from the mass matrix that is inversely proportional to the angular frequency, and another from the stiffness matrix that is proportional to the angular frequency. These two terms and their sum are shown in Fig. 4.11. The damping ratio passes through a minimum value $\zeta_{\min}$ at the frequency $\omega_{\min}$, determined from the parameters α and β as

$$\zeta_{\min} = \sqrt{\alpha\beta} \quad , \quad \omega_{\min} = \sqrt{\alpha/\beta}. \quad (4.76)$$

The parameters have been used to normalize the graph in Fig. 4.11. It is seen that Rayleigh damping implies high damping at low and at high modal frequencies relative to $\omega_{\min}$. Conversely, the parameters α and β can be expressed in terms of $\zeta_{\min}$ and $\omega_{\min}$,

$$\alpha = \zeta_{\min} \omega_{\min} \quad , \quad \beta = \zeta_{\min} / \omega_{\min}. \quad (4.77)$$

In terms of these parameters the modal damping ratio (4.75) formula takes the form

$$2\zeta_j = \zeta_{\min} \left(\frac{\omega_{\min}}{\omega_j} + \frac{\omega_j}{\omega_{\min}} \right), \quad j = 1, 2, \dots, n. \quad (4.78)$$

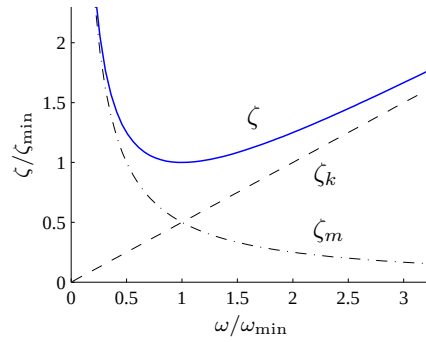


Fig. 4.11: Frequency dependence of the damping ratio in Rayleigh damping.

It is sometimes convenient to represent the Rayleigh parameters and the implied modal damping ratios in terms of the damping ratio ζ_a and ζ_b at the frequencies ω_a and ω_b , respectively. When substituting this pair of values into (4.77) the Rayleigh parameters are obtained as

$$\alpha = 2\omega_a\omega_b \frac{\zeta_a\omega_b - \zeta_b\omega_a}{\omega_b^2 - \omega_a^2}, \quad \beta = 2 \frac{\zeta_a\omega_b - \zeta_b\omega_a}{\omega_b^2 - \omega_a^2}. \quad (4.79)$$

Often the frequencies ω_a and ω_b define an interval of a common desired level of damping, expressed by $\zeta_* = \zeta_a = \zeta_b$. In this case the general result specializes to

$$\alpha = 2 \frac{\omega_a\omega_b}{\omega_a + \omega_b} \zeta_*, \quad \beta = \frac{2}{\omega_a + \omega_b} \zeta_*. \quad (4.80)$$

The minimum damping ratio $\zeta_{\min}$ is attained at the frequency $\omega_{\min}$ located between the frequencies ω_a and ω_b used for the calibration of the Rayleigh parameters. Substitution of the parameters α and β from (4.80) into (4.76) gives

$$\zeta_{\min} = 2 \frac{\sqrt{\omega_a\omega_b}}{\omega_a + \omega_b} \zeta_*, \quad \omega_{\min} = \sqrt{\omega_a\omega_b}. \quad (4.81)$$

Within the frequency interval between ω_a and ω_b the damping ratio varies between ζ_* and $\zeta_{\min}$. Thus, if the frequency interval includes several natural frequencies it may be important to consider the lower limit $\zeta_{\min}$. If for example $\omega_b = 4\omega_a$, then $\zeta_{\min} = 0.8\zeta_*$.

The concept of Rayleigh damping has been generalized by Caughey (1960), who introduced the following series representation,

$$\mathbf{C} = \mathbf{M} \sum_i \alpha_i (\mathbf{M}^{-1}\mathbf{K})^i = \alpha_0 \mathbf{M} + \alpha_1 \mathbf{K} + \alpha_2 \mathbf{K} \mathbf{M}^{-1} \mathbf{K} \dots \quad (4.82)$$

The first two terms of the series correspond to Rayleigh damping, while the remaining terms give orthogonal modal damping, but generalize the frequency dependence. It follows from the definition of the modal damping matrix (4.51) that

$$2\zeta_j\omega_j = \alpha_0 + \alpha_1\omega_j^2 + \alpha_2\omega_j^4 + \dots \quad (4.83)$$

It can be shown by use of the Cayley-Hamilton theorem for powers of matrices that only n of the terms are independent, and thus n consecutive terms describe general modal damping. Thus, the first n terms in (4.82) enable evaluation of the damping matrix corresponding to a given set of n modal damping ratios ζ_j and their corresponding angular frequencies ω_j .

In practice an alternative to the evaluation of modal damping from the damping matrix $\mathbf{C}$ is to estimate the modal damping ratios directly. It should be recognized that damping is usually one of the more uncertain parts of a practical dynamic analysis. The assumptions implied in assuming uncoupling of the undamped modes is equivalent to assuming that energy transport can be neglected, i.e. that energy is dissipated where it occurs without transport through the structure. Clearly this assumption is not representative if there are essential concentrated damping mechanisms in the structure, e.g. in the form of locally mounted dampers. However, it can be shown by an asymp-

otic analysis that the present *orthogonal damping* is dominating, if damping is light, see e.g. G eradin and Rixen (1997) Chapter 3. A dynamic analysis of linear structures with general damping can be performed via modal analysis based on damped complex-valued mode shapes or by direct time integration of the original equations of motion (4.47).

Example 4.9. Rayleigh damping in shear building.

Show figure with external dampers scaling with m_j and internal dampers scaling with k_j .

Fig. 4.12: Rayleigh damping of portal frame.

□

4.5 System reduction

In structural dynamics very large systems are often needed in order to model the geometry of the structure appropriately. This may lead to models in which the frequencies of the higher modes are larger than the frequencies represented in the loading. Furthermore the models often have limited accuracy in the higher modes. As a result it may be desirable to carry out the dynamic analysis on a reduced model. Three model reduction techniques are presented here: master-slave relations, quasi-static reduction, and truncated modal representation with quasi-static correction for high-frequency modes.

4.5.1 Master-slave constraints

The master-slave reduction is particularly useful for structures with a repetitive geometry as illustrated by the simple frame in Fig. 4.13. The frame is represented by beam elements, and thus each of the unsupported nodes contain three degrees of freedom $[u, v, \theta]$. The main interest is in the swaying vibrations in which the displacements of the nodes at the right side can be expressed in terms of the displacement components of the corresponding node at the left side,

$$[u, v, \theta]_{\text{right}} = [u, -v, \theta]_{\text{left}}$$

Let the degrees of freedom at the left side be collected in the vector $\mathbf{q}_d$, with subscript d indicating the dynamic degrees of freedom. The corresponding degrees of freedom at the right side are arranged similarly in the vector $\mathbf{q}_s$. When the master-slave relationship is expressed in terms of the matrix $\mathbf{S}$ the total generalized displacement vector $\mathbf{q}$ is expressed in terms of the dynamic displacements $\mathbf{q}_d$ by the relation

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_d \\ \mathbf{q}_s \end{bmatrix} = \begin{bmatrix} \mathbf{I} \\ \mathbf{S} \end{bmatrix} \mathbf{q}_d \quad (4.84)$$

In the example $\mathbf{q}_d$ contains the degrees of freedom of the left side of the frame, while $\mathbf{q}_s$ contains the corresponding degrees of freedom of the right side.

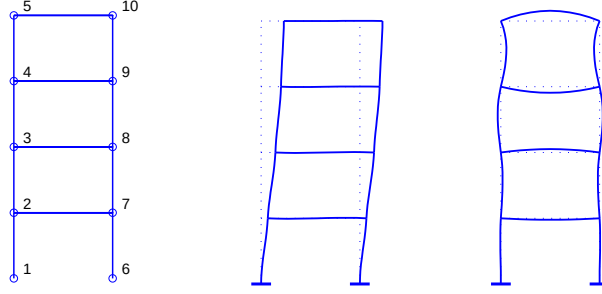


Fig. 4.13: a) Symmetric portal frame, b) Swaying vibration, c) Symmetric vibration.

Let the stiffness mass and damping matrices be given in the block matrix format

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{dd} & \mathbf{K}_{ds} \\ \mathbf{K}_{sd} & \mathbf{K}_{ss} \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} \mathbf{M}_{dd} & \mathbf{M}_{ds} \\ \mathbf{M}_{sd} & \mathbf{M}_{ss} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \mathbf{C}_{dd} & \mathbf{C}_{ds} \\ \mathbf{C}_{sd} & \mathbf{C}_{ss} \end{bmatrix}. \quad (4.85)$$

The stiffness matrix of the reduced system $\mathbf{K}_d$ follows by substitution of the representation (4.84) into the elastic energy $U(\mathbf{q})$,

$$U(\mathbf{q}) = \frac{1}{2} \mathbf{q}^T \mathbf{K} \mathbf{q} = \frac{1}{2} \mathbf{q}_d^T \bar{\mathbf{K}} \mathbf{q}_d, \quad (4.86)$$

with the reduced stiffness matrix

$$\bar{\mathbf{K}} = \mathbf{K}_{dd} + \mathbf{S}^T \mathbf{K}_{sd} + \mathbf{K}_{ds} \mathbf{S} + \mathbf{S}^T \mathbf{K}_{ss} \mathbf{S} \quad (4.87)$$

The reduced mass matrix is obtained similarly from the kinetic energy $T(\dot{\mathbf{q}})$,

$$\bar{\mathbf{M}} = \mathbf{M}_{dd} + \mathbf{S}^T \mathbf{M}_{sd} + \mathbf{M}_{ds} \mathbf{S} + \mathbf{S}^T \mathbf{M}_{ss} \mathbf{S}, \quad (4.88)$$

while the reduced damping matrix follows from the dissipation function $D(\dot{\mathbf{q}})$,

$$\bar{\mathbf{C}} = \mathbf{C}_{dd} + \mathbf{S}^T \mathbf{C}_{sd} + \mathbf{C}_{ds} \mathbf{S} + \mathbf{S}^T \mathbf{C}_{ss} \mathbf{S}, \quad (4.89)$$

It is observed that the reduction has the same form for all three system matrices.

The load is similarly partitioned in the form $\mathbf{f}^T = [\mathbf{f}_d^T, \mathbf{f}_s^T]$, and the load vector of the reduced system then follows from the equivalence of its rate of

work as

$$\bar{\mathbf{f}} = \mathbf{f}_d + \mathbf{S}^T \mathbf{f}_s. \quad (4.90)$$

The role of the term $\mathbf{S}^T \mathbf{f}_s$ is to combine the load contributions on corresponding master and slave degrees of freedom, e.g. from the left and the right part of the frame in Fig. 4.13.

The reduced form of the equations of motion is

$$\bar{\mathbf{M}}\ddot{\mathbf{q}}_d(t) + \bar{\mathbf{C}}\dot{\mathbf{q}}_d(t) + \bar{\mathbf{K}}\mathbf{q}_d(t) = \bar{\mathbf{f}}(t). \quad (4.91)$$

In the case where the reduction implements exact symmetry relations, the solution represents exact response to the corresponding part of the general response, e.g. the swaying motion of the frame in Fig. 4.13.

Example 4.10. Master–slave reduction of symmetric building.

Discuss importance of vertical displacement and rotation of joints!

□

4.5.2 Quasi-static condensation

While the master–slave technique is based on direct kinematic relations between sets of degrees of freedom in a structure, the quasi-static condensation technique is based on a set of quasi-static relations, Guyan (1965) and Irons (1965). Again a partitioning is used in which $\mathbf{q}_d$ contains the degrees of freedom that are retained in the reduced dynamic problem, while $\mathbf{q}_s$ contains the remaining quasi-static degrees of freedom that are eliminated in the reduction. The equation of motion is set up for a frequency dependent problem in terms of the partitioned of the matrices matrices (4.85) and of the external force,

$$\left(\begin{bmatrix} \mathbf{K}_{dd} & \mathbf{K}_{ds} \\ \mathbf{K}_{sd} & \mathbf{K}_{ss} \end{bmatrix} - i\omega \begin{bmatrix} \mathbf{C}_{dd} & \mathbf{C}_{ds} \\ \mathbf{C}_{sd} & \mathbf{C}_{ss} \end{bmatrix} - \omega^2 \begin{bmatrix} \mathbf{M}_{dd} & \mathbf{M}_{ds} \\ \mathbf{M}_{sd} & \mathbf{M}_{ss} \end{bmatrix} \right) \begin{bmatrix} \mathbf{q}_d \\ \mathbf{q}_s \end{bmatrix} = \begin{bmatrix} \mathbf{f}_d \\ \mathbf{f}_s \end{bmatrix}. \quad (4.92)$$

An approximate expression for the quasi-static degrees of freedom $\mathbf{q}_s$ is now obtained by neglecting damping and inertial effects in the last block of equations as well as the load vector $\mathbf{f}_s$. Hereby the last block of equations reduce to the form

$$\mathbf{K}_{sd}\mathbf{q}_d + \mathbf{K}_{ss}\mathbf{q}_s = \mathbf{0} \quad (4.93)$$

with the solution

$$\mathbf{q}_s = -\mathbf{K}_{ss}^{-1}\mathbf{K}_{sd}\mathbf{q}_d. \quad (4.94)$$

It is noted that this is a static set of constraints, based on force equilibrium, and that it does not include the deformation associated with direct loading of

the degrees of freedom contained in $\mathbf{q}_s$. This suggests that degrees of freedom with concentrated loads should not be included in $\mathbf{q}_s$.

The homogeneous quasi-static relation (4.94) corresponds to the previous format with constraint matrix

$$\mathbf{S} = -\mathbf{K}_{ss}^{-1}\mathbf{K}_{sd}. \quad (4.95)$$

The reduced matrices and force vector therefore follow from those obtained previously. The reduced stiffness matrix assumes a special form due to the static form of the present constraint condition (4.93),

$$\bar{\mathbf{K}} = \mathbf{K}_{dd} - \mathbf{K}_{ds}\mathbf{K}_{ss}^{-1}\mathbf{K}_{sd} \quad (4.96)$$

The reduced mass matrix $\bar{\mathbf{M}}$, the reduced damping matrix $\bar{\mathbf{C}}$ and the reduced load vector $\bar{\mathbf{f}}$ are calculated from the previous relations (4.88)–(4.90) with $\mathbf{S}$ given by (4.95). The result is a set of equations of motion in the form (4.91).

The contrast to reduction based on kinematic constraints from e.g. symmetry the quasi-static condensation technique contains a measure of approximation. It is assumed that the degrees of freedom in $\mathbf{q}_s$ can be derived from the dynamic degrees of freedom $\mathbf{q}_d$ by the homogeneous static relation (4.93). Even if the quasi-static degrees of freedom are not loaded directly the omission of dynamic effects is an approximation that is limited to sufficiently ‘low frequencies. For high frequencies the inertial term $\omega^2\mathbf{M}_{ss}$ will dominate the static term $\mathbf{K}_{ss}$. An estimate of the frequency limit for the quasi-static condensation can be obtained by considering vibrations if the generalized displacements were constrained, leaving only the generalized displacement components $\mathbf{q}_s$. The corresponding free undamped vibrations are governed by the eigenvalue problem

$$(\mathbf{K}_{ss} - \omega_s^2\mathbf{M}_{ss})\mathbf{q}_s = \mathbf{0}. \quad (4.97)$$

where damping has been omitted, as the purpose is to obtain an estimate of the magnitude of the limiting frequency. The approximate nature of this upper limit suggests that an estimate of the lowest natural frequency of the eigenvalue problem (4.97) obtained e.g. by Rayleigh’s quotient will be sufficient.

Example 4.11. Quasi-static condensation of beam-rotations.

Elimination of axial displacements and rotations in cantilever beam.

Fig. 4.14: Cantilever represented by 6 elements.

Estimate the frequency of the kinematically constrained system by simple symmetric bending - disregarding irregularity at the support. $\square$

4.5.3 Reduced set of dynamic modes

In a typical dynamic response analysis the load components affecting the global response of the structure are mainly confined to the lower frequencies as illustrated in Fig. 4.15. As a consequence there will be a number of high-frequency modes, which will mainly experience loads at frequencies well below their resonance frequency. Thus the response of these modes will mainly be quasi-static, while the dynamic response will be confined to a number of low-frequency modes. For this type of problem the response can be obtained by a truncated modal analysis, limited to the dynamic low-frequency modes, and then supplemented by a quasi-static contribution from the non-resonant high-frequency modes.

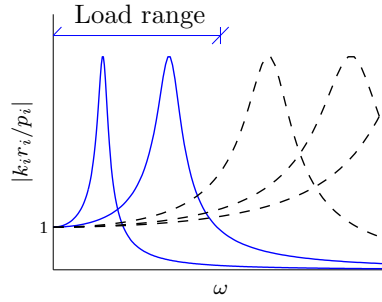


Fig. 4.15: Normalized modal frequency response within or above load frequency interval

In this method it is assumed that the dynamic behavior of the structure is mainly associated with the m lowest modes. These modes are represented by their mode shape vectors $\mathbf{u}_1, \dots, \mathbf{u}_m$, and the response is written in the form

$$\mathbf{q}(t) = \sum_{j=1}^m \mathbf{u}_j r_j(t) + \sum_{j=m+1}^n \mathbf{u}_j r_j(t) = \sum_{j=1}^m \mathbf{u}_j r_j(t) + \mathbf{q}_s(t) \quad (4.98)$$

It is an important point that the quasi-static part of the response $\mathbf{q}^s(t)$ is not going to be expanded in mode-shapes explicitly, and thus only the first m modes need to be calculated.

The dynamic part of the response is expanded in terms of the first m mode-shape vectors,

$$\mathbf{q}_d(t) = [\mathbf{u}_1, \dots, \mathbf{u}_m] \mathbf{r}_d(t) = \underset{n \times m}{\mathbf{U}_d} \underset{m \times 1}{\mathbf{r}_d(t)} \quad (4.99)$$

where the vector $\mathbf{r}_d(t)$ contains the first m modal coordinates. Substitution of the reduced representation (4.99) into the equations of motion (4.47) gives the n equations

$$\mathbf{M}\mathbf{U}_d\ddot{\mathbf{r}}_d + \mathbf{C}\mathbf{U}_d\dot{\mathbf{r}}_d + \mathbf{K}\mathbf{U}_d\mathbf{r}_d = \mathbf{f}(t). \quad (4.100)$$

There are only $m < n$ unknown modal components $r_1(t), \dots, r_m(t)$ in these equations, and the number of equations are therefore reduced by pre-multiplication with the dynamic mode shape matrix $\mathbf{U}_d^T$,

$$\mathbf{U}_d^T\mathbf{M}\mathbf{U}_d\ddot{\mathbf{r}}_d + \mathbf{U}_d^T\mathbf{C}\mathbf{U}_d\dot{\mathbf{r}}_d + \mathbf{U}_d^T\mathbf{K}\mathbf{U}_d\mathbf{r}_d = \mathbf{U}_d^T\mathbf{f}(t) \quad (4.101)$$

For orthogonal damping this results in the m uncoupled modal equations,

$$m_j\ddot{r}_j(t) + c_j\dot{r}_j(t) + k_jr_j(t) = p_j(t), \quad j = 1, \dots, m. \quad (4.102)$$

These equations constitute the first m modal equations of a full modal analysis, and thus correspond to truncation of the full modal analysis. Similarly the initial conditions are the first m of the complete set of modal initial conditions,

$$\mathbf{r}_{d0} = \mathbf{m}_d^{-1}\mathbf{U}_d^T\mathbf{M}\mathbf{q}_0, \quad \dot{\mathbf{r}}_{d0} = \mathbf{m}_d^{-1}\mathbf{U}_d^T\mathbf{M}\dot{\mathbf{q}}_0 \quad (4.103)$$

In these relations $\mathbf{m}_d$ is the diagonal modal mass matrix of the m dynamic modes.

The dynamic response $\mathbf{q}_d(t)$ is the full dynamic response to the part of the load contained in the first m modal loads $p_j(t)$. Thus, the remaining quasi-static part of the response $\mathbf{q}_s(t)$ can be determined as the response to the remaining part of the load, associated with the modes higher than m . The full expansion of the load vector is

$$\mathbf{f}(t) = \sum_{j=1}^n \frac{1}{k_j} (\mathbf{K}\mathbf{u}_j) p_j(t) \quad (4.104)$$

This is easily verified by pre-multiplication by $\mathbf{u}_k^T$, leading to the defining equation for the modal load $p_k(t)$.

The load included in the dynamic response calculation corresponds to the first m terms in this expansion. The quasi-static part of the response $\mathbf{q}^s(t)$ is calculated from the last $n-m$ terms of the load expansion. The quasi-static response can therefore be based the load $\mathbf{f}_s(t)$ found by subtracting the load corresponding to the first m modal load components,

$$\mathbf{f}_s(t) = \mathbf{f}(t) - \sum_{j=1}^m \frac{1}{k_j} (\mathbf{K}\mathbf{u}_j) (\mathbf{u}_j^T \mathbf{f}(t)) \quad (4.105)$$

The quasi-static response to this load is found from the equations of motion, when neglecting the inertial and damping terms,

$$\mathbf{K}\mathbf{q}_s(t) = \mathbf{f}_s(t) \quad (4.106)$$

Substitution of $\mathbf{f}_s(t)$ from (4.105) and pre-multiplication with $\mathbf{K}^{-1}$ gives the quasi-static part of the response as

$$\mathbf{q}_s(t) = \left(\mathbf{K}^{-1} - \sum_{j=1}^m \frac{1}{k_j} \mathbf{u}_j \mathbf{u}_j^T \right) \mathbf{f}(t) \quad (4.107)$$

In practice, this quasi-static correction to the truncated dynamic response will be evaluated by first computing the response corresponding to the full load $\mathbf{f}(t)$, and then subtracting the m terms $k_j^{-1} \mathbf{u}_j p_j(t)$. The quasi-static correction is a computationally small part of the response analysis, and will often provide the local details that may be lost if the modal representation was just truncated.

Example 4.12. System reduction for offshore platform. The accuracy of the quasi-static reduction technique has been illustrated by in Hansteen and Bell (1979)

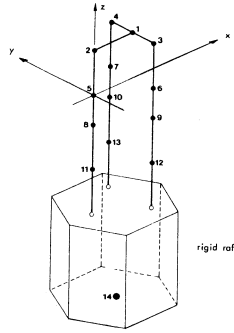


Fig. 4.16: Simplified platform model

The figure shows a simplified model of a concrete gravity offshore platform. A superstructure represented by nodes 1–4 is supported by three shafts, connected to the approximately rigid concrete substructure. The motion of the concrete substructure is represented by node 14. The model contains a total of $14 \times 6 = 84$ degrees of freedom.

The dynamic response was calculated for a sea state described by the Pierson-Moskowitz spectrum with $H_{1/3} = 15$ m and the mean period $T_m = 15$ s. The sea state was simulated by superposition of 35 harmonic waves of unit amplitude with frequencies in the range 0.2–4.0 rad/s.

Three computational models were used

- Full model with all modes.

- 5 mode shapes without corrections.
- 5 mode shapes with quasi-static correction.

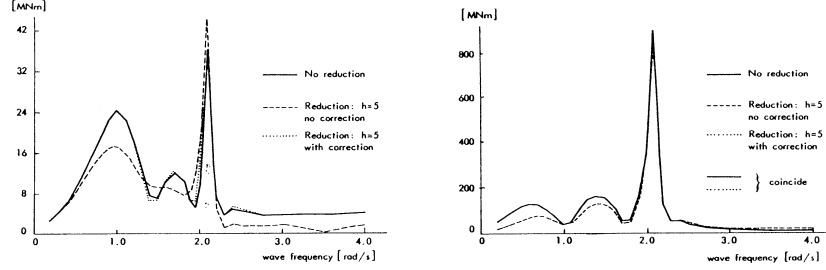
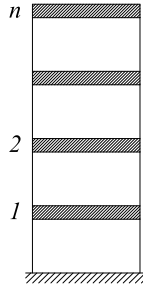


Fig. 4.17: Response spectrum for moment. a) Still water level, node 5, b) Bottom of shaft.

The results show that the horizontal motion of the superstructure is fully represented by all three models, while the moment distribution in the shafts can be represented by use of 5 mode shapes only if the quasi-static correction is included. $\square$

4.6 Exercises

Exercise 4.1. Consider a ‘shear building’ with 5 storeys. The mass of each floor is $m = 40000 \text{ kg}$, and the inter-floor stiffness is $k = 1.6 \cdot 10^6 \text{ N/m}$. Use the analytical results of this Lecture to answer the following.



- Calculate the 5 angular frequencies ω_j and the corresponding natural frequencies f_j .
- Find Rayleigh damping parameters α and β corresponding to a damping ratio around $\zeta \simeq 2 - 3\%$ for the three lowest modes.
- Calculate the five actual damping ratios ζ_j corresponding to the Rayleigh parameters α and β .

Exercise 4.2. The figure shows a simple model of a 2-story building with floor masses m and inter-story stiffness k . A reference frequency is defined as $\omega_0 = \sqrt{k/m}$.

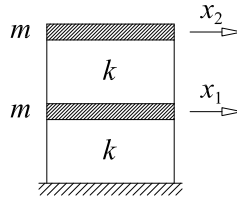


Fig. 4.18: Two story building with floor mass m and inter-floor stiffness k .

- Estimate the first natural vibration frequency by use of Rayleigh's quotient with the mode shape approximation $\mathbf{u}^T = [1, 2]$.
- Estimate the second natural vibration frequency by use of Rayleigh's quotient with the mode shape approximation $\mathbf{u}^T = [1, -1]$.
- Compare with the exact results and discuss the accuracy.

Exercise 4.3. Determine natural frequencies and corresponding mode shape vectors for a 5 story frame with two sets of columns using beam elements. $h = * * *$, $a =$, $E = * * *$ $m =$.

Identify the master-slave reduction matrix S and repeat the analysis on the reduced system. Discuss the results and the 'missing modes'.

Exercise 4.4. Consider a building frame with a 4 story central part and two identical side parts with 3 storys and one additional set of columns. Identify the master-slave reduction matrix S for swaying motion and calculate the first four natural frequencies and mode shapes.

Exercise 4.5. Model a three-span bridge with with beam bending elements, with side spans of length $\ell_1 = * * * * *$ and center span length $\ell_2 = * * * * *$. Find the number of elements needed to determine the 4 lowest natural frequencies with 1 pct.

Use quasi-static reduction to eliminate the rotation degrees of freedom. Determine the error of the 4 lowest natural degrees of freedom for this reduced model.

Summary

- Free vibrations of undamped discrete structures can be described in terms of mode shapes $\mathbf{u}_j$ performing harmonic motion with angular frequencies ω_j .
- The mode shapes $\mathbf{u}_j$ and the corresponding natural frequencies ω_j are determined by substitution of the assumed harmonic representation into the equation of motion, resulting in the generalized eigenvalue problem

$$\left(\mathbf{K} - \omega^2 \mathbf{M} \right) \mathbf{u} = \mathbf{0}$$

where $\mathbf{K}$ is the stiffness matrix and $\mathbf{M}$ is the mass matrix.

- The mode shape vectors $\mathbf{u}_j$ satisfy orthogonality relations with respect to the mass matrix $\mathbf{M}$ and the stiffness matrix $\mathbf{K}$.
- The modal mass and stiffness are defined by the products $m_j = \mathbf{u}_j^T \mathbf{M} \mathbf{u}_j$ and $k_j = \mathbf{u}_j^T \mathbf{K} \mathbf{u}_j$, respectively.
- The square of any natural frequency is given by the ratio of the corresponding modal stiffness and modal mass, $\omega_j^2 = k_j/m_j$, generalizing the similar relation for the single-degree-of-freedom system.
- An approximation to the (lower) natural frequencies can be found from Rayleigh's quotient $\omega^2 = \mathbf{u}^T \mathbf{K} \mathbf{u} / \mathbf{u}^T \mathbf{M} \mathbf{u}$, where the vector $\mathbf{u}$ is an approximate representation of the corresponding mode shape.
- Discrete structures can be analyzed by representing the displacements $\mathbf{q}(t)$ in terms of the *mode-shape vectors* $\mathbf{u}_j$, $j = 1, \dots, n$. This is conveniently done by introducing the mode-shape matrix $\mathbf{U}$ containing the mode shape vectors as columns. The modal representation then takes the form $\mathbf{q}(t) = \mathbf{U} \mathbf{r}(t)$, where $\mathbf{r}(t)$ contains the *modal coordinates*.
- The mechanical energy of the structure is the sum of the mechanical energies of the individual modes, without coupling terms.
- Due to the energy uncoupling the motion of each of the modes can be treated independently, provided the damping matrix satisfies the modal orthogonality relations.
- A convenient way of representing orthogonal damping is the *Rayleigh damping approximation* $\mathbf{C} = \alpha \mathbf{M} + \beta \mathbf{K}$, where the first term represents damping associated with the absolute motion, while the second term represents damping due to deformation of the structure. The approximation is usually only acceptable in a limited frequency interval, and can be replaced by individual modal damping ratios ζ_j , $j = 1, \dots, n$.

- Modal analysis is easily implemented either in the full form, using all mode shapes, or in a reduced form, where only a limited set of mode shapes are used to represent the dynamics of the system. In the reduced analysis improved accuracy can often be obtained by introducing the remaining part of the load via a quasi-static correction.

There is an increasing use of structures or structural elements with low internal damping, such as cables, slender bridges, towers and steel chimneys. In these structures it may be necessary to introduce additional damping by externally mounted devices. These typically have the form of local dampers or tuned mass absorbers. The local damper consists in a damping device that is mounted such that it connects a point of the structure with a fixed point in the surroundings, or connects two different points in the structure. While dampers connecting to the surroundings are typically used in the case of cables, local dampers acting on the relative motion are used e.g. in shear buildings or shear flexible bridge decks. The dampers can either be of passive type, relying on a mechanism such as e.g. viscosity to dissipate energy when the damper extends or contracts, or of active type, with the damper forced controlled electronically based on measurement of appropriate motion data. An intermediate form is the so-called semi-active damper, in which the resulting force can only counteract the relative motion over the damper. The passive dampers are often mechanical and restricted to a specific set of parameter values characterizing the damper. Semi-active dampers require a power supply for the control signal and the control of the force, but typically the necessary power is rather limited, as the forces can be rather small and e.g. have the nature of a clamping force.

Typically the effect of mounting dampers on structures can be analyzed by classic numerical analysis methods, once the appropriate damper properties are known. Thus, the use of concentrated dampers typically involves two steps - an introductory selection of the appropriate damper parameters, often determined by approximate analytical methods, and then subsequent dynamic analyses of the performance of the structure under the required loading conditions by conventional analysis methods, such as the finite element method discussed in Chapter 3. The basic damper calibration problem is illustrated in Section [refsec:cable-damping](#), in which optimal calibration of a concentrated damper on a cable is considered in detail. This problem is important in its own right and lends to a rather elegant analytical solution. In the case of a cable the calibration can be based on an analytical solution of the problem of a vibrating cable with a concentrated damper. However, more complex structures must be analyzed by numerical methods. The calibration problem for

local dampers on structures that are represented by a discrete finite element model is treated in Section 5.2. As it turns out, the characteristics of the cable calibration procedure carries over in an only slightly modified form.

Mechanical damping devices depend on a sufficient magnitude of the motion on which the damping operates. Thus, there may be a need to generate a sufficiently large relative motion within the damping device. In the tuned mass absorber a mass is suspended in a way that generates a resonant motion near the frequency, at which the device is designed to operate. A damper is then applied between the mass and the point of suspension, leading to energy dissipation related to the resonant motion. The basic principles and the the question of optimal calibration are treated in Section ?? . A calibration procedure based on complex resonance frequencies is developed that may also be used in connection with resonant dampers based on active control.

5.1 Damping of Cables

Cables constitute an efficient means of supporting static loads and they find increasing use as stay cables for e.g. bridges and towers. However, the slenderness of the cables makes them prone to vibrations, either induced by direct loads on the cable from wind or a combination of wind and rain, or via motion of the supported structure. The cables are usually made of strands of high strength steel, and this leads to extremely low material damping, typically in the order of 0.001 of critical, Yamaguchi and Fujino (1998).



Fig. 5.1: a) Eiland Bridge, b) MR-damper attached to stay cable.

The light damping of the cable itself leads to a need to provide external damping, and for this transverse dampers have typically been attached at a distance of 2-4 pct. of the span from one of the supports as shown in Fig. 5.1. The motion of the damper is limited by its closeness to the support, and it is of importance to find the optimal combination of damper location and

external damping magnitude. If the external damping is too small its effect will be negligible. In contrast, if it is too large, the cable will vibrate as if it was supported at the location of the damper. In that case the effect of the damper would mainly be a slight increase in the vibration frequency, without any appreciable damping effect on the vibrations. Calibration of the damper therefore consists in finding an optimal compromise. The solution presented below for a viscous damper is based on Krenk (2000), and subsequently extended to more general damper properties.

5.1.1 Taut cable with a local damper

The basic problem is that of an ideal homogeneous taut cable without bending stiffness, fixed at its two ends. The cable has an axial force T , mass per unit length m , and length ℓ as shown in Fig. 5.2. A concentrated damping device is located at the distance $a \ll \ell$ from the left support. For convenience the damping device will first be assumed to be a transverse viscous damper with viscosity coefficient c . The basic solution procedure is fairly insensitive to the details of the damping device, and the effect of more general damping properties, as well as local stiffness and mass is discussed subsequently in Section 5.1.2.

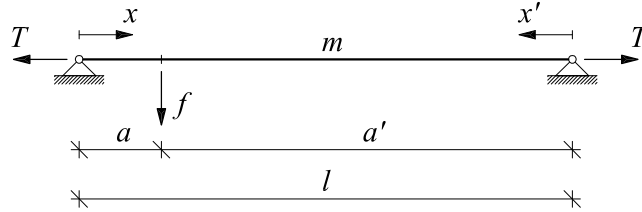


Fig. 5.2: Taut cable with local transverse control force.

The equation of motion of a taut cable without bending stiffness was treated in Section 2.1. In the present case there is a concentrated force from the damper acting at $x = a$. This force is conveniently written by use of the delta function as $f(t)\delta(x - c)$. The equation of transverse motion then takes the well known form

$$T \frac{\partial^2 u}{\partial x^2} - m \frac{\partial^2 u}{\partial t^2} = f \delta(x - a). \quad (5.1)$$

where $u(x, t)$ is the transverse displacement. The partial differential equation is to be solved with the boundary conditions $u(0, t) = u(\ell, t) = 0$, specifying fixed ends.

At the location of the damper there is a discontinuity in the inclination of the cable, providing a transverse force matching the force in the damper. This

relation has the form

$$T \left(\left. \frac{\partial u}{\partial x} \right|_{a+} - \left. \frac{\partial u}{\partial x} \right|_{a-} \right) = f. \quad (5.2)$$

Within a linear setting the damped free vibration response of the cable and the damper force can be represented in terms of complex exponentials as

$$u(x, t) = u_n(x) \exp(i\omega_n t), \quad f(t) = f_n \exp(i\omega_n t), \quad (5.3)$$

where ω_n is a natural eigenfrequency, $u_n(x)$ is the corresponding mode shape, and f_n the control force amplitude. If there were no damping, the natural frequencies and mode shapes would be real-valued, but in the case of damping they will take on complex values. The damping will show up via the imaginary part of the complex natural frequencies ω_n as explained in Chapter 1. Similarly, in the present case of a local damper the mode-shapes $u_n(x)$ will be complex.

It is convenient to describe the cable to the left of the damper by the coordinate x , and the part to the right of the damper by the complementary coordinate $x' = \ell - x$. Similarly the complement to the damper location $x = a$ is $x' = a'$ with $a' = \ell - a$ as shown in Fig. 5.2. The mode-shapes satisfy the homogeneous equation of motion on both sides of the damper,

$$\frac{d^2 u_n}{dx^2} + \beta_n^2 u_n = 0, \quad \begin{cases} 0 < x < a, \\ 0 < x' < a'. \end{cases} \quad (5.4)$$

where the wave number β_n has been introduced as

$$\beta_n = \omega_n \sqrt{\frac{m}{T}}, \quad n = 1, 2, \dots \quad (5.5)$$

The differential equation (5.4) is homogeneous in each of the two intervals between the supports and the damper. The solution is conveniently expressed in the form

$$u_n(x) = \begin{cases} A_n \sin(\beta_n x) & , \quad 0 \leq x \leq a \\ A'_n \sin(\beta_n x') & , \quad 0 \leq x' \leq a' \end{cases} \quad (5.6)$$

where A_n and A'_n are arbitrary constants. Continuity of the shape function at $x = a$ requires

$$u_n(a) = A_n \sin(\beta_n a) = A'_n \sin(\beta_n a'). \quad (5.7)$$

Hereby the solution (5.6) takes the form

$$u_n(x) = \begin{cases} u_n(a) \frac{\sin(\beta_n x)}{\sin(\beta_n a)}, & 0 \leq x \leq a, \\ u_n(a) \frac{\sin(\beta_n x')}{\sin(\beta_n a')}, & 0 \leq x' \leq a'. \end{cases} \quad (5.8)$$

The amplitude $u_n(a)$ at the damper is undetermined in a free vibration, while the wavenumber β_n is determined from force balance at the damper.

Substitution of the mode shape representation (5.8) into the force discontinuity relation (5.2) provides an equation for determination of the complex eigenfrequencies.

$$\cot(\beta_n a) + \cot(\beta_n a') = -\frac{1}{\beta_n T} \frac{f_n}{u_n(a)} = -\frac{H(\omega_n)}{\beta_n T}, \quad (5.9)$$

where $H(\omega)$ is the frequency function of the damper,

$$f = H(\omega) u(a). \quad (5.10)$$

In the case of linear viscous damping the damper force is proportional to the velocity, $f(t) = c \dot{u}(a, t)$, corresponding to a purely imaginary transfer function $H(\omega) = ic\omega$, where c is the viscous damping coefficient. In general, damping is associated with a positive imaginary part of the frequency function $H(\omega)$, leading to complex-valued wavenumber solutions β_n .

The wavenumber equation (5.9) can not be solved in closed form, but is easily solved numerically or asymptotically, when the damper is located close to a support. In both cases it is convenient to reformulate the equation to depend explicitly only on the lengths l and a , but not on the length a' . In the frequency equation (5.9) the term $\cot(\beta a')$ is expressed in terms of $\cot(\beta \ell)$ and $\cot(\beta a)$ by use of the trigonometric relation

$$\cot(\beta a') = \cot(\beta \ell - \beta a) = \frac{1 + \cot(\beta \ell) \cot(\beta a)}{\cot(\beta a) - \cot(\beta \ell)}. \quad (5.11)$$

Substitution of this expression into (5.9) gives the equation

$$\frac{1 + \cot^2(\beta a)}{\cot(\beta \ell) - \cot(\beta a)} = \frac{H}{\beta_n T}. \quad (5.12)$$

The next step is to write this equation in a form that reflects the fact that the effect of moderate damping and small a should appear as a perturbation of the wavenumber equation for the similar undamped cable. The frequency equation for wavenumbers β_n^0 of the undamped cable follows immediately from $H = 0$ and $a = 0$ in the form

$$\tan(\beta_n^0 \ell) = 0 \quad \Rightarrow \quad \beta_n^0 = n \frac{\pi}{\ell}, \quad n = 1, 2, \dots \quad (5.13)$$

The damped wave number equation (5.12) is written in a similar form, by isolating the term $\tan(\beta_n \ell)$ on the left side of the equation, whereby

$$\tan(\beta \ell) = \frac{\frac{H}{\beta_n T} \sin^2(\beta_n a)}{1 + \frac{H}{\beta_n T} \cos(\beta_n a) \sin(\beta_n a)} \quad (5.14)$$

This form of the frequency equation is suitable for solution, either numerically by iteration or in asymptotic form.

An iterative numerical solution is obtained for moderate damping by substituting the current estimate β_n^j on the right side and evaluating a new estimate β_n^{j+1} by inversion of the tangent function. Starting from the undamped wave numbers β_n^0 from (5.13), the iterative scheme is given by

$$\beta_n^{j+1} \ell = n\pi + \arctan \left(\frac{\frac{H}{\beta_n^j T} \sin^2(\beta_n^j a)}{1 + \frac{H}{\beta_n^j T} \cos(\beta_n^j a) \sin(\beta_n^j a)} \right). \quad (5.15)$$

Numerical results from this procedure are used later to validate the asymptotic solution, presented next.

Asymptotic solution

The asymptotic solution is based on the damper being located close to the support, whereby $a/\ell \ll 1$. Assuming the wave numbers β_n of the damped problem to be small perturbations from the undamped values β_n^0 , the tangent function on the left side of the wavenumber equation (5.14) can be approximated by

$$\tan(\beta_n \ell) \simeq \beta_n \ell - n\pi = \Delta\beta_n \ell \quad (5.16)$$

where it has been convenient to introduce the change of wavenumber $\Delta\beta_n = \beta_n - \beta_n^0$ due to the damping device. The assumption $a/\ell \ll 1$ leads to the following representations for the trigonometric functions appearing on the right side of the wavenumber equation,

$$\sin(\beta_n a) \simeq \beta_n a, \quad \cos(\beta_n a) \sin(\beta_n a) \simeq \beta_n a. \quad (5.17)$$

These representations are useful for $\beta_n a < \pi/4$, although perhaps not very accurate around the upper limit. With the estimate $\beta_n \ell \simeq n\pi$ this corresponds to $na/\ell < 1/4$. When the linearized representations of the trigonometric functions are substituted into the wavenumber equation, the following asymptotic form is obtained

$$\frac{\Delta\beta_n}{\beta_n} \simeq \frac{a}{\ell} \frac{\frac{Ha}{T}}{1 + \frac{Ha}{T}} \quad (5.18)$$

It is convenient to rewrite this formula in terms of the angular frequency by use of the relation (5.5). In the present case of a taut cable the wavenumber and the angular frequency are proportional, and thus the frequency form of the relation is

$$\frac{\Delta\omega_n}{\omega_n} \simeq \frac{a}{\ell} \frac{\frac{Ha}{T}}{1 + \frac{Ha}{T}} \quad (5.19)$$

In this formula ω_n is a representative angular frequency that may be taken as either the undamped natural frequency ω_n^0 or the mean value of the angular frequency for a cable without the damper and the corresponding frequency when the damper is sufficiently large to effectively lock the point of attachment, i.e. the mean frequency $\frac{1}{2}(\omega_n^0 + \omega_n^\infty)$.

The asymptotic frequency formula (5.19) may be given a more ‘universal’ form by observing that for infinite damping the magnitude of the frequency response function $|H|$ tends to infinity, giving the limit

$$\frac{\Delta\omega_n^\infty}{\omega_n} = \frac{\omega_n^\infty - \omega_n^0}{\omega_n} \simeq \frac{a}{\ell}. \quad (5.20)$$

Thus, the factor a/ℓ represents the relative increase of the natural frequency ω_n , when the damping increases from zero to infinity. By use of this limiting behavior the frequency formula (5.19) can be expressed in the form

$$\frac{\Delta\omega_n}{\Delta\omega_n^\infty} = \frac{\omega_n - \omega_n^0}{\omega_n^\infty - \omega_n^0} \simeq \frac{\frac{Ha}{T}}{1 + \frac{Ha}{T}} \quad (5.21)$$

In this form the asymptotic relation gives the relative increment of the natural frequency as a function of a single non-dimensional parameter Ha/T , expressing the effective magnitude of the force introduced by the damping device. The formula is asymptotic in nature, but turns out to give a fairly accurate representation of the effect of a concentrated damping device on e.g. cables with sag and certain classes of discrete structures as discussed later.

Linear viscous damper

Many of the general features of the damped cable response can be illustrated by reference to a cable with a linear viscous damper with damping coefficient c . In this case the non-dimensional damping is conveniently con-

veniently expressed in terms of the real-valued parameter

$$\eta_n = \frac{1}{i} \frac{Ha}{T} = \frac{\omega_n^0 a c}{T}. \quad (5.22)$$

It is seen that this parameter is the ratio of the amplitude of the damper force $c \dot{u}_n(a)$ to the tension T in the cable. In terms of this parameter the asymptotic wavenumber formula (5.21) takes the form

$$\frac{\Delta\omega_n}{\Delta\omega_n^\infty} = \frac{\omega_n - \omega_n^0}{\omega_n^\infty - \omega_n^0} \simeq \frac{i\eta_n}{1 + i\eta_n} \quad (5.23)$$

This relation gives the natural complex frequency ω_n of mode n as a function of the non-dimensional damping parameter η_n , and the resulting modal damping of the cable follows from analyzing the natural frequency ω_n in the complex plane.

Separation of the frequency relation (5.23) into real and imaginary parts gives

$$\operatorname{Re}\left[\frac{\Delta\omega_n}{\Delta\omega_n^\infty}\right] = \frac{\eta_n^2}{1 + \eta_n^2}, \quad \operatorname{Im}\left[\frac{\Delta\omega_n}{\Delta\omega_n^\infty}\right] = \frac{\eta_n}{1 + \eta_n^2} \quad (5.24)$$

Elimination of the damping parameter η_n between these two equations leads to the equation

$$\operatorname{Re}[\Delta\omega_n](\operatorname{Re}[\Delta\omega_n] - \Delta\omega_n^\infty) + \operatorname{Im}[\Delta\omega_n]^2 = 0 \quad (5.25)$$

This equation demonstrates that the wave number increment $\Delta\omega_n$ is located on a semi-circle in the complex plane as illustrated in Fig. 5.3a. A geometric interpretation of the non-dimensional damping parameter η_n is obtained by evaluating the argument of the wave number increment $\Delta\omega_n$,

$$\frac{\operatorname{Re}[\Delta\omega_n]}{\operatorname{Im}[\Delta\omega_n]} = \eta_n = \cot \psi \quad (5.26)$$

from which it follows that the semi-circle is swept from left to right by increasing the damping parameter η_n . The variation of the imaginary part, determining the damping of the system, is illustrated in Fig. 5.3b. The maximum value of the imaginary part $\operatorname{Im}[\Delta\omega_n] = \frac{1}{2}\Delta\omega_n^\infty$ corresponds to $\eta_n = 1$, representing optimal tuning of the external damper.

The damping of the system is represented via the complex eigenfrequencies of the modes. The modal damping ratio ζ_n , follows from the representation

$$\omega_n = |\omega_n|(\sqrt{1 - \zeta_n^2} + i\zeta_n) \quad (5.27)$$

The modulus $|\omega_n|$ describes the magnitude of the angular frequency, and for small damping ratio the frequency of the oscillatory part of the solution. The

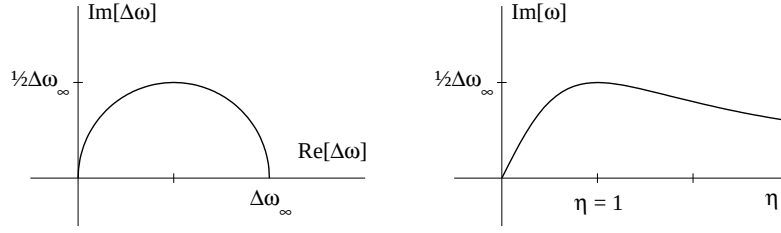


Fig. 5.3: a) Location of $\Delta\omega$ in complex plane, b) 'Universal' curve for $\text{Im}[\omega]$.

damping ratio ζ_n of mode n then is

$$\zeta_n = \frac{\text{Im}[\omega_n]}{|\omega_n|} \simeq \frac{\text{Im}[\omega_n]}{\omega_n^0} \quad (5.28)$$

where the last expression follows from use of the undamped natural frequency ω_n^0 . Use of the frequency analogue of (5.24) gives the damping ratio as function of the non-dimensional damping parameter η_n ,

$$\zeta_n \simeq \frac{\Delta\omega_n^\infty}{\omega_n^0} \frac{\eta_n}{1 + \eta_n^2} = \frac{a}{\ell} \frac{\eta_n}{1 + \eta_n^2} \quad (5.29)$$

This is the so-called 'universal damping curve' for a taut cable, determined numerically by Pacheco et al. (1993) and analytically in the present form by Krenk (2000).

The maximum damping at $\eta_n = 1$ corresponds to

$$\zeta_n^{\max} \simeq \frac{\Delta\omega_n^\infty}{2\omega_n^0} = \frac{a}{2\ell}, \quad c^{\text{opt}} \simeq \frac{T}{\omega_n^0 a} = \frac{\sqrt{mT}}{n\pi} \frac{\ell}{a} \quad (5.30)$$

Thus, the resulting damping ratio for an optimally tuned viscous damper is given solely by the relative increase of the undamped natural frequency caused by locking the damper. For the cable with a local viscous damper this implies that the maximum damping is determined solely by the distance of the damper from the support as $\zeta_n \simeq \frac{1}{2}a/\ell$. Experience suggests that rain-wind induced oscillations can be controlled by a damping ratio around 1 pct. According to the asymptotic formula (5.30a) this corresponds to an optimally tuned viscous damper located at $a = 0.02\ell$. Typical damper locations on stay-cables are 2-4 pct. of the cable span.

In the derivation of the asymptotic relations it was assumed that $na/\ell \ll 1$. Thus, it is to be expected that the accuracy of the asymptotic results decrease with increasing wavenumber for a given location of the damper. This is illustrated in Figs. 5.4a and 5.4b, showing the resulting damping ratio ζ_n of the first five modes for damper location $a = 0.02\ell$ and $a = 0.05\ell$, respectively.

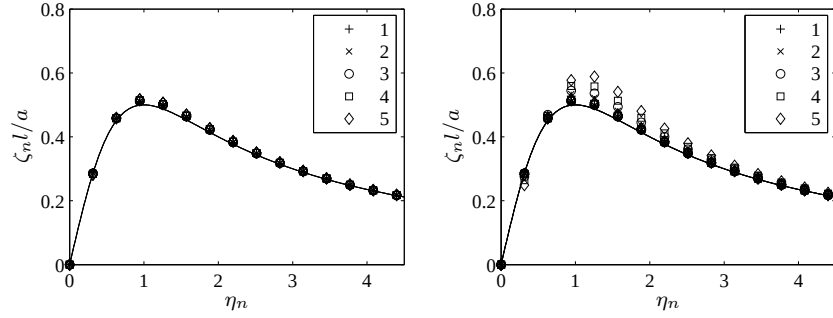


Fig. 5.4: Damping ratio ζ_n of the first 5 modes, a) $a/\ell = 0.02$, b) $a/\ell = 0.05$.

While the asymptotic format presents the results in an invariant format, the numerical results from the iterative procedure (5.15) demonstrate that the damping ratio is somewhat higher than predicted by the asymptotic results when $a = 0.05\ell$. It has been demonstrated in Krenk (2000) that the optimal damping is predicted accurately by the parameter

$$\eta_n^{\text{opt}} = \frac{\pi n a / \ell}{\sin(\pi n a / \ell)} \quad (5.31)$$

It is an interesting fact that the damping curve (5.29) giving the modal damping ratio as function of a maximum value and a non-dimensional damping parameter represents a fairly general asymptotic result with validity for a many other structures. Examples are cables with sag and beams treated in Sections 2.2 and 2.3, and also certain damper arrangements on discrete structures, dealt with in Section 4.

5.1.2 Effect of damper stiffness and mass

5.1.3 Non-linear damper properties

5.1.4 Cable with sag

The linear vibration response of a taut cable depends on the mass per unit length m and the cable force T , which to a good approximation can be assumed to be constant. For cables with sag the situation is different because the cable force will change during symmetric vibrations, and the response therefore depends on the elastic extensibility of the cable as well as the flexibility of the supports. The basic vibration behavior of a cable with sag is illustrated in Fig. 5.5. The cable may vibrate either in a symmetric or an anti-symmetric mode. In the anti-symmetric mode one side of the cable moves up, when the other side moves down, Fig. 5.5a. The motion of the center of the cable is horizontal, but as there is little resistance the cable force remains approximately constant in time. Thus, the antisymmetric vibrations are independent of cable and support flexibility. The symmetric modes, shown in Fig. 5.5b with rigid supports, are different because a downward motion tends to extend the cable, and conversely. The upper figure shows an extensible cable - such as a rubber rope - while the lower figure shows an inextensible cable - such as a chain. In the case of an inextensible cable the parts close to the supports are lifted, when the central part is lowered. This behavior is fundamentally different from that of a taut cable, and for symmetric vibrations of cables with sag it is therefore necessary to include the effect of sag and flexibility. The basic theory for free undamped vibrations of shallow cables was developed by Irvine and Caughey (1974). The basic features of shallow cable vibrations needed for describing the effect of a transverse force is summarized here, following Krenk and Nielsen (2002).

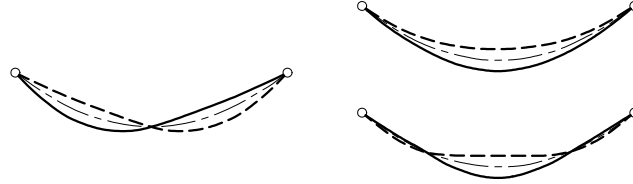


Fig. 5.5: Vibration modes of shallow cable: a) inextensional anti-symmetric vibration, b) extensional symmetric vibrations.

Shallow cable basics

The main features of a cable with sag are shown in Fig. 5.6 illustrating a horizontal cable. The first order effect of cable inclination is to reduce the magnitude of the transverse component of the sag in the equilibrium position. Therefore this effect is not treated explicitly here. For shallow cables the vibration properties can be described by a perturbation $u(x, t)$ of the static

shape $y(x)$, which is described by a parabola. The shallow cable theory is usually applicable for a sag e limited by $e \leq \ell/8$, Irvine (1981), and thus applies to many of the cables used in engineering practice. Here T denotes the horizontal force component, and the sag at mid-span then follows from the assumption of a parabolic equilibrium shape as

$$e = \frac{mg\ell^2}{8T} \quad (5.32)$$

Thus from $e \leq \ell/8$ it follows that $mg\ell \leq T$, i.e. the static horizontal cable force exceeds the weight $mg\ell$ of the cable.

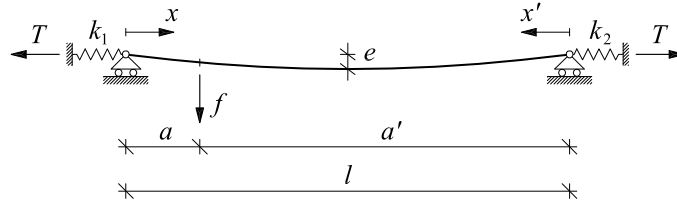


Fig. 5.6: Shallow cable with transverse control force.

The linearized equation of motion for a shallow cable with a transverse force at $x = a$ is

$$T \frac{\partial^2 u}{\partial x^2} + \Delta T \frac{d^2 y}{dx^2} = m \frac{\partial^2 u}{\partial t^2} + f(t) \delta(x-a) \quad (5.33)$$

This equation corresponds to that of a taut cable or string, apart from the second term in which the curvature of the static deflection follows from the shallow cable approximation as

$$T \frac{\partial^2 y}{\partial x^2} = -mg \quad (5.34)$$

while the change of cable force ΔT is obtained from the elastic elongation of the cable and the support springs.

The ends of the cable are supported by springs with stiffness parameters k_1 and k_2 as illustrated in Fig. 5.6. The change of cable force is related to the shape of the vibration mode $u(x, t)$ and the combined flexibility of cable and supports Irvine and Caughey (1974),

$$\Delta T \left(\frac{L_e}{EA} + \frac{1}{k_1} + \frac{1}{k_2} \right) = \frac{8e}{\ell^2} \int_0^\ell u(x, t) dx \quad (5.35)$$

L_e is an effective length, approximately equal to the span ℓ . It is seen that a fully anti-symmetric mode of vibration leads to $\Delta T = 0$, while a nearly anti-

symmetric mode such as that produced by a local damper near the support leads to a negligible change of the cable force, Krenk and Nielsen (2002). Thus, the anti-symmetric vibrations are described by the theory of the taut cable, when T represents the horizontal force component.

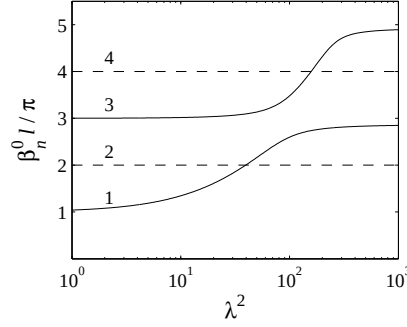


Fig. 5.7: Normalized wavenumber of the first four undamped vibration modes.

The symmetric modes, on the other hand, depend on the total flexibility of the cable and the supports. This dependence may be described by the non-dimensional stiffness parameter λ^2 introduced by Irvine and Caughey (1974),

$$\frac{1}{\lambda^2} = \left(\frac{L_e/\ell}{EA} + \frac{1}{k_1\ell} + \frac{1}{k_2\ell} \right) \frac{T^3}{(mg\ell)^2} \quad (5.36)$$

For a cable with high axial stiffness, i.e. $\lambda^2 \geq 10 - 100$, the symmetric vibration modes develop local zones near the supports in which the motion is reversed as illustrated in the lower part of Fig. 5.5b. This leads to a shorter distance between the nodes of the vibration modes, and thereby to an increased wavenumber. The corresponding wavenumber equation for the symmetric modes is

$$\tan\left(\frac{1}{2}\beta_n^0\ell\right) = \frac{\beta_n^0\ell}{2} - \frac{4}{\lambda^2}\left(\frac{\beta_n^0\ell}{2}\right)^3, \quad n = 1, 3, \dots \quad (5.37)$$

The effect of sag and cable tension is illustrated in Fig. 5.7, showing the wavenumber of the undamped cable as function of the non-dimensional cable stiffness for the first four modes. It is seen that for cables with large axial stiffness the symmetric modes may have higher wavenumber, and thereby higher natural frequency, than the anti-symmetric modes.

Complex wavenumbers for damped symmetric modes

In the present context the interest is in the relative change of the natural frequency, or equivalently the relative change in the wavenumber $\Delta\beta_n^\infty/\beta_n^0$, when a damping force is acting at the point $x = a \ll \ell$. The wavenumber

(frequency) dependence illustrated in Fig. 5.7 does not have an explicit representation, and it does not seem to be possible to obtain an explicit result for the relative frequency increase. However, a quite accurate approximation to the relative increase of the wavenumber of the near-symmetric modes by locking the point $x = a$ has been obtained by Krenk and Nielsen (2002) as

$$\frac{\Delta\beta_n^\infty}{\beta_n^0} \simeq \frac{a}{\ell} \frac{\left(\tan\left(\frac{1}{2}\beta_n^0\ell\right) - \frac{1}{2}\beta_n^0 a\right)^2}{\tan^2\left(\frac{1}{2}\beta_n^0\ell\right) + \frac{12}{\lambda^2}\left(\frac{1}{2}\beta_n^0\ell\right)^2}, \quad n = 1, 3, \dots \quad (5.38)$$

The first factor is the result (5.18) for the taut cable, while the last factor represents the influence of the stiffness parameter λ^2 on the near-symmetric modes. The general behavior resulting from this factor is illustrated by the two vibration mode shapes shown in Fig. 5.5b. The effect of sag is to reduce the motion close to the supports, and for sufficiently high values of λ to reverse the motion. When the point $x = a$ corresponds to a point, where the direction of the mode changes, the frequency becomes insensitive to locking of the point, and thus a force applied to this point will have no effect on this particular mode of vibration. This effect is illustrated in Fig. 5.8, showing the wavenumber increase caused by locking the cable at $a = 0.02\ell$ as function of the non-dimensional cable stiffness parameter λ^2 .

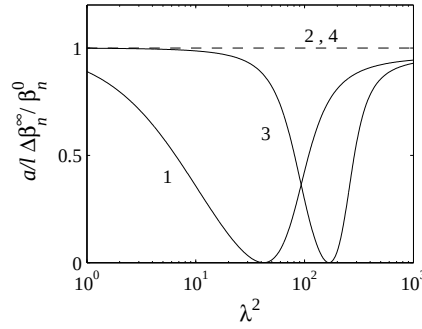


Fig. 5.8: Relative change of wavenumber by locking at $a = 0.02\ell$.

The natural frequencies ω_n are related to the wavenumbers β_n by (5.5) as for the taut cable. Thus a complex frequency analysis can be performed via the complex wavenumber relations. The asymptotic relations (5.22) for the complex wavenumber and (5.23) for the complex natural frequency also apply for cables with sag, and the non-dimensional damping parameter is still given by (5.21) when using the appropriate wavenumber or natural frequency for a cable with sag. The accuracy of the asymptotic format is illustrated in Fig. 5.9a for the first vibration mode of a shallow cable with a viscous damper at $a = 0.02\ell$. The figure shows results for four different values of the stiffness parameter λ^2 . The smallest value, $\lambda^2 = 0.01$ corresponds closely to a taut

cable. With increasing values of λ^2 the radius of the circle decreases until $\lambda^2 \simeq 40$ and then increases again. The particular value $\lambda^2 \simeq 40$, at which the effect of the damper vanishes, corresponds to a free vibration node at the location of the damper.

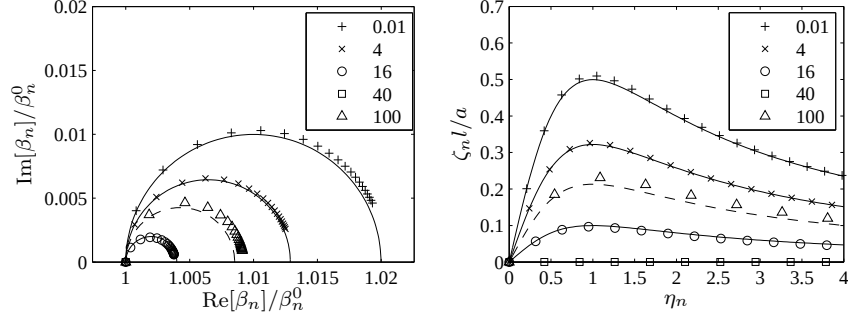


Fig. 5.9: First symmetric mode, $a/\ell = 0.02$, $\lambda^2 = 0.01, \dots, 100$, a) Complex wave number β_1 , b) Damping ratio ζ_1 .

The damping ratio is given by the asymptotic relation (5.29). Substitution of the wavenumber increment from (5.38) leads to the following explicit asymptotic formula for the damping ratio of the near-symmetric modes of a cable with sag

$$\zeta_n \simeq \frac{a}{\ell} \frac{\left(\tan\left(\frac{1}{2}\beta_n^0 \ell\right) - \frac{1}{2}\beta_n^0 a \right)^2}{\tan^2\left(\frac{1}{2}\beta_n^0 \ell\right) + \frac{12}{\lambda^2}\left(\frac{1}{2}\beta_n^0 \ell\right)^2} \frac{\eta_n}{1 + \eta_n^2}, \quad n = 1, 3, \dots \quad (5.39)$$

The damping ratio corresponding to the complex wavenumbers shown in Fig. 5.9a is shown in Fig. 5.9. It is seen that the asymptotic formula reproduces the correct result with high accuracy for damper locations close to the support as in the case of taut cables. For the moderate values of λ^2 used in practice the effect of sag is to reduce the damping of the nearly-symmetric modes, a reduction that is well represented by the middle factor in the asymptotic result (5.39).

5.2 Local Dampers on Structures

5.2.1 Introduction of local damper

When introducing one or more actuators on the structure the system equations are modified. In many cases the actuators are introduced in order to modify only particular features of the response, and it is therefore of interest to develop theories in which the response of the full system appears as a

modification of that of the original system. The principle is here illustrated by damping of a particular mode by one or more dampers, see e.g. Cha and Wong (1999) and Gürgöze (2002).

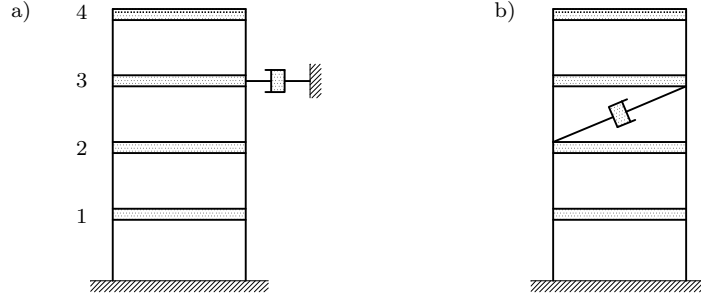


Fig. 5.10: Damper acting on a) absolute motion, b) relative motion.

The effect of an external damper on a structure is described in terms of its influence vector $\mathbf{w}$ and the magnitude of the force $f(t)$,

$$\mathbf{f}(t) = -\mathbf{w}f(t) \quad (5.40)$$

The vector $\mathbf{w}$ identifies the connection of the actuator to the structure as illustrated in Fig. 5.10. The actuator may connect a node to a fixed point outside the structure as illustrated in Fig. 5.10a, corresponding to the vector $\mathbf{w}^T = [0, 0, 1, 0]$. Alternatively the actuator may connect two points of the structure, thereby acting through their relative motion as illustrated in Fig. 5.10b with $\mathbf{w}^T = [0, -1, 1, 0]$. The actuator is assumed to be linear and collocated. This implies that the actuator displacement is $q = \mathbf{w}^T \mathbf{q}$. When expressed in the frequency domain in terms of the transfer function $H(\omega)$ the actuator force is

$$f(\omega) = H(\omega)q(\omega) \quad (5.41)$$

Substitution into (5.40) leads to a full representation of the force vector in terms of the full displacement vector,

$$\mathbf{f}(\omega) = -H(\omega)\mathbf{w}\mathbf{w}^T\mathbf{q}(\omega) \quad (5.42)$$

In this form $H(\omega)\mathbf{w}\mathbf{w}^T$ appears as an additional damping matrix.

The format (5.42) for a single damper can be generalized directly to the case of multiple dampers, simply by adding the force contributions from each damper, resulting in the extended format

$$\mathbf{f}(\omega) = - \sum_k H_k(\omega) \mathbf{w}_k \mathbf{w}_k^T \mathbf{q}(\omega) \quad (5.43)$$

In this relation $\mathbf{q}(\omega)$ is the common displacement vector of the structure, while $\mathbf{w}_k$ and $H_k(\omega)$ are the connectivity vector and the frequency function of device k .

5.2.2 Two-component subspace approximation

The introduction of a local damper enables a gradual transition from the original undamped or lightly damped system to a system where the degrees of freedom associated with the damper become constrained with increasing gain of the damper. For the modes that do not undergo fundamental changes this gradual transition can be represented approximately by a two-component procedure introduced by Main and Krenk (2005). The idea is to project the full displacement space on to the displacement states corresponding to the undamped structure and an equivalent structure, in which the damper has been locked. These mode shapes are real and easily found e.g. by a real-valued modal analysis. The system resulting from the projection will only have two degrees of freedom and the response characteristics are easily analyzed. This two-component representation is exact in the limits for zero and for infinite gain and turns out to be quite accurate over the full gain interval in many cases. The sub-space representation can be written as

$$\mathbf{q}(t) = \mathbf{u}_0 \xi_0(t) + \mathbf{u}_\infty \xi_\infty(t) = \mathbf{S} \boldsymbol{\xi}(t) \quad (5.44)$$

where the real valued shape vector matrix $\mathbf{S} = [\mathbf{u}_0, \mathbf{u}_\infty]$ spans the subspace, while $\boldsymbol{\xi} = [\xi_0, \xi_\infty]^T$ contains the corresponding response variables. The interpolation between the undamped and infinitely damped mode shapes is illustrated in Fig. 5.11 for the first mode of a shear building.

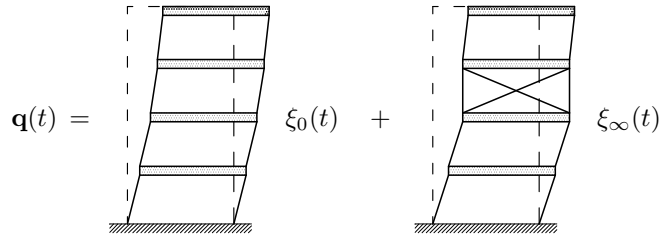


Fig. 5.11: Decomposition via $\mathbf{u}_0$ and $\mathbf{u}_\infty$.

The original structure is assumed to be undamped, and the frequency equation then follows from the equation of motion (4.1) as

$$[\mathbf{K} - \omega^2 \mathbf{M} + \sum_k H_k(\omega) \mathbf{w}_k \mathbf{w}_k^T] \mathbf{q} = \mathbf{f} \quad (5.45)$$

The undamped mode shape $\mathbf{u}_0$ is found as the eigenvector of the undamped eigenvalue problem,

$$(\mathbf{K} - \omega_0^2 \mathbf{M}) \mathbf{u}_0 = \mathbf{0} \quad (5.46)$$

The mode shape $\mathbf{u}_\infty$ is the eigenvector of the equivalent eigenvalue problem arising when all the dampers are locked, i.e. when imposing all the constraints $\mathbf{w}_k^T \mathbf{q} = 0$. Locking of the dampers introduces reaction forces represented by the reaction vector $\mathbf{r}$ in the equation of motion,

$$(\mathbf{K} - \omega_\infty^2 \mathbf{M}) \mathbf{u}_\infty = \mathbf{r} \quad (5.47)$$

In this equation ω_∞ is the corresponding natural frequency of the structure with all dampers clamped. The eigenvalue problem is solved by first eliminating the locked degree of freedom associated with all the vectors $\mathbf{w}_k$. This reduces the number of vibration modes. In the example illustrated in Fig. 5.11 the number of modes is reduced from 4 to 3. Thus, the use of the representation (5.44) in terms of a ‘matching pair’ of mode shapes is limited to modes that retain their identity when clamping the dampers.

Projection of the full equation (5.45) on to the subspace spanned by $\mathbf{u}_0$ and $\mathbf{u}_\infty$ is accomplished by introducing the subspace representation (5.44) and pre-multiplying the equations by the transpose of the subspace matrix $\mathbf{S}$,

$$[\mathbf{S}^T \mathbf{K} \mathbf{S} - \omega^2 \mathbf{S}^T \mathbf{M} \mathbf{S} + \sum_k H_k(\omega) \mathbf{S}^T \mathbf{w}_k \mathbf{w}_k^T \mathbf{S}] \boldsymbol{\xi} = \mathbf{S}^T \mathbf{f} \quad (5.48)$$

The properties of this equation depends on the transformed stiffness and mass matrix, and on the influence vectors $\mathbf{w}_k^T \mathbf{S}$ of the damper in the subspace. By construction the mode shape $\mathbf{u}_\infty$ is orthogonal to the influence vector, while the projection of the undamped mode shape $\mathbf{u}_0$ is denoted u_k ,

$$\mathbf{w}_k^T \mathbf{u}_0 = u_k, \quad \mathbf{w}_k^T \mathbf{u}_\infty = 0 \quad (5.49)$$

It follows that $\mathbf{w}_k^T \mathbf{S} = [u_k, 0]$. The matrices in (5.48) are found by observing that pre-multiplication of (5.46) by $\mathbf{u}_\infty^T$ makes the right hand side vanish. The infinitely damped system is therefore orthogonal in the same way as the undamped system, and the various mode shapes are normalized as follows

$$\begin{aligned} \mathbf{u}_0^T \mathbf{M} \mathbf{u}_0 &= 1, & \mathbf{u}_0^T \mathbf{K} \mathbf{u}_0 &= \omega_0^2 \\ \mathbf{u}_\infty^T \mathbf{M} \mathbf{u}_\infty &= 1, & \mathbf{u}_\infty^T \mathbf{K} \mathbf{u}_\infty &= \omega_\infty^2 \end{aligned} \quad (5.50)$$

The product of the vectors $\mathbf{u}_0$ and $\mathbf{u}_\infty$ with respect to the mass matrix is neither normalized nor orthogonal. The product is here denoted by κ ,

$$\kappa = \mathbf{u}_\infty^T \mathbf{M} \mathbf{u}_0 = 1 - \frac{1}{2} \Delta \mathbf{u}_\infty^T \mathbf{M} \Delta \mathbf{u}_\infty < 1 \quad (5.51)$$

where $\Delta \mathbf{u}_\infty = \mathbf{u}_\infty - \mathbf{u}_0$ denotes the change of the mode shape vector $\mathbf{u}_0$ by locking the damper. It follows that for small changes of the undamped mode vector $\kappa \simeq 1$.

The reduced system consists of the following two-dimensional matrix equation,

$$\begin{bmatrix} \omega_0^2 - \omega^2 + H(\omega) & \kappa(\omega_0^2 - \omega^2) \\ \kappa(\omega_0^2 - \omega^2) & \omega_\infty^2 - \omega^2 \end{bmatrix} \begin{bmatrix} \xi_0 \\ \xi_\infty \end{bmatrix} = \begin{bmatrix} f_0 \\ f_\infty \end{bmatrix} \quad (5.52)$$

where the load vector components are defined by

$$f_0 = \mathbf{u}_0^T \mathbf{f} \quad , \quad f_\infty = \mathbf{u}_\infty^T \mathbf{f} \quad (5.53)$$

The total effect of all external dampers is contained in the term

$$H(\omega) = \sum_k H_k(\omega) u_k^2 \quad (5.54)$$

It is to be noted that in the common case of viscous dampers the resulting effect is simply the weighted average of the individual dampers

$$H(\omega) = i\omega \sum_k c_k u_k^2 = i\omega c_0 \quad (5.55)$$

defining a resulting viscosity coefficient c_0 as a weighted sum of the individual viscosity coefficients.

The root locus behavior associated with the subspace method is defined by vanishing determinant, corresponding to the characteristic equation

$$(\omega^2 - \omega_0^2)[\omega_\infty^2 - \omega_0^2 - (\omega^2 - \omega_0^2)(1 - \kappa^2)] = H(\omega)(\omega_\infty^2 - \omega^2) \quad (5.56)$$

As seen from the inequality in (5.51) κ must always be less than 1, but if the change in the mode shape is modest and $\kappa \simeq 1$. When collecting terms containing the factor $\omega^2 - \omega_0^2$ on the left side of the equation, the following relation is obtained

$$\frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega_0^2} = \frac{\frac{H(\omega)}{\omega_\infty^2 - \omega_0^2}}{1 - (1 - \kappa^2) \frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega_0^2} + \frac{H(\omega)}{\omega_\infty^2 - \omega_0^2}} \quad (5.57)$$

The middle term in the denominator is new relative to the relation (5.21) based on the differential equation of a cable. The form of the equation is suitable for iterative solution, and fairly accurate results can often be obtained by assuming $\kappa = 1$, and using the asymptotic solution discussed in Section 5.1.1. In the present case this corresponds to the frequency relation

$$\frac{\Delta\omega}{\Delta\omega_\infty} \simeq \frac{i\eta}{1 + i\eta} \quad (5.58)$$

with the damper parameter

$$\eta = \frac{1}{i} \frac{\sum_k H_k(\omega) u_k^2}{\omega_\infty^2 - \omega_0^2} \quad (5.59)$$

combining the effect of the dampers. For a set of viscous dampers this can be written in the form

$$\eta = \frac{\omega}{\omega_\infty + \omega_0} \frac{\sum_k c_k u_k^2}{\Delta\omega_\infty} \simeq \frac{\sum_k c_k u_k^2}{2 \Delta\omega_\infty} \quad (5.60)$$

This form generalizes the approximate result for a single damper on a cable discussed in Section 5.1.1 to multiple dampers on a discrete structure. It is seen that provided the dampers are suitably calibrated to work together, the effect is given by the weighted sum $H(\omega)$ given in (5.54). Provided the relative size of the dampers is balanced correctly the optimal damping is obtained for $\eta^{\text{opt}} = 1$, and the corresponding damping ratio follows from (5.30) as

$$\zeta^{\text{max}} \simeq \frac{\omega_\infty - \omega_0}{\omega_\infty + \omega_0}. \quad (5.61)$$

Conditions for the dampers to ‘work together’ is derived in the next section.

5.2.3 Placement and sizing of multiple dampers

In the two-component procedure it is fairly straightforward to include several vibration absorbers designed for damping the same mode. This raises the question of the best placement and the relative sizing of the dampers in order best to control the selected mode. The individual dampers should be placed where the modal amplitude is sufficient to lead to dissipation, but at the same time the damper gains should be balanced to sustain the mode and avoid coupling to other modes. A simple and efficient approach was proposed by Main and Krenk (2005) as described in the following.

The idea is that the force residual introduced by the two-component representation of the damped mode shape (5.44) should be as small as possible in a free vibration. After substitution of the two-component displacement representation into the full set of equations of motion (5.45), they take the form

$$[\mathbf{K} - \omega^2 \mathbf{M} + \sum_k H_k(\omega) \mathbf{w}_k \mathbf{w}_k^T] [\mathbf{u}_0 \xi_0 + \mathbf{u}_\infty \xi_\infty] = \mathbf{f}_{res} \quad (5.62)$$

where $\mathbf{f}_{res}$ denotes the residual force. In this equation the stiffness terms $\mathbf{K}\mathbf{u}_0$ and $\mathbf{K}\mathbf{u}_\infty$ are substituted from (5.46) and (5.47), respectively. This leads to the equation

$$[(\omega_0^2 - \omega^2)\mathbf{M}\mathbf{u}_0 + \sum_k H_k(\omega)u_k \mathbf{w}_k] \xi_0 + [(\omega_\infty^2 - \omega^2)\mathbf{M}\mathbf{u}_\infty + \mathbf{r}] \xi_\infty = \mathbf{f}_{res} \quad (5.63)$$

where u_k is the modal displacement across the damper, defined in (5.49a). The ratio ξ_∞/ξ_0 in a free vibration is determined from the second equation in (5.52) with $f_\infty = 0$,

$$\frac{\xi_\infty}{\xi_0} = \kappa \frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega^2} \quad (5.64)$$

It is easily demonstrated that for viscous damping the top point of the semi-circular root locus contour corresponds closely to $\xi_\infty = i\xi_0$, and thus maximum damping corresponds to the free and the locally locked mode shapes being $\pi/2$ out of phase.

Division of the residual force relation (5.63) by the amplitude ξ_0 , and substitution of the amplitude ratio ξ_∞/ξ_0 gives

$$(\omega_0^2 - \omega^2)\mathbf{M}\mathbf{u}_0 + (\omega^2 - \omega_0^2)\mathbf{M}\mathbf{u}_\infty + \sum_k H_k(\omega)u_k \mathbf{w}_k + \kappa \frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega^2} \mathbf{r} = \mathbf{f}_{res}/\xi_0 \quad (5.65)$$

The two mass terms combine to $(\omega^2 - \omega_0^2)\mathbf{M}\Delta\mathbf{u}$. The occurrence of products of increments suggests that the term is small, and it is therefore omitted in the approximate determination of optimal damper balancing. The objective is to minimize the remaining residual force components, i.e. to satisfy the approximate equation

$$\sum_k H_k(\omega)u_k \mathbf{w}_k + \kappa \frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega^2} \mathbf{r} \simeq \mathbf{0} \quad (5.66)$$

The optimal balancing of the dampers is found from this equation by observing that the reactions $\mathbf{r}$ are generated by constraints introduced by locking the dampers acting through the connection vectors $\mathbf{w}_k$. This implies that the reaction forces can be expressed in terms of these vectors in the form

$$\mathbf{r} = \sum_k \rho_k \mathbf{w}_k \quad (5.67)$$

If the damper connection vectors $\mathbf{w}_k$ are linearly independent, which is typically the case, the coefficients ρ_k are determined uniquely from the reaction force $\mathbf{r}$. Substitution of the reaction force representation into the approximate balance equation (5.66) then gives

$$\sum_k \left[H_k(\omega) u_k + \kappa \frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega^2} \rho_k \right] \mathbf{w}_k \simeq \mathbf{0} \quad (5.68)$$

For non-redundant damper configurations this equation system is solved by equating the coefficients of the vectors $\mathbf{w}_k$ to zero, whereby

$$H_k(\omega) \simeq -\kappa \frac{\omega^2 - \omega_0^2}{\omega_\infty^2 - \omega^2} \frac{\rho_k}{u_k}, \quad k = 1, \dots \quad (5.69)$$

This relation leads to the following rule for distributing the relative gain of dampers acting together on the same mode,

$$\frac{H_k(\omega)}{H_1(\omega)} \simeq \frac{\rho_k/u_k}{\rho_1/u_1}, \quad k = 1, \dots \quad (5.70)$$

Thus, the dampers are sized by the ratio of the corresponding reaction force expansion coefficient ρ_k to the modal displacement u_k spanned by the damper.

Example 5.1. Damping of shear frame.

Consider a shear frame as illustrated in Fig. 5.11, but with $n = 10$ stories. The quality of the approximation complex frequency representation (5.59) is illustrated in Fig. 5.12. The figure shows the complex root locus and the corresponding damping ratio of the lowest mode for a single viscous damper located between floors 0–1, 3–4 and 6–7, respectively. It is observed that the damping effect decreases with increasing height. This is due to the first mode shape which produces decreasing inter-story displacements with floor number as shown in the first line of Table 5.1. As a consequence locking of the inter-story displacement produces decreasing effect for increasing story number as shown in the second line. The last line of the table gives the parameter $1 - \kappa$, which is a measure of the error in the assumption of two-component representation of the damped mode shape. It is seen that this parameter is quite small, suggesting high accuracy as illustrated by the good fit of the analytical approximations to the exact numerically calculated results in Fig. 5.12.

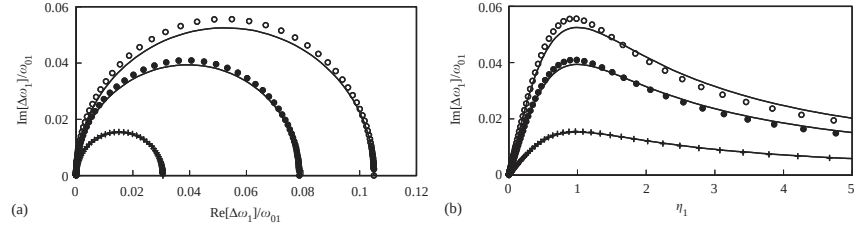


Fig. 5.12: Root locus and damping plots for a single damper connecting stories 0–1 (o), 3–4 (•), and 7–8 (+).

Table 5.1: Parameters of the two-component representation.

Damper location	0–1	3–4	7–8
u_k	0.0650	0.0565	0.0367
$\Delta\omega_\infty/\omega_0$	0.1050	0.0787	0.0306
$1 - \kappa$	0.0054	0.0031	0.0011

The application of two viscous dampers at inter-story positions 0–1 and 4–5, respectively, is illustrated in Fig. 5.12. The dampers are balanced by use of the condition (5.70), giving $c_{4-5} = 1.038c_{0-1}$, and the figure corresponds to optimal damping at $\eta = 1$. The assumption of the two-mode representation is seen to be very good in this case.

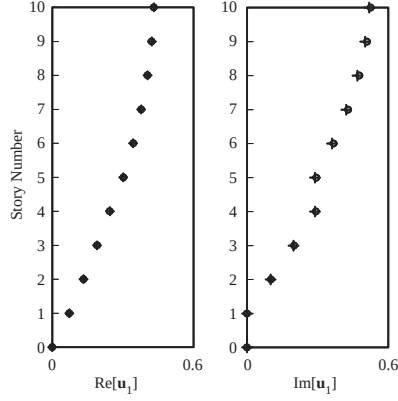


Fig. 5.13: Real and imaginary part of damped mode 1: (o) exact, (+) two-component approximation.

If the dampers are not balanced their combined efficiency will be smaller. This is illustrated in Fig. 5.14, showing two cases, one in which the upper damper is too weak, and one in which it is too strong. Fig. 5.14a shows the nearly balanced case $c_{4-5} = c_{0-1}$, together with three cases of increasing unbalance $c_{4-5} < c_{0-1}$. The case of balanced dampers produce a circular root locus contour as in the case of a single damper. With increasing unbalance the curve develops a tendency to dip towards the frequency corresponding to locking of the strongest damper. For a very large unbalance, i.e. $c_{4-5} \ll c_{0-1}$, the curve consists of two semi-circular parts, the first gradually producing locking of the damper c_{0-1} , and the other part describing the subsequent locking of the weaker damper c_{4-5} . In Fig. 5.14b the roles of the two dampers are reversed, and now the upper damper c_{4-5} locks first at a slightly different intermediate frequency. It is seen that a moderate unbalance of the dampers has only a modest influence on the damping efficiency.

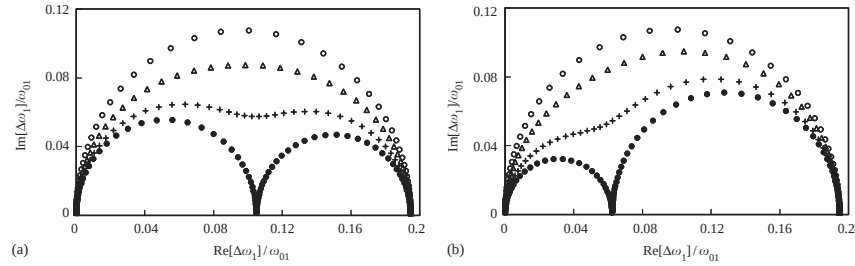


Fig. 5.14: Two dampers. a) $c_{4-5}/c_{0-1} = 1$ (o), 0.3 (Δ), 0.1 (+), $\simeq 0$ ($\bullet$); b) $c_{0-1}/c_{4-5} = 1$ (o), 0.3 (Δ), 0.1 (+), $\simeq 0$ ($\bullet$).

□

5.3 Resonant vibration absorbers

Lightly damped structures may develop large amplitude vibrations for loads acting near the resonance frequency. These vibrations may be reduced by attaching a secondary mass through a suitably selected spring and damper. The ‘tuning’ of the spring and damper to produce optimal reduction is an important feature, and the device is therefore called a tuned mass damper.

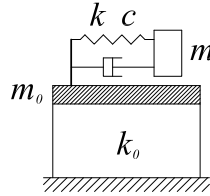


Fig. 5.15: Tuned mass absorber on simple structure.

The principle is illustrated in Fig. 5.15 showing a structural mass m_0 , supported by a spring with stiffness k_0 . These parameters are determined such that m_0 is a representative mass of the structure in relation to the mode to be damped, and the structural stiffness k_0 is determined such that the corresponding natural frequency $\omega_0 = \sqrt{k_0/m_0}$ is the natural angular frequency of the structural vibration mode.



Fig. 5.16: Tuned mass damper from Langelinie Footbridge 2006.

The tuned mass damper consists of a damper mass m attached to the structure by a spring of stiffness k and a viscous damper with parameter c as shown in the figure. These parameters must be chosen to

- i) involve sufficient mass to generate the desired damping,
- ii) produce a suitable amplification by resonance,
- iii) extract the maximal energy by suitable tuning of damping.

The mass often consists of smaller steel blocks that can be used to adjust the resonance frequency of the damper. This is illustrated in Fig. 5.16, showing

a tuned mass damper fitted inside the closed bridge girder of the Langelinie Footbridge in Copenhagen. For lighter bridges tuned mass dampers can often be fitted under the bridge deck.

5.3.1 Basic theory of the tuned mass absorber

The basic two-degree-of-freedom system is shown in Fig. 5.17. The equations of motion are conveniently established by use of the energy balance equation. In addition to providing the equations of motion of the present simple case the energy procedure leads to an important generalization for modal response of flexible structures, discussed in Section 5.4.

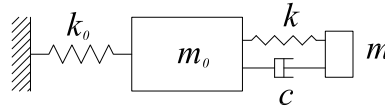


Fig. 5.17: Tuned mass damper.

The energy balance equation discussed in Section 4.1.1 is a statement that the rate of change of the mechanical energy is equal to the external rate of work minus the rate of dissipation,

$$\frac{d}{dt}(E_{kin} + E_{pot}) = W - D \quad (5.71)$$

In the present context it is convenient to describe the motion of the combined system by the motion of the structure x_0 and the *relative* motion of the damper x_d . These variables give simple equations, and they are also the two important design variables: reduced motion of the structure $x_0(t)$ for comfort and use, and limited relative motion of the damper $x_d(t)$ due to design constraints on the damper (limited space, length of springs etc.). With these variables to describe the motion the energy terms take the form,

$$\begin{aligned} E_{kin} &= \frac{1}{2}m_0\dot{x}_0^2 + \frac{1}{2}m(\dot{x}_0 + \dot{x}_d)^2 \\ E_{pot} &= \frac{1}{2}k_0x_0^2 + \frac{1}{2}kx_d^2 \\ D &= c\dot{x}_d^2 \end{aligned} \quad (5.72)$$

The equations of motion follow from the energy balance equation,

$$\begin{aligned} m_0\ddot{x}_0 + m(\ddot{x}_0 + \ddot{x}_d) + k_0x_0 &= f(t) \\ m(\ddot{x}_0 + \ddot{x}_d) + c\dot{x}_d + kx_d &= 0 \end{aligned} \quad (5.73)$$

The structural mass m_0 and stiffness k_0 are assumed known. The task is to select the parameters m , k and c of the tuned mass absorber damper to provide optimal vibration reduction. This problem will be considered in the frequency domain.

5.3.2 Harmonic response

The load and response are assumed harmonic with angular frequency ω ,

$$x(t) = x e^{i\omega t} \quad , \quad f(t) = f e^{i\omega t} \quad (5.74)$$

In the following x_0 and x_d and f then represent the amplitudes of the displacements and force, respectively. Substitution of the representation (5.74) into the equations of motion gives

$$\begin{aligned} [k_0 - \omega^2(m_0 + m)] x_0 - \omega^2 m x_d &= f \\ -\omega^2 m x_0 + [k + i\omega c - \omega^2 m] x_d &= 0 \end{aligned} \quad (5.75)$$

These equations are now solved for the amplification factor $x_0 k_0 / f$ for the structural motion and $x_d k_0 / f$ for the relative motion. Solution of these equations gives the complex displacement amplitudes

$$\begin{aligned} \frac{x_0}{f} &= \frac{k - \omega^2 m + i\omega c}{[k_0 - \omega^2(m_0 + m)][k - \omega^2 m + i\omega c] - (\omega^2 m)^2} \\ \frac{x_d}{f} &= \frac{\omega^2 m}{[k_0 - \omega^2(m_0 + m)][k - \omega^2 m + i\omega c] - (\omega^2 m)^2} \end{aligned} \quad (5.76)$$

In practice the structure is characterized by its mass m_0 and its natural frequency ω_0 , while the damper is characterized by its mass m_d , frequency and damping. It is therefore convenient to introduce the following parameters,

$$\begin{aligned} \omega_0^2 &= \frac{k_0}{m_0} \quad , \quad \mu = \frac{m}{m_0} \\ \omega_d^2 &= \frac{k}{m} \quad , \quad \zeta_d = \frac{c}{2\sqrt{k m}} \end{aligned} \quad (5.77)$$

This gives the amplification factors

$$\frac{x_0}{f/k_0} = \frac{\omega_0^2 [\omega_d^2 - \omega^2 + 2i\zeta_d \omega_d \omega]}{\omega^4 - [\omega_0^2 + (1 + \mu)\omega_d^2] \omega^2 + \omega_0^2 \omega_d^2 + 2i\zeta_d \omega_d \omega [\omega_0^2 - (1 + \mu)\omega_d^2]} \quad (5.78)$$

and

$$\frac{x_d}{f/k_0} = \frac{\omega_0^2 \omega^2}{\omega^4 - [\omega_0^2 + (1 + \mu)\omega_d^2]\omega^2 + \omega_0^2\omega_d^2 + 2i\zeta_d\omega_d\omega[\omega_0^2 - (1 + \mu)\omega^2]} \quad (5.79)$$

where the denominator f/k_0 is the equivalent static displacement of the structure. The mass ratio μ is selected, and the damper system is then characterized by the frequency ratio ω_d/ω_0 and the damping ratio ζ_d . Optimal damper characteristics can now be identified from a frequency analysis of the two amplification factors.

5.3.3 Frequency analysis

If damping is too low, undesirable dynamic amplification will occur near the two undamped resonance frequencies, because the damper does not absorb enough energy. On the other hand, if damping is too high, the damper restricts the relative motion, and thereby creates the effect of a single mass $m_0 + m$ oscillating with frequency

$$\omega_\infty = \sqrt{\frac{k_0}{m_0 + m}} = \frac{\omega_0}{\sqrt{1 + \mu}} \quad (5.80)$$

The transition from two resonance peaks to one with increased damping is illustrated in Fig. 5.18.

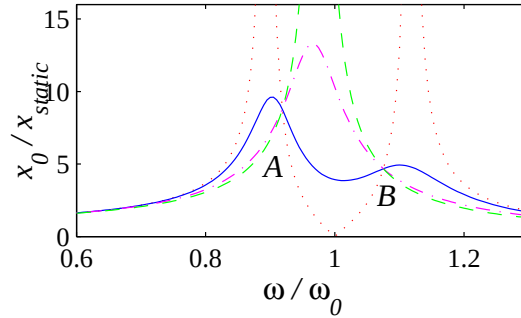


Fig. 5.18: Structure response x_0 for $\mu = 0.05$, $\omega_d = \omega_0$ and ζ : $\cdots$ 0, $—$ 0.1, $- \cdot -$ 0.3, $--$ ∞ .

It is a remarkable fact that there are two frequencies ω_A and ω_B around the natural frequency ω_0 for which the magnitude of the response is independent of the damping parameter c . These points can be used to determine a suitable tuning frequency ω_d of the damper. The idea is to select the damper frequency ω_d such that the dynamic amplification is equal at the two frequencies ω_A and ω_B , Den Hartog (1956). The optimal damping ζ_d can then be determined and the resulting amplification factors for the structure response x_0 and the relative motion of the damper x_d can be calculated. A detailed derivation is

presented in the following. The key results are presented as framed formulae to permit their use without following the details of their derivation.

Optimal frequency tuning

The dynamic amplification factor at ω_A and ω_B are independent of damping, and can be determined by studying the structure of the frequency response formula (5.78), Den Hartog (1956). Both numerator and denominator consist of a real part that is independent of the damping ratio ζ_d and an imaginary term that has the factor $2\zeta_d$. This is expressed in the form

$$\frac{x_0}{f/k_0} = \frac{A + 2i\zeta_d B}{C + 2i\zeta_d D} \quad (5.81)$$

The corresponding dynamic amplification is the absolute value, determined from

$$\left| \frac{x_0}{f/k_0} \right|^2 = \frac{A^2 + (2\zeta_d)^2 B^2}{C^2 + (2\zeta_d)^2 D^2} \quad (5.82)$$

For the magnitude to be independent of ζ_d the limits for $\zeta_d = 0$ and $\zeta_d \rightarrow \infty$ must be equal, i.e.

$$\frac{A^2}{C^2} = \frac{B^2}{D^2} \Rightarrow \frac{A}{C} = \pm \frac{B}{D} \quad (5.83)$$

This corresponds to the equations

$$AD = \pm BC \quad (5.84)$$

Substitution of A , B , C and D , identified from (5.78), gives,

$$\left[1 - \left(\frac{\omega}{\omega_0} \right)^2 \right] \left[1 - \left(\frac{\omega}{\omega_d} \right)^2 \right] - \mu \left(\frac{\omega}{\omega_0} \right)^2 = \pm \left[1 - \left(\frac{\omega}{\omega_d} \right)^2 \right] \left[1 - (1 + \mu) \left(\frac{\omega}{\omega_0} \right)^2 \right] \quad (5.85)$$

after cancelling the common factor ω/ω_d . Use of the plus sign leads to the root $\omega = 0$. This is the static solution where there is no motion, and therefore no damping force. Use of the minus sign leads to

$$\left[1 - \left(\frac{\omega}{\omega_d} \right)^2 \right] \left[2 - (2 + \mu) \left(\frac{\omega}{\omega_0} \right)^2 \right] = \mu \left(\frac{\omega}{\omega_0} \right)^2 \quad (5.86)$$

This corresponds to the following quadratic equation in ω^2 ,

$$(2 + \mu) \left(\frac{\omega}{\omega_0} \right)^2 \left(\frac{\omega}{\omega_d} \right)^2 - 2 \left[\left(\frac{\omega}{\omega_d} \right)^2 + (1 + \mu) \left(\frac{\omega}{\omega_0} \right)^2 \right] + 2 = 0 \quad (5.87)$$

The roots of this quadratic equation are ω_A^2 and ω_B^2 . The roots ω_A^2 and ω_B^2 are not needed explicitly, but only in the form of their sum. The sum of the

roots of a quadratic equation is equal to minus the coefficient of the linear term, divided by the coefficient of the quadratic term. In this case

$$\left(\frac{\omega_A}{\omega_0}\right)^2 + \left(\frac{\omega_B}{\omega_0}\right)^2 = \frac{2}{2+\mu} \left[1 + (1+\mu)\left(\frac{\omega_d}{\omega_0}\right)^2\right] \quad (5.88)$$

This provides one relation of the frequencies ω_A and ω_B .

The other relation is determined by specifying equal magnitude of the dynamic amplification at the frequencies ω_A and ω_B . At these frequencies the response magnitude is independent of ζ_d , and the relevant response can therefore be determined from (5.78) for $\zeta_d \rightarrow \infty$, where

$$\frac{x_0}{f/k_0} = \frac{1}{1 - (1+\mu)\left(\frac{\omega}{\omega_0}\right)^2} \quad (5.89)$$

This response is that of the combined mass $m_0 + m$ moving as a unit. At the frequency ω_A the response is in phase with the load, while at ω_B it is in opposite phase. Thus, equal response magnitude at ω_A and ω_B corresponds to

$$\frac{1}{1 - (1+\mu)\left(\frac{\omega_A}{\omega_0}\right)^2} = - \frac{1}{1 - (1+\mu)\left(\frac{\omega_B}{\omega_0}\right)^2} \quad (5.90)$$

Multiplication with the denominators gives the equation

$$\left(\frac{\omega_A}{\omega_0}\right)^2 + \left(\frac{\omega_B}{\omega_0}\right)^2 = \frac{2}{1+\mu} \quad (5.91)$$

This is the condition that the dynamic amplification is identical at the frequencies ω_A and ω_B .

The optimal frequency ratio ω_d/ω_0 is now determined by elimination of the sum of squares of the frequencies ω_A and ω_B between the equations (5.88) and (5.91),

$$1 + (1+\mu)\left(\frac{\omega_d}{\omega_0}\right)^2 = \frac{2+\mu}{1+\mu} \quad (5.92)$$

This equation determines the optimal tuned mass frequency as

$$\boxed{\frac{\omega_d}{\omega_0} = \frac{1}{1+\mu}} \quad (5.93)$$

Thus, it is seen that optimal frequency tuning of the secondary system corresponds to $\omega_d < \omega_0$.

The frequency tuning (5.93) is optimal in a more strict sense than appears from this classic derivation using damping-independent points. It has been shown by Krenk (2005) that for this frequency tuning the two vibration modes of the combined structure-damper system are equal.

Structural amplification factor

The dynamic amplification of the structure response x_0 at the frequencies ω_A and ω_B is determined by considering the quadratic equation (5.87) for the special tuning $\omega_d/\omega_0 = 1/(1 + \mu)$.

$$(2 + \mu)(1 + \mu)^2 \left(\frac{\omega}{\omega_0} \right)^4 - 2(1 + \mu)(2 + \mu) \left(\frac{\omega}{\omega_0} \right)^2 + 2 = 0 \quad (5.94)$$

The roots of this equation are

$$(1 + \mu) \left(\frac{\omega_{A,B}}{\omega_0} \right)^2 = 1 \pm \sqrt{\frac{\mu}{2 + \mu}} \quad (5.95)$$

Substitution of these frequencies back into the structural response (5.89) for infinite damping gives

$$\frac{x_0}{f/k_0} = \frac{1}{1 - (1 + \mu) \left(\frac{\omega_{A,B}}{\omega_0} \right)^2} = \pm \sqrt{\frac{2 + \mu}{\mu}} \quad (5.96)$$

As the dynamic amplification at these two frequencies is independent of damping the general result is

$$\boxed{\left| \frac{x_0}{f/k_0} \right|_{A,B} = \sqrt{\frac{2 + \mu}{\mu}}} \quad (5.97)$$

In practice the mass of the damper system is most often considerably less than that of the original structure, and thus the mass ratio μ is considerably less than one. It is seen that this limits the reduction of dynamic amplification that can be attained at the frequencies ω_A and ω_B . Thus, a mass ratio of $\mu = 0.02$ leads to a dynamic amplification of the structural response of 10.

Optimal damper tuning

NOTE: SOMETHING ABOUT COMPLEX ROOTS - FROM LECTURE

A suitable damping parameter ζ_d can be selected in several ways. In the classic approach due to Brock (1946) the value of ζ_d^2 was obtained as an arithmetic average of two situations: one with a local maximum of the dynamic amplification at ω_A , and another with a local maximum at ω_B . The result has been used by Den Hartog (1956) and has found widespread use. However, a slightly higher value of the damping parameter appears to be somewhat better, Krenk (2005).

A simple way of obtaining a suitable damping parameter ζ_d consists in observing from the dynamic amplification curves in Fig. 5.18 that when the amplification at ω_A and ω_B have been adjusted to be equal, a fairly flat plateau can be obtained between these frequencies by selecting a suitable frequency in between and selecting the damping to give the same dynamic amplification at this frequency. A good choice of this intermediate frequency is the frequency ω_∞ of the two masses if joined to one. For optimal frequency tuning ω_∞ is the geometric mean of ω_0 and ω_d ,

$$\omega_\infty = \frac{\omega_0}{\sqrt{1+\mu}} = \sqrt{\omega_d \omega_0} \quad (5.98)$$

At this frequency the structural response is

$$\frac{x_0}{f/k_0} = \frac{\mu - 2i\zeta_d\sqrt{1+\mu}}{\mu} \quad (5.99)$$

This corresponds to a dynamic amplification determined by

$$\left| \frac{x_0}{f/k_0} \right|_{\omega_\infty}^2 = \frac{\mu^2 + (2\zeta_d)^2(1+\mu)}{\mu^2} \quad (5.100)$$

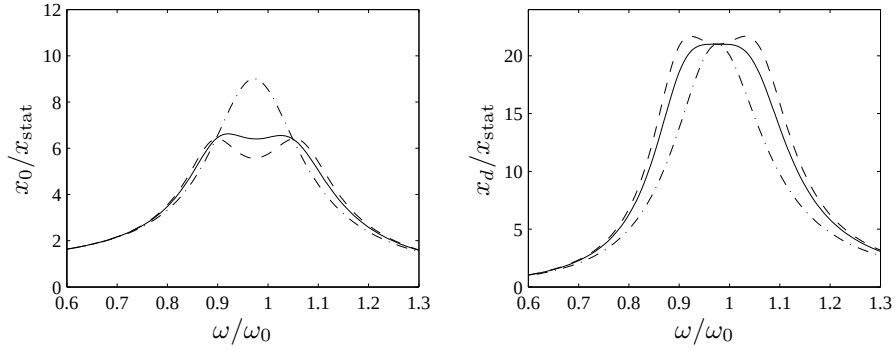


Fig. 5.19: Amplitude of relative damper motion for $\mu = 0.05$. a) structure motion x_0 , b) relative damper motion x_d . Damping parameter: - - $\zeta_{classic}$, — ζ_{opt} , - · - ζ_* .

By equating the dynamic amplification at ω_∞ to that at ω_A and ω_B the damping ratio is determined to

$$\zeta_{\text{opt}}^2 = \frac{1}{2} \frac{\mu}{1 + \mu} \quad (5.101)$$

This is larger than the classic value of Brock (1946), given by Den Hartog (1956),

$$\zeta_{\text{classic}}^2 = \frac{3}{8} \frac{\mu}{1 + \mu} \quad (5.102)$$

The difference in the dynamic amplification of the structural response is illustrated in Fig. 5.19a. It is seen that the classic value leads to a dip between the frequencies ω_A and ω_B , where the amplification is the same by construction. The third graph in the figure refers to a damping ratio $\zeta_* = \sqrt{\mu/(1+\mu)}$, which is the largest damping that can be applied without changing the nature of the damping of the combined system.

There are two reasons, why the value of ζ_d given by (5.101) can be considered preferable relative to the classic value. It has recently been demonstrated that when the damper frequency is given by (5.93) the two modes of the combined structure damper system have the same damping ratio ζ , and that this damping ratio is nearly exactly half that of the device damping ratio ζ_d , Krenk (2005),

$$\zeta \simeq \frac{1}{2} \zeta_d \quad (5.103)$$

The implication is that ζ_{opt} determined by (5.101) leads to about 12 pct. higher damping ratio than the classic value.

The second reason that ζ_d given by (5.101) may be termed optimal is related to the relative motion of the damper. It can be seen from the formula (5.79) that the amplification factor $x_d k_0 / f$ of the relative motion is independent of damping for $\omega = \omega_\infty$,

$$\left| \frac{x_d}{f/k_0} \right|_{\omega_\infty} = \frac{1 + \mu}{\mu} \quad (5.104)$$

For design reasons it is desirable not to exceed this value. It can be shown that the amplification factor of the relative motion is flat in the sense that its first three derivatives with respect to frequency vanish at this particular frequency, when the damping ratio is ζ_d as given by (5.101). This is illustrated in Fig. 5.19. Any value of ζ_d smaller than ζ_{opt} will lead to peaks exceeding the amplification at ω_∞ . In contrast, any value of ζ_d larger than ζ_{opt} will lead to a central peak in the amplification of the structural response.

5.4 Tuned mass absorbers on flexible structures

Tuned mass dampers work by a resonance principle and when applying tuned mass dampers to flexible structures it is therefore natural to use modal analysis to extract the relevant properties of the structure, see e.g. Krenk and Høgsberg (2007).

5.4.1 Modal representation of the structure

When the flexible structure is discretized, the equation of motion has the form

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{C} \dot{\mathbf{x}} + \mathbf{K} \mathbf{x} = \mathbf{f}(t) \quad (5.105)$$

where $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are the mass, the damping and the stiffness matrix, respectively. Typically, in connection with tuned mass dampers individual modes are considered in the design phase. In mode No. j the undamped angular frequency is ω_j and the motion is described in terms of the mode-shape vector $\mathbf{u}_j$ as

$$\mathbf{x}(t) = \mathbf{u}_j r_j(t) \quad (5.106)$$

$r_j(t)$ is the modal coordinate of mode j . It is important to realize that for a given displacement history $\mathbf{x}(t)$ its representation in the form (5.106) depends on the normalization of the mode-shape vector $\mathbf{u}_j$. Clearly, a consistent generalization of the theory of tuned mass dampers must result in properties that are independent of the particular normalization of the mode-shape vector.

When the modal representation (5.106) into the equation of motion the resulting modal equation of motion becomes

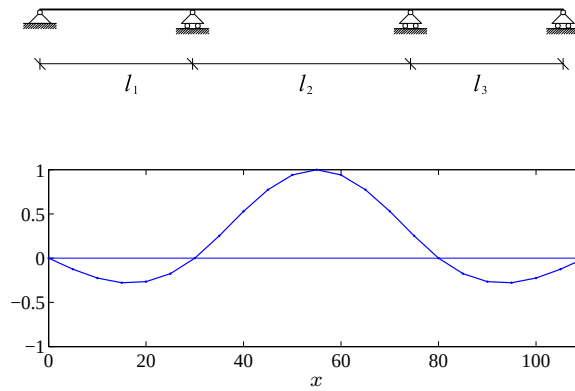


Fig. 5.20: Three-span bridge and first mode-shape.

$$\ddot{r}_j(t) + 2\zeta_j\omega_j \dot{r}_j(t) + \omega_j^2 r_j(t) = \frac{p_j(t)}{m_j}, \quad j = 1, 2, \dots, n \quad (5.107)$$

where the modal mass m_j is defined by

$$m_j = \mathbf{u}_j^T \mathbf{M} \mathbf{u}_j \quad (5.108)$$

and the modal load $p_j(t)$ is

$$p_j(t) = \mathbf{u}_j^T \mathbf{f}(t) \quad (5.109)$$

The role of the normalization is illustrated by the simple 3-span bridge and the first mode-shape shown in Fig. 5.20. In the figure the mode-shape vector is normalized such that the maximum displacement – at the center – is 1. This implies that a concentrated load acting at the center will directly represent the modal load $f_1(t)$, and the modal mass represents the bridge deck mass activated by a motion of magnitude 1 at the center of the bridge. Although this particular normalization is not necessary, it may help the intuitive understanding of the meaning of the modal load and the modal mass. Often the lower mode-shapes can be considered as modifications of a regular sine-function. For that case the modal mass would be half of the bridge deck mass, and the modal mass of the modified mode-shapes, can often be estimated directly in the interval 0.1–0.5 of the total deck mass.

5.4.2 Modal representation of tuned mass absorbers

Now, consider one or more dampers installed to act on mode j . These dampers are tuned to the common damper frequency $\omega_{d,j}$, and the design is based on the situation, where they move in phase. Each absorber mass will move proportional to the motion of the bridge deck at the point where the damper is mounted. Thus the motion of this group of dampers can then be represented by the modal coordinate $r_{d,j}(t)$ as

$$\mathbf{x}_d(t) = \mathbf{u}_j r_{d,j}(t) \quad (5.110)$$

The mass of this group of dampers is described by the diagonal damper mass matrix,

$$\mathbf{M}_d = \begin{bmatrix} \ddots & & \\ & m_d & \\ & & \ddots \end{bmatrix} \quad (5.111)$$

The absorber mass m_d is placed at the location corresponding to the degree-of-freedom, at which the damper is mounted. Thus, a tuned mass damper mounted at the center of the bridge of Fig. 5.20 will have its mass placed at the position corresponding to the displacement u_{center} . If there are more

than one damper for the particular mode j , each mass will be placed in the diagonal of $\mathbf{M}_d$ according to its corresponding degree-of-freedom.

The equations of motion (5.73) for the simple tuned mass problem with a single structural mass were derived from the corresponding energy balance. When the same procedure is followed for the modal representation of the structural motion (5.106) and the damper motion (5.110), the energy is represented in terms of the modal coordinates $r_j(t)$ of the structure and $r_{d,j}(t)$ for the group of damper masses. This implies that the mass of the structure is represented in terms of the modal mass m_j given by (5.113), while the mass of the group of dampers is represented similarly by the modal damper mass

$$m_{d,j} = \mathbf{u}_j^T \mathbf{M}_d \mathbf{u}_j \quad (5.112)$$

As a consequence the important parameter, the mass ratio μ , in this case is defined for the group of dampers as

$$\mu_j = \frac{m_{d,j}}{m_j} = \frac{\mathbf{u}_j^T \mathbf{M}_d \mathbf{u}_j}{\mathbf{u}_j^T \mathbf{M} \mathbf{u}_j} \quad (5.113)$$

It is seen, that this definition of the modal mass ratio is indeed independent of the normalization of the mode-shape vector $\mathbf{u}_j$.

It is seen from the modal damper mass and the corresponding modal mass ratio that dampers are most effective, when mounted at the points of extreme displacement of the mode-shape vector. Thus, for example a damper mounted in the middle of one of the side spans of the bridge of Fig. 5.20 will only experience a modal displacement of around 0.3 times the displacement at the center of the bridge, and thus it will contribute only about 0.1 of the modal mass relative to a position at the center of the mid-span. Thus, it is very important for the efficiency of tuned mass dampers that they are mounted at points of considerable modal amplitude.

5.4.3 Design for resonant loads

Pedestrian loads on footbridges are often considered under the assumption that they act in resonance with the vibration modes in the interval around 1.5-2.5 Hz and with reduced effect in every second period of modes in the interval around 2.5-5 Hz. This permits a very simple analysis.

At resonance the inertial and stiffness terms cancel in the modal equation of motion (5.107), and thus the equation of motion reduces to a balance between the external load and the damping term. When the phase shift is neglected $\dot{r}(t) = \omega_j r_j(t)$, leading to the amplitude relation

$$\omega_j^2 r_j = \frac{p_j}{2\zeta_j m_j} \quad (5.114)$$

This relation in terms of modal load and coordinate is brought back to physical variables by multiplication with the mode-shape vector $\mathbf{u}_j^T$. This gives the acceleration under resonant load as

$$\ddot{\mathbf{x}} = \omega_j^2 r_j \mathbf{u}_j = \frac{p_j}{2\zeta_j m_j} \mathbf{u}_j \quad (5.115)$$

It is seen that this equation is independent of the scaling of the mode-shape vector $\mathbf{u}_j$. However, again the simplest intuitive interpretation is obtained if the mode-shape vector is normalized with a maximum displacement of 1 or the displacement 1 under e.g. the most efficient damper.

Example 5.2. The Langelinie Footbridge. The Langelinie Footbridge was designed in 2004 and built in 2006 in Copenhagen. It connects a part of the city to a park area by a four span 170 m crossing of a railway terrain with heavy traffic, Fig. 5.21. The general layout and cross-section are shown in Fig. 5.22. The design was based on damping of all vibration modes with natural frequency below 5 Hz to an effective damping ratio of around 6–7 pct., Krenk et al. (2005) The bridge has four vertical vibration modes in this interval, and the structural damping of these modes is very low due to the low intrinsic damping in the steel box girder and the ideal supports.



Fig. 5.21: Langelinie Footbridge.

Damping was introduced via tuned mass absorbers. These were placed at the maximum points of the vibration modes as shown in Fig. 5.23. The three lowest modes activate several spans, and are therefore equipped with tuned mass absorbers in two spans, while the fourth mode is local, and therefore is only equipped with a tuned mass absorber in the active span. A typical mass absorber is shown in Fig. 5.16.

The design calculations were carried out in two steps, summarized in the two tables below. The first step consists in a preliminary calculation based on the undamped modes of the original bridge without additional masses. This determines the modal frequencies, the mode shapes and the corresponding modal masses. The modal mass ratio was selected as $\mu_j = 0.05$ for all modes. Due to the mode shapes this corresponds to around 2 pct. of the bridge deck mass in the first three modes and less for the fourth vibration mode. The table shows the initial natural frequencies of the bridge without mass absorbers, and also gives

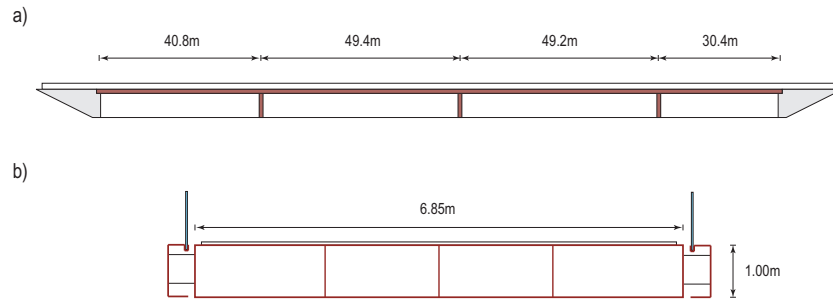


Fig. 5.22: Langelinie Footbridge, a) general view, b) cross-section.

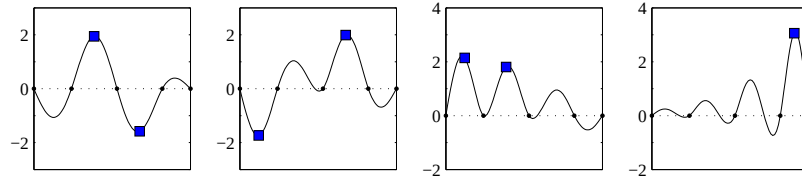


Fig. 5.23: Four lowest modes with vibration absorber positions.

the tuning frequencies according to (5.93). These parameters give damping ratio estimates for the absorbers of around 15 pct. The last two rows give damping ratios as calculated from a complex eigenvalue analysis of the the bridge model, including the preliminary

Step 1: Design based on original undamped bridge deck modes.

Mode	1	2	3	4
$f_{0,j}$ (Hz)	1.67	2.31	3.04	4.59
$f_{d,j}$ (Hz)	1.59	2.20	2.90	4.37
Damper mass (kg)	4450 / 4450	4000 / 4000	3550 / 3550	3000
ζ_{struc} (%)	6.0 / 10.1	8.1 / 9.6	9.5 / 8.1	7.8 / 8.0
	14.1	13.7	14.9	

Step 2: Correction by use of modes including the other damper masses.

Mode	1	2	3	4
$f_{0,j}$ (Hz)	1.58	2.26	3.11	4.60
$f_{d,j}$ (Hz)	1.51	2.16	2.96	4.38
Damper mass (kg)	4775 / 4775	4275 / 4275	3750 / 3750	2925
ζ_{struc} (%)	7.8 / 8.4	8.6 / 8.9	8.5 / 9.1	7.8 / 7.9
	14.3	13.7	15.1	

dampers. It is seen that the damping ratio of the two modes generated from mode 4 are approximately equal and half of the applied absorber damping ratio $\zeta_d = 0.15$ as predicted from (5.103). The three lowest modes each have two dampers, and this leads to three modes: two in which the absorber masses are in the same phase and thereby obey the theory developed for single mass systems, and an extra mode with the full damping of the absorber. This latter mode corresponds to opposing motion of the two absorber masses. In the present case the two masses contribute approximately equally, and therefore they do not really excite the structure. These modes are therefore not likely to be excited in practice. The full design was based on a recalculation accounting for the effect of the suspended masses Krenk et al. (2005). $\square$

5.5 Exercises

Most structural analysis is based on a discretized model of the structure. In a model with n degrees of freedom the displacement components are denoted $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T$, and time derivatives are indicated by a dot, e.g. $\dot{\mathbf{x}}(t)$ for the velocity and $\ddot{\mathbf{x}}(t)$ for acceleration. The equations of motion can be given in the form the following system of second order differential equations

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{g}(\mathbf{x}, \dot{\mathbf{x}}) = \mathbf{f}(t) \quad (6.1)$$

where $\mathbf{M}$ is the mass matrix, $\mathbf{g}(\mathbf{x}, \dot{\mathbf{x}})$ is damping and internal structural forces, and $\mathbf{f}(t)$ is the external load. The function $\mathbf{g}(\mathbf{x}, \dot{\mathbf{x}})$ may contain kinematic as well as material non-linearities. If the structure is linear, i.e. if the the displacements are small to moderate and the material and damping are linear, the equations of motion take the familiar form

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{C} \dot{\mathbf{x}} + \mathbf{K} \mathbf{x} = \mathbf{f}(t) \quad (6.2)$$

where the structure is described by the mass matrix $\mathbf{M}$, the viscous damping matrix $\mathbf{C}$, and the stiffness matrix $\mathbf{K}$. The load history is given by the vector $\mathbf{f}(t)$.

In addition to the equations of motion, solution of the dynamic response problem also requires specification of two sets of initial conditions, here denoted

$$\mathbf{x}(0) = \mathbf{x}_0 \quad , \quad \dot{\mathbf{x}}(0) = \dot{\mathbf{x}}_0 \quad (6.3)$$

Note, that a system of n second order differential equation requires $2n$ initial conditions, the displacement and velocity of each degree of freedom.

The state of a linear structure at any particular time t is described completely by the load vector $\mathbf{f}(t)$ and the *state vector* $(\mathbf{x}(t), \dot{\mathbf{x}}(t))$ containing the displacements and velocities at time t . The state variables may contain additional internal variables. The state vector at time t_0 can serve as initial conditions for the system response for $t > t_0$. The simplest direct numerical integration procedures for the equations of motion (6.2) are the one-step algorithms, in which the state vector $(\mathbf{x}_{n+1}, \dot{\mathbf{x}}_{n+1})$ at time t_{n+1} is calculated by use of the state vector $(\mathbf{x}_n, \dot{\mathbf{x}}_n)$ at the previous time t_n and the load vector at both these times, $\mathbf{f}_n$ and $\mathbf{f}_{n+1}$. While the mental picture of the time history

of the response of the structure may be as shown in Fig. 6.1a, the numerical solution by one-step time integration corresponds to forming increments in the $2n$ -dimensional state space as shown in Fig. 6.1b.

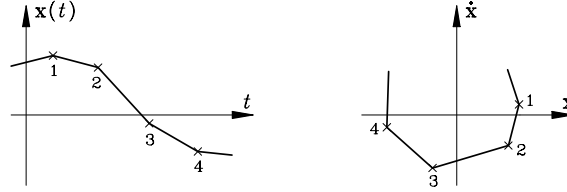


Fig. 6.1: Time history and state-space diagrams.

For a single-degree-of-freedom system with linear variation of the load over the time intervals the equations of motion can be integrated explicitly, and an exact one-step formula is obtained. Linear systems with many degrees of freedom can be reduced to a corresponding set of uncoupled equations by modal analysis. However, a full modal analysis may require extensive computations, and if the damping matrix is non-modal complex-valued modal analysis will be required for exact representation of the original equations of motion. Thus, efficient direct numerical integration techniques for multi-degree-of-freedom systems play an important role in dynamics of linear as well non-linear structures. The following sections describe the theoretical background, numerical implementation, and accuracy and stability properties of the Newmark family of one-step integration procedures. This family of integration procedures was originally developed by Newmark (1959) and later generalized by Hilber et al. (1977), Wood et al. (1981), and Chung and Hulbert (1993).

6.1 Newmark integration of linear structures

The objective of the Newmark integration procedure is the computation of the state vector $(\mathbf{x}_{n+1}, \dot{\mathbf{x}}_{n+1})$ at time $t_{n+1} = t_n + h$, given the state vector $(\mathbf{x}_n, \dot{\mathbf{x}}_n)$ at the previous time t_n and the load vector at both these times, $\mathbf{f}_n$ and $\mathbf{f}_{n+1}$. The procedure is established in two steps: first the increment of $\mathbf{x}(t)$ and $\dot{\mathbf{x}}(t)$ are expressed in terms of integrals of the acceleration $\ddot{\mathbf{x}}$ over the time interval $[t_n, t_{n+1}]$, and then the acceleration is evaluated by use of the equations of motion. While the first step is rather unambiguous, the equations of motion can be introduced in several different ways. In the original formulation of Newmark (1959), the equations of motion were imposed at the forward time t_{n+1} , while later generalizations used a weighted mean between times t_n and t_{n+1} . First the original Newmark integration algorithm for linear structures is presented. It is then generalized to the weighted α -format for linear structures in Section 6.2.1, and for non-linear structures in Section 6.2.2.

6.1.1 The basis of the Newmark procedure

It follows immediately from integral calculus that the value of the vector $\mathbf{x}(t_{n+1})$ at time t_{n+1} can be expressed as its value $\mathbf{x}(t_n)$ at time t_n plus an increment evaluated as the integral of the time derivative $\dot{\mathbf{x}}(\tau)$ over the time interval,

$$\mathbf{x}(t_n + h) = \mathbf{x}(t_n) + \int_{t_n}^{t_{n+1}} \dot{\mathbf{x}}(\tau) d\tau \quad (6.4)$$

Now, a factor 1 can be introduced under the integral sign, and the integral evaluated via integration by parts. When the integral of the factor 1 is selected as $\tau - t_{n+1}$, the result is

$$\mathbf{x}(t_{n+1}) = \mathbf{x}(t_n) - [(t_{n+1} - \tau) \dot{\mathbf{x}}(\tau)]_{t_n}^{t_{n+1}} + \int_{t_n}^{t_{n+1}} (t_{n+1} - \tau) \ddot{\mathbf{x}}(\tau) d\tau \quad (6.5)$$

Note, that the upper limit in the square brackets does not contribute due to the factor $(t_{n+1} - \tau)$.

In the Newmark procedure formulae are needed for the new velocity $\dot{\mathbf{x}}_{n+1}$ and the new displacement $\mathbf{x}_{n+1}$ in terms of integrals of the acceleration. The first formula is obtained from (6.4), when replacing $\mathbf{x}$ and $\dot{\mathbf{x}}$ with $\dot{\mathbf{x}}$ and $\ddot{\mathbf{x}}$, respectively. The second formula follows from (6.5) by evaluating the term in the square brackets.

$$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_n + \int_{t_n}^{t_{n+1}} \ddot{\mathbf{x}}(\tau) d\tau \quad (6.6)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \dot{\mathbf{x}}_n + \int_{t_n}^{t_{n+1}} (t_{n+1} - \tau) \ddot{\mathbf{x}}(\tau) d\tau \quad (6.7)$$

where $h = t_{n+1} - t_n$ is the length of the time interval.

In the present use of the integration formula (6.6)–(6.7) the acceleration will not be known throughout the interval, and the integrals will have to be evaluated approximately from the value of the acceleration vector at the interval end-points. The appropriate formulae are

$$\int_{t_n}^{t_{n+1}} \ddot{\mathbf{x}}(\tau) d\tau \simeq (1 - \gamma) h \ddot{\mathbf{x}}_n + \gamma h \ddot{\mathbf{x}}_{n+1} \quad (6.8)$$

$$\int_{t_n}^{t_{n+1}} (t_{n+1} - \tau) \ddot{\mathbf{x}}(\tau) d\tau \simeq \left(\frac{1}{2} - \beta\right) h^2 \ddot{\mathbf{x}}_n + \beta h^2 \ddot{\mathbf{x}}_{n+1} \quad (6.9)$$

where the parameters $0 < \gamma < 1$ and $0 < \beta < \frac{1}{2}$ determine the degree of forward weighting. The values $\gamma = 0$ and $\beta = 0$ correspond to full backward weighting, while $\gamma = 1$ and $\beta = \frac{1}{2}$ correspond to full forward weighting. Note,

that in (6.8) the sum of the integration weights is h , while in (6.9) the sum is $\frac{1}{2}h$ due to the linear factor in the integrand. See e.g. G  rardin and Rixen (1997) p. 374 for a detailed derivation.

When the approximate integration formulae (6.8) and (6.9) are substituted into (6.6) and (6.7) the following discrete relations are obtained between the displacement, velocity and acceleration vectors at the two interval end-points.

$$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_n + (1 - \gamma) h \ddot{\mathbf{x}}_n + \gamma h \ddot{\mathbf{x}}_{n+1} \quad (6.10)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \dot{\mathbf{x}}_n + \left(\frac{1}{2} - \beta\right) h^2 \ddot{\mathbf{x}}_n + \beta h^2 \ddot{\mathbf{x}}_{n+1} \quad (6.11)$$

where γ and β are parameters of the algorithm.

The Newmark integration algorithm is obtained by satisfying the equations of motion *at time* t_{n+1} , using the representation (6.10)–(6.11) for the velocity and displacement vector. Substitution of velocity and displacement vector from (6.10) and (6.11) into (6.2) gives the following equation for the acceleration vector $\ddot{\mathbf{x}}_{n+1}$.

$$\begin{aligned} \left(\mathbf{M} + \gamma h \mathbf{C} + \beta h^2 \mathbf{K} \right) \ddot{\mathbf{x}}_{n+1} = \mathbf{f}_{n+1} - \mathbf{C} \left(\dot{\mathbf{x}}_n + (1 - \gamma) h \ddot{\mathbf{x}}_n \right) \\ - \mathbf{K} \left(\mathbf{x}_n + h \dot{\mathbf{x}}_n + \left(\frac{1}{2} - \beta\right) h^2 \ddot{\mathbf{x}}_n \right) \end{aligned} \quad (6.12)$$

These equations permit calculation of the acceleration vector $\ddot{\mathbf{x}}_{n+1}$ at time t_{n+1} , and the velocity and displacement can then be calculated from (6.10) and (6.11). However, it is convenient to organize the computations in a slightly different way as explained in the next section.

6.1.2 Numerical implementation

It is convenient to organize the computation of a time step in the Newmark algorithm in the form of a prediction step followed by a correction step.

In the prediction step preliminary values $\dot{\mathbf{x}}_{n+1}^*$ and $\mathbf{x}_{n+1}^*$ of velocity and displacement are evaluated from (6.10) and (6.11), without the last term containing the as yet unknown acceleration $\ddot{\mathbf{x}}_{n+1}$.

$$\dot{\mathbf{x}}_{n+1}^* = \dot{\mathbf{x}}_n + (1 - \gamma) h \ddot{\mathbf{x}}_n \quad (6.13)$$

$$\mathbf{x}_{n+1}^* = \mathbf{x}_n + h \dot{\mathbf{x}}_n + \left(\frac{1}{2} - \beta\right) h^2 \ddot{\mathbf{x}}_n \quad (6.14)$$

These predicted values appear directly in the equations (6.12) for the acceleration $\ddot{\mathbf{x}}_{n+1}$. With the notation

$$\mathbf{M}_* = \mathbf{M} + \gamma h \mathbf{C} + \beta h^2 \mathbf{K} \quad (6.15)$$

Table 6.1: The Newmark Algorithm.

1) System matrices:
$\mathbf{K}, \mathbf{C}, \mathbf{M}$
$\mathbf{M}_* = \mathbf{M} + \gamma h \mathbf{C} + \beta h^2 \mathbf{K}$
2) Initial conditions:
$\mathbf{x}_0, \dot{\mathbf{x}}_0$
$\ddot{\mathbf{x}}_0 = \mathbf{M}^{-1}(\mathbf{f}_0 - \mathbf{C} \dot{\mathbf{x}}_0 - \mathbf{K} \mathbf{x}_0)$
3) Increment time:
$t_{n+1} = t_n + h$
4) Prediction step:
$\dot{\mathbf{x}}_{n+1}^* = \dot{\mathbf{x}}_n + (1 - \gamma) h \ddot{\mathbf{x}}_n$
$\mathbf{x}_{n+1}^* = \mathbf{x}_n + h \dot{\mathbf{x}}_n + (\frac{1}{2} - \beta) h^2 \ddot{\mathbf{x}}_n$
5) Correction step:
$\ddot{\mathbf{x}}_{n+1} = \mathbf{M}_*^{-1}(\mathbf{f}_{n+1} - \mathbf{C} \dot{\mathbf{x}}_{n+1}^* - \mathbf{K} \mathbf{x}_{n+1}^*)$
$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_{n+1}^* + \gamma h \ddot{\mathbf{x}}_{n+1}$
$\mathbf{x}_{n+1} = \mathbf{x}_{n+1}^* + \beta h^2 \ddot{\mathbf{x}}_{n+1}$
6) Return to 3) or stop.

these equations take the simplified form

$$\mathbf{M}_* \ddot{\mathbf{x}}_{n+1} = \mathbf{f}_{n+1} - \mathbf{C} \dot{\mathbf{x}}_{n+1}^* - \mathbf{K} \mathbf{x}_{n+1}^* \quad (6.16)$$

These equations are solved for the acceleration $\ddot{\mathbf{x}}_{n+1}$.

The correction consists in adding the last term in (6.10) and (6.11) to the predicted velocity and displacement vector,

$$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_{n+1}^* + \gamma h \ddot{\mathbf{x}}_{n+1} \quad (6.17)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_{n+1}^* + \beta h^2 \ddot{\mathbf{x}}_{n+1} \quad (6.18)$$

thus completing the time step.

The implementation of the Newmark algorithm is summarized in Table 6.1. For undamped structures the algorithm can be formulated in explicit form, if the mass matrix is diagonal, and $\beta = 0$. In that case (6.12) can be solved directly for $\ddot{\mathbf{x}}_{n+1}$, without matrix inversion. This particular case is only conditionally stable and requires the time increment h to be within an upper limit, identified in the next section.

6.1.3 Numerical stability

The accuracy and stability of the Newmark integration algorithm depends on the parameters β and γ , as well as on the size of the time step h . The use of correct parameters, and their influence on the computed results are important aspects of numerical integration. The analysis is rather cumbersome, although simple in principle. The general results are summarized in the stability diagram in Fig. 6.2, and the properties of the more common parameter combinations listed in Table 6.2. The present derivation of the stability criteria for the Newmark family follows the presentation of the complex root case given by G  rardin and Rixen (1997). A complete analysis, along slightly different lines, including also the effect of modal damping has been presented by Hughes (1983).

The properties of the algorithm can be analyzed by expressing the state vector $(\mathbf{x}_{n+1}, \dot{\mathbf{x}}_{n+1})$ explicitly in terms of the previous state vector $(\mathbf{x}_n, \dot{\mathbf{x}}_n)$ and the load vectors $\mathbf{f}_{n+1}$ and $\mathbf{f}_n$. The relation is obtained by multiplying the equations (6.10) and (6.11) with the mass matrix $\mathbf{M}$ and then eliminating the acceleration terms $\mathbf{M}\ddot{\mathbf{x}}_{n+1}$ and $\mathbf{M}\ddot{\mathbf{x}}_n$ from the equation of motion at time t_{n+1} and t_n , respectively. When the acceleration terms are eliminated (6.10) and (6.11) become

$$\begin{aligned} \mathbf{M}\dot{\mathbf{x}}_{n+1} &= \mathbf{M}\dot{\mathbf{x}}_n + (1 - \gamma)h(\mathbf{f}_n - \mathbf{C}\dot{\mathbf{x}}_n - \mathbf{K}\mathbf{x}_n) \\ &\quad + \gamma h(\mathbf{f}_{n+1} - \mathbf{C}\dot{\mathbf{x}}_{n+1} - \mathbf{K}\mathbf{x}_{n+1}) \end{aligned} \quad (6.19)$$

$$\begin{aligned} \mathbf{M}\mathbf{x}_{n+1} &= \mathbf{M}\mathbf{x}_n + h\mathbf{M}\dot{\mathbf{x}}_n + \left(\frac{1}{2} - \beta\right)h^2(\mathbf{f}_n - \mathbf{C}\dot{\mathbf{x}}_n - \mathbf{K}\mathbf{x}_n) \\ &\quad + \beta h^2(\mathbf{f}_{n+1} - \mathbf{C}\dot{\mathbf{x}}_{n+1} - \mathbf{K}\mathbf{x}_{n+1}) \end{aligned} \quad (6.20)$$

These equations can be rearranged to give the new state vector in terms of the old by collecting terms relating to $\mathbf{x}_{n+1}$ and $\dot{\mathbf{x}}_{n+1}$ on the left side of the equation. In block matrix format the result is

$$\begin{aligned} \begin{bmatrix} \mathbf{M} + \beta h^2 \mathbf{K} & \beta h^2 \mathbf{C} \\ \gamma h \mathbf{K} & \mathbf{M} + \gamma h \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \dot{\mathbf{x}} \end{bmatrix}_{n+1} &= \begin{bmatrix} \left(\frac{1}{2} - \beta\right)h^2 & \beta h^2 \\ (1 - \gamma)h & \gamma h \end{bmatrix} \begin{bmatrix} \mathbf{f}_n \\ \mathbf{f}_{n+1} \end{bmatrix} \\ &\quad + \begin{bmatrix} \mathbf{M} - \left(\frac{1}{2} - \beta\right)h^2 \mathbf{K} & h\mathbf{M} - \left(\frac{1}{2} - \beta\right)h^2 \mathbf{C} \\ -(1 - \gamma)h \mathbf{K} & \mathbf{M} - (1 - \gamma)h \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \dot{\mathbf{x}} \end{bmatrix}_n \end{aligned} \quad (6.21)$$

where the loading term is written first for typographical reasons.

The recursive equations (6.21) apply directly to the equations of motion. However, in the analysis of the algorithm it is convenient to consider the development of each natural mode of vibration. The following analysis will be restricted to the undamped system, and thus there is no additional restriction

in considering an arbitrary natural vibration mode. Let the modal amplitude be denoted $r(t)$ and introduce the modal stiffness k , the modal mass m , and the modal load $mf(t)$. The undamped natural frequency is $\omega = \sqrt{k/m}$. It turns out that the time increment h only appears in the non-dimensional combination $\alpha = \omega h$. The recurrence equations (6.21) for the selected natural mode of the undamped system can then be written in the following non-dimensional form

$$\begin{bmatrix} 1 + \beta\alpha^2 & 0 \\ \gamma\alpha^2 & 1 \end{bmatrix} \begin{bmatrix} r \\ h\dot{r} \end{bmatrix}_{n+1} = \begin{bmatrix} 1 - (\frac{1}{2} - \beta)\alpha^2 & 1 \\ -(1 - \gamma)\alpha^2 & 1 \end{bmatrix} \begin{bmatrix} r \\ h\dot{r} \end{bmatrix}_n + h^2 \begin{bmatrix} (\frac{1}{2} - \beta) & \beta \\ (1 - \gamma) & \gamma \end{bmatrix} \begin{bmatrix} f_n \\ f_{n+1} \end{bmatrix} \quad (6.22)$$

The meaning the parameters β and γ as relative forward weighting of integrals over the time interval is clear from the load term.

An integration algorithm is stable for a time increment h , if the unloaded system with an arbitrary initial state vector $(\mathbf{x}, \dot{\mathbf{x}})_0$ leads to state vectors $(\mathbf{x}, \dot{\mathbf{x}})_n$ at any later time t_n bounded by the initial state vector. In the present linear case stability of the algorithm for the full system of equations is equivalent with stability of all natural modes of vibration.

In the absence of external load the recurrence relation (6.22) takes the form

$$\begin{bmatrix} r \\ h\dot{r} \end{bmatrix}_{n+1} = \mathbf{A} \begin{bmatrix} r \\ h\dot{r} \end{bmatrix}_n \quad (6.23)$$

with the matrix

$$\mathbf{A} = \begin{bmatrix} 1 + \beta\alpha^2 & 0 \\ \gamma\alpha^2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 - (\frac{1}{2} - \beta)\alpha^2 & 1 \\ -(1 - \gamma)\alpha^2 & 1 \end{bmatrix} \quad (6.24)$$

Now, let $\mathbf{y} = (r, h\dot{r})^T$ be an eigenvector of the matrix $\mathbf{A}$ corresponding to the eigenvalue λ . The development of this vector with time then follows from the relation (6.23).

$$\mathbf{y}_n = \mathbf{A} \mathbf{y}_{n-1} = \lambda \mathbf{y}_{n-1} = \lambda \mathbf{A} \mathbf{y}_{n-2} = \lambda^2 \mathbf{y}_{n-2} = \cdots = \lambda^n \mathbf{y}_0 \quad (6.25)$$

Thus, each time increment is equivalent with multiplication with the factor λ . Clearly, stability of an eigenvector corresponds to $|\lambda| \leq 1$. Stability of the algorithm with respect to the full set of equations therefore requires $|\lambda| \leq 1$ for each of the modes of vibration.

It has now been established that stability of the Newmark algorithm hinges on the eigenvalues of the matrix $\mathbf{A}$. It follows from (6.24) that

$$\mathbf{A} = \begin{bmatrix} 1 - \frac{\frac{1}{2}\alpha^2}{1 + \beta\alpha^2} & \frac{1}{1 + \beta\alpha^2} \\ \frac{\frac{1}{2}\gamma\alpha^4}{1 + \beta\alpha^2} - \alpha^2 & 1 - \frac{\gamma\alpha^2}{1 + \beta\alpha^2} \end{bmatrix} \quad (6.26)$$

The eigenvalues λ are determined from the characteristic equation

$$\lambda^2 - \left(2 - (\gamma + \frac{1}{2})\xi^2\right)\lambda + \left(1 - (\gamma - \frac{1}{2})\xi^2\right) = 0 \quad (6.27)$$

where it is a remarkable fact that the parameters β and ωh only appear in the combination

$$\xi^2 = \frac{(\omega h)^2}{1 + \beta(\omega h)^2} \quad (6.28)$$

with the reciprocal

$$\frac{1}{\xi^2} = \beta + \frac{1}{(\omega h)^2} \quad (6.29)$$

Thus, a stability criterion formulated in terms of ξ^{-2} implicitly contains the parameters β and $(\omega h)^2$ via (6.29).

A condition on the parameter γ may be obtained directly from the quadratic equation (6.27) without actually solving it. The constant term in the equation is the product of the two roots, λ_1 and λ_2 . The stability boundary corresponds to $|\lambda_1| = 1$ and $|\lambda_2| \leq 1$, or conversely. The constant term in the equation is the product of the two roots,

$$|\lambda_1||\lambda_2| = 1 - (\gamma - \frac{1}{2})\xi^2 \quad (6.30)$$

The parameter ξ^2 is necessarily positive, and for $|\lambda_1| = 1$ it then follows that the stability requirement $|\lambda_2| \leq 1$ gives the necessary condition

$$\gamma \geq \frac{1}{2} \quad (6.31)$$

This condition is shown as a vertical line in the stability diagram in Fig. 2.

If the roots are real, the stability limit corresponds to $\lambda_1 = -1$. This introduces a condition on the parameter ξ^2 . This condition is most easily established by observing that the coefficient of the linear term in (6.26) is $\lambda_1 + \lambda_2$, while the constant term is $\lambda_1\lambda_2$. This gives the following two expressions for λ_2 at the stability limit.

$$\lambda_2 = 3 - (\gamma + \frac{1}{2})\xi^2 = -1 + (\gamma - \frac{1}{2})\xi^2 \quad (6.32)$$

Solving for γ gives the following stability condition

$$\frac{1}{2}\gamma = \frac{1}{\xi^2} = \beta + \frac{1}{(\omega h)^2} \quad (6.33)$$

This condition is shown as an inclined line in the stability diagram in Fig. 6.2. The condition can be satisfied in two ways. If $\beta > \frac{1}{2}\gamma$, the condition is satisfied independent of the time step. This is termed *unconditional stability*. If β is too small to satisfy the unconditional stability condition, stability may be obtained by selecting a sufficiently small time interval. This so-called *conditional stability* requires the time step to satisfy a stability limit. In the present case it is seen from (6.33) that unconditional stability requires

$$\gamma \geq \frac{1}{2} \quad \text{and} \quad \beta \geq \frac{1}{2}\gamma \quad (6.34)$$

while conditional stability is obtained for

$$\gamma \geq \frac{1}{2} \quad \text{and} \quad \omega h \leq \frac{1}{\sqrt{\frac{1}{2}\gamma - \beta}} \quad (6.35)$$

This completes the analysis for real roots.

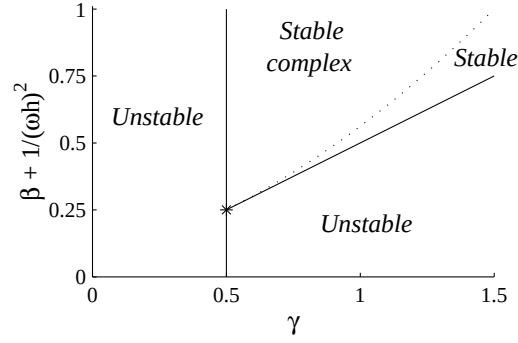


Fig. 6.2: Stability diagram for the Newmark algorithm.

In the case of complex conjugate roots there is no additional condition on the parameter ξ^2 . However, a solution with real eigenvalues will not produce a very accurate solution to an oscillation problem, and it is therefore of practical interest to identify the limit between real roots and complex conjugate roots. It follows from the discriminant of the quadratic equation (6.27) that this limit is identified by the equation

$$\frac{1}{4}(\gamma + \frac{1}{2})^2 = \frac{1}{\xi^2} = \beta + \frac{1}{(\omega h)^2} \quad (6.36)$$

This condition is shown as a parabola in the stability diagram in Fig. 6.2. It is always above the stability line (6.33), except at the common point $(\gamma, \xi^{-2}) = (\frac{1}{2}, \frac{1}{4})$. The more stringent condition of unconditional stability with complex conjugate eigenvalues is

$$\gamma \geq \frac{1}{2} \quad \text{and} \quad \beta \geq \frac{1}{4}(\gamma + \frac{1}{2})^2 \quad (6.37)$$

while conditional stability with complex conjugate roots is obtained for

$$\gamma \geq \frac{1}{2} \quad \text{and} \quad \omega h \leq \frac{1}{\sqrt{\frac{1}{4}(\gamma + \frac{1}{2})^2 - \beta}} \quad (6.38)$$

For $\gamma = \frac{1}{2}$ the general stability conditions (6.34)–(6.35) and the corresponding conditions with complex conjugate roots (6.37)–(6.38) are identical, while for $\gamma > \frac{1}{2}$ the complex root conditions are more stringent.

It is important to note, that when using a numerical integration algorithm on the full system of equations of motion, *all* modes of vibration must be stable, and thus the conditional stability condition (6.35) or (6.38) must be satisfied for the highest natural frequency of the system, $\omega = \omega_{max}$. In finite element models it may be shown that the highest natural frequency of any individual element provides an upper bound on ω_{max} . In beam and plate vibration problems the use of conditionally stable integration algorithms may lead to very small time steps, because the axial and in-plane modes of vibration have much higher frequencies than the transverse vibration modes, that are usually the main object of the analysis. On the other hand some conditionally stable methods, like the central difference algorithm corresponding to $\gamma = \frac{1}{2}$ and $\beta = 0$ permit implementation that is explicit for diagonal mass and damping matrices, see e.g. G  radin and Rixen (1997) Section 7.4. The increased computational efficiency of the explicit formulation makes this the method of choice e.g. in wave propagation or metal forming problems using elements of similar size.

6.1.4 Numerical accuracy

Numerical stability of an algorithm is necessary, but in addition to being stable a good algorithm should also be ‘accurate’. A necessary condition for accuracy is consistency, which means that the difference equations of the algorithm reproduce the original differential equations exactly in the limit $h \rightarrow 0$. It is easily seen that the Newmark representation (6.10) and (6.11) of the velocity and displacement increments is consistent, and substitution into the equation of motion therefore gives a consistent algorithm.

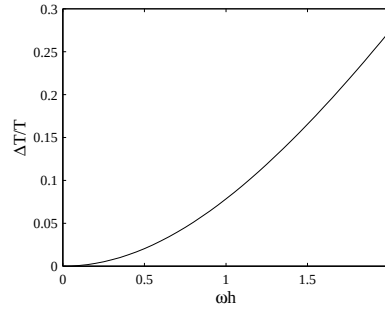
At finite time increments the accuracy is evaluated from the ability of the algorithm to keep the amplitude and the period of an undamped free oscillation. Clearly the ability to reproduce an oscillation requires complex roots λ , and thus the algorithm parameters should satisfy the stringent requirements (6.37) and (6.38). When the roots are complex conjugate their moduli are equal and then follow from (6.30) as

Table 6.2: Algorithms from the Newmark family.

Algorithm	γ	β	$(\omega h)_{cr}$	$ \lambda - 1$	$\Delta T/T$
Fully Explicit:	0	0	0	$\frac{1}{4}(\omega h)^2$	$-\frac{1}{96}(\omega h)^2$
Central Difference:	$\frac{1}{2}$	0	2	0	$-\frac{1}{24}(\omega h)^2$
Fox-Goodwin:	$\frac{1}{2}$	$\frac{1}{12}$	$\sqrt{6} \simeq 2.45$	0	$O((\omega h)^3)$
Linear acceleration:	$\frac{1}{2}$	$\frac{1}{6}$	$2\sqrt{3} \simeq 3.46$	0	$\frac{1}{24}(\omega h)^2$
Average Acceleration:	$\frac{1}{2}$	$\frac{1}{4}$	∞	0	$\frac{1}{12}(\omega h)^2$

$$|\lambda| = \sqrt{1 - (\gamma - \frac{1}{2})\xi^2} \quad (6.39)$$

An oscillation with amplitude 1 in step n will have amplitude $|\lambda|$ in step $n+1$, and thus the relative amplitude error, accumulated in one step, is $|\lambda| - 1$, with $|\lambda|$ given by (6.39). It is seen that there is no amplitude error in Newmark algorithms with $\gamma = \frac{1}{2}$. Essentially this corresponds to energy conservation in the undamped system, a very desirable property. The energy balance is discussed in more detail by G eradin and Rixen (1997) Section 7.3.6.

Fig. 6.3: Algorithmic period error as function of ωh .

The phase change in one time step is given by the argument φ of the complex number $\lambda = |\lambda|e^{i\varphi}$. The argument is determined from the ratio between the imaginary and real part of the complex number λ via the relation

$$\tan \varphi = \frac{\text{Im}(\lambda)}{\text{Re}(\lambda)} = \frac{\xi \sqrt{1 - \frac{1}{4}(\gamma + \frac{1}{2})^2 \xi^2}}{1 - \frac{1}{2}(\gamma + \frac{1}{2})\xi^2} \quad (6.40)$$

The phase increment in a time step is predicted by the numerical algorithm as φ , while the exact result is ωh . Thus the phase increment of the algorithm

relative to the exact result is $\varphi/(\omega h)$. The ratio between the period of oscillation predicted by the algorithm to the exact is the reciprocal of this. This gives the period error of the algorithm as

$$\frac{\Delta T}{T} = \frac{\omega h}{\varphi} - 1 \quad (6.41)$$

Figure 6.3 shows the period error for $\gamma = \frac{1}{2}$ and $\beta = \frac{1}{4}$.

An explicit asymptotic result for the period error for small time increments can be obtained from (6.40) by a second order expansion of the phase angle. After some manipulations the result is found to be

$$\frac{\Delta T}{T} = \frac{\omega h}{\varphi} - 1 \simeq \frac{1}{2} \left[\beta - \frac{1}{12} - \frac{1}{4} \left(\frac{5}{2} - \gamma \right) \left(\gamma - \frac{1}{2} \right) \right] (\omega h)^2 \quad (6.42)$$

It is seen that for algorithms with $\gamma = \frac{1}{2}$ the period error is a linear function of β and of higher order for $\beta = \frac{1}{12}$. This is the so-called Fox-Goodwin algorithm. It is only conditionally stable. The properties of this and other common members of the Newmark family are summarized in Table 6.2.

In some applications it is advantageous to use a numerical integration algorithm that retains the full response below a certain cut-off frequency, but introduces numerical damping at higher frequencies. There are two common causes for this. Often the discretized model may contain an erroneous representation of the underlying physical system in the high frequency domain, and it may therefore be desirable to have damping of the high frequency components to prevent them to retain or even build up their part of the system energy, thereby adding noise to the solution and possibly creating convergence problems in the iterative solution of nonlinear problems. A second class of problems is those with very high frequency components - e.g. in the modeling of mechanisms, where rigid links may correspond to infinite frequencies. These very high frequency components may be misrepresented by the very time integration algorithm, and numerical damping may be used to remove spurious high frequency oscillations, see e.g. Cardona and Geradin (1989).

Numerical damping can be introduced into the Newmark integration algorithm, by introducing a small positive parameter α , and selecting the Newmark parameters as

$$\gamma = \frac{1}{2} + \alpha \quad , \quad \beta = \frac{1}{4} \left(\frac{1}{2} + \gamma \right)^2 = \frac{1}{4} (1 + \alpha)^2 \quad (6.43)$$

It follows from (6.37) that this choice of parameters leads to an unconditionally stable algorithm with complex eigenvalues for any value of $\alpha > 0$. In fact, (6.43b) gives the smallest value of β satisfying this condition, and for small values of α it is seen from (6.42) that this gives the minimum period error

for given γ . In fact, the value of the parameter α has no appreciable effect on the period error shown in Fig. 6.3.

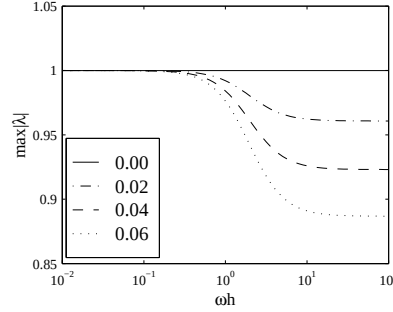


Fig. 6.4: Spectral radius of the Newmark algorithm with α -damping.

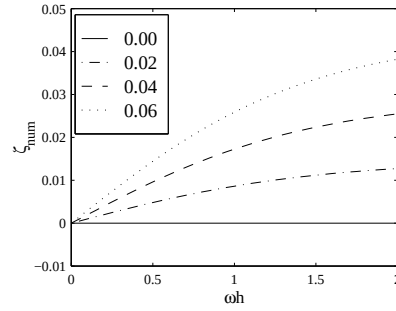


Fig. 6.5: Damping ratio of Newmark algorithm.

The attenuation of a free oscillation, as predicted by the algorithm, will be governed by the spectral radius, $\max |\lambda|$. Figure 6.4 shows the spectral radius for $\alpha = 0, 0.02, 0.04, 0.06$ as function of the frequency parameter ωh . It is seen that a positive value of α leads to a decrease of the spectral radius from its value 1 at $\omega h = 0$ to a value

$$|\lambda_\infty| = \sqrt{1 - \alpha/\beta} = \frac{1 - \alpha}{1 + \alpha} \quad (6.44)$$

for $\omega h > 10$. The spectral radius is a measure of the decrease of free undamped response due to the algorithm. This effect represents numerical damping in the algorithm. The spectral radius gives the decrease in amplitude over the time increment h , and thus the graphs in Fig. 6.4 do not fully illustrate the numerical damping over finite time intervals. This is better captured by the damping ratio ζ that represents the rate of decrease in amplitude per unit phase angle.

$$\zeta_{num} = - \frac{\ln(\max |\lambda|)}{\varphi} \quad (6.45)$$

The damping ratio is shown in Fig. 6.5 for the same values of α used in Fig. 6.4. It is seen that the numerical damping ratio ζ_{num} increases nearly in proportion to α right from the low frequency limit $\omega h \simeq 0$. In the Newmark algorithms it is therefore difficult to impose a controlled damping with a well defined cut-off frequency.

Remarks

The Newmark family of time integration algorithms are commonly used in structural dynamics, either in the general implicit form or in an explicit formulation of the central difference scheme as described e.g. by Geradin & Rixen (1997) Section 7.4. In explicit algorithms stability must always be secured by limiting the time increment. In some problems, e.g. metal forming, artificial dynamic properties are often introduced to enable solution by an explicit algorithm.

Amplitude decrease can be considered as numerically introduced damping. Most discrete models in structural dynamics are only representative below a certain frequency limit, that is usually much less than the maximum natural frequency of the discretized equations of motion. This means that a certain amount of controlled damping of the high frequency response can be desirable. In the Newmark family damping can be introduced by selecting the parameter γ (slightly) larger than $\frac{1}{2}$. However, improved damping properties can be obtained by an essential modification of the Newmark algorithms, whereby the equation of motion is satisfied via a weighted mean between the two times t_n and t_{n+1} . In the so-called Hilber-Hughes-Taylor algorithm the acceleration at t_{n+1} is matched with a weighted mean value of the forces at t_n and t_{n+1} , see e.g. Hilber et al. (1977) and G eradin and Rixen (1997). A further generalization was introduced by Chung and Hulbert (1993), who matched a weighted mean of the acceleration with a weighted mean of the forces, using different weights for acceleration and force terms. These so-called α -algorithms, named after the new weight factor, have improved damping properties in the high frequency regime, while retaining the basic algorithmic simplicity of the Newmark algorithms.

6.2 Generalized α -procedure

Improved numerical damping properties may be introduced into the basic Newmark scheme by changing the basic assumption that the equation of motion was satisfied *at time* t_{n+1} . This type of modification was introduced in two different ways: forming all force terms as a weighted mean between the values at t_n and t_{n+1} , while retaining the acceleration term at t_{n+1} , Hilber

et al. (1977), or alternatively forming the acceleration term as a weighted mean, while retaining the force terms corresponding to time t_{n+1} , Wood et al. (1981). Both of these schemes lead to improved numerical damping properties, but have largely been superseded by a combination, called the generalized α method, in which both acceleration and force terms are formed by weighted means between t_n and t_{n+1} , but with different weights, Chung and Hulbert (1993). As there is no essential simplification from using only one weighted average, we shall address the generalized α method directly here first for linear and then for non-linear problems.

6.2.1 Generalized α -procedure for linear problems

The basis of the generalized α method is the two Newmark representation formulae (6.10) and (6.11) and the following weighted mean value formulation of the equations of motion.

$$\begin{aligned} \alpha_m \mathbf{M} \ddot{\mathbf{x}}_n + (1 - \alpha_m) \mathbf{M} \ddot{\mathbf{x}}_{n+1} + \alpha_f (\mathbf{C} \dot{\mathbf{x}}_n + \mathbf{K} \mathbf{x}_n) \\ + (1 - \alpha_f) (\mathbf{C} \dot{\mathbf{x}}_{n+1} + \mathbf{K} \mathbf{x}_{n+1}) = \alpha_f \mathbf{f}_n + (1 - \alpha_f) \mathbf{f}_{n+1} \end{aligned} \quad (6.46)$$

The parameter α_m represents the relative weight of the old inertial term, while α_f represents the relative weight of the old force terms from damping, stiffness and external load.

Like in the Newmark method $\dot{\mathbf{x}}_{n+1}$ and $\mathbf{x}_{n+1}$ are eliminated by use of the representation (6.10) and (6.11). However, due to the α -weights it is convenient in this case to formulate the resulting equation in terms of the acceleration increment

$$\Delta \ddot{\mathbf{x}} = \ddot{\mathbf{x}}_{n+1} - \ddot{\mathbf{x}}_n \quad (6.47)$$

The representation formulae (6.10) and (6.11) are then split into a 'prediction' part without the acceleration *increment* plus the 'correction' contribution from the acceleration increment.

$$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_n + \Delta \dot{\mathbf{x}}_* + \gamma h \Delta \ddot{\mathbf{x}} \quad (6.48)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta \mathbf{x}_* + \beta h^2 \Delta \ddot{\mathbf{x}} \quad (6.49)$$

where the 'predictor' parts are

$$\Delta \dot{\mathbf{x}}_* = h \ddot{\mathbf{x}}_n, \quad \Delta \mathbf{x}_* = h \dot{\mathbf{x}}_n + \frac{1}{2} h^2 \ddot{\mathbf{x}}_n \quad (6.50)$$

corresponding to a truncated Taylor expansion at t_n . It should be realized that in linear problems the separation of the updated velocity and displacement into a predictor and a corrector part is largely a matter of convenience and does not really correspond to a prediction. Thus the use of different 'pre-

Table 6.3: The generalized α -algorithm.

1) System matrices:
$\mathbf{K}, \mathbf{C}, \mathbf{M}$
$\mathbf{M}_* = (1 - \alpha_m) \mathbf{M} + (1 - \alpha_f)(\gamma h \mathbf{C} + \beta h^2 \mathbf{K})$
2) Initial conditions:
$\mathbf{x}_0, \dot{\mathbf{x}}_0$
$\ddot{\mathbf{x}}_0 = \mathbf{M}^{-1}(\mathbf{f}_0 - \mathbf{C} \dot{\mathbf{x}}_0 - \mathbf{K} \mathbf{x}_0)$
3) Increment time:
$t_{n+1} = t_n + h$
4) Increment predictors:
$\Delta \dot{\mathbf{x}}_* = h \ddot{\mathbf{x}}_n$
$\Delta \mathbf{x}_* = h \dot{\mathbf{x}}_n + \frac{1}{2} h^2 \ddot{\mathbf{x}}_n$
$\Delta \mathbf{f} = \mathbf{f}_{n+1} - \mathbf{f}_n$
5) Acceleration increment:
$\Delta \ddot{\mathbf{x}} = \mathbf{M}_*^{-1} \left(\mathbf{f}_n - (\mathbf{M} \ddot{\mathbf{x}}_n + \mathbf{C} \dot{\mathbf{x}}_n + \mathbf{K} \mathbf{x}_n) \right. \\ \left. + (1 - \alpha_f)(\Delta \mathbf{f} - \mathbf{C} \Delta \dot{\mathbf{x}}_* - \mathbf{K} \Delta \mathbf{x}_*) \right)$
6) State vector update:
$\ddot{\mathbf{x}}_{n+1} = \ddot{\mathbf{x}}_n + \Delta \ddot{\mathbf{x}}$
$\dot{\mathbf{x}}_{n+1} = \dot{\mathbf{x}}_n + \Delta \dot{\mathbf{x}}_* + \gamma h \Delta \ddot{\mathbf{x}}$
$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta \mathbf{x}_* + \beta h^2 \Delta \ddot{\mathbf{x}}$
7) Return to 3) or stop.

dictors' here than previously in the Newmark method simply corresponds to a slightly different grouping of the terms.

When the acceleration increment is isolated at the left side of (6.46) they define the modified mass matrix

$$\mathbf{M}_* = (1 - \alpha_m) \mathbf{M} + (1 - \alpha_f)(\gamma h \mathbf{C} + \beta h^2 \mathbf{K}) \quad (6.51)$$

It is seen, that the introduction of the α -factors reduce all three terms in $\mathbf{M}_*$, and that the introduction of two weight factors permit different reduction on the mass relative to the combined effect of damping and stiffness.

The weighted equation of motion of the generalized α -method can now be written in the form

$$\begin{aligned} \mathbf{M}_* \Delta \ddot{\mathbf{x}} = & \mathbf{f}_n - (\mathbf{M} \ddot{\mathbf{x}}_n + \mathbf{C} \dot{\mathbf{x}}_n + \mathbf{K} \mathbf{x}_n) \\ & + (1 - \alpha_f)(\Delta \mathbf{f} - \mathbf{C} \Delta \dot{\mathbf{x}}_* - \mathbf{K} \Delta \mathbf{x}_*) \end{aligned} \quad (6.52)$$

where the force increment is defined as

$$\Delta \mathbf{f} = \mathbf{f}_{n+1} - \mathbf{f}_n \quad (6.53)$$

It is seen that the parameter α_m is fully absorbed in the modified mass matrix $\mathbf{M}_*$, while $(1 - \alpha_f)$ appears as an explicit factor on the ‘predicted’ force increments. The first two terms on the right represent the force unbalance at time t_n . In the Newmark method these terms vanish, because the equation of motion is matched at t_n, t_{n+1} etc.

An implementation of the generalized α algorithm for linear analysis is shown in Table 6.3. It is seen to be very similar to the Newmark algorithm shown in Table 6.1. The only essential difference between the two algorithms as shown here is the formulation of the α algorithm in terms of state vector increments. The Newmark algorithm can also be implemented in this format, and indeed appears as a special case of the α algorithm for $\alpha_m = \alpha_f = 0$.

Parameters and properties

The properties of the generalized α -algorithm are analyzed by considering modal response with amplitude $r(t)$ and natural frequency $\omega = \sqrt{k/m}$. The generalized α -algorithm gives the vector $\mathbf{y} = (r, h\dot{r}, h^2\ddot{r})$ at time t_{n+1} from the value of the vector at time t_n . The basic form of the algorithm is a matrix relation between the two vectors, consisting of the two Newmark representation relations (6.10), (6.11) and the α -averaged equation of motion (6.46).

$$\begin{bmatrix} 0 & 1 & -\gamma \\ 1 & 0 & -\beta \\ (1-\alpha_f)(\omega h)^2 & 0 & 1-\alpha_m \end{bmatrix} \begin{bmatrix} r \\ h\dot{r} \\ h^2\ddot{r} \end{bmatrix}_{n+1} = \begin{bmatrix} 0 & 1 & 1-\gamma \\ 1 & 1 & \frac{1}{2}-\beta \\ -\alpha_f(\omega h)^2 & 0 & -\alpha_m \end{bmatrix} \begin{bmatrix} r \\ h\dot{r} \\ h^2\ddot{r} \end{bmatrix}_n \quad (6.54)$$

Note, that in general the dimension of the problem is three, while the Newmark algorithm could be analyzed as a two, dimensional problem, because the last equation in (6.54) permits explicit elimination of the acceleration, when $\alpha_f = \alpha_m = 0$.

The relation (6.54) is of the form

$$\mathbf{B} \mathbf{y}_{n+1} = \mathbf{A} \mathbf{y}_n \quad (6.55)$$

Modes associated with the amplification factor λ satisfy the relation $\mathbf{y}_{n+1} = \lambda \mathbf{y}_n$, and substitution into (6.55) then leads to the generalized eigenvalue problem

$$(\mathbf{A} - \lambda \mathbf{B}) \mathbf{y}_n = \mathbf{0} \quad (6.56)$$

In order to permit non-trivial solutions the determinant must vanish, giving the characteristic equation,

$$\begin{aligned} & [(1 - \alpha_m) + (1 - \alpha_f)\beta(\omega h)^2](1 - \lambda)^3 \\ & - [1 + \beta(\omega h)^2 + (1 - \alpha_f)(\tfrac{1}{2} + \gamma)(\omega h)^2](1 - \lambda)^2 \\ & + [(\tfrac{1}{2} + \gamma) + (1 - \alpha_f)](\omega h)^2(1 - \lambda) - (\omega h)^2 = 0 \end{aligned} \quad (6.57)$$

The first task is to determine requirements in the parameters γ , β , α_f and α_m such that the algorithm is stable. This requirement is equivalent to the condition that the roots λ are all within the unit circle in the complex plane.

It is not easy to investigate the properties of the roots λ_j directly from the equation (6.57). The complex variable λ is therefore replaced by a new variable z via the Möbius transformation

$$z = \frac{\lambda + 1}{\lambda - 1} \quad (6.58)$$

This transformation represents a mapping of the complex plane, whereby the inside of the unit circle in the λ -plane is mapped on the left half of the z -plane, as shown in Fig. 6.6. The stability condition $|\lambda| \leq 1$ then takes the form $\text{Re}(z) \leq 0$. This condition can be checked by the Routh-Hurwitz conditions, Meirovitch (1986). Note, that the mapping introduced by (6.58) changes the orientation of the complex plane, and maps the point $\lambda=1$ to infinity.

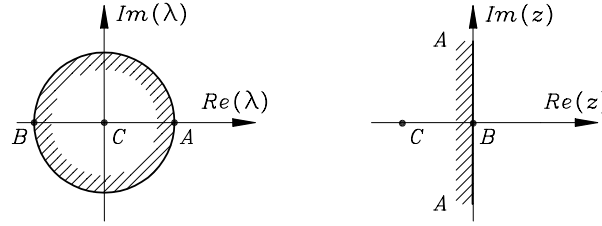


Fig. 6.6: Möbius transformation of the λ -plane.

In terms of the new variable z the cubic equation (6.57) takes the form

$$\begin{aligned} z^3 + 2[\gamma - \alpha_f]z^2 + 4\left[\frac{1}{(\omega h)^2} + \beta - \tfrac{1}{2}\gamma + (\tfrac{1}{2} - \alpha_f)(\gamma - \tfrac{1}{2})\right]z \\ + 8\left[\frac{\tfrac{1}{2} - \alpha_m}{(\omega h)^2} + (\tfrac{1}{2} - \alpha_f)(\beta - \tfrac{1}{2}\gamma)\right] = 0 \end{aligned} \quad (6.59)$$

The general cubic equation is

$$a_0 z^3 + a_1 z^2 + a_2 z + a_3 = 0 \quad (6.60)$$

In order for the roots of the cubic equation to satisfy the condition $\text{Re}(z) \leq 0$ it is necessary that none of the coefficients are of different sign. In the case of (6.59), where $a_0 = 1$, this implies that none of the coefficients can be negative. In order for the coefficient a_0 to be non-negative for all values of ωh , the following inequalities must be satisfied,

$$\alpha_m \leq \frac{1}{2} \quad , \quad \alpha_f \leq \frac{1}{2} \quad , \quad \beta \geq \frac{1}{2}\gamma \quad (6.61)$$

Note, that the condition $\beta \geq \frac{1}{2}\gamma$ is identical to (6.34b) in the case of the Newmark algorithm.

According to the Routh-Hurwitz criterion the condition $\text{Re}(z) \leq 0$ will be satisfied, if the condition on the sign of the coefficients is supplemented by the following determinant condition

$$8 \begin{vmatrix} a_1 & a_0 \\ a_3 & a_2 \end{vmatrix} = \frac{\gamma - (\frac{1}{2} + \alpha_f - \alpha_m)}{(\omega h)^2} + (\gamma - \frac{1}{2}) \left[\beta - \frac{1}{2}\gamma + (\frac{1}{2} - \alpha_f)(\gamma - \alpha_f) \right] \geq 0 \quad (6.62)$$

When this inequality is imposed for all values of ωh , it follows that

$$\gamma \geq \frac{1}{2} + \alpha_f - \alpha_m \quad , \quad \gamma \geq \frac{1}{2} \quad (6.63)$$

Combination of these inequalities further imply that

$$\alpha_f \geq \alpha_m \quad (6.64)$$

A close analogy with the Newmark algorithm is seen, when α is replaced by $\alpha_f - \alpha_m$.

In order for the numerical solution to be able to represent oscillatory behaviour it is desirable that two of the roots, λ_1, λ_2 are complex. The desired qualitative behaviour of the complex roots λ_1, λ_2 is, that they start at $\lambda_1 = \lambda_2 = 1$ for $\omega h = 0$, follow the unit circle closely for increasing ωh , and eventually leave the unit circle to meet at the negative real axis at some point $\lambda_{1,2}^\infty$, $-1 < \lambda_{1,2}^\infty < 0$. The coefficients of the high frequency limit of the cubic equation (6.59) are expressed in terms of the three variable combinations $\beta - \frac{1}{2}\gamma$, $\gamma - \frac{1}{2}$ and $\frac{1}{2} - \alpha_f$. The requirement of a double root corresponds to a relation between these variable combinations. In the present case the relation gives β as a function of γ . The result is most easily established by observing that the high frequency limit of the cubic equation has the roots

$$z_1^\infty = z_2^\infty = -(\gamma - \frac{1}{2}) \quad , \quad z_3 = -(1 - 2\alpha_f) \quad (6.65)$$

provided that the parameter β is given by

$$\beta = \frac{1}{2}\gamma + \frac{1}{4}(\gamma - \frac{1}{2})^2 = \frac{1}{4}(\gamma + \frac{1}{2})^2 \quad (6.66)$$

Note, that this condition is identical to the condition (6.37) for complex conjugate roots in the Newmark algorithm.

The parameter γ is now selected as small as possible in accordance with the stability requirement (6.63),

$$\gamma = \frac{1}{2} + \alpha_f - \alpha_m \quad (6.67)$$

The double root equation (6.65a) then establishes the following expression

$$\alpha_f - \alpha_m = -z_{1,2}^\infty \quad (6.68)$$

Finally, the high frequency limit λ_3^∞ of the real root is chosen identical to the double root $\lambda_{1,2}^\infty$, whereby the low frequency damping is minimized. Thus, all parameters of the algorithm are determined by the high frequency limit $\lambda_\infty = -|\lambda_\infty|$ of the amplification factor. Combination of (6.68) and (6.65b), and use of the transformation (6.58), leads to

$$\alpha_f = \frac{|\lambda_\infty|}{|\lambda_\infty| + 1} \quad , \quad \alpha_m = \frac{2|\lambda_\infty| - 1}{|\lambda_\infty| + 1} \quad (6.69)$$

The parameters γ and β then follow from (6.67) and (6.66), respectively. Note, that these formulae are identical to (6.43a) and (6.43b) for the Newmark algorithm, when α is replaced by $\alpha_f - \alpha_m$. However, the resulting algorithms are different, because the parameters α_f and α_m appear explicitly in the generalized α -algorithm.

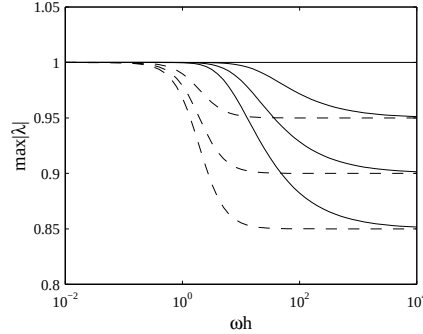


Fig. 6.7: Spectral radius of the generalized α (—) and Newmark (---) algorithms for $|\lambda_\infty| = 1.0, 0.95, 0.90, 0.85$.

The properties of the generalized α algorithm with parameters determined by (6.66), (6.67) and (6.69) can be characterized by the development of the period error $\Delta T/T$, the spectral radius $\max |\lambda|$ and the numerical damping ratio ζ_{num} as function of the non-dimensional frequency parameter ωh . The period error is as shown in Fig. 6.3 for the Newmark algorithm, without

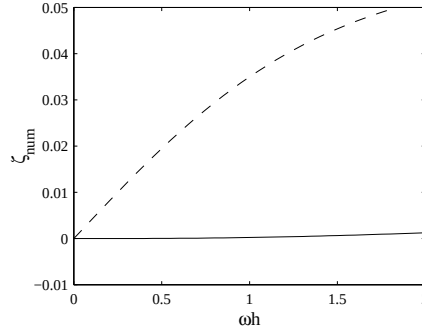


Fig. 6.8: Damping ratio of generalized α (—) and Newmark (---) algorithms for $|\lambda_\infty| = 0.85$.

appreciable influence from the α_f, α_m parameters. The improvement obtained in the generalized α -algorithm shows up in the development of the spectral radius. Figure 6.7 shows the spectral radius of the generalized α and Newmark algorithms for $|\lambda_\infty| = 1.0, 0.95, 0.90, 0.85$. The spectral radius is obtained by solving the cubic equation (6.59) for z and then finding λ from (6.58). It is seen that the numerical damping occurs at much higher values of ωh for the generalized α -algorithm. The generalized α -algorithm leaves the low frequency domain nearly without damping, as shown in Fig. 6.8. Thus, the generalized α -algorithm exhibits superior numerical damping characteristics, when compared to the Newmark algorithm with α -damping. The magnitude of the imposed high frequency damping is conveniently controlled by the spectral radius at infinity $|\lambda_\infty|$ via the formulae (6.69), originally derived by Chung and Hulbert (1993).

6.2.2 Non-linear generalized α -procedure

In non-linear structural mechanics problems the nonlinearity is often associated with the displacement, and it is therefore convenient to rearrange the solution method in such a way that the prediction concerns the velocity $\dot{\mathbf{x}}$ and the acceleration $\ddot{\mathbf{x}}$, while the displacement $\mathbf{x}$ is solved for in the iterative solution of the equation of motion.

Let the non-linear equation of motion be given in the form

$$\mathbf{M} \ddot{\mathbf{x}} + \mathbf{g}(\mathbf{x}, \dot{\mathbf{x}}) = \mathbf{f}(t) \quad (6.70)$$

where $\mathbf{g}(\mathbf{x}, \dot{\mathbf{x}})$ denotes the non-linear internal forces. If the problem involves material constitutive equations in incremental form, the nonlinear internal force at t_{n+1} is calculated from the internal force at t_n via the relation

$$\mathbf{g}_{n+1} = \mathbf{g}_n + \Delta \mathbf{g}(\mathbf{x}, \dot{\mathbf{x}}, \Delta \mathbf{x}) \quad (6.71)$$

where the increment $\Delta \mathbf{g}$ may depend on current displacement and velocity as well as the displacement increment. In the common case of plasticity theory it is convenient to use the displacement increment $\Delta \mathbf{x}$ as the unknown state variable, that is solved for by iteration. Hereby the dynamic problem takes the form of a modification of the corresponding static problem.

The non-linear equation of motion may be expressed in the form of a residual vector

$$\mathbf{r} = \mathbf{M} \ddot{\mathbf{x}} + \mathbf{g}(\mathbf{x}, \dot{\mathbf{x}}) - \mathbf{f}(t) \quad (6.72)$$

In principle the residual vector should vanish identically, but in the numerical integration it is made to vanish by iteration at one point within each time step. In the generalized α -method, the residual is formulated as a weighted average of the corresponding values at the times t_n and t_{n+1} with a weight factor α_m associated with the mass term and the weight factor α_f associated with the internal and external forces. Thus, in the generalized α -method the residual for the time interval (t_n, t_{n+1}) is

$$\mathbf{r} = \alpha_m \mathbf{M} \ddot{\mathbf{x}}_n + (1 - \alpha_m) \mathbf{M} \ddot{\mathbf{x}}_{n+1} + \alpha_f (\mathbf{g}_n - \mathbf{f}_n) + (1 - \alpha_f) (\mathbf{g}_{n+1} - \mathbf{f}_{n+1}) \quad (6.73)$$

The residual takes a somewhat clearer and convenient form when expressed in terms of increments,

$$\mathbf{r} = (\mathbf{M} \ddot{\mathbf{x}}_n + \mathbf{g}_n - \mathbf{f}_n) + (1 - \alpha_m) \mathbf{M} \Delta \ddot{\mathbf{x}} + (1 - \alpha_f) (\Delta \mathbf{g} - \Delta \mathbf{f}) \quad (6.74)$$

The first parenthesis contains the residual at time t_n . Because of the use of weighted averages, this residual does not vanish. The last terms are the increments with appropriate weights. In plasticity problems it is convenient to use this latter form as the plasticity algorithm evaluates the increment $\Delta \mathbf{g}$, and not the total internal force $\mathbf{g}_{n+1}$.

The equations $\mathbf{r} = \mathbf{0}$ are non-linear and must be solved iteratively. Assume that an estimate of the vectors $\Delta \mathbf{x}$, $\Delta \dot{\mathbf{x}}$, $\Delta \ddot{\mathbf{x}}$ is available. Corrections $\delta \mathbf{x}$, $\delta \dot{\mathbf{x}}$, $\delta \ddot{\mathbf{x}}$ are then determined by using the linearized system of equations

$$\mathbf{r} + \left(\frac{\partial \mathbf{r}}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \mathbf{r}}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} + \frac{\partial \mathbf{r}}{\partial \ddot{\mathbf{x}}} \delta \ddot{\mathbf{x}} \right) + \dots = \mathbf{0} \quad (6.75)$$

These equations are supplemented with the incremental form of the Newmark representation formulae (6.10)–(6.11), taking the form

$$\delta \dot{\mathbf{x}} = \gamma h \delta \ddot{\mathbf{x}} \quad (6.76)$$

$$\delta \mathbf{x} = \beta h^2 \delta \ddot{\mathbf{x}} \quad (6.77)$$

These linear equations enable the elimination of the increments $\delta \dot{\mathbf{x}}$ and $\delta \ddot{\mathbf{x}}$ from the linearized residual equation (6.75). By elimination the linear term becomes

Table 6.4: Non-linear generalized α -algorithm.

1) Initial conditions:

$$\begin{aligned}\mathbf{x}_0, \dot{\mathbf{x}}_0 \\ \ddot{\mathbf{x}}_0 = \mathbf{M}^{-1}(\mathbf{f}_0 - \mathbf{g}(\mathbf{x}_0, \dot{\mathbf{x}}_0))\end{aligned}$$

2) Increment time:

$$t_{n+1} = t_n + h$$

3) Increment predictors:

$$\begin{aligned}\Delta \mathbf{x} &= h \dot{\mathbf{x}}_n + \frac{1}{2} h^2 \ddot{\mathbf{x}}_n \\ \Delta \dot{\mathbf{x}} &= h \ddot{\mathbf{x}}_n \\ \Delta \ddot{\mathbf{x}} &= \mathbf{0} \\ \Delta \mathbf{f} &= \mathbf{f}_{n+1} - \mathbf{f}_n\end{aligned}$$

4) Residual calculation:

$$\begin{aligned}\Delta \mathbf{g} &= \mathbf{g}(\mathbf{x}_n + \Delta \mathbf{x}, \dot{\mathbf{x}}_n + \Delta \dot{\mathbf{x}}) - \mathbf{g}(\mathbf{x}_n, \dot{\mathbf{x}}_n) \\ \mathbf{r} &= (\mathbf{M} \ddot{\mathbf{x}}_n + \mathbf{g}_n - \mathbf{f}_n) \\ &\quad + (1 - \alpha_m) \mathbf{M} \Delta \ddot{\mathbf{x}} + (1 - \alpha_f)(\Delta \mathbf{g} - \Delta \mathbf{f})\end{aligned}$$

5) System matrices and increment correction:

$$\begin{aligned}\mathbf{K} &= \partial \mathbf{g} / \partial \mathbf{x} \quad , \quad \mathbf{C} = \partial \mathbf{g} / \partial \dot{\mathbf{x}} \quad , \quad \mathbf{M} \\ \mathbf{K}_* &= (1 - \alpha_f)(\mathbf{K} + \frac{\gamma h}{\beta h^2} \mathbf{C}) + (1 - \alpha_m) \frac{1}{\beta h^2} \mathbf{M} \\ \delta \mathbf{x} &= \mathbf{K}_*^{-1} \mathbf{r} \\ \Delta \mathbf{x} &= \Delta \mathbf{x} + \delta \mathbf{x} \\ \Delta \dot{\mathbf{x}} &= \Delta \dot{\mathbf{x}} + \frac{\gamma h}{\beta h^2} \delta \mathbf{x} \\ \Delta \ddot{\mathbf{x}} &= \Delta \ddot{\mathbf{x}} + \frac{1}{\beta h^2} \delta \mathbf{x}\end{aligned}$$

If $\|\mathbf{r}\| > \varepsilon_{\mathbf{r}}$ or $\|\delta \mathbf{x}\| > \varepsilon_{\mathbf{x}}$ return to 4) for new iteration.

6) State vector update:

$$\begin{aligned}\mathbf{x}_{n+1} &= \mathbf{x}_n + \Delta \mathbf{x} \\ \dot{\mathbf{x}}_{n+1} &= \dot{\mathbf{x}}_n + \Delta \dot{\mathbf{x}} \\ \ddot{\mathbf{x}}_{n+1} &= \ddot{\mathbf{x}}_n + \Delta \ddot{\mathbf{x}}\end{aligned}$$

7) Return to 2) for new time step or stop.

$$\begin{aligned}d\mathbf{r} &= \frac{d\mathbf{r}}{d\mathbf{x}} d\mathbf{x} = \frac{\partial \mathbf{r}}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \mathbf{r}}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} + \frac{\partial \mathbf{r}}{\partial \ddot{\mathbf{x}}} \delta \ddot{\mathbf{x}} \\ &= \left(\frac{\partial \mathbf{r}}{\partial \mathbf{x}} + \frac{\gamma h}{\beta h^2} \frac{\partial \mathbf{r}}{\partial \dot{\mathbf{x}}} + \frac{1}{\beta h^2} \frac{\partial \mathbf{r}}{\partial \ddot{\mathbf{x}}} \right) \delta \mathbf{x}\end{aligned}\tag{6.78}$$

The partial derivatives of the residual vector are expressed in terms of the tangent stiffness matrix and the tangent damping matrix, defined as

$$\mathbf{K} = \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \quad , \quad \mathbf{C} = \frac{\partial \mathbf{g}}{\partial \dot{\mathbf{x}}} \quad (6.79)$$

The total derivative of the residual vector with respect to the displacement vector $\mathbf{x}$ defines the modified tangent stiffness matrix

$$\mathbf{K}_* = \frac{d\mathbf{r}}{d\mathbf{x}} = (1 - \alpha_f) \left(\mathbf{K} + \frac{\gamma h}{\beta h^2} \mathbf{C} \right) + (1 - \alpha_m) \frac{1}{\beta h^2} \mathbf{M} \quad (6.80)$$

When formulated in terms of the displacement increments $\delta \mathbf{x}$, the residual equation (6.75) takes the more familiar form

$$\mathbf{K}_* \delta \mathbf{x} = -\mathbf{r} \quad (6.81)$$

where $\mathbf{K}_*$ is the modified tangent stiffness matrix (6.59).

The non-linear α -algorithm is shown in Table 6.4. The link to the linear formulation is established by observing that $\mathbf{M}_* = \beta h^2 \mathbf{K}_*$. Thus, it is largely a matter of terminology and tradition that the linear problem is formulated as a modified mass matrix problem, while the non-linear problem is formulated as a modified stiffness problem.

6.3 The Newmark method revisited

The analysis of the Newmark algorithm given above is limited to the linear form and is traditionally carried out in terms of a frequency analysis as demonstrated here. More recently attention has turned to the formulation of an energy balance equation associated with the algorithm. The issues relating to the existence and actual formulation of energy relations for finite difference based time integration methods are typically not trivial. However, if an energy relation, expressing the development of an energy over a time increment, has been established many of the basic characteristics of the underlying algorithm can be seen fairly directly from the specific form of this energy relation. The rest of this chapter is devoted to a summary of the energy balance equation for the basic Newmark algorithm, when applied to undamped systems. The results are based on Krenk (2006), which also contains results for damped systems and for the α -version of the algorithm.

6.3.1 Energy relations of the Newmark method

The linear equation of motion is given by the second-order differential equation

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f}(t) \quad (6.82)$$

The corresponding energy balance equation is obtained by pre-multiplication with the velocity vector $\dot{\mathbf{u}}^T$,

$$\frac{d}{dt} \left[\frac{1}{2} \dot{\mathbf{u}}^T \mathbf{M} \dot{\mathbf{u}} + \frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u} \right] = \dot{\mathbf{u}}^T \mathbf{f} - \dot{\mathbf{u}}^T \mathbf{C} \dot{\mathbf{u}} \quad (6.83)$$

The square brackets on the left side contains the mechanical energy, consisting of the sum of the kinetic energy and the elastic energy. The energy balance equation gives the rate of change of the mechanical energy as expressed by the right side of the equation, consisting of the rate of external work and the rate of energy dissipation expressed as a quadratic form of the damping matrix $\mathbf{C}$.

An energy relation of the homogeneous Newmark algorithm in terms of finite increments can be obtained via elimination of increments by the Taylor approximations, Krenk (2006). After a somewhat lengthy reformulation the final energy balance is found to be

$$\begin{aligned} & \left[\frac{1}{2} \dot{\mathbf{u}}^T \mathbf{M} \dot{\mathbf{u}} + \frac{1}{2} \mathbf{u}^T \left(\mathbf{K} + (\beta - \frac{1}{2}\gamma) h^2 \mathbf{K} \mathbf{M}^{-1} \mathbf{K} \right) \mathbf{u} \right]_n^{n+1} \\ & = -(\gamma - \frac{1}{2}) \Delta \mathbf{u}^T \left(\mathbf{K} + (\beta - \frac{1}{2}\gamma) h^2 \mathbf{K} \mathbf{M}^{-1} \mathbf{K} \right) \Delta \mathbf{u} \end{aligned} \quad (6.84)$$

It is seen from this finite-increment energy balance the the original stiffness matrix $\mathbf{K}$ is replaced by an *equivalent stiffness* given by

$$\mathbf{K}_{eq} = \mathbf{K} + (\beta - \frac{1}{2}\gamma) h^2 \mathbf{K} \mathbf{M}^{-1} \mathbf{K} \quad (6.85)$$

It now turns out that many of the results for the Newmark algorithm, derived above by use of the classic procedure in terms of frequency analysis, can in fact be obtained fairly directly by use of the energy relation (6.85) as demonstrated in the following.

6.3.2 Stability of the Newmark method

The criteria for unconditional stability of the Newmark method can be identified directly from the energy balance equation (6.84) by observing that the equivalent energy, defined by the left side of the equation, must be positive definite, and that the dissipation must be non-negative. The first requirement implies that the factor $(\beta - \frac{1}{2}\gamma)$, must be positive, and the second that the factor $(\gamma - \frac{1}{2})$ on the dissipation term on the left side of the equation must be non-negative. Thus, unconditional stability requires

$$\gamma \geq \frac{1}{2} \quad \text{and} \quad \beta \geq \frac{1}{2}\gamma \quad (6.86)$$

These are the conditions found above by the somewhat more elaborate frequency analysis procedure. They correspond to the two full lines in the stability diagram in Fig. 6.2. It is noted that it follows directly from the balance equation (6.84) that $\gamma > \frac{1}{2}$ associated with damping, but also that damping is expressed in terms of loss of the equivalent energy, expressed by the equivalent stiffness $\mathbf{K}_{eq}$.

Conditional stability corresponds to the situation, in which the factor $(\beta - \frac{1}{2}\gamma)h^2$ on the second term in the definition (6.85) of the equivalent stiffness is negative, but still results in a positive definite equivalent stiffness. This imposes a limit on the magnitude of the time step h for stability, and as seen from the expression this limit will be in terms of ωh , where ω is a natural vibration frequency of the system.

The detailed balance between the two terms in the equivalent stiffness $\mathbf{K}_{eq}$ is clarified by representing the response in terms of the linear undamped modes,

$$\mathbf{u}(t) = \sum_{j=1}^m u_j(t) \mathbf{u}_j \quad (6.87)$$

The free response of an individual mode then is

$$\mathbf{u}(t) = \mathbf{u}_j e^{i\omega_j t} \quad (6.88)$$

The mode shapes $\mathbf{u}_j$ and frequencies ω_j found from generalized eigenvalue problem

$$(\mathbf{K} - \omega_j^2 \mathbf{M}) \mathbf{u}_j = \mathbf{0} \quad (6.89)$$

whereby the modes satisfy the orthogonality conditions

$$\mathbf{u}_j^T \mathbf{M} \mathbf{u}_k = \delta_{jk} \quad , \quad \mathbf{u}_j^T \mathbf{K} \mathbf{u}_k = \omega_j^2 \delta_{jk} \quad (6.90)$$

Substitution of mode j into the energy balance equation,

$$\begin{aligned} \left[\frac{1}{2} \dot{u}_j^2 + \frac{1}{2} \left(1 + (\beta - \frac{1}{2}\gamma)(\omega_j h)^2 \right) \omega_j^2 u_j^2 \right]_n^{n+1} = \\ - (\gamma - \frac{1}{2}) \left(1 + (\beta - \frac{1}{2}\gamma)(\omega_j h)^2 \right) \omega_j^2 (\Delta u_j)^2 \end{aligned} \quad (6.91)$$

leading to the stability condition

$$\beta_* = \beta + (\omega_j h)^{-2} \geq \frac{1}{2}\gamma \quad (6.92)$$

For the explicit method $\beta = 0$ with $\gamma = \frac{1}{2}$ this corresponds to $h \leq h_{max} = 2/\omega_{max} = T_{min}/\pi$, i.e. about 3 points per period.

6.3.3 Effect of modified stiffness

The character of the main errors introduced by time integration by the Newmark method can be characterized by analyzing the form of the equivalent energy appearing in the balance equation (6.84) for the equivalent energy.

- Explicit methods: $\beta = 0 < \frac{1}{2}\gamma \Rightarrow$ stiffness reduction.
- Implicit methods: $\beta \geq \frac{1}{2}\gamma \Rightarrow$ possible stiffness increase.

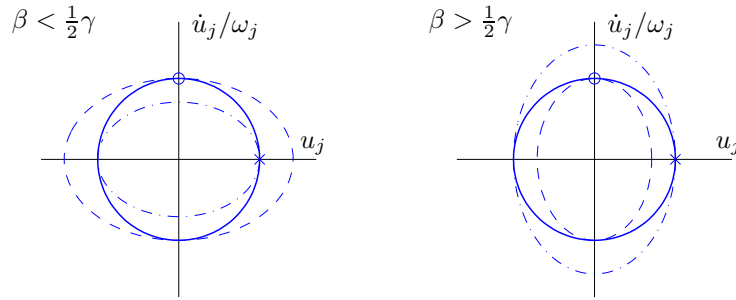


Fig. 6.9: a) Reduced stiffness $\beta < \frac{1}{2}\gamma$, b) Increased stiffness $\beta > \frac{1}{2}\gamma$.

It is to be noted that the stability criterion of explicit methods corresponds to complete loss of stiffness, whereby the algorithmic energy ceases to be positive definite. Intuitively speaking, there will exist a mode without stiffness, and when excited the response associated with this mode will grow towards infinity – much like a free mass.

The energy relations as illustrated by the phase plane diagrams in Fig. (6.84) also explain the phenomenon of response 'overshoot' that may be observed in connection with Newmark based time integration of problems defined in terms of initial conditions. If started with initial velocities, marked by $\circ$ in the figure, the mechanical energy exceeds the initial value for $\beta < \frac{1}{2}\gamma$ as shown in Fig. 6.84a, whereas start from initial displacements, marked by $\times$ in the figure, leads to excess mechanical energy as shown in Fig. 6.84b.

Example 6.1. Explicit integration.

Consider the response of the eight-story idealized shear building shown in the Fig. 6.10. The results are given in terms of the natural frequencies from Table 6.5.

The free response is evaluated for the undamped structures by explicit Newmark integration for $h = 0.5$. This corresponds to a ratio between the smallest period and the integration increment of $T_{min}/h \simeq 6.4$, giving about 6 time increments within the fastest period of oscillation.

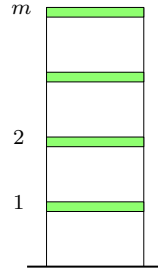


Fig. 6.10: Portal frame with m storeys.
Table 6.5: Modal parameters in 8-storey portal building.

	1	2	3	4	5	6	7	8
ω_i	0.185	0.547	0.892	1.205	1.478	1.700	1.865	1.966
T_i	34.048	11.480	7.048	5.213	4.251	3.695	3.369	3.196
$1 - \frac{1}{4}(\omega h)^2$	0.998	0.981	0.950	0.909	0.864	0.819	0.783	0.758
$E_j/E_0 _{v_8}$	0.233	0.218	0.189	0.150	0.107	0.065	0.031	0.008
$E_j/E_0 _{x_8}$	0.008	0.065	0.150	0.218	0.233	0.189	0.107	0.031

Two sets of initial conditions are considered: a unit velocity $v_8 = 1$ at the top floor, and a unit displacement $u_8 = 1$ at the top floor. The modal energies are given in the last three rows of Table 6.5. The first of these rows gives the reduction of the equivalent stiffness introduced by the algorithm, and it is seen that the highest mode has a reduction factor of 0.758 – reducing the effective elastic energy to about the quarter of its intended value. The last two rows give the distribution of modal energies for the two load-cases. It is seen that the initial velocity case has rather large energy in the two lowest modes, while the initial displacement case concentrates the larger energies in the middle modes.

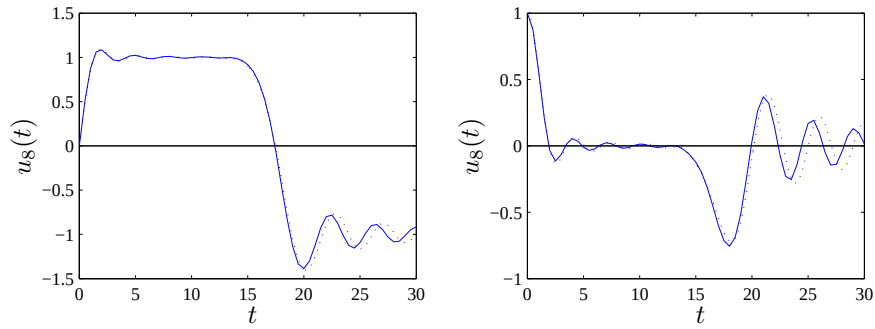


Fig. 6.11: Displacement $u_8(t)$. a) Initial condition $v_8(0) = 1$, b) Initial condition $u_8(0) = 1$.

The displacement time-history of the two different initial conditions are shown in Fig. 6.11a and Fig. 6.11b, respectively. The numerically integrated results are shown in full line, while

an accurate evaluation is shown as a dotted curve. It is seen that a moderate phase difference builds up during the response time shown.

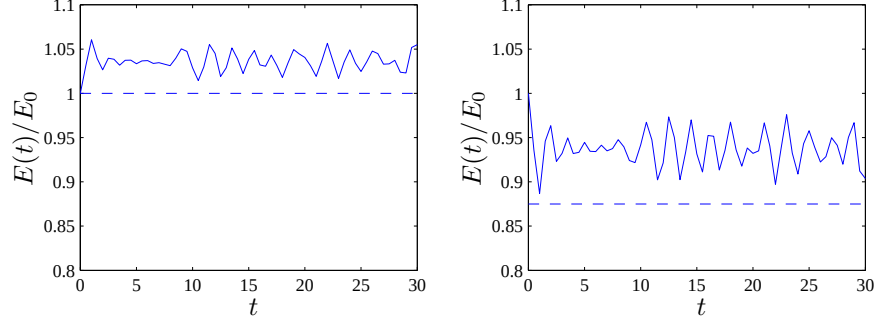


Fig. 6.12: Energy (—) and equivalent energy (---). a) Initial velocity $v_8(0) = 1$, b) Initial displacement $u_8(0) = 1$.

Finally, Figure 6.12 shows the energy development during the analyzed response time interval. The full curves show the calculated mechanical energy, whereas the algorithmic energy – defined by the algorithmic balance equation (6.84) – is shown by the dashed lines. As predicted by the algorithmic energy balance equation the algorithmic energy is conserved, whereas the actual mechanical energy exhibits rather substantial fluctuations. The general deviation between the computed mechanical energy and the energy resulting from the initial conditions can be identified from the phase diagram in Fig. 6.9a. The figure shows that for explicit Newmark integration initial velocity conditions lead to prediction of higher energy, whereas initial displacement conditions lead to prediction of lower mechanical energy. for velocity initial conditions the algorithmic energy is represented correctly, whereas initial displacement conditions lead to a lower algorithmic energy, due to the reduced algorithmic stiffness. $\square$

Example 6.2. Elastic pendulum – Newmark algorithm.

The family of Newmark algorithms are based on satisfying the equation of motion at a sequence of points in time, so-called collocation. This basis makes it easy to extend the algorithms to cases with nonlinear system properties. However, it turns out that the inconsistencies relating to the lack of proper energy balance typically increase for nonlinear systems. This example illustrates the problem, and points towards a different basis for time integration algorithms in which the momentum change over the integration increments take a fundamental role. The basic idea of these momentum based algorithms is described in the next chapter.

The present example illustrates the characteristics of the problem with reference to the simple ideal pendulum shown in Fig. 6.13. The pendulum of initial length l_0 and point mass m swings under the action of gravity with acceleration g in the x -direction. The equations of motion follow from the kinetic energy T and potential energy U ,

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \quad , \quad U = \frac{1}{2}l_0 EA \varepsilon^2 - mgx \quad (6.93)$$

in terms of the non-linear Green strain

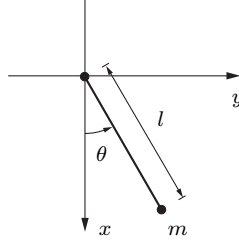


Fig. 6.13: Elastic pendulum with concentrated mass.

$$\varepsilon = \frac{l^2 - l_0^2}{2l_0^2} \quad (6.94)$$

The equations of motion follow from the time derivative of the total energy,

$$\frac{d}{dt}(T + U) = (m\ddot{x} + \frac{\partial U}{\partial x})\dot{x} + (m\ddot{y} + \frac{\partial U}{\partial y})\dot{y} = 0 \quad (6.95)$$

Identical satisfaction of this equation requires the factors to $\dot{x}$ and $\dot{y}$ vanish identically. This defines the components of equations of motion in terms of the mass matrix, the internal and the external force vectors

$$\mathbf{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}, \quad \mathbf{g} = \frac{N}{l_0} \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} mg \\ 0 \end{bmatrix} \quad (6.96)$$

The tangent stiffness matrix $\mathbf{K} = \partial \mathbf{g} / \partial \mathbf{u}$ consists of geometric and constitutive component

$$\mathbf{K} = \frac{N}{l_0} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{EA}{l_0^3} \begin{bmatrix} x^2 & xy \\ yx & y^2 \end{bmatrix} \quad (6.97)$$

These properties define the non-linear equations of motion.

In this example the pendulum is started from rest at a stretched horizontal position defined by $(x_0, y_0) = (0, 1.1 l_0)$. The response is shown in Fig. 7.6 with the left part showing the response in cartesian components (x, y) and the right part the polar components consisting of the length l and the angle φ .

It is observed in Fig. 6.15 that there are fluctuations in the total energy, when integrating the response by the Newmark method. $\square$

6.4 Exercises

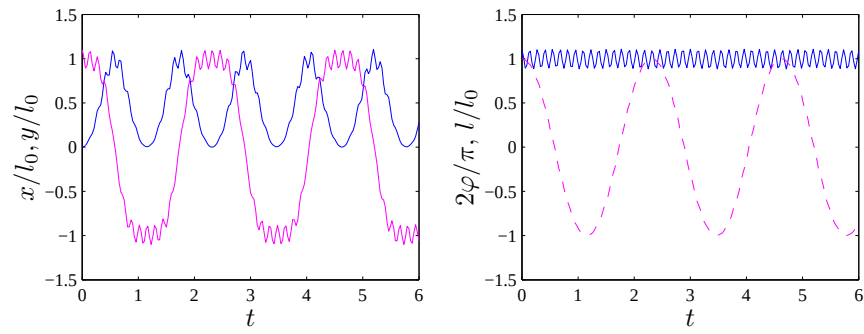


Fig. 6.14: a) Coordinates $x(t)$ and $y(t)$, b) Length l/l_0 and angle $2\varphi/\pi$.

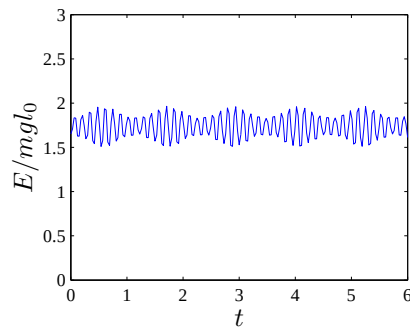


Fig. 6.15: Energy development in Newmark integration with $\gamma = \frac{1}{2}$, $\beta = \frac{1}{4}$.

Summary

- Numerical integration in structural dynamics is dominated by ‘single step - single solve’ algorithms, where the response is updated one step at a time by solving only one system of equations of the size of the similar static equations. The classical form of these algorithms is the Newmark algorithm.
- The Newmark algorithm is formed by setting up an explicit weighted interpolation for the velocities and displacements in terms of the forward accelerations. The forward accelerations are then found by solving the equations of motion at the forward time t_{n+1} .
- The Newmark algorithm contains two parameters γ and β . Stability requires $\gamma \geq \frac{1}{2}$ and β ‘sufficiently large’. β may be chosen large enough to render the algorithm unconditionally stable, while smaller values of β leads to conditionally stable algorithms, where there is an upper limit on the time step, $h \leq h_{max}$.
- Algorithmic damping of the high-frequency components may be introduced by selecting $\alpha = \gamma - \frac{1}{2} > 0$. This leads to undesirable damping of the low-frequency components also. Improved, so-called α -methods have been developed, that lead to improved low-frequency behavior by satisfying the equations of motion at an intermediate time. The generalized α -method is as easy to implement as the original Newmark algorithm and is preferable, when algorithmic damping is desired.
- The time integration algorithms discussed here can be applied to non-linear problems by solving the equations of motion by iteration. However, the simple stability criteria for linear systems are no longer valid for non-linear systems.

Traditionally time integration in structural dynamics has been based on collocation methods, i.e. methods in which the equations of motion are satisfied at a sequence of points in time. In many cases this leads to quite satisfactory results, but typically the stability and accuracy criteria are expressed somewhat indirectly and turn out to be difficult to generalize to non-linear systems. For the equations of motion of mechanical systems there exists an alternative approach to setting up the relevant integration equations. The alternative consists in considering the development of the momentum of the system over a time integration increment, instead of just matching the equations at specific points in time. This approach is quite close in spirit to the original formulation of the equations of motion, which is based on the statement that the increment of the momentum equals the time integral of the involved forces. The crucial point in establishing these so-called momentum-based time integration procedures is the evaluation of the increment of the momentum and the representation of the appropriate force time integrals. In addition to the momentum basis of these methods they also typically exhibit a simple and direct energy balance equation that can be used to guarantee stability and to design appropriate damping schemes. The present chapter gives a brief introduction to momentum-based time integration and the associated incremental energy balance equation.

7.1 Momentum Conserving Algorithm

The procedure of setting up a momentum-based time integration algorithm typically involves setting up the equation of motion in state space format, integration to finite increment form, establishment of the incremental energy balance equation, and the consideration of suitable damping mechanisms and their properties. The procedure is first presented for a linear system.

7.1.1 Discrete state-space format

For a linear system with displacement vector $\mathbf{u}(t)$ the equation of motion is

$$\mathbf{M} \ddot{\mathbf{u}} + \mathbf{C} \dot{\mathbf{u}} + \mathbf{K} \mathbf{u} = \mathbf{f}(t) \quad (7.1)$$

where the system properties are described by the stiffness matrix $\mathbf{K}$, the mass matrix $\mathbf{M}$, and the damping matrix $\mathbf{C}$. The stiffness and mass matrices are symmetric and positive definite, while the damping matrix is symmetric and non-negative definite for non-gyroscopic systems.

The initial conditions are defined in terms of displacement vector $\mathbf{u}$ and the velocity vector $\dot{\mathbf{u}}$. These can be combined into the system state-space vector $[\mathbf{u}^T, \dot{\mathbf{u}}^T]$. In state-space time integration methods the velocity vector is represented by a separate vector $\mathbf{v} = \dot{\mathbf{u}}$. For continuous systems, this may seem like a minor detail, but when turning to finite increment formulations, in which time integration of the variables is not directly available, this is an important point. First the equation of motion and the definition of the velocity vector $\mathbf{v}$ are combined into the block-matrix equation format

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{u}} \\ \dot{\mathbf{v}} \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \quad (7.2)$$

These equations are now integrated over the time increment $[t_n, t_{n+1}]$ with length h ,

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} \int \mathbf{u} dt \\ \int \mathbf{v} dt \end{bmatrix} = \begin{bmatrix} \int \mathbf{f}(t) dt \\ \mathbf{0} \end{bmatrix} \quad (7.3)$$

In principle, these equations are exact, but the development of the state-space vector $[\mathbf{u}^T, \mathbf{v}^T]$ over the time increment is not known exactly, and the corresponding integrals must therefore be replaced by an approximation. A second order approximation is obtained if the time integrals are expressed by the mean value of the integrand at the interval end points. The mean value is denoted $\bar{\mathbf{u}} = \frac{1}{2}(\mathbf{u}_{n+1} + \mathbf{u}_n)$ and similar for $\mathbf{v}$ and $\mathbf{f}$. Hereby the discretized state-space equations take the form

$$\boxed{\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + h \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{u}} \\ \bar{\mathbf{v}} \end{bmatrix} = h \begin{bmatrix} \bar{\mathbf{f}} \\ \mathbf{0} \end{bmatrix}} \quad (7.4)$$

This is a simple difference format in the state-space variables $\mathbf{u}$ and $\mathbf{v}$. Before turning to the algorithmic format it is advantageous to establish the energy balance equation and issues relating to damping.

7.1.2 Energy balance equation

The incremental energy balance equation is obtained by pre-multiplication with $[\Delta \mathbf{u}^T, -\Delta \mathbf{v}^T]$,

$$[\Delta \mathbf{u}^T, -\Delta \mathbf{v}^T] \left(\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + h \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{u}} \\ \bar{\mathbf{v}} \end{bmatrix} \right) = h \Delta \mathbf{u}^T \bar{\mathbf{f}} \quad (7.5)$$

The terms involving the mean values $\bar{\mathbf{u}}$ and $\bar{\mathbf{v}}$ are reformulated by using the identity

$$\left[\frac{1}{2} \mathbf{v}^T \mathbf{M} \mathbf{v} \right]_n^{n+1} = \Delta \mathbf{v}^T \mathbf{M} \bar{\mathbf{v}} \quad (7.6)$$

and its equivalent in terms of $\mathbf{u}$. Hereby the energy balance equation takes the form

$$\left[\frac{1}{2} \mathbf{v}^T \mathbf{M} \mathbf{v} + \frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u} \right]_n^{n+1} = \Delta \mathbf{u}^T \bar{\mathbf{f}} - \frac{1}{h} \Delta \mathbf{u}^T \mathbf{C} \Delta \mathbf{u} \quad (7.7)$$

The left side is the increment of the mechanical energy, while the right side contains the approximate form of the external work and the dissipation via the damping matrix $\mathbf{C}$.

The free response amplification λ for viscous damping is illustrated for a single mode with natural frequency ω in Fig. 7.1 damping ratio $\zeta = 0.1, 0.2, 0.4$, respectively. It is seen that for an integration time interval h with $\omega h < 2$ the amplification factor decreases with increasing frequency as expected from the theory for the original continuous formulation. However, beyond that limit the amplification factor increases again. Thus, high-frequency components with $\omega h < 2$ may not be much damped, leading to the possibility of numerically generated high-frequency noise. This behavior associated with the viscous damping matrix $\mathbf{C}$ suggests a closer look at the possibility of introducing high-frequency damping directly into the algorithm.

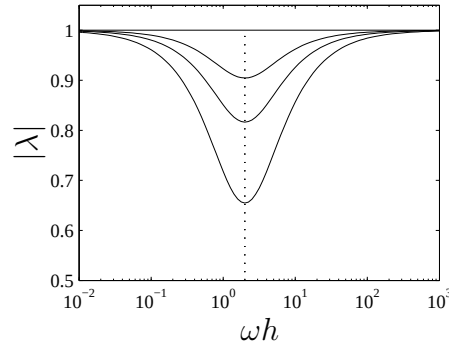


Fig. 7.1: Amplification factor λ for viscous damping with $\zeta = 0.1, 0.2, 0.4$.

7.1.3 Damping in state space algorithms

The state space format (7.4) suggests that algorithmic damping can be introduced via diagonal terms. Typically only the stiffness and mass matrices are known via the initial model, and dimensional considerations then suggest the following form of algorithmic damping

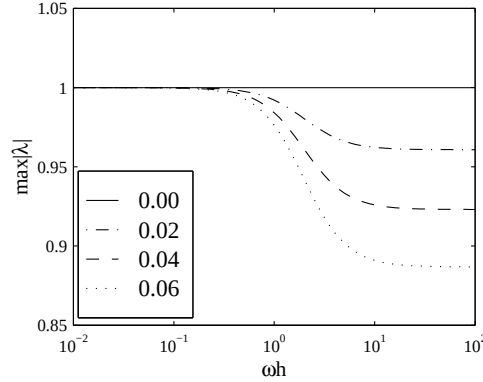
$$\begin{bmatrix} \frac{1}{2}\alpha h \mathbf{K} & \mathbf{M} \\ \mathbf{M} & -\frac{1}{2}\alpha h \mathbf{M} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} h \bar{\mathbf{u}} \\ h \bar{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} h \bar{\mathbf{f}} \\ \mathbf{0} \end{bmatrix} \quad (7.8)$$

The parameter α controls the magnitude of the damping, and the common factor on the two damping terms is justified by the form of the resulting energy dissipation term. Note, that the introduction of the extra term in the second equation leads to a reinterpretation of the relation between $\mathbf{v}$ and $\mathbf{u}$, giving extra flexibility in the formulation of the algorithm.

The energy balance equation is obtained from the incremental equations of motion by pre-multiplication with $[\Delta \mathbf{u}^T, -\Delta \mathbf{v}^T]$,

$$\left[\frac{1}{2} \mathbf{v}^T \mathbf{M} \mathbf{v} + \frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u} \right]_n^{n+1} = \Delta \mathbf{u}^T \bar{\mathbf{f}} - \alpha \left[\frac{1}{2} \Delta \mathbf{v}^T \mathbf{M} \Delta \mathbf{v} + \frac{1}{2} \Delta \mathbf{u}^T \mathbf{K} \Delta \mathbf{u} \right] \quad (7.9)$$

It is seen that the stiffness matrix contribution to the rate of dissipation gives a quadratic form in the displacement increment $\Delta \mathbf{u}$, whereas the mass term gives a quadratic form in the velocity increment $\Delta \mathbf{v}$. When considering free modal response the velocity contains the frequency factor ω , compared to the corresponding displacement, and thus the rate of dissipation in (7.9) increases with ω in the same way as the kinetic energy, leading to a finite limit of the amplification factor as illustrated in the figure.



High-frequency limit

$$\lambda_{\infty} = \frac{1 - \alpha}{1 + \alpha} \quad (7.10)$$

Low-frequency damping ratio

$$\zeta \simeq \frac{1}{2} \alpha \omega h \quad (7.11)$$

ALGORITHM 7.1. Linear state-space algorithm.

(1) System matrices $\mathbf{K}$, $\mathbf{C}$, $\mathbf{M}$,

$$\mathbf{K}_* = \kappa \left[\mathbf{K} + \frac{2}{\kappa h} \mathbf{C} + \left(\frac{2}{\kappa h} \right)^2 \mathbf{M} \right]$$

(2) Initial conditions $\mathbf{u}_0$, $\mathbf{v}_0$

(3) Displacement increment:

$$\Delta \mathbf{u} = \mathbf{K}_*^{-1} \left[\mathbf{f}_{n+1} + \mathbf{f}_n - 2\mathbf{K}\mathbf{u}_n + \frac{4}{\kappa h} \mathbf{M}\mathbf{v}_n \right]$$

(4) State vector update:

$$\begin{aligned} \mathbf{u}_{n+1} &= \mathbf{u}_n + \Delta \mathbf{u} \\ \mathbf{v}_{n+1} &= \mathbf{v}_n + \left[\frac{2}{\kappa h} \Delta \mathbf{u} - \frac{2}{\kappa} \mathbf{v}_n \right] \end{aligned}$$

(5) Return to (3) for new time step, or stop

7.1.4 Linear algorithm with energy dissipation

The algorithm with consistent energy dissipation can now be expressed as Algorithm 7.1 for linear systems including structural damping $\mathbf{C}$. The state-space form follows from (7.8) as

$$\begin{bmatrix} \mathbf{C} + \frac{1}{2}\alpha h \mathbf{K} & \mathbf{M} \\ \mathbf{M} & -\frac{1}{2}\alpha h \mathbf{M} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{bmatrix} h\bar{\mathbf{u}} \\ h\bar{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} h\bar{\mathbf{f}} \\ \mathbf{0} \end{bmatrix} \quad (7.12)$$

It is most efficient to solve these equations by eliminating the velocity increment $\Delta \mathbf{v}$ from the first equation. The formulation takes on a particularly compact form when introducing the dissipation parameter

$$\kappa = 1 + \alpha. \quad (7.13)$$

The second state-space equation then gives the forward velocity $\mathbf{v}_{n+1}$ via the simple vector relation

$$\kappa \Delta \mathbf{v} = \frac{2}{h} \Delta \mathbf{u} - 2\mathbf{v}_n. \quad (7.14)$$

It is observed that the effect of the parameter α is to reduce the velocity increment $\Delta \mathbf{v}$ in the algorithm. In the Newmark algorithm this effect is accomplished by a correction involving the acceleration, while here the effect is obtained by a simple weighting.

The algorithm is most conveniently formulated in terms of the increments $\Delta \mathbf{u}$ and $\Delta \mathbf{v}$. Substitution of $\Delta \mathbf{v}$ from (7.14) into the first of the state-space equations (7.12) and multiplication by $2/h$ gives the following equation for the displacement increment $\Delta \mathbf{u}$:

$$\kappa \left[\mathbf{K} + \frac{2}{\kappa h} \mathbf{C} + \frac{4}{(\kappa h)^2} \mathbf{M} \right] \Delta \mathbf{u} = (\mathbf{f}_{n+1} + \mathbf{f}_n) - 2\mathbf{K}\mathbf{u}_n + \frac{4}{\kappa h} \mathbf{M} \mathbf{v}_n. \quad (7.15)$$

In this equation the displacement is determined by the effective stiffness matrix

$$\mathbf{K}_* = \kappa \left[\mathbf{K} + \frac{2}{\kappa h} \mathbf{C} + \frac{4}{(\kappa h)^2} \mathbf{M} \right]. \quad (7.16)$$

It is seen that the terms are scaled as if extending the time increment from h to κh , whereby the dynamic terms get less weight.

7.2 Nonlinear Dynamics

A very important aspect of momentum-based algorithms is the relative ease with which they generalize to systems with non-linear stiffness. Let the internal force vector $\mathbf{g}(\mathbf{u})$ be given as the gradient of an elastic potential $G(\mathbf{u})$,

$$\mathbf{g}(\mathbf{u}) = \nabla_{\mathbf{u}} G(\mathbf{u}) \quad (7.17)$$

The equation of motion then takes the form

$$\boxed{\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \nabla_{\mathbf{u}} G(\mathbf{u}) = \mathbf{f}(t)} \quad (7.18)$$

Pre-multiplication with $\dot{\mathbf{u}}$ then gives the energy balance equation

$$\frac{d}{dt} \left(\frac{1}{2} \dot{\mathbf{u}}^T \mathbf{M} \dot{\mathbf{u}} + G(\mathbf{u}) \right) = \dot{\mathbf{u}}^T \mathbf{f} - \dot{\mathbf{u}}^T \mathbf{C} \dot{\mathbf{u}} \quad (7.19)$$

It is seen that the energy balance equation correctly replaces the quadratic form in $\mathbf{u}$ in the linear case (7.7) with the more general formulation in terms of the elastic potential $G(\mathbf{u})$. When using models for linear elastic structures in terms of the non-linear Green strain, this formulation leads to an energy in the form of a fourth degree polynomial in the displacement components, and the algorithm developed in the following will increment the momentum correctly and conserve energy, irrespective of the magnitude of the displacements. A simple case is illustrated in Fig. 7.2 in which a frame translates to the right, while sinking under the action of gravity. The frame has an initial deformation, and the analysis demonstrates conservation of momentum and total energy.

7.2.1 Discrete state-space format

As in the linear case a separate velocity vector $\mathbf{v} = \dot{\mathbf{u}}$ is introduced as an independent variable. Hereby the state-space format of the equations of motion takes the form

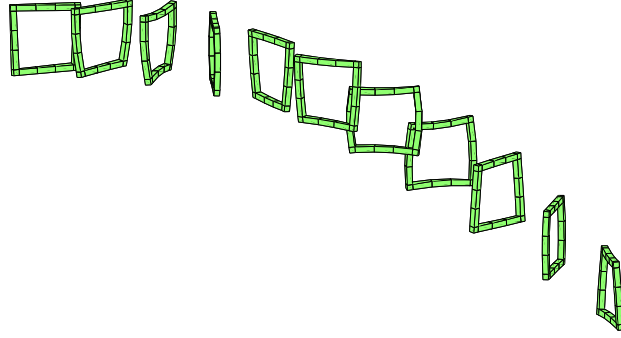


Fig. 7.2: Motion of quadratic frame with solid isoparametric elements.

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{u}} \\ \dot{\mathbf{v}} \end{bmatrix} + \begin{bmatrix} \nabla_{\mathbf{u}} G(\mathbf{u}) \\ -\mathbf{M} \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{f}(t) \\ \mathbf{0} \end{bmatrix} \quad (7.20)$$

The next step in the derivation of the algorithm is integration over the time increment $[t_n, t_{n+1}]$,

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + \begin{bmatrix} \int \nabla_{\mathbf{u}} G(\mathbf{u}) dt \\ -\mathbf{M} \int \mathbf{v} dt \end{bmatrix} = \begin{bmatrix} \int \mathbf{f}(t) dt \\ \mathbf{0} \end{bmatrix} \quad (7.21)$$

With a view to obtaining a second order accurate algorithm, the integrals of the velocity and the external load are expressed in terms of the mean value of the integrand.

If the time integral of the non-linear internal force were expressed by a simple mean value, the procedure would essentially reduce to a simple collocation scheme, and the possibility of energy conservation would be lost. The specific form of the representation of the integral of the internal force is therefore left open, to be determined later. Introduce a ‘representative mean’ $\mathbf{g}_*$ of non-linear internal force,

$$\int_n^{n+1} \nabla_{\mathbf{u}} G(\mathbf{u}) dt \simeq h \nabla_{\mathbf{u}} G_* = h \mathbf{g}_* \quad (7.22)$$

In terms of this ‘representative mean’ of the internal forces the following set of discrete state-space equations is obtained,

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + \begin{bmatrix} h \mathbf{g}_*(\mathbf{u}) \\ -h \mathbf{M} \bar{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} h \bar{\mathbf{f}} \\ \mathbf{0} \end{bmatrix} \quad (7.23)$$

The specific form of the representative internal force $\mathbf{g}_*$ is determined via a requirement of energy conservation as demonstrated in the following.

Energy conservation condition

The discrete energy balance equation is established by pre-multiplication of the equation of motion (7.23) with $[\Delta \mathbf{u}^T, -\Delta \mathbf{v}^T]$,

$$[\Delta \mathbf{u}^T, -\Delta \mathbf{v}^T] \left(\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta \mathbf{v} \end{bmatrix} + \begin{bmatrix} h \nabla_{\mathbf{u}} G_* \\ -h \mathbf{M} \bar{\mathbf{v}} \end{bmatrix} \right) = h \Delta \mathbf{u}^T \bar{\mathbf{f}} \quad (7.24)$$

This gives the incremental energy balance equation

$$\left[\frac{1}{2} \mathbf{v}^T \mathbf{M} \mathbf{v} \right]_n^{n+1} + \Delta \mathbf{u}^T \nabla_{\mathbf{u}} G_* = \Delta \mathbf{u}^T \bar{\mathbf{f}} - \frac{1}{h} \Delta \mathbf{u}^T \mathbf{C} \Delta \mathbf{u} \quad (7.25)$$

This relation will represent the correct energy balance, provided that the internal force is represented such that

$$\boxed{[G(\mathbf{u})]_n^{n+1} = \Delta \mathbf{u}^T \nabla_{\mathbf{u}} G_* = \Delta \mathbf{u}^T \mathbf{g}_*} \quad (7.26)$$

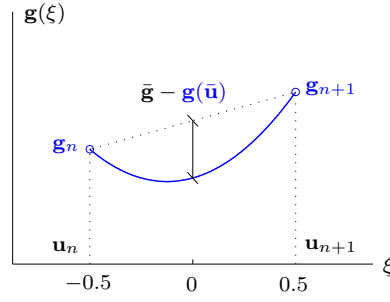
This condition is equivalent to defining the internal force $\mathbf{g}_*$ by the finite derivative $\nabla_{\mathbf{u}} G_*$. However, to make this definition useful it must be expressed in an explicit form.

Internal Force Representations

The internal force integral appearing in the discretized equations (7.23) defines the increment of the elastic potential,

$$\Delta G = \int_n^{n+1} \mathbf{g}(\mathbf{u})^T d\mathbf{u} = \Delta \mathbf{u}^T \underbrace{\int_{-1/2}^{1/2} \mathbf{g}(\bar{\mathbf{u}} + \xi \Delta \mathbf{u}) d\xi}_{\text{Internal force } \mathbf{g}_*} \quad (7.27)$$

The integral can be taken along any path, and the last form is in terms of an integral along the displacement increment $\Delta \mathbf{u}$. It is a simple matter to represent the integral by any approximate quadrature rule, e.g. the Simpson rule. The background of the Simpson rule is illustrated in Fig. 7.3. The figure shows a parabolic approximation to the integrand matching at the end points and the center. In addition to this there may be a cubic function that vanishes at these points, but due to anti-symmetry about the interval center this contribution does not contribute to the value of the integral.

Fig. 7.3: Integration of parabolic representation in terms of ξ .

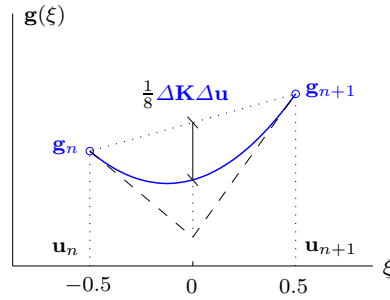
The integral of the parabolic representation is evaluated directly as the sum of the area of the parallelopiped defined by the end-points and that of the parabola with center height $\bar{\mathbf{g}} - \mathbf{g}(\bar{\mathbf{u}})$. The result is the so-called ‘Simpson’s rule’,

$$\mathbf{g}_* = \frac{1}{6}[\mathbf{g}_n + \mathbf{g}_{n+1}] + \frac{2}{3}\mathbf{g}(\bar{\mathbf{u}}) + O(h^4) \quad (7.28)$$

It is noted that this evaluation is exact for any quartic internal energy potential $G(\mathbf{u})$, e.g. that associated with linear elastic isoparametric elements in terms of the quadratic Green strain. However, the formula requires the additional evaluation of the internal force $\mathbf{g}(\mathbf{u})$ at the increment mid-point.

Stiffness matrix increment

An alternative formula can be obtained in terms of the increment of the stiffness matrix, avoiding the extra evaluation of the internal force. The derivation is illustrated in Fig. 7.4 showing the internal force $\mathbf{g}(\xi\Delta\bar{\mathbf{u}})$ as a function of the normalized path-length variable ξ . The inclination of this curve is given by the ξ -derivative of the internal force,

Fig. 7.4: Integration of parabolic representation in terms of ξ .

$$\mathbf{g}'(\mathbf{u}) = \frac{d\mathbf{g}}{d\xi} = \frac{d\mathbf{g}}{d\mathbf{u}} \Delta\mathbf{u} = \mathbf{K}(\mathbf{u}) \Delta\mathbf{u} \quad (7.29)$$

The difference between the slope at the two ends determine the height of the parabola shown in Fig. 7.4. The result is easily established directly by considering an affine mapping of the figure, in which the parabola becomes symmetric, whereby the extensions of the end tangents intersect the vertical center line at $\frac{1}{4}\Delta\mathbf{K}$. The height of the parabola is half of this, i.e. $\frac{1}{8}\Delta\mathbf{K}$, as shown in the figure. Using this height of the parabola the integral determines the effective internal force

$$\mathbf{g}_* = \frac{1}{2}[\mathbf{g}_{n+1} + \mathbf{g}_n] - \frac{1}{12}\Delta\mathbf{K} \Delta\mathbf{u} + O(h^4) \quad (7.30)$$

The advantage of this result relative to the Simpson formula (7.28) is that the increment of the stiffness matrix is already available as part of the iterative analysis for the time increment, and thus no additional evaluations are necessary. Like the previous result, the quadrature formula (7.30) is exact for systems with a quintic elastic potential $G(\mathbf{u})$, a class of problems involving many of theoretical and practical interest.

Conservation format for quartic energy

A quite simple algorithm with exact energy conservation for elastic potential up to degree four is obtained by use of the stiffness matrix increment formula (7.30). The ‘representative internal force’ $\mathbf{q}_*$ is represented by the explicit part of (7.30),

$$\mathbf{g}_q = \frac{1}{2}[\mathbf{g}_{n+1} + \mathbf{g}_n] - \frac{1}{12}\Delta\mathbf{K} \Delta\mathbf{u} \quad (7.31)$$

The presence of the factor $\Delta\mathbf{u}$ in the last term suggests placing this term in the top left corner of the first matrix in the state space equations (7.23) that then take the form

$$\begin{bmatrix} \mathbf{C} - \frac{1}{12}h\Delta\mathbf{K} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta\mathbf{u} \\ \Delta\mathbf{v} \end{bmatrix} + \frac{h}{2} \begin{bmatrix} \mathbf{g}(\mathbf{u}_{n+1}) + \mathbf{g}(\mathbf{u}_n) \\ -\mathbf{M}\mathbf{v}_{n+1} - \mathbf{M}\mathbf{v}_n \end{bmatrix} = \begin{bmatrix} h\bar{\mathbf{f}} \\ \mathbf{0} \end{bmatrix} \quad (7.32)$$

This is a second order momentum-based time integration algorithm with energy conservation for elastic potential of up to fourth degree. Energy conservation is a very important property; not only because it reflects a property of the original system, but also because it guarantees unconditional stability for systems with a positive definite elastic potential.

In the implementation of the algorithm, shown in Algorithm 7.2, the velocity $\mathbf{v}_{n+1}$ is condensed out of the non-linear equations. These are then solved for

ALGORITHM 7.2. Nonlinear energy conserving algorithm.

-
- 1) Initial conditions:
 $\mathbf{u}_0, \mathbf{v}_0$
 - 2) Increment time:
 $t_{n+1} = t_n + h$
 $\Delta \mathbf{u} = h \mathbf{v}_n$
 - 3) Residual calculation:
 $\mathbf{u}_{n+1} = \Delta \mathbf{u} + \mathbf{u}_n$
 $\mathbf{r} = \mathbf{f}_{n+1} + \mathbf{f}_n - (\mathbf{g}_{n+1} + \mathbf{g}_n) + (4/h)\mathbf{M} \mathbf{v}_n$
 $- [(2/h)^2 \mathbf{M} + (2/h)\mathbf{C} - \frac{1}{6}\Delta \mathbf{K}] \Delta \mathbf{u}$
 - 4) Displacement subincrement:
 $\mathbf{K}_* = [\mathbf{K}_c + \frac{1}{6}\Delta \mathbf{K}_* + (2/h)^2 \mathbf{M} + (2/h)\mathbf{C}]$
 $\delta \mathbf{u} = \mathbf{K}_*^{-1} \mathbf{r}$
 $\Delta \mathbf{u} = \Delta \mathbf{u} + \delta \mathbf{u}$
 If $\|\mathbf{r}\| > \varepsilon_{\mathbf{r}}$ or $\|\delta \mathbf{u}\| > \varepsilon_{\mathbf{u}}$ repeat from 3).
 - 5) State vector update:
 $\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta \mathbf{u}$
 $\mathbf{v}_{n+1} = (2/h)\Delta \mathbf{u} - \mathbf{v}_n$
 - 6) Return to 2) for new time step, or stop.
-

the displacement $\mathbf{u}_{n+1}$, followed by a linear update of the velocity $\mathbf{v}_{n+1}$. Energy conservation is obtained by the stiffness increment term in the residual.

It is of interest to give the structure of the algorithm a closer look. If the increment of the stiffness matrix $\Delta \mathbf{K}$ were omitted from the algorithm it would correspond precisely to the central difference form of a second order collocation-based algorithm. In other terms, the collocation formulation is obtained from the energy conserving algorithm ‘by adding’ an extra damping term $\frac{1}{12}h\Delta \mathbf{K}$. Typically this term changes sign repeatedly during dynamic response, and thus intermittently adds or subtracts energy from the system. It is this mechanism that prevents energy conservation and possible instability.

General secant format

The algorithm shown as Algorithm 7.2 is based on the state space equations (7.32) with a representation of the effective internal force that is exact for a quartic potential. The energy conservation property is important in its own right and for guaranteeing unconditional stability, and it is therefore

of interest to consider extending this property beyond systems with quartic potential.

Energy conservation hinges on the finite gradient property (7.26), conveniently stated as

$$\Delta \mathbf{u}^T \mathbf{g}_* = \Delta G \quad (7.33)$$

Energy conservation can be obtained by adding a higher order term to the quartic approximation

$$\mathbf{g}_q = \frac{1}{2} [\mathbf{g}_{n+1} + \mathbf{g}_n] - \frac{1}{12} \Delta \mathbf{K} \Delta \mathbf{u} \quad (7.34)$$

A general and convenient for is the secant type representation

$$\mathbf{g}_* = \mathbf{g}_q + \underbrace{\frac{\Delta G - \Delta \mathbf{u}^T \mathbf{g}_q}{\Delta \mathbf{u}^T \mathbf{A} \Delta \mathbf{u}} \mathbf{A} \Delta \mathbf{u}}_{O(h^4)} \quad (7.35)$$

in which $\mathbf{A}$ is a positive definite matrix. It is easily established that the additional term is of higher order, and that the effective internal force $\mathbf{g}_*$ satisfies the energy increment relation (7.33). The formula (7.35) is generic and does not determine the specific form of the matrix $\mathbf{A}$. Typical choices are

$$\mathbf{A} \Delta \mathbf{u} = \begin{cases} \bar{\mathbf{K}} \Delta \mathbf{u} \\ \Delta \mathbf{g} \\ \Delta \mathbf{u} \end{cases} \quad (7.36)$$

The implementation of the additional term in the residual is rather straightforward as it enters similar to $\Delta \mathbf{K} \Delta \mathbf{u}$. Typically the term is of modest magnitude, and the iterative solution can be based on the previous iteration matrix.

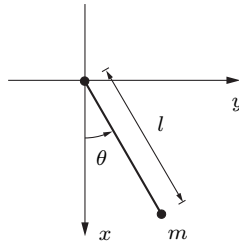
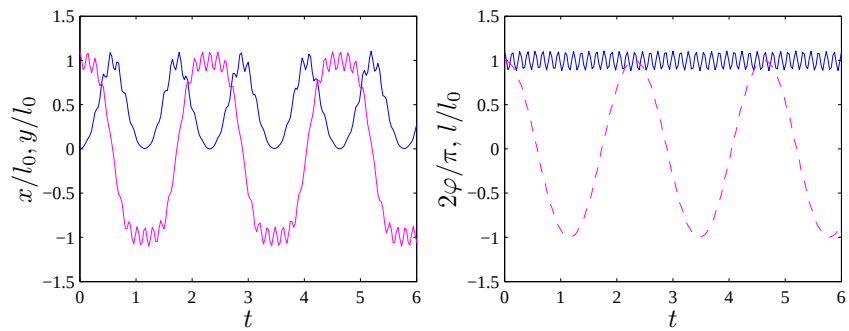
Example 7.1. Elastic pendulum.

Fig. 7.5: Elastic pendulum with concentrated mass.

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \quad , \quad U = \frac{1}{2}l_0 EA \varepsilon^2 - mgx \quad (7.37)$$

Fig. 7.6: a) Coordinates $x(t)$ and $y(t)$, b) Length l/l_0 and angle $2\varphi/\pi$.

NOTE: Energy conserved to within computational accuracy.

□

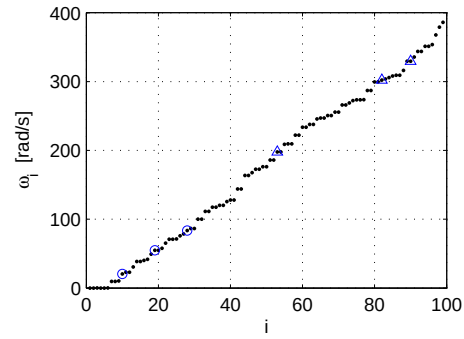
Example 7.2. Free motion of space truss.

Fig. 7.7: Natural angular frequencies.

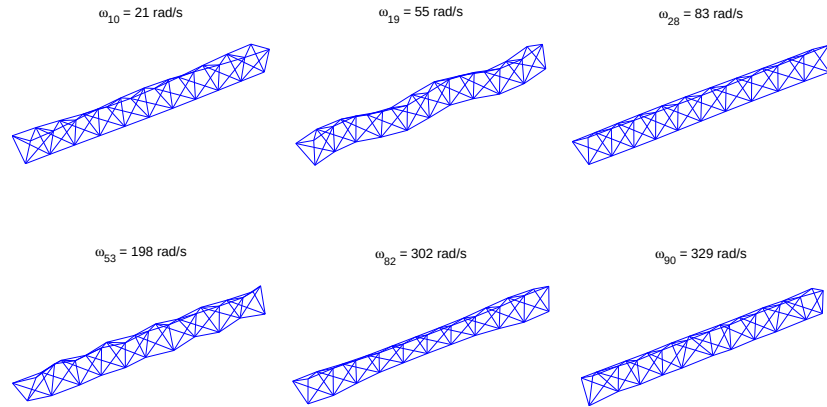


Fig. 7.8: Selected vibration modes of truss.

□

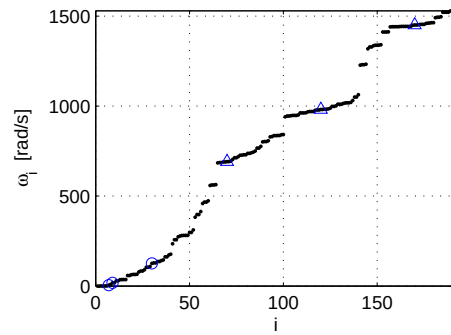
Example 7.3. Free motion of solid frame.

Fig. 7.9: Natural angular frequencies.

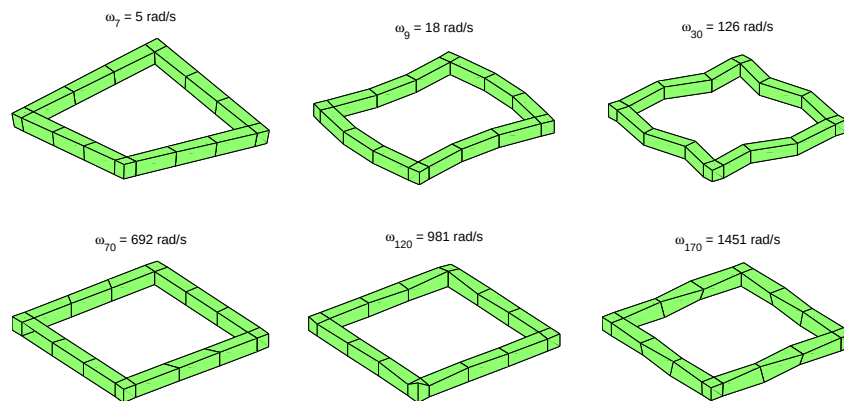


Fig. 7.10: Angular frequencies and selected vibration modes.

□

7.2.2 Non-linear algorithm with energy dissipation

Algorithmic damping can be introduced into the momentum based algorithm presented in (7.32) and Algorithm 7.2 by extending the principles used for the linear algorithm in section 7.1.3. The equivalent additional terms in the non-linear case are

- $\frac{1}{2}\alpha h \mathbf{K} \delta \mathbf{u} \simeq \frac{1}{2}\alpha h \Delta \mathbf{g}$ in the first state space equation.
- $-\frac{1}{2}\alpha h \mathbf{M} \Delta \mathbf{v}$ in the second state-space equation.

After division by $\frac{1}{2}h$ the first equation takes the form

$$\begin{aligned} \frac{2}{h} \left[\mathbf{M} \Delta \mathbf{v} + \left(\mathbf{C} - \frac{1}{12} h \Delta \mathbf{K} \right) \Delta \mathbf{u} \right] + \\ (1+\alpha) \mathbf{g}_{n+1} + (1-\alpha) \mathbf{g}_n = \mathbf{f}_{n+1} + \mathbf{f}_n \end{aligned} \quad (7.38)$$

Elimination of the forward velocity from the second equation gives

$$\begin{aligned} \left[\frac{1}{1+\alpha} \frac{4}{h^2} \mathbf{M} + \frac{2}{h} \mathbf{C} - \frac{1}{6} \Delta \mathbf{K} \right] \Delta \mathbf{u} + (1+\alpha) \mathbf{g}(\mathbf{u}_{n+1}) \\ = \mathbf{f}_{n+1} + \left[\mathbf{f}_n - (1-\alpha) \mathbf{g}(\mathbf{u}_n) + \frac{1}{1+\alpha} \frac{4}{h} \mathbf{M} \mathbf{v}_n \right] \end{aligned} \quad (7.39)$$

For $\alpha = 0$ this equation reproduces the energy conserving equation of motion.

The residual of the non-linear equation of motion is chosen as the difference between the right and left hand sides,

$$\begin{aligned} \mathbf{r} = \mathbf{f}_{n+1} + \mathbf{f}_n - \left[(1+\alpha) \mathbf{g}(\mathbf{u}_{n+1}) + (1-\alpha) \mathbf{g}(\mathbf{u}_n) \right] \\ - \left[\frac{1}{1+\alpha} \frac{4}{h^2} \mathbf{M} + \frac{2}{h} \mathbf{C} - \frac{1}{6} \Delta \mathbf{K} \right] \Delta \mathbf{u} + \frac{1}{1+\alpha} \frac{4}{h} \mathbf{M} \mathbf{v}_n \end{aligned} \quad (7.40)$$

The solution is found by iterations using the linearized increment $\delta \mathbf{r}$ of the residual. When written as $\delta \mathbf{r} = -\mathbf{K}_\alpha \delta \mathbf{u}$,

$$\mathbf{K}_\alpha \delta \mathbf{u} = \mathbf{r} \quad (7.41)$$

The effective stiffness matrix $\mathbf{K}_\alpha$ is non-symmetric, but the symmetric counterpart may be used.

ALGORITHM 7.3. Non-linear energy dissipating algorithm.

1) Initial conditions:

$$\mathbf{u}_0, \mathbf{v}_0$$

2) Prediction step:

$$\Delta \mathbf{u} = h \mathbf{v}_n$$

3) Residual calculation:

$$\begin{aligned} \mathbf{u}_{n+1} &= \Delta \mathbf{u} + \mathbf{u}_n \\ \mathbf{r} &= \mathbf{f}_{n+1} + \mathbf{f}_n - [(1+\alpha)\mathbf{g}_{n+1} + (1-\alpha)\mathbf{g}_n] \\ &\quad - [(1+\alpha)^{-1}(2/h)^2\mathbf{M} + (2/h)\mathbf{C} - \frac{1}{6}\Delta\mathbf{K}]\Delta\mathbf{u} \\ &\quad + (1+\alpha)^{-1}(4/h)\mathbf{M}\mathbf{v}_n \end{aligned}$$

4) Displacement sub-increment:

$$\begin{aligned} \mathbf{K}_\alpha &= [\mathbf{K}_\alpha^c + \frac{1}{6}\Delta\mathbf{K}_* + (1+\alpha)^{-1}(2/h)^2\mathbf{M} + (2/h)\mathbf{C}] \\ \delta\mathbf{u} &= \mathbf{K}_\alpha^{-1} \mathbf{r} \\ \Delta\mathbf{u} &= \Delta\mathbf{u} + \delta\mathbf{u} \end{aligned}$$

If $\|\mathbf{r}\| > \varepsilon_r$ or $\|\delta\mathbf{u}\| > \varepsilon_u$ repeat from 3).

5) State vector update:

$$\begin{aligned} \mathbf{u}_{n+1} &= \mathbf{u}_n + \Delta\mathbf{u} \\ \mathbf{v}_{n+1} &= [(2/h)\Delta\mathbf{u} - (1-\alpha)\mathbf{v}_n]/(1+\alpha) \end{aligned}$$

6) Return to 2) for new time step, or stop.

Example 7.4. Elastic pendulum with dissipation.

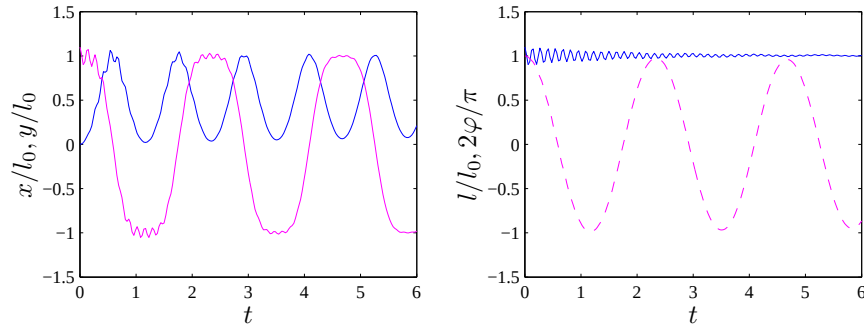


Fig. 7.11: Energy dissipating integration of pendulum vibrations with $\alpha = 0.02$ and $h = 0.03$. a) Coordinates $x(t)$ and $y(t)$, b) Length l/l_0 and angle $2\varphi/\pi$.

Energy development for different time steps:

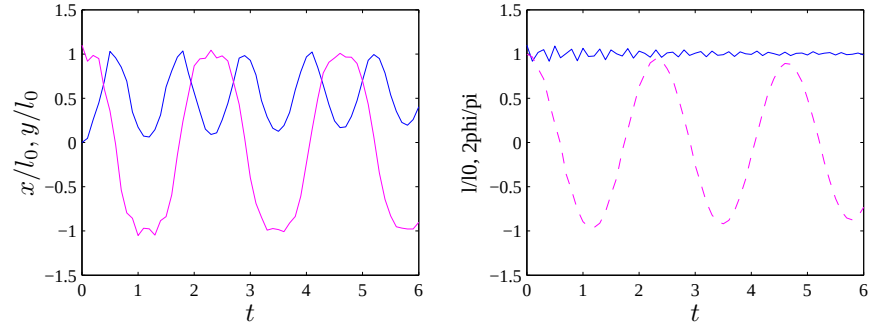


Fig. 7.12: Energy conserving integration of pendulum vibrations with $h = 0.1$. a) Coordinates $x(t)$ and $y(t)$, b) Length l/l_0 and angle $2\phi/\pi$.

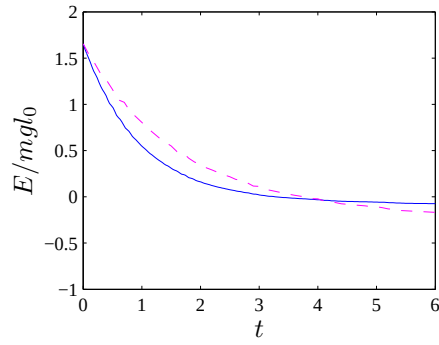


Fig. 7.13: Energy development for $\alpha = 0.02$. $h = 0.03$: —; $h = 0.1$: ---.

NOTE: The larger time step increases the cut-off frequency and therefore also damps the pendulum motion.

NOTE: Important to obtain sharper filter characteristics to permit the low-frequency domain to remain virtually undamped, see e.g. Krenk (2008). □

7.3 Exercises

Summary

- Momentum formulation obtained in terms of the state-space variables $\mathbf{u}$ and $\mathbf{v}$. Closely related to energy conservation.
- Time integration algorithm by integrating the momentum equations over the time increment $h = [t_{n+1} - t_n]$.
- For non-linear structures a momentum and energy conserving algorithm can be obtained by introduction of a more detailed representation of the internal force containing the increment of the stiffness matrix and, if necessary a higher order term defined by the elastic potential.
- Exact representation of the energy potential guarantees unconditional stability.
- Balanced algorithmic damping can be introduced into the energy conserving algorithm by combining $\mathbf{K}$ and $\mathbf{M}$ damping terms in the first and second state space equations, respectively.
- The format of the discretized damped state space equations of motion identifies that central difference collocation corresponds to omitting an increment of the stiffness matrix - possibly causing conditional instability.

References

- A. Cardona and M. Geradin. Time integration of the equations of motion in mechanism analysis. *Computers and Structures*, 33:801–820, 1989.
- P. D. Cha and W. C. Wong. A novel approach to determine the frequency equations of combined dynamical systems. *Journal of Sound and Vibration*, 219:689–706, 1999.
- J. Chung and G. M. Hulbert. A time integration algorithm for structural dynamics with improved numerical dissipation: The generalized α method. *Journal of Applied Mechanics*, 60:371–375, 1993.
- M. Géradin and D. Rixen. *Mechanical Vibrations*. Wiley, Chichester, U.K., 2nd edition, 1997.
- G. H. Golub and C. F. van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, MD., 2nd edition, 1989.
- M. Gürgöze. Proportionally damped systems subject to damping modifications by several viscous dampers. *Journal of Sound and Vibration*, 255:407–412, 2002.
- R. J. Guyan. Reduction of stiffness and mass matrices. *AIAA Journal*, 3:380, 1965.
- O. E. Hansteen and K. Bell. On the accuracy of mode superposition analysis in structural dynamics. *Earthquake Engineering and Structural Dynamics*, 7:405–411, 1979.
- H. M. Hilber, T. J. R. Hughes, and R. L. Taylor. Improved numerical dissipation for time integration algorithms in structural dynamics. *Earthquake Engineering and Structural Dynamics*, 5:282–292, 1977.
- T. J. R. Hughes. Analysis of transient algorithms with particular reference to stability behavior. In T. Belytschko and T. J. R. Hughes, editors, *Computational Methods for Transient Analysis*, pages 67–155, Amsterdam, 1983. Elsevier.
- T. J. R. Hughes. *The Finite Element Method*. Prentice Hall, Englewood Cliffs, N.J., 1987.

- T. J. R. Hughes, K. S. Pister, and R. L. Taylor. Implicit-explicit finite elements in nonlinear transient analysis. *Computer Methods in Applied Mechanics and Engineering*, 17/18:159–182, 1979.
- B. M. Irons. Structural eigenvalue problems: Elimination of unwanted variables. *AIAA Journal*, 3:961–962, 1965.
- H. M. Irvine. *Cable Structures*. MIT Press, Cambridge, Mass., 1981.
- H. M. Irvine and T. Caughey. The linear theory of free vibrations of a suspended cable. *Proceedings of the Royal Society, London, A* 341:299–315, 1974.
- S. Krenk. Vibrations of a taut cable with an external damper. *Journal of Applied Mechanics*, 67:772–776, 2000.
- S. Krenk. Energy conservation in Newmark based time integration algorithms. *Computer Methods in Applied Mechanics and Engineering*, 195:6110–6124, 2006.
- S. Krenk. Extended state-space time integration with high-frequency energy dissipation. *International Journal for Numerical Methods in Engineering*, 73:1767–1787, 2008.
- S. Krenk and S. R. K. Nielsen. Vibrations of a shallow cable with a viscous damper. *Proceedings of the Royal Society, London, Series A*, 458:339–357, 2002.
- S. Krenk, A. Brønden, and A. Kristensen. Dynamic actions and reactions of pedestrians. In *FOOTBRIDGE 2005, 2nd International Conference*, Venice, Italy, 2005. CD-Rom.
- J. A. Main and S. Krenk. Efficiency and tuning of viscous dampers on discrete systems. *Journal of Sound and Vibration*, 286:97–122, 2005.
- L. Meirovitch. *Elements of Vibration Analysis*. McGraw-Hill, New York, 1986.
- N. M. Newmark. A method of computation for structural dynamics. *Journal of the Engineering Mechanics Division, ASCE*, 85:67–94, 1959.
- B. N. Pacheco, Y. Fujino, and A. Sulekh. Estimation curve for modal damping in stay cables with viscous damper. *Journal of Structural Engineering*, 119:1961–1979, 1993.
- T. T. Soong and G. F. Dargush. *Passive Energy Dissipation Systems in Structural Engineering*. Wiley, Chichester, UK, 1997.
- W. L. Wood, M. Bossak, and O. C. Zienkiewicz. An alpha modification of Newmark's method. *International Journal for Numerical Methods in Engineering*, 64:1562–1566, 1981.

- H. Yamaguchi and Y. Fujino. Stayed cable dynamics and its vibration control.
In A. Larsen and S. Eisdahl, editors, *Bridge Aerodynamics*, pages 235–253,
Rotterdam, 1998. Balkma.

Index

- Angular frequency, 84
- Beam
 - bending vibration, 55
- Cable
 - vibrations, 45
- Damping matrix, 82
- dissipation parameter, 191
- effective stiffness matrix, 192
- Energy balance equation, 82, 83
- energy-dissipating time integration
 - displacement increment, 191
 - velocity increment, 191
- energy-dissipating time integration algorithm, 191
- equation of motion
 - state-space increment format, 191
- Equations of motion
 - multi degrees of freedom, 80
- Generalized eigenvalue problem, 84
- Mass matrix
 - discretized cable, 88
- Modal analysis, 79
- Modal coordinates, 93
- Modal mass, 87
- Modal matrix, 93
- Modal stiffness, 87
- Phase angle, 54
- Rayleigh's quotient, 94
- Separation of variables
 - beams, 56
 - cable, 45
- Stiffness matrix
 - discretized cable, 88
- Vibration
 - of beams, 55
 - of cable, 45
- Virtual work
 - beams, 72
 - cables, 64
- Wave equation, 43
- Wave-number, 46